IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

**IECEx Bulletin –
Section 1: Introduction**

IECEx Bulletin:2011(E)

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About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

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The IEC on-line Catalogue enables you to search by a variety of criteria (reference number, text, technical committee,…). It also gives information on projects, withdrawn and replaced publications.

- IEC Just Published: www.iec.ch/online_news/justpub
Stay up to date on all new IEC publications. Just Published details twice a month all new publications released. Available on-line and also by email.

- Electropedia: www.electropedia.org
The world's leading online dictionary of electronic and electrical terms containing more than 20 000 terms and definitions in English and French, with equivalent terms in additional languages. Also known as the International Electrotechnical Vocabulary online.

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IECEx Bulletin – Section 1: Introduction

INTERNATIONAL
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IECEx Bulletin 4.0 Edition 2011

The latest edition (4.0) of the IECEx Bulletin contains 4 sections. These are as follows;

Section 1: Introduction	***Section 2***: Matrix
Section 3: National Differences to IEC Standards	***Section 4***: IECEx Certification Bodies (ExCBs) and IECEx Testing Laboratories (ExTLs)

These 4 Sections have been divided into separate files in order to allow for quick and easy access.

Section 1: Introduction {*one single electronic file*}

Section 1: Introduction - Elements
IECEx Bulletin Contents Page
Preface
Introduction to the IECEx System o *IECEx System Objectives* o *What is an Ex area?* o *Where do you commonly find Ex Equipment?* o *IECEx International Certification System* o *1. The IECEx Certified Equipment Scheme* o *2. The IECEx Certified Services Facilities Scheme* o *3. The Ex Mark of Conformity Scheme* o *4. The IECEx Certified Persons Scheme* o *Applications*
Executive of the IECEx System - Current list of Contact details
IECEx Member Countries and their Representatives - Current list of Contact details

Section 2: Matrix {*one single electronic file*}

This section contains a comprehensive matrix of all member countries and their adoption method of IEC Standards.
The format of the matrix is in descending order of Standard editions with the latest edition first and highlighted with Grey shading for quick reference.
The matrix also uses a key (1-5) to identify member countries adoption of IEC Standards. The key is located at the bottom left corner of the page.
The matrix is provided as a quick index for Section 3: National Differences.

Section 3: National Differences {*multiple files, one for each Standard*}

This section is the largest of the Bulletin. In order to make the information as assessable as possible each IEC Standard is presented as separate files with the file name corresponding to the Standard.

Each Standard file contains;

- Latest Edition of Standard first and then previous editions follow in descending order
- Reference table of member countries at the beginning of each edition
- List of countries without National Differences
- List of Countries with National Differences
- Details of the National Differences listed in alphabetical country order

This section includes headers on each page to highlight the different editions of the Standards.

Section 4: IECEx CBs (Certified Bodies) and IECEx TLs (Testing Laboratories) {*one single electronic file*}

The final section of the Bulletin contains a table of all current ExCBs and ExTLs and their current Contact details. The table is presented in alphabetical country order.

Included at the end of this section are instructions on how to access ExCB and ExTL information from the IECEx website. This was added for future reference for updated contact details.

IECEx Bulletin 4.0, 2011 Contents

PREFACE

The IECEx Secretariat, from information supplied by the National Member Bodies, ExCBs and ExTLs, compiled the information contained in this bulletin. Neither the IEC nor the IECEx Secretariat is therefore liable for the accuracy of the information.

New editions of the bulletin will be issued as new information becomes available.

Document History:

Edition	Date	Comments
1	December 2000	First Edition
2	May 2003	Second Edition
3	February 2006	Third Edition
3.1	August 2006	Minor changes
4.0	Feburary 2011	Major update

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IECEx Bulletin and other IECEx Publications may be purchased directly from the IEC Web store: http://www.iec.ch

Section 1
Introduction

IECEx System Introduction

IECEx System Objective

The objective of the IECEx System is to facilitate international trade in equipment and services for use in explosive atmospheres, while maintaining the required level of safety:

- reduced testing and certification costs to manufacturer
- reduced time to market
- international confidence in the product assessment process
- one international database listing
- maintaining International Confidence in equipment and services covered by IECEx Certification

What is an Ex area?

Ex areas can be known by different names such as "Hazardous Locations", "Hazardous Areas" "Explosive Atmospheres", and the like and relate to areas where flammable liquids, vapours, gases or combustible dusts are likely to occur in quantities sufficient to cause a fire or explosion.

The modern day automation of industry has meant an increased need to use equipment in Ex areas. Such equipment is termed "Ex equipment"

Where do you commonly find Ex equipment?

Ex equipment in such areas include:

Automotive refuelling stations or petrol stations
Oil refineries, rigs and processing plants
Chemical processing plants
Printing industries, paper and textiles
Hospital operating theatres
Aircraft refuelling and hangars
Surface coating industries
Underground coalmines
Sewerage treatment plants
Gas pipelines and distribution centres
Grain handling and storage
Woodworking areas
Sugar refineries
Metal surface grinding, especially aluminium dusts and particles

IECEx International Certification System

In addition to the preparation of International Standards, the IEC facilitates the operation of Conformity Assessment Systems. One such System is the IECEx System.

The IECEx System comprises the following

1. The IECEx Certified Equipment Scheme
2. The IECEx Certified Service Facilities Scheme
3. The IECEx Conformity Mark Licensing System
4. The IECEx Certified Persons Scheme

Section 1
Introduction
1. The IECEx Certified Equipment Scheme

This IECEx Scheme is an International Certification Scheme covering product that meets the requirements of International Standards, e.g. IEC Standards prepared by TC 31.

The IECEx Certified Equipment Scheme provides both:

a) A single International Certificate of Conformity that requires manufacturers to successfully complete:-

- Testing and Assessment of samples for compliance with Standards
- Assessment and auditing of manufacturers premises
- On-going surveillance audits of manufacturers premises

or

b) A "fast-track" process for countries where regulations still require the issuing of national Ex Certificates or approval. This is achieved by way of global acceptance of IECEx equipment Test and Assessment Reports.

Certificates issued by the IECEx Certified Equipment Scheme are issued as "Electronic Certificates" and are live on the IECEx Website. This enables full public access for viewing and printing. Visit the IECEx "On-Line Certificate" System

A further feature of the IECEx Equipment Scheme is the IECEx Unit Verification Certification. This type of certification fulfills the need within the industry to issue IECEx Certificates that cover a defined number of products/items that are produced under a single production run. Unit Verification may also cover repaired items.

Rules of the IECEx Certified Equipment Scheme are detailed in IECEx publication, IECEx 02

2. The IECEx Certified Service Facilities Scheme

This IECEx Scheme is an International Certification Scheme that covers the assessment and the on-site audit of organizations that provide services such as Repair and Overhaul service to the Ex industry.

Due to the very high capital investment made by industry in most Ex equipment it is much more economical to repair and overhaul equipment rather than to replace it with new. This also has obvious environmental benefits.

The challenge for industry is to ensure that all the very unique Ex safety features, included in the design and manufacturing of Ex equipment, are not compromised during the repair process.

Ex Repair and Overhaul Facilities and Workshops, certified under the IECEx Certified Service Facilities Scheme, provide industry with the assurance that repairs and overhaul to Ex equipment will be undertaken according to the strict requirements of IECEx Scheme to the International Standard IEC 60079-19

Like the IECEx Certified Equipment Scheme, only "Electronic Certificates" are issued via the "On-Line" system thereby giving industry full access to both the viewing and printing of certificates.

Rules of the IECEx Certified Service Facility Scheme are detailed in IECEx publication, IECEx 03

Section 1

Introduction

3. The Ex Mark of Conformity System

The IECEx Conformity Mark License provides for the granting of a license to manufacturers to display the IECEx Mark of Conformity for products covered by an IECEx Certificate of Conformity manufactured under systems that are under ongoing surveillance by ExCBs.

It will help governments, safety regulators, and industry to have greater assurance that the equipment being operated or supplied for use in areas where flammable gases and vapours and combustible dusts (termed explosive atmospheres) are present, meet the world's most respected and vigorous safety standards.

The Mark shall only be placed on products or on packaging and promotional material covered by a valid IECEx Certificate of Conformity issued in accordance with the IECEx System rules.

Rules of the IECEx Conformity Mark License System are detailed in IECEx publication, IECEx 04

4. IECEx Certified Persons Scheme

This IECEx Scheme is an International Conformity Scheme that provides the global Ex industries with a single system for the assessment and qualification of persons meeting the competency prerequisites needed to properly implement the safety requirements based on the suite of IEC International Standards covering explosive atmospheres, e.g. the IEC 60079 and IEC 61241 series of standards.

The Certified Persons Scheme provides the international Ex industries with a qualification system that is transportable across borders.

Rules of the IECEx Certified Person Scheme are detailed in IECEx publication, IECEx 05

Applications

Ex equipment Manufacturers, Ex Service Providers and persons seeking IECEx certification can apply to IECEx Certification Bodies (termed ExCBs), in any country.

Countries and Certification Bodies can apply to become members of the IECEx System with Certification bodies able to issue IECEx Certificates once accepted according to the application procedures.

Rules of the IECEx System including Rules of Procedures are available, free of charge, from the IECEx website

National Member Bodies of the IEC System make application for their country to participate in the IECEx System on a Standard-by-Standard basis. The application is made to the Secretary of the Ex Management Committee.

Certification bodies and testing laboratories wishing to be accepted into the IECEx System must reside in a participating country. Their application for acceptance is made through their National Member Body of the IECEx System for the country in which they reside.

Certification bodies and testing laboratories are accepted into the IECEx System after satisfactory assessment of their competence. Assessors are selected who will provide confidence to regulatory authorities, users, manufacturers and certification bodies. ISO/IEC Standards and Guides 17025, 65 and IECEx Technical Guidance Documents are used as part of the IECEx assessment process.

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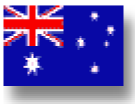
(AU) **Australia** 	Standards Australia Mr Andrew McKay Standards Australia Level 10, 20 Bridge Street Sydney 2000 GPO box 476 Sydney NSW 2001	Tel: +61 2 92376079 Fax: +61 2 92376010 E-mail: andrew.mckay@standards.org.au
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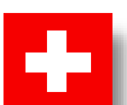
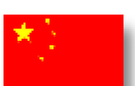
(CH) **Switzerland** 	Electrosuisse Mr Fritz Beglinger Luppmenstrasse 1 CH-8320 FEHRALTORF SWITZERLAND	Tel: +41 44 956 1325 Fax: +41 44 956 1825 E-mail: fritz.beglinger@electrosuisse.ch
(CN) **China** 	Certification and Accreditation Administration of the People's Republic of China CNCA Chinese National Committee of the IEC Ms. Bo Yumin Director General Department for International Cooperation of CNCA and	Ms Bo Yumin Tel: +86 10 82262749 Fax: +86 10 82260741 Email: boym@cnca.gov.cn Ms Du Chunjing Tel : 86-10-8226 2669 Fax :86-10-8226 0819 E-mail : dchj@cnca.gov.cn

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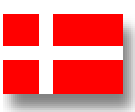
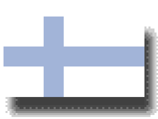
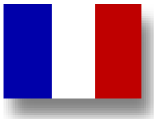
(DK) Denmark 	Dansk Standard Mr R Nielsen Kollegievej 6 DK- 2920 CHARLOTTENLUND DENMARK	Tel: +45 39 966 101 Fax: +45 39 966 102 E-mail: rn@ds.dk
(FI) Finland 	SESKO Standardization in Finland Mr Risto Sulonen VTT Technical Research Centre of Finland Otakaari 7B, Espoo P.O box 1000 FI-02044 VTT/Finland	Tel: +358 20 722 6415 Fax: +358 20 722 7042 E-mail: Risto.Sulonen@vtt.fi

(FR) ***France*** 	Laboratories Central des Industries Electriques LCIE On behalf of: Union Technique de L'Electricite et de La Communication (UTE) Mr Michel Brenon 33 Avenue du General Leclerc FR - 92262 FONTENAY-AUX - ROSES France	Tel: +33 1 40 95 5519 Fax: +33 1 40 95 5520 E-mail: michel.brenon@lcie.fr
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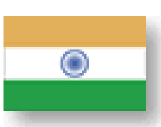
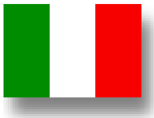
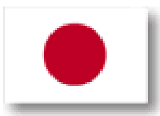
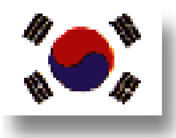
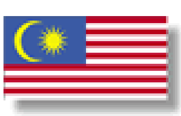
(IN) ***India*** 	Bureau of Indian Standards Mr P K Gambhir Deputy Director General (Marks) 9 Bahadur Shah Zafar Marg NEW DELHI 110002 INDIA	Tel: +91 11 2323 1120 Fax: +91 11 2323 9842 Email: ddgm@bis.org.in

(IT) ***Italy*** 	CEI - Comitato Elettrotecnico Italiano Mr Carlo Masetti Via Saccardo, 9 20134 Milano ITALIA	Tel: +39 02 2100 6213 Fax: +39 02 2100 6210 E-mail: masetti@ceiweb.it
(JP) ***Japan*** 	Japanese Industrial Standards Committee JISC Ministry of Economy, Trade and Industry in Japan Mr Toshiyuki Nawata 1-3-1 Kasumigaseki, Chiyoda-ku, Tokyo 100-8901, JAPAN	Tel: +81 3 3501 9473 Fax +81 3 3580 8598 E-mail:
(KR) ***Korea*** 	Korean Agency for Technology and Standards KATS Mr. Yang- Hoi SONG Director of Electrical and Electronics Standards Division 2 Joongang-dong Kwachon Kyunggi-do 427-010 REPUBLIC OF KOREA	Tel: + 82 2 503 7949 Fax: +82 2 509 7416 E-mail: elec@kats.go.kr
(MY) ***Malaysia*** 	Department of Standards Malaysia (STANDARDS MALAYSIA) Mr. Hussalmizzar Hussain Level 1&2, Block 2300, Century Square Jalan Usahawan, Cyberjaya 63000 Selangor MALAYSIA	1) Mr. Hussalmizzar Hussain Tel: +603 83191445 Fax: +603 83191511 E-mail: mizzar@standardsmalaysia.gov.my 2) Mr. Feris Frederick Tel: +603 83191475 Fax: +603 83191511 E-MAIL: FERIS@STANDARDSMALAYSIA.GOV.MY
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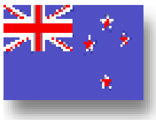
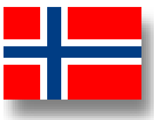
(NZ) ***New Zealand*** 	Standards New Zealand Mr John Kelly Private Bag 2439 Wellington NEW ZEALAND	Tel: + 64 4 498 5990 Fax: +64 4 498 5994 E-mail: nzec@standards.co.nz
(NO) ***Norway*** 	Mr Bjørn Spongsveen Strandveien 18 PO Box 280 N-1326 Lysaker NORWAY	Tel: +47 67 83 31 11 Fax: +47 67 83 31 01 E:mail: bjorn.spongsveen@nek.no
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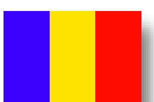
(RO) ***Romania*** 	INSEMEX PETROSANI Mr Constantin LUPU Str. General Vasile Milea nr.32-34 Petrosani 2675 ROMANIA	Tel: +40 2 5454 6277 Fax: +40 2 5454 6277 E-mail: Constantin.Lupu@insemex.ro
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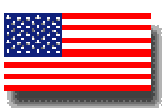
(US) ***United States of America*** 	US National Committee of the IECEx Mr Joel Solis NEMA 1300 North 17th Street Suite 1847 Rosslyn VA USA 22209 UNITED STATES OF AMERICA	Tel: +1 703 841 3267 Fax: +1 703 841 3367 E-mail: Joel_Solis@nema.org
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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 2: Matrix

IECEx Bulletin:2011(E)

About the IEC

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 2: Matrix

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

SECTION 2: Adoption of IEC Standards at National Level

IEC Publication Numbers	Country	AU	BR	CA	CH	CN	CZ	DE	DK	FI	FR	GB	HR	HU	IN	IT	JP	KR	MY	NL	NO	NZ	PL	RO	RU	SE	SG	SI	TR	US	ZA
60079-0 5th ed 2007 Page:04		1	1	2	2	2	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	2	2	1	2	3	2	1
60079-0 4th ed 2004 Page:38		1	4	2	1	3	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	2	1	1	1	3	2	1
60079-0 3.1 ed 2000 Page:66		1	4	2	3	3	3	3	3	3	3	3	3	3	1	3	1	2	1	3	3	1	3	3	4	3	1	3	3	4	1
60079-0 3rd ed 1998 Page:68		1	4	NA	2	2	2	2	2	2	2	2	2	2	4	2	1	2	4	2	2	1	2	2	2	2	1	2	3	4	1
60079-0 2.2 ed 1991 Page:79		1	4	4	3	NA	3	3	3	3	3	3	3	3	4	3	1	3	4	3	3	1	3	3	4	3	4	3	3	4	1
60079-0 2.1 ed 1987 Page:79		1	4	4	3	NA	3	3	3	3	3	3	3	3	4	3	1	3	4	3	3	1	3	3	4	3	4	3	3	4	1
60079-1 6th ed 2007 Page:04		1	1	3	1	2	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	2	1	1	1	3	2	1
60079-1 5th ed 2003 Page:24		1	4	2	1	3	1	1	1	1	1	1	NA	1	4	1	1	3	1	1	1	1	NA	1	3	1	1	1	3	2	1
60079-1 4th ed 2001 Page:44		1	4	1	3	3	3	3	3	3	3	3	NA	3	1	3	1	3	1	3	3	1	NA	3	2	3	1	3	3	4	1

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60079-1 3.2 ed 1998 Page:46		1	4	NA	3	2	3	3	3	3	3	3	3	3	4	3	1	3	4	3	3	1	3	3	4	3	4	3	3	4	4
60079-1 3.1 ed 1993 Page:46		1	4	NA	2	2	2	2	2	2	2	2	2	2	4	2	1	2	4	2	2	1	2	2	4	2	4	2	3	4	4
60079-1 3rd ed 1990 Page:46		1	4	NA	3	2	3	3	3	3	3	3	3	3	4	3	1	3	4	3	3	1	3	3	4	3	4	3	3	4	4
60079-2 5th ed 2007 Page:04		1	1	2	1	2	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	1	1	1	1	3	4	1
60079-2 4th ed 2001 Page:08		1	4	2	1	3	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	2	1	1	1	3	2	1
60079-2 3rd ed 1983 Page:16		1	4	2	2	3	2	2	2	2	2	2	2	2	4	2	1	2	4	2	2	1	2	2	4	2	4	2	3	4	1
60079-5 3rded 2007 Page:04		1	1	2	1	2	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	1	1	1	1	3	2	1
60079-5 2.1 ed 2003 Page:11		1	4	1	4	1	4	4	4	4	4	4	NA	4	1	4	1	1	1	4	4	1	NA	4	2	4	1	4	3	2	1
60079-5 2nd ed 1997 Page:11		1	4	1	2	1	2	2	2	2	2	2	NA	2	1	2	1	1	1	2	2	1	NA	2	2	2	1	2	3	2	1

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60079-6 3rd ed 2007 Page:04		1	1	2	1	2	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	3	1	1	1	3	2	1
60079-6 2nd ed 1995 Page:09		1	4	1	2	1	2	2	2	2	2	2	NA	2	1	2	1	2	1	2	2	1	NA	2	1	2	1	2	3	2	1
60079-7 4th ed 2006 Page:04		1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	1	1	1	1	3	2	1
60079-7 3rd ed 2001 Page:07		1	4	3	1	3	1	1	1	1	1	1	NA	1	1	1	1	3	1	1	1	1	NA	1	2	1	1	1	3	2	1
60079-7 2.2 ed 1993 Page:13		1	4	1	4	2	4	4	4	4	4	4	NA	4	4	4	1	4	4	4	4	1	NA	4	4	4	4	4	3	4	1
60079-7 2.1 ed 1991 Page:13		1	4	1	4	2	4	4	4	4	4	4	NA	4	4	4	1	4	4	4	4	1	NA	4	4	4	4	4	3	4	1
60079-7 2nd ed 1990 Page:13		1	4	1	2	2	2	2	2	2	2	2	NA	2	4	2	1	2	4	2	2	1	NA	2	4	2	4	2	3	4	1
60079-11 5th ed 2006 Page:04		1	1	2	1	2	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1	1	1	1	1	3	2	1
60079-11 4th ed 1999 Page:19		1	4	2	2	2	2	2	2	2	2	2	NA	2	1	2	1	3	1	2	2	1	NA	2	2	2	1	2	3	2	1

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60079-11 3rd ed 1991 Page:46		1	4	4	2	3	2	2	2	2	2	2	NA	2	4	2	1	2	4	2	2	1	NA	2	4	2	4	2	3	4	1
60079-15 4th ed 2010 Page:04		1	4	3	1	2	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	3	1	1	1	3	4	1
60079-15 3rd ed 2005 Page:06		1	1	2	1	2	1	1	1	1	1	1	NA	1	1	1	1	3	1	1	1	1	NA	1	1	1	4	1	3	2	1
60079-15 2nd ed 2001 Page:21		4	4	1	2	3	2	2	2	2	2	2	NA	2	4	2	1	3	1	2	2	4	NA	2	1	2	4	2	3	4	1
60079-15 1st ed 1987 Page:26		1	4	4	4	3	4	4	4	4	4	4	NA	4	4	4	1	2	4	4	4	1	NA	4	4	4	4	4	3	2	1
60079-18 3rd ed 2009 Page:04		1	4	2	1	2	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	1	1	1	1	3	4	1
60079-18 2nd ed 2004 Page:08		1	1	3	1	1	1	1	1	1	1	1	NA	1	1	1	1	3	1	1	1	1	NA	1	3	1	1	1	3	2	1
60079-18 1st ed 1992 Page:16		1	4	2	2	3	2	2	2	2	2	2	NA	2	1	2	1	2	1	2	2	1	NA	2	2	2	1	2	3	4	1
60079-19 2nd ed 2006 Page:04		1	1	3	1	3	1	1	1	1	1	1	1	1	1	1	1	3	3	1	1	1	1	1	3	1	1	1	3	4	1

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60079-25 2nd ed 2009 Page:04		1	1	2	1	1	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	1	1	1	1	3	4	1
60079-25 1st ed 2003 Page:06		1	1	3	1	3	1	1	1	1	1	1	NA	1	1	1	1	3	1	1	1	1	NA	1	3	1	4	1	3	4	1
60079-26 2nd ed 2006 Page:04		1	1	2	1	3	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	1	1	1	1	3	4	1
60079-26 1st ed 2004 Page:07		1	4	3	2	3	2	2	2	2	2	2	NA	2	4	2	1	3	1	2	2	1	NA	2	4	2	4	2	3	2	1
60079-27 2nd ed 2008 Page:04		1	1	2	1	1	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	2	1	4	1	3	4	1
60079-27 1st ed 2004 Page:08		1	4	3	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	3	1	4	1	3	2	1
60079-28 1st ed 2006 Page:04		1	1	3	1	3	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	3	1	1	1	3	3	1
60079-29-1 1st ed 2007 Page:04		1	1	2	2	3	2	2	2	2	2	2	2	2	1	2	1	3	1	2	2	1	2	2	3	2	1	2	3	4	4
60079-29-4 1st ed 2009 Page: 04		1	3	3	2	3	2	2	2	2	2	2	2	2	3	2	1	3	4	2	2	1	2	2	3	2	1	2	3	3	3

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60079-30-1 1st ed 2007 Page:04		1	1	2	1	3	1	1	1	1	1	1	1	1	4	1	1	3	1	1	1	1	1	1	1	1	1	1	3	2	1
60079-31 1st ed 2008 Page:04		1	1	2	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	3	1	1	1	3	2	1
61241-0 1st ed 2004 Page:04		1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	3	4	1	1	1	1	1	1	1	4	1	3	2	1
61241-1 1st ed 2004 Page:04		1	3	3	1	1	1	1	1	1	1	1	1	1	1	1	1	3	4	1	1	1	1	1	3	1	4	1	3	2	1
61241-4 1st ed 2001 Page:04		1	3	2	1	1	1	1	1	1	1	1	1	1	1	1	1	3	4	1	1	1	1	1	3	1	4	1	3	2	1
61241-11 1st ed 2005 Page:04		1	3	2	1	1	1	1	1	1	1	1	1	1	1	1	1	3	4	1	1	1	1	1	3	1	1	1	3	2	1
61241-18 1st ed 2004 Page:04		1	3	4	1	1	1	1	1	1	1	1	1	1	1	1	1	3	4	1	1	1	1	1	3	1	4	1	3	2	1
61241-1-1 2nd ed 1999 Page:04		1	3	2	2	1	2	2	2	2	2	2	2	2	4	2	1	2	4	2	2	1	2	2	3	2	1	2	3	4	1
61241-1-1 1st ed 1993 Page:10		1	3	3	4	1	4	4	4	4	4	4	4	4	3	4	1	3	4	4	4	1	4	4	3	4	1	4	3	4	1

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61779-1 1st ed 1998 Page:04		1	3	1	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	3	2	1	2	3	2	4
61779-2 1st ed 1998 Page:04		1	3	3	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	3	2	4	2	3	2	1
61779-3 1st ed 1998 Page:04		1	3	3	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	3	2	4	2	3	2	4
61779-4 1st ed 1998 Page:04		1	3	1	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	3	2	4	2	3	2	4
61779-5 1st ed 1998 Page:04		1	3	1	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	3	2	4	2	3	2	4
62013-1 2nd ed 2005 Page:04		1	3	4	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	2	1	1	1	3	4	4
62013-1 1st ed 1999 Page:07		1	3	4	2	3	2	2	2	2	2	2	2	2	4	2	1	3	4	2	2	1	2	2	2	2	4	2	3	4	4
62013-2 2nd ed 2005 Page:04		1	3	4	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	2	1	1	1	3	4	4
62013-2 1st ed 2000 Page:07		1	3	4	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	2	1	4	1	3	4	4

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62086-1 1st ed 2001 Page:04		1	3	4	1	3	1	1	1	1	1	1	1	1	4	1	1	3	4	1	1	1	1	1	1	1	4	1	3	2	1

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-0

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-0

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 60079-0 5th Edition 2007 Electrical apparatus for explosive gas atmospheres – Part 0: General requirement			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-0:11
CH Switzerland	**YES**		EN 60079-0:2009
CN China		**YES**	GB 3836.1 –2010 Explosive atmospheres Part 1: Equipment - General requirements (IEC60079-0: 2007, MOD）
CZ Czech Republic	**YES**		EN 60079-0:2009
DE Germany	**YES**		EN 60079-0:2009
DK Denmark	**YES**		EN 60079-0:2009
FI Finland	**YES**		EN 60079-0:2009
FR France	**YES**		EN 60079-0:2009
GB United Kingdom	**YES**		EN 60079-0:2009
HR Croatia	**YES**		EN 60079-0:2009
HU Hungary	**YES**		EN 60079-0:2009
IN India		**N/R**	To be adopted
IT Italy	**YES**		EN 60079-0:2009
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 60079-0:2009
NO Norway	**YES**		EN 60079-0:2009
NZ New Zealand		**NO**	
PL Poland	**YES**		EN 60079-0:2009
RO Romania	**YES**		EN 60079-0:2009
RU Russia		**YES**	GOST R IEC 60079.0-2007
SE Sweden	**YES**		EN 60079-0:2009
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 60079-0:2009
TR Turkey*		**TBA**	
US United States*		**YES**	ANSI/ISA 60079-0:2009 ANSI/UL 60079-0:2009
ZA South Africa		**NO**	SANS 60079-0:2007

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed. Please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-0 5th Edition 2007 Without National Differences
AU Australia
BR Brazil
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-0 5th Edition 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-0:11
Explosive atmospheres –
Part 0: Equipment – General requirements

This is the second edition of CAN/CSA-C22.2 No. 60079-0, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-0 (Edition 5.0, 2007-10). It replaces the CAN/CSA-C22.2 No.60079-0:07 (adopted CEI/IEC 60079-0: Fourth edition, 2004-01: Entitled *Electrical apparatus for explosive gas atmospheres – Part 0: General requirements*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

Canadian deviations

1 Scope

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Electrical equipment of **Group I**, (intended for use in mines susceptible to firedamp), is under the jurisdiction of the Canadian Federal Government. Such types of equipment, (for example, caplights), shall be referred to the respective AHJ (authority having jurisdiction).

2 Normative references

[Add the following]

CSA (Canadian Standards Association)
Where reference is made to CSA publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following publications and the years shown indicate the latest editions available at the time of printing:

C22.1-06
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

4.3 Group III

[Add the following explanatory paragraph or as a "note"]

Current Canadian (CSA) classification of explosive dust atmosphere is referred to: *Class II, Group E, Group F and Group G; and Class III.* The closest equivalency is as follow:

- Group IIIA: combustible flyings – (CSA Class III)
- Group IIIB: non-conductive dust – (CSA Class II, Groups G and/or F)
- Group IIIC: conductive dust – (CSA Class II, Groups E and/or F)

6.1 General

*[Add the following "Note" to **Sub-clause (a)**]*

Not all the standards listed in Clause 1 are formally adopted by CSA, with (or without) Canadian deviations. Where found appropriated, CSA publications will take precedence.

9 Fasteners
9.1 General

[Add a paragraph to the following effect]

Fasteners conforming to the national standard of the respective country are also permitted; provided that their effectiveness is equal to, or exceed, the effectiveness and criteria (e.g., strength, tolerance and configuration) stipulated in Clauses 9.2 and 9.3. In this case, such fasteners shall be specified in the marking or instruction manual; and the equipment shall be marked with an "X" in accordance with item e) of 29.2.

28 Manufacturer's responsibility

[Add a sub-clause to the following effect]

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The following requirements shall not be compromised by any clause in the CAN/CSA-C22.2 No. 60079 series Standards:

(a) The safety level of design and installation of equipment shall be in accordance with the requirements and rules of the *Canadian Electrical Code, Part I*.

(b) It shall be the responsibility of the manufacturer to ensure that the equipment meets the applicable standards for general electrical safety and usage in accordance with the regulatory requirements.

For mines and quarry applications, see also requirements under Standard CAN/CSA-M421.

29.3 Ex marking for explosive gas atmospheres

[Replace existing clause with the following]

i) Specific installation instructions or reference to a specific installation document when it is necessary to indicate special conditions for safe use;

Note: The manufacturer should ensure that the requirements of the special conditions for safe use are passed to the purchaser together with any other relevant information.

29.4 Ex marking for explosive gas atmospheres

[Add a "Note" under Sub-clause (a) to the following effect]

Currently, the *Canadian Electrical Code, Part I* also permits the symbol "*EEx*" as an alternative to "*Ex*".

29.11 Warning markings

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

30 Instructions

[Add a paragraph to the following effect]

Instructions, relevant to the safe installation, usage and operation of the equipment, shall be in both English and French.

A.3.1.2 Cable glands with clamping by filling compound

[Add a paragraph to the following effect]

Due to the extreme weather in the Canadian geography; and, the preparation of the compound, the pouring and curing procedures are so weather and labour dependent, *clamping by filling compound* is **not** an acceptable method in Canada.

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CN **China**

IEC Standard Number IEC 60079-0:2007

IEC Clause or Annex Number	Details of Differences
1	Add the following extra requirement to this Clause: Add note7: here flameproof enclosure is the same meaning with type "d"
8.1.1	Add the following extra requirement to this Clause. *details of extra requirement(s): the enclosure of the electric drill, portable apparatus and luminaries can be made of light alloy with special strength and should pass the friction spark test according to GB/T 13813-2008*
14.2	Add the following extra requirement to this Clause. *details of extra requirement(s): for Group I equipment, the inner face of the metallic termination compartment should be coated with arc- resistant paint.*
26.9	Add the following extra requirement to this Clause. *details of extra requirement(s): require the interval between thermal endurance to heat and cold, after former test wait for 24h.*
28.2	Add the following extra requirement to this Clause. *details of extra requirement(s): add the procedure for applying the certificate specified in Annex D.*

18.4 b) 2), 19 This requirement of the waring label does not apply to Group I equipment.
Annex C (informative) Replace the Clause with Annex G (informative)
Annex D (informative) Replace the Clause with Annex E (informative)
Annex E (informative) Replace the Clause with Annex F (informative)

Add the additional Annex C (normative)as note.

special requirement for Group I equipment, wet heat test should be conducted, defined the test condition and method comply with GB/T2423.4-2008; defined the material of the plastic enclosure and should stand burning test and the requirement of the test result.

Add the additional Annex D (normative): *procedure for applying the certificate*.

Group Difference for: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia

European Group differences

There are no technical differences but Annex ZY (reproduced below), introduced by Cenelec, provides additional marking requirements to satisfy the requirements of the ATEX Directive 94/9/EC

Annex ZY
(informative)
Equipment groups and marking examples

Equipment groups

In all cases Equipment Protection Levels (EPL) as defined by EN 60079-0 are related to the corresponding Equipment Groups and Equipment Categories according to the following table. The same applies if a standard makes reference to the intended use of equipment in Zones according to the definitions in EN 60079-10.

EN 60079-0		Directive 94/9/EC		EN 60079-10-x
EPL	**Group**	Equipment Group	Equipment Category	**Zones**
Ma	**I**	**I**	**M1**	NA
Mb			M2	
Ga	**II**	**II**	1G	0
Gb			2G	1
Gc			3G	2
Da	**III**		1D	20
Db			2D	21
Dc			3D	22

Instructions

The manufacturer or his authorized representative in the Community is to draw up the instructions for use in the required Community languages.

Marking

The marking according to this standard is to be supplemented by the marking according to Directive 94/9/EC. Examples are given below.

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European marking examples

Directive part	Standard part	Equipment example
I M2	**Ex d I Mb**	Mining equipment,

type of protection “Flameproof Enclosure” d

Directive part	Standard part	Equipment example
II 2G	**Ex e IIB T4 Gb**	Gas explosion protected equipment

type of protection, “Increased Safety” e

Directive part	Standard part	Equipment example
II 1D	**Ex ma IIIC 120°C Da**	Dust explosion protected equipment,

type of protection “Encapsulation” ma

NOTE Attention is drawn to the requirement in 29.8:
The Ex marking for explosive gas atmospheres and explosive dust atmospheres shall be separate and not combined:

II 1 G - Ex ia IIB T4

II 1 D - Ex ia IIIC T120°C

or alternatively:

II 1 GD

Ex ia IIB T4 Ga

Ex ia IIIC T120°C Da

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RU Russia

IEC 60079-0 2007 5th ed.	GOST R IEC 60079.0-2007	MOD	1) in 3.18.1 add the note: «*NOTE – In electrical equipment with the level of protection Ma additional means of explosion protection specified in the standards for the types of protection are used as compared with the equipment with the level of protection Mb* »; 2) in 3.18.2 add the note: «*NOTE – In electrical equipment with the level of protection Mb the explosion protection is provided both in normal operation and during expected malfunctions depending on the operating conditions, except for explosion protection means failure*»; 3) in 3.18.3 add the note: «*NOTE – In electrical equipment with the level of protection Ga additional means of explosion protection specified in the standards for the types of protection are used as compared with the equipment with the level of protection Gb* »; 4) in 3.18.4 add the following text: «*…and which is unlikely to become a source of ignition within the time from the moment of explosive atmosphere occurrence to the moment of power-off»,* and the note: *«NOTE – In electrical equipment with the level of protection Gb the explosion protection is provided both in normal operation and during expected malfunctions depending on the operating conditions, except for explosion protection means failure».* 5) in 3.18.5 add the notes: «*NOTES* *1 Electrical equipment operates in explosive atmosphere during the time from the moment of explosive atmosphere occurrence to the moment of power-off.* *2 In electrical equipment with the level of protection Gc explosion protection is only provided in established normal operation»;.* 6) in 3.18.3 add the note: «*NOTE – In electrical equipment with the level of protection Da additional means of explosion protection specified in the standards for the types of protection are used as compared with the equipment with the level of protection Db* »; 7) in 3.18.7 add the following text: «*…and which is unlikely to become a source of ignition within the time from the moment of explosive atmosphere*

			occurrence to the moment of power-off», and the note: *«NOTE – In electrical equipment with the level of protection Db the explosion protection is provided both in normal operation and during expected malfunctions depending on the operating conditions, except for explosion protection means failure»;* 8) in 3.18.8 add the notes: *«NOTES* *1 Electrical equipment operates in explosive atmosphere during the time from the moment of explosive atmosphere occurrence to the moment of power-off.* *2 In electrical equipment with the level of protection Dc explosion protection is only provided in established normal operation»;* 9) In section 4 add Note 2, which determines quantitative characteristics of methane (firedamp), released in underground workings of coal mines: **«***Note 2. Methane in underground workings means firedamp, which in addition to methane contains gas hydrocarbons – methane homologs C_2 - C_5 - no more than 0.1 % (volume fraction), and hydrogen in gas samples taken from blastholes immediately after drilling - no more than 0.002 % (volume fraction) of the total volume of flammable gases***»;** 10) in 9.2 add Notes 2 and 3, which extend the requirements to fasteners for Group I electrical equipment I: *«NOTES* *2 The diameter of bolts, screws and pins intended to fasten parts of enclosures of Group I electrical equipment, shall be at least 6 mm. To fasten parts of enclosures of control equipment and automation devices, fastening bolts, screws and pins of at least 5 mm in diameter may be used.* *The requirements to the minimum diameter of fastening bolts, screws and pins do not apply to the enclosures of instruments and devices for personal use, if these fasteners are not intended to be unscrewed under mine conditions, for example, if they are glued or sealed.* *3 Bolts, screws, nuts and other fasteners shall be prevented from spontaneous weakening by the method specified in the technical documentation»;* 11) in 9.2 add Notes 2 and 3, *2 The diameter of bolts, screws and pins intended to fasten parts of enclosures of Group I electrical equipment, shall be at least 6 mm. To fasten parts*

			of enclosures of control equipment and automation devices, fastening bolts, screws and pins of at least 5 mm in diameter may be used. *The requirements to the minimum diameter of fastening bolts, screws and pins do not apply to the enclosures of instruments and devices for personal use, if these fasteners are not intended to be unscrewed under mine conditions, for example, if they are glued or sealed.* *3 Bolts, screws, nuts and other fasteners shall be prevented from spontaneous weakening by the method specified in the technical documentation.* 12) in Section 10 add notes 2 and 3, which extend the requirements to warnings notices if interlocking devices are not used: *«Note:* *2 The requirement to use interlock shall be specified by the standards for individual types of protection or electrical devices.* *3 On the covers of enclosures of electrical equipment without interlock, where it is not possible to determine if the equipment is energized without the cover removal required during this equipment use for its preventive repair and inspection, the following warning markings shall be affixed: «Do not open when energized», or «Do not open in explosive atmosphere», or «Do not open in the mine»;* 13) in 14.1 add the notes which extend the requirements to terminals of Group I electrical equipment: «*Notes* *1 Electrical equipment intended for connection to external circuits shall include connection facilities, with the exception of electrical equipment that is manufactured with a cable permanently connected to it. Marking of electrical equipment of all types manufactured with a cable permanently connected to it, shall include the symbol «X» to indicate special conditions of free cable end connection.* *2 Terminals shall have a marking if its absence can result in incorrect connection. Marking may appear on the terminal, next to it or on the label attached to it.* *3 Current-carrying parts of terminals shall be connected in a way that the electrical contact at the connection point does not deteriorate because of heating under varying temperature conditions, changes of insulating parts dimensions and vibration during long-term operation. The transfer of contact pressure to*

			electrical connections through insulation materials is not permitted, except for cases when the pressure is transferred through porcelain, steatite or other materials with analogous thermal and mechanical properties, and at the same time the differences in thermal expansion of insulating and live parts shall be taken into account. *4 Current-carrying parts of terminals in Group I electrical equipment shall be made of corrosion-resistant highly conductive materials (such as copper and brass). Non- current-carrying parts of terminal (pressure screws) may be made of steel, if the corresponding anticorrosive coating is provided. The diameter of contact screws (bolts, pins) for connection of external conductors and cable cores of Group I electrical equipment shall be at least 6 mm.* *5 In control, monitoring and automatic devices contact screws of less than 6 мм in diameter may be used. For measuring devices the minimum diameter of contact screws is not fixed. In communication, automatic and signaling devices the diameter of contact screws shall be at least 4 mm*»; 14) in 17.4 add the note that extends the requirements to clearances between rotating and fixed elements of external fans of rotational electrical machines: «*Note – The requirements to clearances between rotating and fixed elements may not be observed, if external fans are made of materials which are safe from the viewpoint of frictional sparking (for example, brass and zinc alloy for Group II electrical equipment, and brass, zinc alloy and steel for Group I electrical equipment)»;* 15) in 23.1 add the second paragraph: *«The requirements to traction accumulators and accumulator batteries are stated in section 20 of GOST R 51330.0»;* 16) in 29.2, e) replace " the symbol «X» shall be placed after the certificate reference" by "the symbol «X» shall be placed after the Ex marking" to read: «e) *If it is necessary to indicate special conditions of use, the symbol «X» in accordance with 29.3 or 29.4 shall be placed after the Ex*; 17) in section 29.8, g) replace " the symbol «U» shall be placed after the certificate reference" by "the symbol «U» shall be placed after the Ex marking" to read: «g) *The symbol «U» after the designation (symbol) of the electrical equipment Group of Ex-component*». Added text is italicised.

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US United States of America

Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
General	The U.S. version of IEC 60079-0 does not include requirements for Group I electrical apparatus.
General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
General	To align with the NEC, references to specific EPLs have been replaced with the appropriate Zone.
1	To align with the recognized NEC terms, text added to explain that “explosive atmospheres” are referred to in the NEC as specific locations as follows: ……electrical equipment and Ex components intended for use in explosive atmospheres. Explosive atmospheres are identified by the National Electrical Code®, ANSI/NFPA 70 as hazardous (classified) locations and include the following specified locations: • Class I, Zone 0 • Class I, Zone 1 • Class I, Zone 2 • Zone 20 • Zone 21 • Zone 22
1	To recognized the US standard for Trace Heating, the following was added: This standard is supplemented or modified by the following apparatus standards: …………………….. IEEE 515, Standard for Testing, Design, Installation, and Maintenance of Electrical Resistance Heat Tracing for Industrial Applications NOTE — Components such as power connections, splices, end terminations, as well as cable entries, fittings, and seals are to be evaluated according to the type of protection employed.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
2	Additional references added to align with US practice as follows: ANSI/NFPA 70:2008, *National Electrical Code®.* ANSI/UL 347, *High Voltage Industrial Control Equipment* ANSI/UL 486E, *Equipment Wiring Terminals for use with Aluminum and/or Copper COnduictors* UL 508, Industrial Control Equipment ANSI/UL 746C, *Standard for Polymeric Materials – Use in Electrical Equipment Evaluations*
3.11.1.1	Definition revised to align with US practice as follows: **3.11.1.1** **conductive dust** ~~combustible dust with electrical resistivity equal to or less than 103 Ω · m~~ metal dust ~~NOTE IEC 61241-2-2 contains the test method for determining the electrical resistivity of dusts.~~
3.11.1.2	Definition revised to align with US practice as follows: **3.11.1.2** **non-conductive dust** ~~combustible dust with electrical resistivity greater than 103 Ω · m~~ dust other than metal dust
3.18	Note to definition revised to clarify the use of the EPL marking in the US as follows: **3.18** **equipment protection level** **EPL** level of protection assigned to equipment based on its likelihood of becoming a source of ignition and distinguishing the differences between explosive gas atmospheres, explosive dust atmospheres, and the explosive atmospheres in mines susceptible to firedamp NOTE ~~The equipment protection level may optionally be employed as part of a complete risk assessment of an installation, see IEC 60079-14.~~ Although the marking identifying the equipment

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	protection level (EPL) may appear on equipment evaluated against this standard, the 2008 NEC does not recognize the concept of employing the equipment protection level in a complete risk assessment of an installation.
3.20	**3.20** Definition revised to clarify that the suffix "U" is not used in the US as follows: **Ex component** part of electrical apparatus or a module (other than an Ex cable gland), ~~marked with the symbol "U",~~ which is not intended to be used alone and requires additional consideration when incorporated into electrical apparatus or systems for use in explosive gas atmospheres.
3.37	Definition deleted as the symbol "U" is not used in the US.
3.38	Definition deleted as the symbol "X" is not used in the US.
4.3	Note 2 added to clarify the NEC identification of "Group III" as follows: NOTE 2 The 2008 NEC does not recognize the identification of locations or equipment as "Group IIIA, IIIB, or IIIC", but identifies equipment suitable for Zone 20, 21, or 22 by the use of equipment marking where a "D" is appended to the type of protection, for example "iaD" and no separate differentiation is made of combustible dusts or ignitable fibers.
5.1.1	Text revised as follows to clarify that the NEC only permits a "special" temperature range to be *outside* of the "normal" ambient temperature range as follows: Electrical equipment designed for use in a normal ambient temperature range of –20 °C to +40 °C does not require marking of the ambient temperature range. However, electrical equipment designed for use ~~in other than~~ outside this normal ambient temperature range is considered to be special. The marking shall then include either the symbol *T*a or *T*amb together with both the upper and lower ambient temperatures~~ or, if this is impracticable, the symbol "X" shall be used to indicate specific conditions of use that include the upper and lower ambient temperatures.~~ See ~~item d)of 29.2 and~~ Table 1. NOTE The ambient temperature range may be an expanded ~~reduced~~ range, e.g. $-55°C \le T_{amb} \le 150°C$.
5.3.2.3.2	Delete section as the NEC does not recognize surface temperatures based on dust layer thicknesses.
5.3.3	Add text to clarify that these temperature classifications do not apply to cases where catalytic or other chemical reactions can result. Text modified as follows: b) where no catalytic or other chemical reactions can result, for T4 and Group I classification, small components shall conform to Table 3a and Table 3b; or

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	c) where no catalytic or other chemical reactions can result, for T5 classification, the surface temperature of a component with a surface area smaller than 1 000 mm^2 (excluding lead wires) shall not exceed 150 °C.
6.1.b	Revise text to highlight that the US requires conformance with the electrical safety requirements for "normal" electrical equipment. Text revised as follows: b) ~~be constructed in accordance with the applicable safety requirements of the relevant industrial standards.~~ comply with the applicable requirements for similar apparatus for use in unclassified locations. NOTE 2 ~~It is not a requirement of this standard that a certification body check compliance with this requirement. The manufacturer should indicate compliance by marking the apparatus or component in accordance with Clause 29 (and by stating the basis of compliance in the documentation, see Clause 28).~~ Requirements for safety of electrical equipment in ordinary (unclassified) locations can be found in ANSI Standards, NEMA Standards, Federal Regulations, etc. Apparatus listed by a Nationally Recognized Testing Laboratory is considered to meet the applicable requirements found in these standards. A list of commonly applied standards is shown in informative Annex D.
6.6.1	Text added to clarify that the user settings of interest are those that would *increase* the power levels. Text revised as follows: pulsed transmissions whose pulse durations exceed the thermal initiation time shall not exceed the values shown in Table 4. Programmable or software control intended for setting by the user that would increase the power thresholds above these levels shall not be permitted.
6.6.2	Requirements transferred from IEC 60079-28 as the US has not yet adopted that standard. Text revised as follows: The output parameters of lasers or other continuous wave sources of electrical equipment of EPL Ga, Gb, or Gc shall not exceed the following values. • 5 mW/mm^2 or 35 mW for continuous wave lasers and other continuous wave sources ~~NOTE The values for Ga, Gb, and Gc can be found in IEC 60079-28.~~ NOTE ISA 60079-28 is under development
7.1.4.d	Text referring to RTI was deleted as this rating is not applicable to elastomers. Note added to US standard for elastomeric seals. Text revised as follows: d) the continuous operating temperature (COT). ~~As an alternative to the COT, the relative thermal index (RTI – mechanical impact) may be determined in accordance with ANSI/UL 746B.~~ NOTE The COT may be determined using ANSI/UL 157.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
7.3	Added as an alternative, the US standard for resistance to light as follows: The resistance to light of the enclosures, or parts of enclosures, of non-metallic materials shall be satisfactory (see 26.10). As an alternative, the resistance to light may be determined in accordance with ANSI/UL 746C.
8.1.2	Revise text to highlight known concerns with copper-based enclosures as follows: NOTE Caution should be observed when copper or copper alloys are used for enclosures in atmospheres containing acetylene due to the potential formation of acetylides on the surface that can be ignited by friction or impact. The risk of ignition can be reduced by coating the copper or copper alloy with tin, nickel, or by other coatings, or by limiting the maximum copper content of the alloy to 30 percent.
15.1.2	To align with the NEC installation requirements, where the external terminal is not required, but is permitted, the text is revised as below: **15.1.2 External** An optional additional external connection facility for an equipotential bonding conductor ~~shall~~ may be provided for electrical equipment with a metallic enclosure~~, except for electrical equipment which is designed to be:~~ ~~a) moved when energized and is supplied by a cable incorporating an earthing or equipotential bonding conductor; or~~ ~~b) installed only with wiring systems not requiring an external earth connection, for example, metallic conduit or armoured cable.~~ ~~The manufacturer shall provide details on any earthing or equipotential bonding required for the installation under conditions a) or b) above in the instructions provided in accordance with Clause 30.~~ If provided, The additional external connection facility shall be electrically in contact with the connection facility required in 15.1.1. NOTE The expression "electrically in contact" does not necessarily involve the use of a conductor.
15.3	To align with the NEC requirements for equipment grounding conductors, the text has been revised as follows: **15.3 Size of conductor connection**

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	Protective earthing connection facilities shall allow for the effective connection of at least one conductor with a cross-sectional area ~~given in Table 9~~ as given in ANSI/NFPA 70, National Electrical Code, Section 250.122. **[Table 9 Deleted]** When provided, the supplemental Equipotential bonding connection facilities on the outside of electrical equipment shall provide effective connection of a conductor with a cross-sectional area of at least 4 mm^2 (10 AWG) Effective connection shall be verified by compliance with the pullout and secureness tests in ANSI/UL 486E. The terminal shall be marked in accordance with 29.15. NOTE The requirements for equipment grounding facilities and bonding are specified in the appropriate standard for electrical equipment for use in unclassified locations (see Clause 1) and may be more stringent than the requirements shown in 15.3. Refer to Annex F for additional information.
15.6	To align with the US practice of not permitting insulators to contribute to the security of an electrical connection, the text has been revised as follows: **15.5 Secureness of electrical connections** Connection facilities shall be designed so that the electrical conductors cannot be readily loosened or twisted. Contact pressure on the electrical connections shall be maintained and not be affected by dimensional changes of insulating materials in service, due to factors such as temperature or humidity. ~~For non-metallic walled enclosures provided with an internal earth continuity plate, the test of 26.12 shall be applied.~~ NOTE The material and dimensions of the earth continuity plate should be appropriate for the anticipated fault current.
16.5	To align with the default conductor temperature specified int eh NEC, the text has been revised as follows: When the temperature under rated conditions is higher than 60 °C ~~70 °C~~ at the entry point or 60 °C ~~80 °C~~ at the branching point of the conductors, information shall be marked on the equipment exterior to provide guidance to the user on the proper selection of cable and cable gland or conductors in conduit.
18.2	To align with the US standards for disconnect switches, the text has been revised as follows: Where switchgear includes a disconnector, it shall disconnect all poles. The switchgear shall be designed so that either – the position of the disconnector contacts is visible, or – their open position is reliably indicated (see ~~IEC 60947-1~~ ANSI/UL 60947-1, ANSI/UL347, or ANSI/UL 508; as applicable).

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
22.2	To align with the NEC requirements for flexible cords, the text has been revised as follows: **22.2 Group II and Group III caplights and handlights** Leakage of the electrolyte shall be prevented in all positions of the equipment. Where the source of light and the source of supply are housed in separate enclosures, which are not mechanically connected other than by an electric cable, the cable glands and the connected cable shall be tested according to the specific type of protection ~~A.3.1 or A.3.2, as appropriate~~. The test shall be carried out using the cable which is to be used for connecting both parts. The type, dimensions and other relevant information about the cable which is to be used shall be specified in the manufacturer's documentation. The cable shall be a cord which is listed for extra-hard usage.
26.1	To highlight that the test of an Ex Component must be carried out by a body acceptable to the US authorities, the text has been revised as follows: It is not necessary to repeat the tests that have already been carried out by an accepted source on an Ex component.
26.5.1.1	To acknowledge that the NEC does not recognize surface temperature based on specified dust layers, the test has been deleted as follows: ~~For electrical equipment of Group III evaluated with a dust layer in accordance with 5.3.2.3.2, the equipment to be tested shall be mounted in accordance with the instructions and surrounded on all available surfaces by a dust thickness at least equal to the specified layer depth *L*. The measurement for the maximum surface temperature shall be determined using a test dust having a thermal conductivity of no more than 0,10 W/(m.K) measured at (100 ± 5) ºC.~~
26.5.1.3	To reflect the minimum default upper ambient specified by the NEC, the text has been revised as follows: The result shall be corrected for either 40°C or the maximum ambient temperature specified in the rating, whichever is higher. The measurement of temperatures as prescribed in this standard, and in the specific standards for the types of protection concerned, shall be made in still ambient air, with the electrical equipment mounted in its normal service position.

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Clause	**US Differences to IEC 60079-0, 5th Ed, 2007-10**
26.8	To incorporate the clarification of IS-H 01 to Edition 5 of IEC 60079-0, the text has been revised as follows: **26.8 Thermal endurance to heat** The thermal endurance to heat shall be determined by submitting the enclosures or parts of enclosures in non-metallic materials, on which the integrity of the type of protection depends, to continuous storage for ~~four weeks~~ $672\,^{0}_{+30}$ h at (90 ± 5) % relative humidity at a temperature of (20 ± 2) K above the maximum service temperature, but at least 80 °C. In the case of a maximum service temperature above 75 °C, the period of ~~four weeks~~ $672\,^{0}_{+30}$ h specified above shall be replaced by a period of ~~two weeks~~ $336\,^{0}_{+30}$ h at (95 ± 2) °C and (90 ± 5) % relative humidity followed by a period of ~~two weeks~~ $336\,^{0}_{+30}$ h in an air oven at a temperature of (20 ± 2) K higher than the maximum service temperature.
26.9	To incorporate the clarification of IS-H 01 to Edition 5 of IEC 60079-0, the text has been revised as follows: The thermal endurance to cold shall be determined by submitting the enclosures and parts of enclosures of non-metallic materials, on which the type of protection depends, to storage for $24\,^{0}_{+2}$ h in an ambient temperature corresponding to the minimum service temperature reduced according to 26.7.2.
26.12	To align with the US practice of not permitting insulators to contribute to the security of an electrical connection, the test of 26.12 has been deleted.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
29.2	To align with the NEC marking requirements, the text has been revised as follows: **29.2 General** The marking shall include the following: a) the name of the manufacturer or his registered trade mark; b) the manufacturer's type identification; c) a serial number, except for – connection accessories (cable glands, blanking element, thread adaptor and bushings); – very small electrical equipment on which there is limited space; (The batch number can be considered to be an alternative to the serial number.) d) the name or mark of the certificate issuer and the certificate reference ~~in the following form: the last two figures of the year of the certificate followed by a "." followed by a unique four character reference for the certificate in that year;~~ ~~NOTE 1 For some regional third-party certification, the separating character "." may be replaced by another separating designator such as "ATEX".~~ e) **Identification of Specific Conditions of Use** - Specific installation instructions or reference to a specific installation document when~~if~~ it is necessary to indicate specific conditions of use~~, the symbol "X" shall be placed after the certificate reference. An advisory marking may appear on the equipment as an alternative to the requirement for the "X" marking;~~ NOTE 2 The manufacturer should ensure that the requirements of the specific conditions of use are passed to the purchaser together with any other relevant information. f) the specific Ex marking for explosive gas atmospheres, see 29.3, or for explosive dust atmospheres, see 29.4. The Ex marking for explosive gas atmospheres and explosive dust atmospheres shall be separate and not combined; g) any additional marking prescribed in the specific standards for the types of protection concerned, as in Clause 1. NOTE 3 Additional marking may be required by the applicable industrial safety standards for construction of the electrical equipment. h) flameproof enclosures with a factory-installed conduit sealing device at the field wiring entry, not requiring a field-installed conduit sealing device, shall be permanently marked "Seal Not Required," "Leads Factory Sealed," "Factory Sealed," or the equivalent.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
29.3	To align with the NEC marking requirements, the text has been revised as follows: The marking shall include the following: <u>US-1) Class I</u> <u>US-2) the applicable Zone marking — i.e., Zone 0, Zone 1, or Zone 2;</u> a) the symbol <u>A</u>Ex, which indicates that the electrical apparatus corresponds to one or more of the types of protection which are the subject of the specific standards listed in clause 1; b) the symbol for each type of protection used: – "d": flameproof enclosure, (for EPL Gb or Mb) – "e": increased safety, (for EPL Gb or Mb) – "ia": intrinsic safety, (for EPL Ga or Ma) – "ib": intrinsic safety, (for EPL Gb or Mb) – "ic": intrinsic safety, (for EPL Gc) <u>NOTE The 2008 NEC does not recognize type of protection “ic”</u> – "ma": encapsulation, (for EPL Ga or Ma) – "mb": encapsulation, (for EPL Gb or Mb) *– "mc": encapsulation, (for EPL Gc) – Under Consideration* <u>NOTE The 2008 NEC does not recognize type of protection “mc”</u> – "nA": non-sparking, (for EPL Gc) – "nC": protected sparking, (for EPL Gc) – "nR": restricted breathing, (for EPL Gc) ~~– "nL": energy limited, (for EPL Gc)~~ – "o": oil immersion, (for EPL Gb) – "px": pressurization, (for EPL Gb or Mb) – "py": pressurization, (for EPL Gb) – "pz": pressurization, (for EPL Gc) – "q": powder filling, (for EPL Gb or Mb) ……………………… ~~e) the equipment protection level, “Ga”, “Gb”, “Gc”, “Ma”, or “Mb” as appropriate~~ <u>The equipment protection level, “Ga”, “Gb”, “Gc”, as appropriate, may also be marked on the equipment.</u> <u>NOTE Although the marking identifying the equipment protection level (EPL) may appear on equipment evaluated against this standard, the 2008 NEC does not recognize the concept of employing the equipment protection level in a complete risk assessment of an installation.</u> ………………………

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
29.4	To align with the NEC marking requirements, the text has been revised as follows: US-1) the applicable Zone marking – i.e. Zone 20, Zone 21, or Zone 22 a) the symbol AEx, which indicates that the electrical equipment corresponds to one or more of the types of protection which are the subject of the specific standards listed in Clause 1; b) the symbol for each type of protection used: – "t": protection by enclosure, (for EPL Db) – "ta": protection by enclosure, (for EPL Da) NOTE The 2008 NEC does not recognize type of protection "ta" – "tb": protection by enclosure, (for EPL Db) NOTE The 2008 NEC does not recognize type of protection "tb" – "tc": protection by enclosure, (for EPL Dc) NOTE The 2008 NEC does not recognize type of protection "tc" – "ia": intrinsic safety, (for EPL Da) – "ib": intrinsic safety, (for EPL Db) *– "ic": intrinsic safety, (for EPL Dc) – Under Consideration* NOTE The 2008 NEC does not recognize type of protection "ic" – "ma": encapsulation, (for EPL Da) – "mb": encapsulation, (for EPL Db) *– "mc": encapsulation, (for EPL Dc) – Under Consideration* NOTE The 2008 NEC does not recognize type of protection "mc" – "p": pressurization, (for EPL Db or Dc) c) the symbol of the group: – IIIA, IIIB or IIIC for electrical equipment for places with an explosive dust atmosphere; Alternatively, the "Group IIIA, IIIB, or IIIC is not marked and a "D" is appended to the type of protection shown in b), for example "iaD". NOTE 1 Equipment marked "IIIB" is suitable for applications requiring Group IIIA equipment. Similarly, equipment marked "IIIC" is suitable for applications requiring Group IIIA and Group IIIB equipment. NOTE 2 The 2008 NEC does not recognize the identification of locations or equipment as "Group IIIA, IIIB, or IIIC", but identifies equipment suitable for Zone 20, 21, or 22 by the use of equipment marking where a "D" is appended to the type of protection, for example "iaD" and no separate differentiation is made of combustible dusts or ignitable fibers. d) the maximum surface temperature in degrees Celsius and the unit of measurement °C preceded with the letter "T", (e.g. T 90 °C). ~~Where appropriate according to 5.3.2.3, the maximum surface temperature T_L shall be shown as a temperature value in degrees Celsius and the unit of measurement °C, with the layer depth $_L$ indicated as a subscript in mm, (e.g. T_{500} 320 °C) or marking shall include the symbol "X" to indicate this condition of use according to item e) of 29.2.~~ Where appropriate according to 5.1.1, the marking shall include either the symbol T_a or T_{amb} together with the range of ambient temperature or the symbol "X" to indicate this specific condition of use according to item e) of 29.2. Ex cable glands, Ex blanking elements, and Ex thread adapters need not be marked with a maximum surface temperature; ~~e) the equipment protection level, "Da", "Db", or "Dc", as appropriate;~~ The equipment protection level, "Da", "Db", or "Dc", as appropriate, may also be marked on the equipment. NOTE Although the marking identifying the equipment protection level (EPL) may appear on equipment evaluated against this standard, the 2008 NEC does not recognize the concept of employing the equipment protection level in a complete risk assessment of an installation. f) the degree of protection (e.g. IP54).

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
29.7	To align with the NEC, all of the text of 29.7 has been deleted.
29.8	To align with the NEC marking requirements, the text has been revised as follows: **29.8 Ex components** Ex components, according to Clause 13, shall be legibly marked and the marking shall include the following: a) the name or the registered trade mark of the manufacturer; b) the manufacturer's type identification; c) the symbol AEx; d) the symbol for each type of protection used; e) the symbol of the group of the electrical equipment of the Ex component; f) the name or mark of the issuer of the certificate~~, and the number of the certificate;~~ g) ~~the symbol "U"; and~~ ~~NOTE 1 The symbol "X" is not used.~~ h) the additional marking prescribed in the specific standard for the types of protection concerned, as in Clause 1. NOTE 2 Additional marking may be required by the standards for construction of the electrical equipment. i) As much of the remaining marking information per 29.3 or 29.4, as applicable, as can be accommodated. j) Cl I, Zn 0; Cl I, Zn 1; Cl I, Zn 2; Zn 20; Zn 21; or Zn 22; as appropriate k) II, IIA, IIB, IIC, chemical formula, IIIA, IIIB, or IIIC; as appropriate l) For equipment for explosive gas atmospheres, the symbol indicating the temperature class or the maximum surface temperature in degrees Celsius, or both. When the marking includes both, the temperature class shall be given last in parentheses. m) For equipment for explosive dust atmospheres, the maximum surface temperature in degrees Celsius and the unit of measurement °C preceded with the letter "T", (e.g. T 90 °C). The Ex marking for explosive gas atmospheres and explosive dust atmospheres shall be separate and not combined.
29.9	To align with the NEC marking requirements, the text has been revised as follows: **29.9 Small equipment and small Ex components** On small electrical equipment and on Ex components where there is limited space, a reduction in the marking is permitted. The following lists the minimum marking that is required on the equipment or Ex component: a) the name or registered trademark of the manufacturer; b) the manufacturer's type identification. The type identification is permitted to be abbreviated if the complete type identification is included on the smallest

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	unit package ~~or omitted if the certificate reference allows identification of the specific type;~~ c) the name or mark of the issuer of the certificate, and the number of the certificate; and ~~d) the symbol "X" or "U" (if appropriate);~~ ~~NOTE The symbols "X" and "U" are never used together.~~ e) As much of the remaining marking information per 29.3 or 29.4, as applicable, as can be accommodated. f) Cl I, Zn 0; Cl I, Zn 1; Cl I, Zn 2; Zn 20; Zn 21; or Zn 22; as appropriate g) II, IIA, IIB, IIC, chemical formula, IIIA, IIIB, or IIIC; as appropriate h) For equipment for explosive gas atmospheres, the symbol indicating the temperature class or the maximum surface temperature in degrees Celsius, or both. When the marking includes both, the temperature class shall be given last in parentheses. i) For equipment for explosive dust atmospheres, the maximum surface temperature in degrees Celsius and the unit of measurement °C preceded with the letter "T", (e.g. T 90 °C).
29.10	The examples have been revised to align with the NEC marking requirements as follows: **29.14 Examples of marking** ~~Electrical equipment with type of protection flameproof enclosure "d" (EPL Mb) for use in mines susceptible to firedamp:~~ ~~BEDELLE S.A~~ ~~TYPE A B 5~~ ~~Ex d I Mb alternate Ex db I~~ ~~No. 325~~ ~~ABC 02.1234~~ ~~.................~~ ~~.................~~ Ex component, with type of protection flameproof enclosure "d" (EPL Gb) with intrinsically safe "ia" (EPL Ga) output circuit, for explosive gas atmospheres other than in mines susceptible to firedamp, gas of subdivision C, manufactured by H. RIDSTONE and Co. Ltd. Type KW 369: Class I, Zone 1, AEx d [ia Ga] IIC Gb alternate Class I, Zone 1, AEx db [ia] IIC DEF 02.0536 ~~U~~ ....

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	HR .... Electrical equipment, utilizing types of protection increased safety "e" (EPL Gb) and pressurized enclosure "px" (EPL Gb), maximum surface temperature of 125 °C, for explosive gas atmospheres other than mines susceptible to firedamp, with gas of ignition temperature greater than 125 °C ~~and with specific conditions of use indicated in the certificate~~. H. ATHERINGTON Ltd TYPE 250 JG 1 Class I, Zone 1, AEx e px IIC 125 °C (T4) Gb alternate Class I, Zone 1, AEx eb pxb IIC 125 °C (T4) No. 56732 GHI 02.0076 ~~X~~ .... Electrical equipment, utilizing types of protection flameproof enclosure "d" (EPL Mb and Gb) and increased safety "e" (EPL ~~Mb and~~ Gb) for use in ~~mines susceptible to firedamp and~~ explosive gas atmospheres other than mines susceptible to firedamp with gas of subdivision B and ignition temperature greater than 200 °C. A.R. ACHUTZ A.G. TYPE 5 CD ~~Ex d e I Mb~~ ~~alternate Ex db eb I~~ Class I, Zone 1, AEx d e IIB T3 Gb alternate Class I, Zone 1, AEx db eb IIB T3 No. 5634 JKL 02.0521 .... Electrical equipment with type of protection flameproof enclosure "d" (EPL Gb) for explosive gas atmospheres other than mines susceptible to firedamp on the basis of ammonia gas only. WOKAITERT SARL TYPE NT 3

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	vEx d II (NH_3) Gb alternate Class I, Zone 1, AEx db II (NH_3) No. 6549 MNO 02.3102 .... Electrical equipment with type of protection encapsulation “ma” (EPL Da) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 120 °C. ABC company Type RST Serial No. 123456 Zone 20, AEx ma IIIC T120 °C Da alternate Zone 20, AEx ma IIIC T120 °C IP68 N.A. 01.9999 .... Electrical equipment with type of protection "ia" (EPL Da) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 120 °C. ABC company Type XYZ Serial No. 123456 Zone 20, AEx ia IIIC T120 °C Da alternate Zone 20, AEx ia IIIC T120 °C IP20 N.A. 01.9999 .... Electrical equipment with type of protection "p" (EPL Db) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 120 °C. ABC company Type KLM Serial No. 123456 Zone 20, AEx p IIIC T120 °C Db alternate Zone 20, AEx pb IIIC T120 °C

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	IP65 N.A. 01.9999 .... Electrical equipment with type of protection "t" (EPL Db) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 225 °C ~~and less than 320 °C when tested with a 500 mm dust layer~~. ABC company Type RST Serial No. 987654 Zone 21, AEx tb IIIC T225 °C ~~T_{500} 320 °C~~ Db alternate Zone 21, AEx tb IIIC T225 °C ~~T_{500} 320 °C~~ IP65 N.A. 02.1111 .... Electrical equipment with type of protection "t" (EPL Db) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 175 °C with an extended ambient temperature range of -40 °C to +120 °C. ABC company Type RST Serial No. 987654 Zone 21, AEx tb IIIC T175 °C Db alternate Zone 21, AEx tb IIIC T175 °C IP65 -40ºC ≤ T_{amb} ≤ 120 °C N.A. 02.1111 .... Electrical equipment with type of protection encapsulation "ma" (EPL Ga) for explosive gas atmospheres of Group IIC with a maximum surface temperature of less than 135 °C and with type of protection encapsulation "ma" (EPL Da) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 120 °C. A single certificate has been prepared. ABC company Type RST Serial No. 123456

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Clause	**US Differences to IEC 60079-0, 5th Ed, 2007-10**
	Class I, Zone 0, AEx ma IIC T4 Ga alternate Class I, Zone 0, AEx ma IIC T4 Zone 20, AEx ma IIIC T120 °C Da alternate Zone 20, AEx ma IIIC T120 °C IP67 N.A. 01.9999 .... Electrical equipment with type of protection encapsulation "ma" (EPL Ga) for explosive gas atmospheres of Group IIC with a maximum surface temperature of less than 135 °C and with type of protection encapsulation "ma" (EPL Da) for explosive dust atmospheres containing conductive dusts of Group IIIC with a maximum surface temperature of less than 120 °C. Two independent Certificates have been prepared. ABC company Type RST Serial No. 123456 Class I, Zone 0, AEx ma IIC T4 Ga alternate Class I, Zone 0, AEx ma IIC T4 N.A. 01.1111 Zone 20, AEx ma IIIC T120 °C Da alternate Zone 20, AEx ma IIIC T120 °C IP54 N.B. 01.9999 ....
29.15 NEW	Additional marking requirements for the permitted supplemental external grounding terminal have been added as follows: **29.15 External grounding or bonding terminal** If a supplementary external grounding or bonding terminal is identified by being either coloured green or by being marked "G," "GR," "Ground," "Grounding," "Protective Earth," "PE," or "⏚"; the instructions provided with the equipment shall indicate that the internal grounding terminal shall be used for the equipment grounding connection and that the external terminal is for a supplementary bonding connection where local codes or authorities permit or require such connection.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
29.16 NEW	To align with the NEC permission for alternate marking, the additional permitted markings have been introduced as shown below: **29.16 Class I, Division 2, Equivalency Marking** **29.16.1 Class I, Division 2, Group D** Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIA requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group D* – along with the appropriate temperature class. **29.16.2 Class I, Division 2, Group C** Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIB requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group C* – along with the appropriate temperature class. Equipment marked *Group C* may also be marked *Group C, D* or any combination thereof. **29.16.3 Class I, Division 2, Group B** Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIB + H2 requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group B* – along with the appropriate temperature class. Equipment marked *Group B* may also be marked *Group B, C, D*, or any combination thereof. **29.16.4 Class I, Division 2, Group A** Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIC requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group A* – along with the appropriate temperature class. Equipment marked *Group A* may also be marked *Group A, B, C, D*, or any combination thereof.
29.17 NEW	To align with the NEC permission for alternate marking, the additional permitted markings have been introduced as shown below: **29.17 Class I, Division 1, Equivalency Marking** **29.17.1 Class I, Division 1, Group D** Electrical equipment complying with the Group IIA requirements for types of protection "ia" or "ma", are permitted to additionally be marked *Class I, Division 1, Group D* – along with the appropriate temperature class. **29.17.2 Class I, Division 1, Group C** Electrical equipment complying with the Group IIB requirements for types of protection "ia" or "ma" are permitted to additionally be marked *Class I, Division*

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Clause	**US Differences to IEC 60079-0, 5th Ed, 2007-10**
	1, Group C – along with the appropriate temperature class. Equipment marked *Group C* may also be marked *Group C, D* or any combination thereof. **29.17.3 Class I, Division 1, Group B** Electrical equipment complying with the Group IIB + H2 requirements for types of protection "ia" or "ma" are permitted to additionally be marked *Class I, Division 1, Group B* – along with the appropriate temperature class. Equipment marked *Group B* may also be marked *Group B, C, D*, or any combination thereof. **29.17.4 Class I, Division 1, Group A** Electrical equipment complying with the Group IIC requirements for types of protection "ia" or "ma" are permitted to additionally be marked *Class I, Division 1, Group A* – along with the appropriate temperature class. Equipment marked *Group A* may also be marked *Group A, B, C, D*, or any combination thereof.
29.18 NEW	To permit abbreviated markings where space is limited, new text has been introduced as shown below: **29.18 Abbreviated Markings** On very small electrical apparatus and on components where there is limited space, the following abbreviations are permitted to be used: Class – Cl Division – Div Group – Gr or Gp or Grp
Annex A	To align with the NEC permitted wiring methods, the compelte text of Annex A has been deleted and replace with the text shown below: Cable glands or cord connectors, whether integral or separate, for type of protection "e" or "d" shall comply with ANSI/UL 2225. NOTE 1 The ordinary location requirements for cable glands or cord connectors can be found in UL 514B, "Conduit, Tubing, and Cable Fittings." NOTE 2 The requirements for cable glands or cord connectors, type of protection "i", can be found in UL 514B, "Conduit, Tubing, and Cable Fittings."
Annex D	Add note to clarify the US position on the use of EPLs as follows: NOTE Although the marking identifying the equipment protection level (EPL) may appear on equipment evaluated against this standard, the 2008 NEC does not recognize the concept of employing the equipment protection level in a complete risk assessment of an installation.

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
Annex F	Insert Annex: **Annex F (Informative) – Common standards – Safety requirements for electrical equipment** The following are common standards used to verify conformance with safety requirements for electrical equipment. This list is not comprehensive. ANSI/ISA-61010-1 (82.02.01)-2004 — Electrical Safety Requirements for Measurement, Control, and Laboratory Use – Part 1: General Requirements ISA-82.03-1988 — Safety Standard for Electrical and Electronic Test, Measuring, Controlling, and Related Equipment (Electrical and Electronic Process Measurement and Control Equipment) UL 20 — General-Use Snap Switches UL 50 — Enclosures for Electrical Equipment UL 67 — Panelboards UL 94 — Tests for Flammability of Plastic Materials for Parts in Devices and Appliances UL 153 — Portable Electric Lamps UL 298 — Portable Electric Hand Lamps UL 347 — High Voltage Industrial Control Equipment UL 427 — Refrigerating Units UL 429 — Electrically Operated Valves UL 464 — Audible Signal Appliances UL 486A-B — Wire Connectors UL 486C — Splicing Wire Connectors UL 486E — Equipment Wiring Terminals for Use with Aluminum and/or Copper A Conductors UL 489 — Molded-Case Circuit Breakers and Circuit-Breaker Enclosures UL 498 — Attachment Plugs and Receptacles UL 508 — Industrial Control Equipment UL 512 — Fuseholders UL 514A — Metallic Outlet Boxes UL 514B — Conduit, Tubing, and Cable Fittings UL 514C — Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers UL 746A — Polymeric Materials – Short Term Property Evaluations UL 746B — Polymeric Materials – Long Term Property Evaluations UL 746C — Polymeric Materials – Use in Electrical Equipment Evaluations UL 746D — Polymeric Materials – Fabricated Parts UL 746E — Polymeric Materials – Industrial Laminates, Filament Wound Tubing UL 840 — Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment

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Clause	US Differences to IEC 60079-0, 5th Ed, 2007-10
	UL 864 Control Units and Accessories for Fire Alarm Systems UL 873 Temperature-Indicating and -Regulating Equipment UL 924 Emergency Lighting and Power Equipment UL 943 Ground-Fault Circuit-Interrupters UL 984 Hermetic Refrigerant Motor-Compressors UL 1004 Electric Motors UL 1053 Ground-Fault Sensing and Relaying Equipment UL 1059 Terminal Blocks UL 1097 Double Insulation Systems for Use in Electrical Equipment UL 1104 Marine Navigation Lights UL 1480 Speakers for Fire Protective Signaling Systems UL 1481 Power Supplies for Fire Protective Signaling Systems UL 1562 Transformers, Distribution, Dry-Type – Over 600 Volts UL 1585 Class 2 and Class 3 Transformers UL 1598 Luminaires UL 1598A Supplemental Requirements for Luminaires for Installation on Marine Vessels UL 1638 Visual Signaling Appliances – Private Mode Emergency and General Utility Signaling UL 1682 Plugs, Receptacles, and Cable Connectors of the Pin and Sleeve Type UL 1740 Industrial Robots and Robotic Equipment UL 1950 Safety of Information Technology Equipment, Including Electrical Business Equipment (Second Edition) UL 1971 Signaling Devices for the Hearing Impaired UL 2111 Overheating Protection for Motors UL 2225 Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified)
Annex H	**Annex H (INFORMATIVE) — Equipment Grounding** The following requirements are generally acknowledged as the minimum acceptable provisions for equipment grounding facilities on industrial and process equipment and are based on ANSI/UL 508, Industrial Control Equipment. F.1 Acceptable means for grounding shall be provided as follows: a) Motor control equipment shall be provided with a means of attachment of a terminal for connecting an equipment grounding conductor as specified in Article 430-144 of the National Electric Code, ANSI/NFPA 70-1999. The terminal shall be sized to receive a grounding conductor as specified in Section 250-122 in ANSI/NFPA 70-1999;

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	b) Pendant, cord-connected equipment shall be provided with a terminal for connecting one conductor of a multiple-conductor cord to the enclosure; c) Portable equipment shall be provided with a power-supply cord with a grounding conductor. The grounding conductor shall be connected to the grounding blade of a grounding attachment plug and shall be connected to the frame or enclosure of the equipment. The surface of the insulation on the grounding conductor shall be green with or without one or more yellow stripes; d) A proximity switch, limit switch, and similar end-of-the-line devices shall be provided with a means for mounting all exposed dead metal parts to a metal frame, or shall be provided with a terminal mounted to exposed dead metal, or the equivalent, to receive an equipment grounding conductor; or e) Other equipment requiring grounding shall be provided with a terminal. The grounding means may be in the form of a kit. F.2 A wire binding screw intended for the connection of a field-installed equipment grounding conductor shall have a green colored head and shall be No. 8 or larger. *Exception: A No. 6 screw may be used at a terminal intended only for connection of a No. 14 AWG (2.1 mm²) conductor..* F.3 For wiring device type equipment, the wire binding screw shall have a hexagonal head. The head may or may not be slotted. F.4 A pressure wire connector intended for connection to a field-installed equipment grounding conductor shall be green-colored or plainly identified, such as being marked “G,” “GR,” “GRD,” “GND,” “GND,” “GRND,” “Ground,” “Grounding,” or the like. The ground symbol from IEC Publication 417, Symbol 5019 may be used. F.5 A terminal plate tapped for a wire-binding screw shall be of metal not less than 0.030 inch (0.76 mm) thick for a No. 14 AWG (2.1 mm²) or smaller wire, and not less than 0.050 inch (1.27 mm) thick for a wire larger than No. 14 AWG. There shall be at least two full threads in the plate. *Exception: Two full threads are not required if fewer threads result in a secure connection in which the threads will not strip upon application of a 20 pound-inch (2.3 N·m) tightening torque.* F.6 A terminal plate formed from stock having the required thickness specified in 27.5.12 may have the metal extruded at the tapped hole for the binding screw to provide two full threads. F.7 A wire-binding screw shall thread into metal. F.8 A terminal to which wiring is to be connected shall be a soldering lug or pressure wire connector.

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	Exception: A terminal to which No. 10 AWG (5.3 mm²) or smaller wiring connections are to be made may consist of a clamp or binding screw with a terminal plate having upturned lugs or the equivalent to hold the wire in position. F.9 If leads, wire binding screw, or pressure wire connectors are not provided on the equipment as shipped, the equipment shall be marked stating which pressure wire connector or component terminal kits are acceptable for use with the equipment. A wire connector of the type mentioned in the marking may be installed in the equipment at the factory with instructions, if necessary, to effect proper connection of the conductor. A terminal kit shall carry an identifying marking, wire size, and manufacturer's name or trademark.

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IEC Standard 60079-0 4th Edition 2004 Electrical apparatus for explosive gas atmospheres – Part 0: General requirement			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-0 2005
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-0:07
CH Switzerland	**NO**		EN 60079-0 2004
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-0 2004
DE Germany	**NO**		EN 60079-0 2004
DK Denmark	**NO**		DS/EN 60079-0 2004
FI Finland	**NO**		EN 60079-0 2004
FR France	**NO**		EN 60079-0 2004
GB United Kingdom	**NO**		EN 60079-0 2004
HR Croatia	**NO**		
HU Hungary	**NO**		MESZ EN 60079-0 2004
IN India		**NO**	Adopted as IS /IEC 60079-0:2004
IT Italy	**NO**		EN 60079-0 2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-0:2004
NL The Netherlands	**NO**		EN 60079-0 2004
NO Norway	**NO**		EN 60079-0 2004
NZ New Zealand		**NO**	AS/NZS 60079-0 2005
PL Poland	**NO**		
RO Romania	**NO**		EN 60079-0 2004
RU Russia		**YES**	GOST R 52350.0 2005
SE Sweden	**NO**		SS-EN 60079-0 2004
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-0 2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-0 2005 ANSI/ISA 60079-0 2005
ZA South Africa		**NO**	SANS 60079-0:2004

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed. Please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Countries that have declared adoption of IEC 60079-0 4th Edition 2004 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-0 4th Edition 2004 With National/Group Differences- Differences follow.

CA Canada

Canadian Deviations

1 Scope

(Delete the following reference)

- IEC 62013-1

(Add the following paragraph)

As the requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only, the CAN/CSA-C22.2 No. 60079 series of CSA Standards (based on the corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

2 Normative references

(Add the following)

CSA (Canadian Standards Association)
Where reference is made to CSA publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following publications, and the years shown indicate the latest editions available at the time of printing:

C22.1-06
Canadian Electrical Code, Part 1

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CAN/CSA-M421-00 (R2005)
Use of electricity in mines

22 Supplementary requirements for caplights and handlights
(Delete this clause)

28 Manufacturer's responsibility
(Add the following clause)

28.2A Additional requirements
The following requirements shall not be superseded by any other clause of Standards in the CAN/CSA-C22.2 No. 60079 series:

(a) This standard shall apply to the safety of such equipment designed to be installed and used in accordance with the Rules of the Canadian Electrical Code, Part 1.
(b) It shall be the responsibility of the manufacturer to ensure that the equipment meets the applicable standards for general electrical safety and usage in accordance with the regulatory requirements.

For mines and quarry applications, see also CAN/CSA-M421.

29 Marketing

29.2 General
(Add the following note to Item (c))

Note 1A: The Canadian Electrical Code, Part 1, also permits the symbol "EEx" as an alternative to "Ex"

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US United States of America
DETAILS OF UNITED STATES NATIONAL DIFFERENCES:

Clause	**US Differences to IEC 60079-0, 4th Ed, 2004**
General	The U.S. version of IEC 60079-0 does not include requirements for Group I electrical apparatus.
General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
1	~~This standard does not specify requirements for safety, other than those directly related to the explosion risk.~~
1	This standard is supplemented or modified by the following apparatus standards: ~~IEC 60079-25;~~ ~~IEC 60079-26;~~ ~~IEC 62013-1;~~ ~~IEC 62086-1.~~ IEEE 515, Standard for Testing, Design, Installation, and Maintenance of Electrical Resistance Heat Tracing for Industrial Applications NOTE — Components such as power connections, splices, end terminations, as well as cable entries, fittings, and seals are to be evaluated according to the type of protection employed.
2	ANSI/API RP505, *Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, or Zone 2*

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	ANSI/NFPA 70:2005, National Electrical Code®. ANSI/NFPA 497, Recommended Practice for the Classification of Flammable Liquids, Gases or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas**.** UL 50, *Enclosures for Electrical Equipment* UL 347, *High Voltage Industrial Control Equipment* UL 508, *Industrial Control Equipment* ANSI/UL 746C, Standard for Polymeric Materials – Use in Electrical Equipment Evaluations ~~ISO 178, Plastics - Determination of flexural properties of rigid plastics~~ ~~ISO 262, ISO general purpose metric screw threads - Selected sizes for screws, bolts and nuts~~ ~~ISO 273, Fasteners - Clearance Holes for bolts and screws~~ ~~ISO 286-2, ISO system of limits and fits - Part 2 Tables of standard tolerance grades and limit deviations for holes and shafts~~ ~~ISO 965-1, ISO general purpose metric screw threads – Tolerances - Part 1: Principles and basic data~~ ~~ISO 965-2, ISO general purpose metric screw threads – Tolerances - Part 2:Limits of sizes for general purpose bolt and~~

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	~~nut threads - medium quality~~ ~~ISO 1817, Rubber, vulcanized - Determination of the effect of liquids~~ ~~ISO 4014, Hexagon head bolts - Product grades A and B~~ ~~ISO 4017, Hexagon head screws - Product grades A and B~~ ~~ISO 4026, Hexagon socket set screws with flat point~~ ~~ISO 4027, Hexagon socket set screws with cone point~~ ~~ISO 4028, Hexagon socket set screws with dog point~~ ~~ISO 4029, Hexagon socket set screws with cup point~~ ~~ISO 4032, Hexagon nuts, style 1 - Product grades A and B~~ ~~ISO 4762, Hexagon socket head cap screws - Product grade A~~
3.14	**3.14** **Ex component** part of electrical apparatus or a module (other than an Ex cable gland), ~~marked with the symbol "U",~~ which is not intended to be used alone and requires additional consideration when incorporated into electrical apparatus or systems for use in explosive gas atmospheres.

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Clause	**US Differences to IEC 60079-0, 4th Ed, 2004**
3.23	Definition Deleted.
3.24	Definition Deleted.
4.2.1	For the types of protection "d", "i", and “nC”~~, and “nL”~~, electrical apparatus of Group II is subdivided into IIA, IIB and IIC, as required in the specific standards concerning these types of protection.
5.1.1	Electrical apparatus designed for use in a different range of ambient temperatures is considered to be special, and the ambient temperature range shall then be stated by the manufacturer. The marking shall then include either the symbol T_a or T_{amb} together with the special range of ambient temperatures ~~or, if this is impracticable, the symbol “X” shall be used to indicate special conditions of use that include a special range of ambient temperature~~. See ~~item i) of 29.2 and~~ Table 1.
5.1.1	In Table 1, Delete “or the symbol X”
6.1.b	b) ~~be constructed in accordance with the applicable safety requirements of the relevant industrial standards.~~ comply with the applicable requirements for similar apparatus for use in unclassified locations. NOTE 2 ~~It is not a requirement of this standard that a certification body check compliance with this requirement. The manufacturer should indicate compliance by marking the apparatus or component in accordance with Clause 29 (and by stating the basis of compliance in the documentation, see Clause 28).~~ Requirements for safety of electrical equipment in ordinary (unclassified) locations can be found in ANSI Standards, NEMA Standards, Federal Regulations, etc. Apparatus listed by a Nationally Recognized Testing Laboratory is considered to meet the applicable requirements found in these standards. A list of commonly applied standards is shown in informative Annex D.

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7.1.1	The requirements given in this clause and in 26.7 shall apply to non-metallic enclosures and non-metallic parts of enclosures, on which the type of protection depends. ~~However, for sealing rings (see 3.5.3) on which the type of protection depends, the proof furnished according to A.3.3 shall be sufficient.~~
7.2	Add text: <u>As an alternative to the TI, the relative thermal index (RTI-mechanical impact) may be determined in accordance with UL 746B.</u>
Table 4	Revise text in Header: **Zone** ~~(as defined in IEC 60079-10)~~
7.3.2 d)	Revise text: For hand-held apparatus only, the inability to store a dangerous charge <u>as verified</u> by measurement of capacitance when tested in accordance with the test method in 26.15; or
7.3.2 e)	Revise text: For electrical apparatus intended for fixed installations, the precautions to avoid risk from electrostatic discharge may form part of the intended installation or be a feature of the process in which the apparatus is mounted. In this case, the

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	apparatus shall be marked ~~"X"~~ in accordance with item i) of 29.2 and the documentation shall indicate all the necessary information to ensure the installation minimizes the risk from electrostatic discharge. Where practicable, the apparatus shall also be marked with the electrostatic charge warning given in item g) of 29.8.
8.1.2	Revise text: Materials used in the construction of enclosures of Group II electrical apparatus for the different zones ~~(as defined in IEC 60079-10)~~ shall not contain, by mass, more than: Where the above compositions are exceeded, the apparatus shall be marked ~~with an "X"~~ in accordance with item i) of 29.2 and the special conditions for safe use shall contain sufficient information to enable the user to determine the suitability of the apparatus for the particular application, for example, to avoid an ignition hazard due to impact or friction. NOTE Caution should be observed when copper or copper alloys are used for enclosures in atmospheres containing acetylene due to the potential formation of acetylides on the surface that can be ignited by friction or impact. The risk of ignition can be reduced by coating the copper or copper alloy with tin, nickel, or by other coatings, or by limiting the maximum copper content of the alloy to 30 percent.
9.2	Text and Note Deleted.
9.3	Text Deleted.
9.3.1	Text Deleted.

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
9.3.2	Text Deleted.
9.3.3	Text Deleted.
15.2	Text and Note Deleted.
15.3	Where there is no requirement for earthing or bonding, for example, in some types of electrical apparatus having double or reinforced insulation, or for which supplementary earthing is not necessary, an internal ~~or external~~ earthing or bonding facility need not be provided.
15.4	15.4 Earthing ~~or equipotential bonding~~ (equipment grounding) conductor connection facilities provided inside terminal compartments, or supplemental grounding or bonding terminals provided on the outside of enclosures, shall allow for the effective connection of at least one conductor with a cross-sectional area ~~given in table 5~~ as given in ANSI/NFPA-70, National Electrical Code, Section 250-122 . **[Table 5 Deleted]** When provided, the supplemental ~~In addition,~~ earthing or bonding connection facilities on the outside of electrical apparatus shall provide effective connection of a conductor with a cross-sectional area of at least 4 mm^2 (10 AWG). Effective connection shall be verified by compliance with the pullout and secureness tests in UL 486E. The terminal shall be marked in accordance with 29.11. NOTE The requirements for equipment grounding facilities and bonding are specified in the appropriate standard for electrical equipment for use in unclassified locations (see Clause 1) and may be more stringent than the requirements

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	shown in 15.4. Refer to Annex F for additional information. .
15.6	Connection facilities shall be designed so that the electrical conductors cannot be readily loosened or twisted. Contact pressure on the electrical connections shall be maintained and not be affected by dimensional changes of insulating materials in service, due to factors such as temperature or humidity. ~~For non-metallic walled enclosures provided with an internal earth continuity plate, the test of 26.12 shall be applied.~~ ~~NOTE The material and dimensions of the earth continuity plate should be appropriate for the anticipated fault current.~~
16.2	The manufacturer shall specify, in the documents submitted according to Clause 24, the entries, their position on the apparatus and the maximum number permitted. The thread form (for example, metric or NPT) of threaded entries shall be marked on the apparatus ~~or shall appear in the installation instructions (See Clause 30)~~. ~~NOTE It is not intended that individual entries be marked, unless required by the specific type of protection.~~
16.3	Revise last paragraph and notes: Cable glands, whether integral or separate, shall meet the relevant requirements of the specific type of protection ~~Annex A~~. NOTE 1 The ordinary location requirements for cable glands can be found in UL 514B, “Conduit, Tubing, and Cable Fittings.” NOTE 2 The requirements for cable glands, type of protection “i”, can be found in UL 514B, “Conduit, Tubing, and Cable Fittings.”

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	NOTE 3 The requirements for cable glands, type of protection "d" or "e", can be found in UL 2225, "Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified) Locations."
16.5	When the temperature under rated conditions is higher than 60 °C ~~70 °C~~ at the entry point or 60 °C ~~80 °C~~ at the branching point of the conductors, the electrical apparatus shall be appropriately marked to provide guidance to the user on the proper selection of cable gland and cable or conductors.
18.2	Where switchgear includes a disconnector, the latter shall disconnect all poles and shall be designed so that the position of the disconnector contacts is visible, or their open position is reliably indicated; see ~~IEC 60947-1~~ ANSI/UL 508 or ANSI/UL 347, as applicable. Any interlock between such disconnector and the cover or door of the switchgear shall allow this cover or door to be opened only when the separation of the disconnector contacts is effective.
20.1	Plugs and sockets shall be ~~either~~ a) interlocked mechanically, or electrically, or otherwise designed so that they cannot be separated when the contacts are energized and the contacts cannot be energized when the plug and socket are separated~~, or~~. b) ~~fixed together by means of special fasteners according to 9.2 and the apparatus marked with the separation marking as required by item d) of 29.8.~~
22.2	Leakage of the electrolyte shall be prevented in all positions of the apparatus. Where the source of light and the source of supply are housed in separate enclosures, which are not

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	mechanically connected other than by an electric cable, the cable glands and the connecting cable shall be tested according to the specific type of protection ~~A.3.1 or A.3.2 as appropriate~~. The test shall be carried out using the cable which is to be used for connecting both parts. The type, dimensions and other relevant information about the cable which is to be used shall be specified in the manufacturer's documentation. The cable shall be a cord which is listed for extra-hard usage.
24	The manufacturer shall prepare documents that give a full and correct specification of the ~~explosion~~ safety aspects of the electrical apparatus.
26.4.2	Revise text: When an electrical apparatus is submitted to tests corresponding to the low risk of mechanical danger, it shall be marked ~~with the symbol "X" to indicate this special condition of use~~ in accordance with item i) of 29.2.
26.4.5.2	Add text: When enclosures are marked with enclosure-type designations, the tests shall be in accordance with ANSI/UL 50, "Enclosures for Electrical Equipment."
26.5.1	Revise Text: ….. For electrical apparatus which can normally be used in different positions, the temperature in each position shall be determined and the highest temperature shall be considered. When the temperature is determined for certain positions only, the electrical apparatus shall be marked ~~with the symbol "X"~~ to indicate this special condition of use according to item i) of 29.2.

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
26.10.1	Revise text: If the apparatus is protected from light (for example, daylight or light from luminaires) when installed, and, in consequence, the test is not carried out, the apparatus shall be marked ~~by the symbol "X" to indicate this special condition of use~~ according to item i) of 29.2.
26.10.2	Revise text: The test is to be conducted as follows unless the material is otherwise evaluated in accordance with the resistance to ultraviolet light tests of UL 746C. The test shall be made on six test bars of standard size 50 mm × 6 mm × 4 mm according to ISO 179. The test bars shall be made under the same conditions as those used for the manufacture of the enclosure concerned; these conditions are to be stated in the test report of the electrical apparatus. The test shall be made in accordance with ISO 4892-1 in an exposure chamber using a xenon lamp and a sunlight simulating filter system, at a black panel temperature of (55 ± 3) °C. The exposure time shall be 1,000 h. Where preparations of test samples in accordance with ISO 179 are not practical due to the nature of the non-metallic material, an alternative test shall be permitted with the justification stated in the test report for the electrical apparatus.
26.12	Text Deleted
27	Revise text: The manufacturer shall carry out ~~the routine verifications and tests necessary to ensure that the electrical apparatus produced complies with the documentation. The manufacturer shall also carry out~~ any routine verifications and tests required by any of the standards listed in Clause 1 which were used for the examination and testing of the apparatus.

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29.2	The marking shall include the following: a) the name of the manufacturer or his registered trade mark; b) the manufacturer's type identification b1) Class I b2) the applicable Zone marking — i.e., Zone 0, Zone 1, or Zone 2; c) the symbol AEx, which indicates that the electrical apparatus corresponds to one or more of the types of protection which are the subject of the specific standards listed in clause 1; d) the symbol for each type of protection used: – "d": flameproof enclosure – "e": increased safety – "ia": intrinsic safety, level of protection "ia" – "ib": intrinsic safety, level of protection "ib" – "m": encapsulation ~~– "ma": encapsulation, level of protection "ma"~~ ~~– "mb": encapsulation, level of protection "mb"~~ – "nA": Type n, method of protection "nA" – "nC": Type n, method of protection "nC" ~~– "nL": Type n, method of protection "nL"~~

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	– "nR": Type n, method of protection "nR" – "o": oil immersion – "px": pressurization, level of protection "px" – "py": pressurization, level of protection "py" – "pz": pressurization, level of protection "pz" – "q": powder filling. For associated apparatus suitable for installation in a hazardous area, the symbols for the type of protection shall be enclosed within square brackets, for example, AEx d[ia] IIC T4. For associated apparatus not suitable for installation in a hazardous area, both the symbol Ex and the symbol for the type of protection shall be enclosed within the same square brackets, for example, [AEx ia] IIC. NOTE ~~1~~ For associated apparatus not suitable for installation in a hazardous area, a temperature class is not included. ~~NOTE 2 Electrical apparatus which does not fully comply with this standard and other relevant parts of IEC 60079 where equivalent safety is claimed, should be marked with the symbol "s".~~ e) the symbol of the group: ~~– I for electrical apparatus for mines susceptible to firedamp;~~ – II, IIA, IIB, or IIC for electrical apparatus for places with an explosive gas atmosphere ~~other than mines susceptible to firedamp.~~ The letters A, B, C shall be used if the specific standard for the type of protection concerned requires this, or if required for compliance with 6.3, 7.3.2 b), 7.3.2 c), 7.3.2 d), or 7.3.2 e). When the electrical apparatus is for use only in a particular gas, the symbol II shall be followed by the chemical

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	formula or the name of the gas in parentheses. When the electrical apparatus is for use in a particular gas in addition to being suitable for use in a specific group of electrical apparatus, the chemical formula shall follow the group and be separated with the symbol "+", for example, IIB + H_2. NOTE 3 Apparatus marked "IIB" is suitable for applications requiring Group IIA apparatus. Similarly, apparatus marked "IIC" is suitable for applications requiring Group IIA and Group IIB apparatus. f) for Group II electrical apparatus, the symbol indicating the temperature class. Where the manufacturer wishes to specify a maximum surface temperature between two temperature classes, he may do so by marking that maximum surface temperature in degrees Celsius alone, or by marking both that maximum surface temperature in degrees Celsius and, in parentheses, the next highest temperature class, for example, T1 or 350 °C or 350 °C (T1). Group II electrical apparatus, having a maximum surface temperature greater than 450 °C, shall be marked only with the maximum surface temperature in degrees Celsius, for example, 600 °C. Group II electrical apparatus, marked for use in a particular gas, need not have a temperature class or maximum surface temperature marking. Where appropriate according to 5.1.1, the marking shall include either the symbol T_a or T_{amb} together with the range of ambient temperature ~~or the symbol "X" to indicate this special condition of use according to item i) of 29.2~~. Cable glands need not be marked with a temperature class or maximum surface temperature in degrees Celsius. g) a serial number, except for – connection accessories (cable and conduit entries, blanking plates, adaptor plates, and bushings); very small electrical apparatus on which there is limited space; (The batch number can be considered to be an alternative to the serial number.) h) the name or mark of the certificate issuer and the certificate reference~~ in the following form: the last two figures of the~~

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	~~year of the certificate followed by the serial number of the certificate in that year~~; i) specific installation instructions or reference to a specific installation document when ~~if~~ it is necessary to indicate special conditions for safe use~~, the symbol "X" shall be placed after the certificate reference. A warning marking may be marked on the apparatus as an alternative to the requirement for the "X" marking~~;. NOTE 4 The manufacturer should ensure that the requirements of the special conditions for safe use are passed to the purchaser together with any other relevant information. j) Any additional marking prescribed in the specific standards for the types of protection concerned, as in Clause 1. NOTE 5 Additional marking may be required by the applicable industrial safety standards for construction of the electrical apparatus. k) flameproof enclosures with a factory-installed lead seal, not requiring a field-installed lead seal, shall be permanently marked "Seal not Required," "Leads Factory Seals," "Factory Sealed," or the equivalent.
29.5	Ex components according to Clause 13 shall be legibly marked and the marking shall include the following: the name or the registered trade mark of the manufacturer; the manufacturer's type identification; the symbol AEx; the symbol for each type of protection used; the symbol of the group of the electrical apparatus of the Ex component; the name or mark of the issuer of the certificate~~, and the number of the certificate, if the Ex component has one to indicate it complies with this standard;~~ ~~the symbol "U"; and~~

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	~~NOTE 1 The symbol “X” is not used.~~ the additional marking prescribed in the specific standard for the types of protection concerned, as in Clause 1. NOTE ~~2~~ Additional marking may be required by the standards for construction of the electrical apparatus. Cl I Zn 0, Zn 1, or Zn 2; as appropriate II, IIA, IIB, IIC, or chemical formula; as appropriate The symbol indicating the temperature class or the maximum surface temperature in degrees Celsius, or both. When the marking includes both, the temperature class shall be given last in parentheses.
29.6	On small electrical apparatus and on Ex components where there is limited space, a reduction in the marking is permitted. The following lists the minimum marking that is required on the apparatus or Ex component: the name or registered trademark of the manufacturer; The manufacturer’s type identification. The type identification is permitted to be abbreviated or omitted if the certificate reference allows identification of the specific type; the symbol AEx and the symbol of each type of protection; the name or mark of the issuer of the certificate~~, and the number of the certificate, if the Ex Component has one to indicate it complies with this standard; and~~ ~~the symbol “X” or “U” (if appropriate)~~ ~~NOTE The symbols “X” and “U” are never used together.~~ Cl I

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	Zn 0, Zn 1, or Zn 2; as appropriate II, IIA, IIB, IIC, or chemical formula; as appropriate the symbol indicating the temperature class or the maximum surface temperature in degrees Celsius, or both. When the marking includes both, the temperature class shall be given last in parentheses.
29.10	~~Flameproof electrical apparatus for use in mines susceptible to firedamp:~~ ~~BEDELLE S.A~~ ~~TYPE A B 5~~ ~~Ex d I~~ ~~No. 325~~ ~~ABC 02.12345~~ Ex component, flameproof with intrinsically safe output circuit, for places in explosive gas atmospheres other than in mines susceptible to firedamp, gas of subdivision C, manufactured by H. RIDSTONE and Co. Ltd. Type KW 369: Class I, Zone 1, AEx d[ia] IIC DEF 02.536 ~~U~~ HR

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	Electrical apparatus, utilizing type~~s~~ of protection increased safety ~~and pressurized enclosure “px”~~, maximum surface temperature of 125 °C, for explosive gas atmospheres other than mines susceptible to firedamp, with gas of ignition temperature greater than 125 °C ~~and with special conditions for safe use indicated in the certificate~~. H. ATHERINGTON Ltd TYPE 250 JG 1 Class I, Zone 1, AEx e~~px~~ II 125 °C (T4) No. 56732 GHI 02.076 ~~X~~ . Electrical apparatus, utilizing flameproof enclosure and increased safety types of protection for use in ~~mines susceptible to firedamp and~~ explosive gas atmospheres other than mines susceptible to firedamp with gas of subdivision B and ignition temperature greater than 200 °C. A.R. ACHUTZ A.G. TYPE 5 CD ~~Ex de I~~ Class I, Zone 1, AEx de IIB T3 No. 5634 JKL 02.521 Flameproof electrical apparatus for explosive gas atmospheres other than mines susceptible to firedamp on the basis of ammonia gas only. WOKAITERT SARL

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Clause	**US Differences to IEC 60079-0, 4th Ed, 2004**
	TYPE NT 3 Class I, Zone 1, AEx d II (NH3) No. 6549 MNO 02.3102 .
29.11 NEW	If a supplementary external grounding or bonding terminal is identified by being either coloured green or by being marked “G”,”GR”,”Ground”,”Grounding”, “Protective Earth”, “PE”, or “⏚”; the instructions provided with the equipment shall indicate that the internal grounding terminal shall be used for the equipment grounding connection and that the external terminal is for a supplementary bonding connection where local codes or authorities permit or require such connection.
29.12 NEW	29.12 Class I, Division 2 Equivalency Marking 29.12.1 Class I, Division 2, Group D Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIA requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group D* – along with the appropriate temperature class. 29.12.2 Class I, Division 2, Group C Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIB requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group C* – along with the appropriate temperature class. Equipment marked *Group C* may also be marked *Group C, D* or any combination thereof.

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	29.12.3 Class I, Division 2, Group B Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIB + H2 requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group B* – along with the appropriate temperature class. Equipment marked *Group B* may also be marked *Group B, C, D*, or any combination thereof. 29.12.4 Class I, Division 2, Group A Electrical equipment complying with all applicable Class I, Zone 0, Zone 1, or Zone 2, Group IIC requirements for any of the Class I, Zone 0, Zone 1, or Zone 2 types of protection are permitted to additionally be marked *Class I, Division 2, Group A* – along with the appropriate temperature class. Equipment marked *Group A* may also be marked *Group A, B, C, D*, or any combination thereof. 29.12.5 Abbreviated Markings On very small electrical apparatus and on components where there is limited space, the following abbreviations are permitted to be used: *Class – Cl* *Division – Div* *Group – Gp or Grp*

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
Annex A	Text Deleted
Annex B	Delete references to 9.2, 9.3, & 15.2.
Annex D	Insert Annex: **Annex D (Informative) – Common standards – Safety requirements for electrical equipment** The following are common standards used to verify conformance with safety requirements for electrical equipment. This list is not comprehensive. ANSI/ISA-61010-1 (82.02.01)-2004 — Electrical Safety Requirements for Measurement, Control, and Laboratory Use – Part 1: General Requirements ISA-82.03-1988 — Safety Standard for Electrical and Electronic Test, Measuring, Controlling, and Related Equipment (Electrical and Electronic Process Measurement and Control Equipment) UL 20 — General-Use Snap Switches UL 50 — Enclosures for Electrical Equipment UL 67 — Panelboards UL 94 — Tests for Flammability of Plastic Materials for Parts in Devices and Appliances UL 153 — Portable Electric Lamps UL 298 — Portable Electric Hand Lamps UL 347 — High Voltage Industrial Control Equipment UL 427 — Refrigerating Units

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	UL 429	Electrically Operated Valves
	UL 464	Audible Signal Appliances
	UL 486A-B	Wire Connectors
	UL 486C	Splicing Wire Connectors
	UL 486E	Equipment Wiring Terminals for Use with Aluminum and/or Copper A Conductors
	UL 489	Molded-Case Circuit Breakers and Circuit-Breaker Enclosures
	UL 498	Attachment Plugs and Receptacles
	UL 508	Industrial Control Equipment
	UL 512	Fuseholders
	UL 514A	Metallic Outlet Boxes
	UL 514B	Conduit, Tubing, and Cable Fittings
	UL 514C	Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
	UL 746A	Polymeric Materials – Short Term Property Evaluations
	UL 746B	Polymeric Materials – Long Term Property Evaluations
	UL 746C	Polymeric Materials – Use in Electrical Equipment Evaluations
	UL 746D	Polymeric Materials – Fabricated Parts
	UL 746E	Polymeric Materials – Industrial Laminates, Filament Wound Tubing
	UL 840	Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment
	UL 864	Control Units and Accessories for Fire Alarm Systems
	UL 873	Temperature-Indicating and -Regulating Equipment
	UL 924	Emergency Lighting and Power Equipment
	UL 943	Ground-Fault Circuit-Interrupters
	UL 984	Hermetic Refrigerant Motor-Compressors
	UL 1004	Electric Motors
	UL 1053	Ground-Fault Sensing and Relaying Equipment
	UL 1059	Terminal Blocks
	UL 1097	Double Insulation Systems for Use in Electrical Equipment
	UL 1104	Marine Navigation Lights

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	UL 1480 Speakers for Fire Protective Signaling Systems UL 1481 Power Supplies for Fire Protective Signaling Systems UL 1562 Transformers, Distribution, Dry-Type – Over 600 Volts UL 1585 Class 2 and Class 3 Transformers UL 1598 Luminaires UL 1598A Supplemental Requirements for Luminaires for Installation on Marine Vessels UL 1638 Visual Signaling Appliances – Private Mode Emergency and General Utility Signaling UL 1682 Plugs, Receptacles, and Cable Connectors of the Pin and Sleeve Type UL 1740 Industrial Robots and Robotic Equipment UL 1950 Safety of Information Technology Equipment, Including Electrical Business Equipment (Second Edition) UL 1971 Signaling Devices for the Hearing Impaired UL 2111 Overheating Protection for Motors UL 2225 Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified)
Annex F	**Annex F (INFORMATIVE) — Equipment Grounding** The following requirements are generally acknowledged as the minimum acceptable provisions for equipment grounding facilities on industrial and process equipment and are based on ANSI/UL 508, Industrial Control Equipment. F.1 Acceptable means for grounding shall be provided as follows: b) Motor control equipment shall be provided with a means of attachment of a terminal for connecting an equipment grounding conductor as specified in Article 430-144 of the National Electric Code, ANSI/NFPA 70-

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	1999. The terminal shall be sized to receive a grounding conductor as specified in Section 250-122 in ANSI/NFPA 70-1999; b) Pendant, cord-connected equipment shall be provided with a terminal for connecting one conductor of a multiple-conductor cord to the enclosure; c) Portable equipment shall be provided with a power-supply cord with a grounding conductor. The grounding conductor shall be connected to the grounding blade of a grounding attachment plug and shall be connected to the frame or enclosure of the equipment. The surface of the insulation on the grounding conductor shall be green with or without one or more yellow stripes; d) A proximity switch, limit switch, and similar end-of-the-line devices shall be provided with a means for mounting all exposed dead metal parts to a metal frame, or shall be provided with a terminal mounted to exposed dead metal, or the equivalent, to receive an equipment grounding conductor; or e) Other equipment requiring grounding shall be provided with a terminal. The grounding means may be in the form of a kit. F.2 A wire binding screw intended for the connection of a field-installed equipment grounding conductor shall have a green colored head and shall be No. 8 or larger. *Exception: A No. 6 screw may be used at a terminal intended only for connection of a No. 14 AWG (2.1 mm²) conductor..* F.3 For wiring device type equipment, the wire binding screw shall have a hexagonal head. The head may or may not be slotted.

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Clause	US Differences to IEC 60079-0, 4th Ed, 2004
	F.4 A pressure wire connector intended for connection to a field-installed equipment grounding conductor shall be green-colored or plainly identified, such as being marked “G,” “GR,” “GRD,” “GND,” “GND,” “GRND,” “Ground,” “Grounding,” or the like. The ground symbol from IEC Publication 417, Symbol 5019 may be used. F.5 A terminal plate tapped for a wire-binding screw shall be of metal not less than 0.030 inch (0.76 mm) thick for a No. 14 AWG (2.1 mm²) or smaller wire, and not less than 0.050 inch (1.27 mm) thick for a wire larger than No. 14 AWG. There shall be at least two full threads in the plate. *Exception: Two full threads are not required if fewer threads result in a secure connection in which the threads will not strip upon application of a 20 pound-inch (2.3 N·m) tightening torque.* F.6 A terminal plate formed from stock having the required thickness specified in 27.5.12 may have the metal extruded at the tapped hole for the binding screw to provide two full threads. F.7 A wire-binding screw shall thread into metal. F.8 A terminal to which wiring is to be connected shall be a soldering lug or pressure wire connector. *Exception: A terminal to which No. 10 AWG (5.3 mm²) or smaller wiring connections are to be made may consist of a clamp or binding screw with a terminal plate having upturned lugs or the equivalent to hold the wire in position.* F.9 If leads, wire binding screw, or pressure wire connectors are not provided on the equipment as shipped, the equipment shall be marked stating which pressure wire connector or component terminal kits are acceptable for use with the equipment. A wire connector of the type mentioned in the marking may be installed in the equipment at the factory with instructions, if necessary, to effect proper connection of the conductor. A terminal kit shall carry an identifying marking, wire size, and manufacturer’s name or trademark.

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National Differences

IEC Standard 60079-0, Consolidated edition 3.1, 2000 Electrical apparatus for explosive gas atmospheres Part 0: General requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-0 2000
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-E60079-0-2001
CH Switzerland*	**TBA**		
CN China*		**TBA**	
CZ Czech Republic*	**TBA**		
DE Germany*	**TBA**		
DK Denmark*	**TBA**		
FI Finland*	**TBA**		
FR France*	**TBA**		
GB United Kingdom*	**TBA**		
HR Croatia*	**TBA**		
HU Hungary*	**TBA**		
IN India		**NO**	Adopted as IS 13346:2004/IEC 60079-0:2000
IT Italy*	**TBA**		
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labour Ministry Notice1998-65
MY Malaysia		**NO**	MS IEC 60079-0:2003 (ALREADY WITHDRAWN)
NL The Netherlands*	**TBA**		
NO Norway*	**TBA**		
NZ New Zealand		**NO**	AS/NZS 60079-0 2000
PL Poland*	**TBA**		
RO Romania*	**TBA**		
RU Russia		**N/R**	
SE Sweden*	**TBA**		
SG Singapore		**NO**	
SI Slovenia*	**TBA**		
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-0:2000

(N/R Not recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details

Section 3
National Differences

Countries that have declared adoption of IEC 60079-0 Consolidated edition 3.1, 2000 Without National Differences
AU Australia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-0 Consolidated edition 3.1, 2000 With National/Group Differences- Differences follow.

CA Canada

Clause 1 - Scope
[Add the following text]

As the requirements of IEC 60079 Series Standards cover the protection techniques with respect to explosion hazard only, the CAN/CSA-E60079 Series of CSA Standards (based on the corresponding IEC Standards) must be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

[Delete the following reference]

"IEC 60079-22: caplights for mines susceptible to firedamp (under consideration)".

Clause 2 - Normative references
[Add the following CSA Standard list following the IEC/ISO list]

CSA International Publications
Where reference is made to CSA Publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following Standards, and the years shown indicate the latest editions available at the time of printing:

CSA Standards
C22.1-98,
Canadian Electrical Code, Part I;

CAN/CSA-C22.2 No. 0-M91 (C1997),
General Requirements — Canadian Electrical Code, Part II;

CAN/CSA-M421-93,
Use of Electricity in Mines;

Clause 22 - Supplementary requirements for caplights, caplamps and handlamps
[Delete this Clause]

Clause 25 - Manufacturer's responsibility
[Add the following text (which is not to be superseded by any other Clause of Standards in this Series)]

This standard applies to the safety of such equipment designed to be installed and used in accordance with the rules of the *Canadian Electrical Code (CEC), Part I*. It is the responsibility of the manufacturer to ensure that the equipment meets the applicable standards for general electrical safety and usage in accordance with the regulatory requirements.

For mines and quarry applications, see also CSA Standard CAN/CSA-M421.

Clause 26 - Verification and tests on modified or repaired electrical apparatus
[Add the following text]

Modifications, repairs and re-builds of electrical equipment shall be governed by Section 2 of the Canadian Electrical Code (CEC) - Part I.

Sub-Clause 27.2 (c) - Markings
[Add the following text]

Note: CEC, Part I also permits the symbol “EEx” as an option to “Ex”.

KR Korea

National differences are the same as IEC Publication 60079-0:1998 except for adding subclause 8.2

Section 3
National Differences

IEC Standard 60079-0, 3rd edition, 1998 and Amd 1 2000 04 Electrical apparatus for explosive gas atmospheres Part 0: General requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-0 2000
BR Brazil		**N/R**	
CA Canada		**Refer to IEC 60079-0:2000-06 Edition 3.1**	
CH Switzerland	**YES**		EN 50014 1992
CN China		**YES**	GB3836.1 2000
CZ Czech Republic	**YES**		ČSN EN 50014:1995
DE Germany	**YES**		EN 50014 1992
DK Denmark	**YES**		EN 50014 1992
FI Finland	**YES**		EN 50014 1992
FR France	**YES**		EN 50014 1992
GB United Kingdom	**YES**		EN 50014 1992
HR Croatia	**YES**		
HU Hungary	**YES**		EN 50014 1992
IN India		**N/R**	
IT Italy	**YES**		EN 50014 1992
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 50014 1992
NO Norway	**YES**		EN 50014 1992
NZ New Zealand		**NO**	AS/NZS 60079-0 2000
PL Poland	**YES**		
RO Romania	**YES**		EN 50014 1992
RU Russia		**YES**	GOST R 51330.0-99
SE Sweden	**YES**		EN 50014 1992
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 50014 1992
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-0:1998

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details

Section 3
National Differences

Countries that have declared adoption of IEC 60079-0 3rd edition, 1998 and Amd 1 2000 04 Without National Differences
AU Australia
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-0 3rd edition, 1998 and Amd 1 2000 04 With National/Group Differences - Differences follow.

CN China

IEC Clause or Annex Number **Details of Differences IEC 60079-0, 3rd edition, 1998**

6.1 Add the following extra requirement to the note of this Clause:

The specific testing conditions of humidity protection for apparatus of Group I are specifically prescribed in Annex C (normative) of GB3836.1-2000, including the temperature, humidity, testing period and testing method.

Add the following extra requirement to this Clause:

7.4 The plastics materials for enclosure Group I shall have characteristics against flame propagation. The details requirements, including test method, technical specifications, are described in Annex E (normative) of GB3836.1-2000. The plastics materials for enclosure Group I shall have characteristics against flame propagation
.

8.3 Replace the Clause with the following:

Enclosures of electrical hand drilling machine (include adherent plugs and sockets), measuring instruments carried by persons and luminaries for Group I may be made of light metals which have tensile strength no less than 120 MPa and accordance with GB13813.

Annex A of IEC 60079-0:1998 has been changed as Annex B of GB3836.1-2000.

Annex B of IEC 60079-0:1998 has been changed as Annex D of GB3836.1-2000.

Annex C of IEC 60079-0:1998 has been changed as Annex F of GB3836.1-2000.

Annex D of IEC 60079-0:1998 has been changed as Annex G of GB3836.1-2000.

Section 3
National Differences

Add Annex A   Procedure of testing. The annex describe requirements of documents,

prototype and some management stipulations of testing and certification in detail.

Add Annex C (normative) Requirement for humidity protection for Group I apparatus.

Add Annex E (normative) Specific requirement for plastic enclosure of electrical
Apparatus Group I.

KR Korea

IEC Clause
Or Annex
Number
Details of Differences IEC 60079-0, 3rd edition, 1998

	The requirements for Group 1 do not apply.
	This requirement does not apply.
3.25	This requirement does not apply.
7.1.2	The requirement of d) does not apply.
7.2	The requirement of the first paragraph does not apply.
13	This requirement does not apply.
23.4.7.5	This requirement does not apply.
27.7.1	This requirement does not apply.
27.7.2	This requirement does not apply.
27.7.4	This requirement does not apply.

ANEX C These requirements do not apply.

RU Russia

IEC Clause Or Annex Number	Details of Differences IEC 60079-0, 3rd edition, 1998

1 Supplemented with:
- Special explosion protection "s".

3.1. Replaced by:
Explosion-proof electrical equipment: To GOST 18311.

3.10.2 Supplemented with:
3.10.2 Level of explosion protection of electrical equipment.

Level of explosion protection of electrical equipment under the conditions specified in normative documents.

3.10.3 Electrical equipment with increased explosion-proof safety.
Explosion –proof electrical equipment for which explosion protection is only achieved in established normal operating conditions.
Note – The established normal operating conditions for electrical equipment are stated in the standards for the types of explosion protection of electrical equipment, as appropriate.

3.10. 4 Explosion-proof electrical equipment

Explosion-protected electrical equipment where explosion protection is provided both in normal operating conditions and in case of admitted probable faults dependent on operating conditions except for the damage of explosion protection means.
Note – The admitted probable faults of electrical equipment are stated in the standards for the types of explosion protection of electrical equipment, as appropriate.

3.10.5 Extra explosion-proof electrical equipment

Explosion-protected electrical equipment where, as compared to explosion-proof electrical equipment, additional explosion protection means provided for by the standards for the types of explosion protection are used.

4.4 Supplemented with:
4.4 Explosion-proof electrical equipment of Group I and Group II dependent on the level of explosion protection is subdivided into:
- electrical equipment with increased explosion-proof safety;
- explosion-proof electrical equipment;
- extra explosion-proof electrical equipment.

Supplemented with:
6.6 Levels of explosion protection of electrical equipment

6.6.1 Electrical equipment with increased explosion-proof safety can be provided by the following:
- explosion protection “i” with the intrinsically safe circuit level “ ic” and over;
- type of protection “p” with an inadmissible pressure drop warning device;
- explosion protection «q»;
- increased safety "e";
- encapsulation "m”;
- flameproof enclosures «d» for electrical equipment with increased explosion-proof safety;
- oil fillup for electrical equipment of GroupII and nonflammable fluid fillup of enclosures of electrical equipment of Group I meeting the requirements of type of protection “o”;
- explosion protection "s”.

6.6.2 Explosion-proof electrical equipment can be provided by:
- explosion protection “i” with the intrinsically safe circuit level not less than "iв";
- type of protection “p” with a warning device and automatic cut off of supply voltage, except for intrinsically safe
circuits of "ia" level at inadmissible pressure drop;
- flameproof enclosures «d» for explosion-proof electrical equipment;

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- special explosion protection "s".
- type of protection "e" contained in explosion-proof enclosure;
- placing into enclosure envisaged for the type of protection «p» with a warning device of inadmissible pressure drop for electrical equipment of Group II with type of protection "e";

6.6.3 Extra explosion-proof electrical equipment can be provided by:
- explosion protection "i" with the intrinsically safe circuit level "ia";
- special explosion protection "s";

- explosion-proof electrical equipment with additional explosion-protection means (for example, by placing of potted or immersed into liquid or granular dielectric material sparking parts into an explosion-proof enclosure, or by blowing out the explosion-proof enclosure by pure air under overpressure in the presence of devices of pressure control, signaling and automatic cut off of voltage at inadmissible pressure drop or explosion-proof enclosure damage).At the same time, the outgoing connections should have intrinsical safety of "ia" level.

8.1 Replaced by:

8.1 Materials that contain light metals used for the production of enclosures for electrical equipment of Group I and Group II should provide frictional intrinsical safety. The methods of frictional intrinsical safety testing of materials are presented in Appendix E. It is advisable that the **materials used for production of enclosures** of electrical equipment of Group I do not contain (by weight):
- more than 15% (in total) of aluminum, magnesium and titanium;
- more than 6% (in total) of magnesium and titanium.

Frictional intrinsical safety of enclosures made of light alloys may be provided by the application of protective coatings.

8.2 Supplemented with:
Materials that meet this requirement should be specified in the technical documentation of the product.

8.3 Supplemented with:
not more than 3 kg in weight

23.4.3.4 Supplemented with the paragraph:

23.4.3.4 Frictional intrinsical safety testing of materials is performed for electrical equipment if during its operation frictional sparks may occur (for example, portable electrical equipment used at sites with a risk of blow by foreign bodies, and electrical equipment that has moving and fixed parts).
Testing is performed according to the methods presented in Appendix E.

27.2. The text in item d) is supplemented with the following:
special type of explosion protection;

Appendix E Supplemented with:

Appendix ***E***

(Normative)

Frictional intrinsical safety testing of materials

This standard presumes making enclosures of Group I and Group II electrical equipment of light alloys which under certain conditions can represent a danger of frictional

sparking. The standard recommends that the materials used for the production of enclosures for electrical equipment of Group I do not contain (by weight):
a) more than 15 % (in total) of aluminum, magnesium and titanium;
b) more than 6 % (in total) of magnesium and titanium.

The materials used for making enclosures of Group II electrical equipment should not contain more than 7.5 % of magnesium by weight.

The availability of these recommendations makes it easier for the maker to choose source materials when designing enclosures, but does not exclude the necessity of their frictional intrinsical safety testing.

At present the IEC does not have the methods of frictional intrinsical safety testing of materials. Therefore, below are presented the methods of testing developed in the Russian Federation which are mandatory for frictional intrinsical safety testing of materials in the country.

E.1 Intrinsical safety test of materials and individual assemblies of electrical equipment is carried out using a dropped weight apparatus with rotating disk or other devices allowing to reproduce (simulate) real sparking processes in explosive mixtures of a specified composition.

E.2 Simulation of sparking process on a dropped weight apparatus for the specified pair of materials is provided for by the exterior form of weight (cylinder, cone, sphere), energy and relative traverse speed of parts at the time of impact. The energy E of impact depends on the height of drop and mass of weight

$$\mathbf{E = mgh} \qquad \text{(E.1)}$$

where m- mass of weight, kg;
h – height of drop, m.

During the tests the weight mass and height of drop that determine relative rate of parts movement at the moment of impact should have the highest values achievable in real conditions.

Simulation of sparking process on a rotating disk apparatus for the specified pair of materials is provided for by the shape of attrition faces of parts, relative slip velocity and hold-down pressure of attrition parts for the mechanisms with dampers.

Slip velocity is found from the formula:

$$V = \frac{\pi df}{60} \qquad \text{(E.2)}$$

where f – rotational speed of the element, min^{-1};
d - diameter of a rotating attrition part, m.

E.3 When frictional intrinsical safety testing of highly oxidizing materials of enclosures or individual assemblies of electrical equipment the following gas-air mixtures are used:
- for explosion-proof electrical equipment of Group I and Group IIA - CH4 (5.5-6.5)%;
- for explosion-proof electrical equipment of Group IIB and IIC - H2. (10-13)% .

E.4 Ignition capability of frictional sparks produced on friction or impact of aluminum and its alloys without or with protective coatings with rusty steel, and also that of frictional sparks of difficult to oxidize materials of enclosures is determined in the following combustible mixtures:
- for explosion-proof electrical equipment of Group I and Group IIA - CH_4 (6,5-7,5)%;
- for electrical equipment of Group IIB and IIC - H_2 (17-20)%.

E.5 Ignition capability of frictional sparks is determined by statistical method. The probability of ignition is found as a relation of the number of ignitions to that of impacts

$$P = \frac{m}{n} \qquad (E.3)$$

where m - number of ignitions;
n – number of weight drops.
The number of impacts on the rotating disc unit is calculated from the formula

$$n = \frac{fk\sum t}{60} \qquad (E.4)$$

where f – rotational speed of the element, min^{-1};
k – number of hitting elements on rotating device;
$\sum t$- total time of the device operation.
Sliding path equal to 0.5 m is assumed as one impact.

E.6 Safety of frictional sparks is assessed by one of the following methods.
Assessment of frictional intrinsical safety of parts of electrical equipment being subject to single impacts by the method of oxygen addition

The method is used to assess the ignition capability of actively oxidizing particles (for example, of steels) at impact energy, slip velocity and shape of parts simulating sparking process.

10 tests are carried out in combustible mixtures in accordance with E.3 and 32 tests in the same oxygen enriched mixtures (up to *(25±0.5)%*). Frictional sparks are considered as non- hazard if:
- during 10 tests in combustible mixtures to E.3 no ignition occurred;
- during 32 tests in explosive atmospheres to E.3 enriched with oxygen to *(25±0.5)%* not more than eight ignitions occurred.

The materials that have undergone the test successfully are safe for use in the corresponding parts of enclosures of explosion-proof equipment.

Assessment of frictional intrinsical safety of parts of electrical equipment being subject to single impacts by the method of change of impact energy

The method is used to test the pairs of materials the impact of which does not produce actively oxidizing particles or gives rise to exothermic reaction between their chemical elements (for example, fusion reaction between aluminum and ferric and ferrous oxides).

32 tests are carried out at maximum permitted traverse speed and two times higher energy of impact. Frictional sparks are considered as non-hazard if in explosive mixtures to E.4 no ignition occurred.

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The materials that have undergone the test successfully are safe for use in the corresponding parts of enclosures of explosion-proof equipment.

<u>Assessment of safety of frictional sparks resulting from friction and quickly alternating impacts of parts of electrical equipment by the method of oxygen addition</u>

16000 impacts are performed at maximum permitted traverse speed, hold-down pressure and shape of attrition surfaces simulating sparking process in combustible mixtures in accordance with E.3 and 16000 impacts in these mixtures enriched with oxygen to (*(25±0.5)%*

Frictional sparks are considered as non-hazard if at maximum permitted traverse speed and pressure:
- during 16000 impacts in explosive mixtures stated in **E.3** no ignition occurred;
- during 16000 impacts in explosive mixtures **to E.3** enriched with oxygen to (*(25±0.5)%*
- not more than eight ignitions occurred.

The materials that have undergone the test successfully are safe for use in the corresponding parts of enclosures of explosion-proof equipment.

E.7 Tests in combustible mixtures not enriched with oxygen can be omitted at discretion of the testing organization.
If previously performed frictional intrinsical safety tests of materials (coatings) and individual assemblies of electrical equipment have shown that intrinsical safety is provided repeated tests can be omitted.

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National Differences

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number **Details of Differences**

1 List of standards for the types of protection refers to the equivalent EN and, where relevant, the differing title of the EN:

IEC 60079-1	EN 50018	Flameproof enclosure "d"
IEC 60079-2	EN 50016	Pressurisation "p"
IEC 60079-5	EN 50017	Powder filling "q"
IEC 60079-6	EN 50015	Oil immersion "o"
IEC 60079-7	EN 50019	Increased Safety “e"
IEC 60079-11	EN 50020	Intrinsic Safety "i"
	EN 50039	Intrinsically safe electrical systems "i"
IEC 60079-18	EN 50028	Encapsulation "m"
IEC 60079-22	EN 50033	Cap lights for mines susceptible to firedamp

Notes 1 and 2 are not included.

2 In addition to the standards for the types of protection given above, the list of normative references includes the CENELEC version of IEC Standards, where published. These documents are either identical or closely related.

3.23 Replace with: Ex component
a part of electrical apparatus or a module (other than an Ex cable entry), marked with the symbol "U", which is not intended to be used alone and requires additional certification when incorporated into electrical apparatus for use in potentially explosive atmospheres

3.26 Definition is not included

6.1 Add: shall ...... b) be constructed in accordance with the principles of good engineering practice in safety matters. The manufacturer shall under his own responsibility indicate compliance by marking the electrical apparatus (in accordance with clause 25), and the testing station is not required to verify compliance.

7.1.2 After b) add: NOTE A standard ISO number should be used where possible.

8.1 Delete: NOTE - Values for Group II are under consideration.
Add: The materials used in the construction of enclosures of electrical apparatus of Group II shall not contain, by mass, more than 6% of magnesium.

8.3 Delete: Group I surveying instruments
Add: Group I measuring instruments.

(Note provided by comparison compiler: Discussions in CENELEC TC31 have confirmed that the IEC wording reflects the original intent of the EN wording.)

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National Differences

9.3.1 Replace with: Holes for threaded fasteners of 9.2 shall be threaded for a distance at least equal to the height of a full nut of the relevant size to accept the associated fastener

12.1 Delete: ... for the minimum and maximum temperatures

Add: ... for the maximum temperature
Delete second paragraph

Add: The thermal stability is considered adequate if the limiting value for the material exceeds this maximum temperature by at least 20K.

15.2 Delete: which is designed to be moved when energised

Add….which can be moved when energised

15.3 Add:…such as apparatus with metallic enclosures used with metallic conduit systems.

Table 3 *(Note provided by comparison compiler: The last line in Table 3 of IEC 60079-0 is incorrect and should be the same as in EN 50014, i.e. S > 35 : 0,5S)*

15.5 Delete second paragraph

Figure 3b *(Note provided by comparison compiler: IEC 60079-0 shows item 2 (branching point) within the body of the sealing compound. EN 50014 shows item 2 as the point where the conductors emerge from the sealing compound. The EN 50014 illustration is consistent with Figure 3a and is preferred.)*

18.3 The reference to IEC 60947-1 is not included.

18.6 Between b) and c) delete: ... minimise the risk of explosion ....

Add: ... minimise the risk to maintenance personnel ...

In d) (second indent) delete: ... and provides a degree of protection of at least IP30, according to IEC 60529:

Add: ... and provides a degree of protection of at least IP20, according to IEC 60529, so arranged that a tool cannot contact the energised parts through any openings;

21.2 Between b) and c) delete: ...minimise the risk of explosion .
Add: ... minimise the risk to maintenance personnel..
In d) delete the first indent

In d) (third indent) add: ..., so arranged that a tool cannot contact the energised parts through any openings;

22.1 Delete: are under consideration.

Add:. are contained in EN 50033.

23.4.7.5.2 Delete sub-clause

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National Differences

23.4.7.6Delete fourth indent (- the two other samples shall remain ...)

Add new fourth indent: The two other samples shall remain for (24 ± 2) hours in a hydraulic liquid of Group HFC (aqueous solution of polymer in 35% water) at a temperature of 50ºC, according to "Sixth Report on the Specifications and Testing Conditions relating to Fire-resistant Hydraulic Fluids Used for Power Transmission (Hydrostatic and Hydrokinetik) in Mines" Commission of the European Communities Safety and Health Commission for Mining and Extractive Industries, Luxembourg 1983 (Document Number 2786/8/81).

23.4.7.8 Add a further sentence to the fourth paragraph: In certain cases this requires the use of batteries or accumulators.

Add a sixth paragraph: Possible methods are indicated in annex E.

(Note provided by comparison compiler: CENELEC TC 31 has agreed that annex E is not helpful and that it (and the reference in 23.4.7.8) will be deleted by amendment. It is not present in IEC 60079-0 and is not reproduced in this comparison.)

26 Delete second paragraph.

27.2 c) Replace "Ex" with "EEx". Replace "specific standards" with "specific European Standards".
27.2 d) Delete the Note and the following final paragraph.

27.5 c) Replace "Ex" with "EEx".

27.6 b) Replace "Ex" with "EEx".

27.7 In all examples, replace "Ex" with "EEx".

B.1 *(Note provided by comparison compiler: In the note to clause B.1, EN 50014 incorrectly refers to the maximum diameter of cable. The wording in IEC 60079-0 is correct.)*

Table C.1 Against clause 27 delete the reference to Note 2. Delete note 2

Section 3
National Differences

IEC Standard 60079-0 1983 2nd ed, Amd 1 1987 02 Amd 2 1991 10 Electrical apparatus for explosive gas atmospheres Part 0: General requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS 2380.1 1989
BR Brazil		**N/R**	
CA Canada		**N/R**	
CH Switzerland*	**TBA**		
CN China		**N/A**	
CZ Czech Republic*	**TBA**		
DE Germany*	**TBA**		
DK Denmark*	**TBA**		
FI Finland*	**TBA**		
FR France*	**TBA**		
GB United Kingdom*	**TBA**		
HR Croatia*	**TBA**		
HU Hungary*	**TBA**		
IN India		**N/R**	
IT Italy*	**TBA**		
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands*	**TBA**		
NO Norway*	**TBA**		
NZ New Zealand		**NO**	
PL Poland*	**TBA**		
RO Romania*	**TBA**		
RU Russia		**N/R**	
SE Sweden*	**TBA**		
SG Singapore		**N/R**	
SI Slovenia*	**TBA**		
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-0:1983

(N/R "Not Recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details

Section 3
National Differences

Countries that have declared adoption of IEC 60079-0 1983 2nd ed, Amd 1 1987 02 Amd 2 1991 10 Without National Differences
AU Australia
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-0 1983 2nd ed, Amd 1 1987 02 Amd 2 1991 10 With National/Group Differences - Differences follow.

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IECEx Bulletin

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-1

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 60079-1: 6th Ed. 2007 Electrical apparatus for explosive gas atmospheres – Part 1: Flameproof enclosures 'd'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-1:2007
CN China		**YES**	GB 3836.2-2010 Explosive Atmospheres Part 2: Equipment protection by flameproof enclosures "d" (IEC60079-1: 2007, MOD)
CZ Czech Republic	**NO**		EN 60079-1:2007
DE Germany	**NO**		EN 60079-1:2007
DK Denmark	**NO**		EN 60079-1:2007
FI Finland	**NO**		EN 60079-1:2007
FR France	**NO**		EN 60079-1:2007
GB United Kingdom	**NO**		EN 60079-1:2007
HR Croatia	**NO**		EN 60079-1:2007
HU Hungary	**NO**		EN 60079-1:2007
IN India		**NO**	Adopted as IS /IEC60079-1: 2007
IT Italy	**NO**		EN 60079-1:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-1:2008
NL The Netherlands	**NO**		EN 60079-1:2007
NO Norway	**NO**		EN 60079-1:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-1:2007
RO Romania	**NO**		EN 60079-1:2007
RU Russia		**YES**	GOST R IEC 60079.1-2008
SE Sweden	**NO**		EN 60079-1:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-1:2007
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-1 2009 ANSI/UL 60079-1 2009
ZA South Africa		**NO**	SANS 60079-1:2007

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Countries that have declared adoption of IEC 60079-1: 6th Ed. 2007 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-1: 6th Ed. 2007 With National/Group Differences- Differences follow

CN China

IEC Standard Number IEC 60079-1:2007

IEC Clause or Annex Number	**Details of Differences**
Annex F	Add the following additional Annex F “additional requirements for group Ⅰ” F.1 Material of flameproof enclosures F.2 Direct entries for flameproof enclosures

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RU Russia

60079-1 2007 6th ed.	GOST R IEC 60079.1-2008	MOD	No differences, except for the references to GOST R standards instead of IEC ones

US United States of America

Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
General	The U.S. version of IEC 60079-1 does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
1	Modification of paragraph as follows to identify the specific applicability of this technique: This ~~part of IEC 60079~~ standard contains specific requirements for the construction and testing of electrical apparatus with type of protection flameproof enclosure "d", intended for use in Class I, Zone 1 explosive gas atmospheres.
2	Additional references to align with US practice and the National Electrical Code: ANSI/ASME B46.1, *Standard for Surface Texture* ANSI/UL 94, *Tests for Flammability of Plastic Materials for Parts in Devices and Appliances* ANSI/UL 2225, *Metal-Clad Cables and Cable-Sealing Fittings for use in Hazardous (Classified) Locations*
5.1	Modification of text as follows to indicate Specific Conditions of Use aligning with ANSI/ISA 60079-0: The dimensions given in 5.2 to 5.5 inclusive specify the minimum or maximum values that may be applied to the essential parameters of

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
	flamepaths. In instances where a dimension of a flameproof joint is other than the relevant minimum or maximum (for example, in order to comply with the test for non-transmission of an internal ignition), the equipment shall be marked ~~"X"~~ according to 29.2 item i) of ~~IEC 60079-0~~ ANSI/ISA-60079-0 and the ~~specific conditions of use on the certificate~~ instructions provided shall be in accordance with one of the following:
5.2.2	Modification of second paragraph as shown below to align with US practice: The surfaces of joints shall be such that their average roughness R_a (derived from ISO 468) does not exceed 6.3 μm (250 microinches in accordance with ANSI/ASME B46.1).
NEW 5.2.9	New sub-clause as shown below to align with US practice: **Labyrinth joints** A labyrinth joint shall consist of not less than 3 adjacent segments where the path changes direction not less than two times. Labyrinth joints need not comply with the requirements of Tables 1 and 2 but shall satisfy the test requirements of 15.2, with the test length reduced to not more than 75% of the manufacturer's specified design minimum length.
5.3	Revised text as shown below to align with US practice: Female threads shall gauge at "flush" to "3½ ~~2~~ turns large" using an L1 plug-gauge
6.2	Correct error in IIC document as shown. The test is only possible with the static method. Cemented joints are only permitted to ensure the sealing of the flameproof enclosure of which they form a part. Arrangements shall be made in the construction so that the mechanical strength of the assembly does not depend upon the adhesion of the cement alone. Cemented joints shall comply with a test having overpressure and time in accordance with 15.1.3.1, with compliance criteria in accordance with C.3.1.1.
8.1.2	Modification of Clause as follows to permit general application of this construction not limited to motor shaft glands: **Labyrinth joints** ~~Labyrinth joints which do not comply with the requirements of Tables 1 and 2 may nevertheless be considered as complying with the requirements of this standard if the tests specified in Clauses 14 through 16 are satisfied.~~

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Clause	**US Differences to IEC 60079-1, 6th Ed., 2007-04**
	Labyrinth joints shall comply with 5.2.9.
10.9.2.2	Modification of sub-clause as follows to align with US practice: Breathing and draining devices as Ex components shall be subjected to the thermal tests when connected to ~~based on the maximum intended flameproof enclosure volume, but no less than the volume of~~ the test rig in Figure 21. ~~NOTE When using the test rig in Figure 21, the maximum volume can be calculated to be approximately 2,5 l.~~ ~~Breathing and draining devices intended for multiple use in any single flameproof enclosure shall be tested additionally with the enclosure.~~
10.9.3	Modification of sub-clause as follows to align with US practice: **10.9.3 Ex component installation instructions ~~certificate~~** The Ex component installation instructions ~~certificate~~ shall record all details necessary to properly select a breathing or draining device for attachment to a type tested flameproof enclosure. The Ex component installation instructions ~~certificate~~ shall show In addition, the ~~Ex component certificate~~ installation instructions shall require that each Ex component or package of Ex components be accompanied by a copy of the ~~certificate, together with the~~ manufacturer's declaration stating ~~– compliance with the certificate conditions,~~

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
11.1	Modification of sub-clause as follows to align with US practice: Fasteners accessible from the outside and necessary for the assembly of the parts of a flameproof enclosure shall comply with 11.2 through 11.8, as applicable. ~~for Group I, be special fasteners complying with the requirements of IEC 60079-0, with the head shrouded or provided in counter bored holes, or inherently protected by the equipment construction;~~ ~~for Group II, be special fasteners complying with the requirements of IEC 60079-0.~~
11.3 (b)	Modification of sub-clause as follows to align with US practice: b) specified in the relevant ~~certificate~~ instructions.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
11.9	Modification of sub-clause as follows to align with US practice: **11.9** If apertures provided in a flameproof enclosure (for example, for cable gland or conduit entry) are not used, they shall be closed by blanking elements so that the flameproof properties of the enclosure are maintained (see Figure 22 for examples). Blanking elements shall comply with Annex C. ~~The blanking element may be made so that it can be fitted or removed from either the outside or the inside of the wall of the flameproof enclosure.~~ ~~The mechanically or frictionally locked blanking element shall comply with one or more of the requirements of 11.9.1 to 11.9.3.~~ ~~**11.9.1** If it is removable from the outside, this shall be possible only after disengagement of a retaining device inside the enclosure (see Figure 22a).~~ **11.9.2** It ~~may~~ shall be so designed that it can be fitted or removed only by the use of a tool ~~(see Figure 22b)~~. ~~**11.9.3** It may be of a special construction in which insertion is carried out by a method other than that used for removal. Removal shall only be by one of the methods specified in 11.9.1 or 11.9.2 or by a special technique (see Figure 22c).~~ **11.9.4** A blanking element shall not be used with an adapter.
11.10	Delete sub-clause, delete Figure 22; as follows to align with US practice: ~~**11.10** Threaded doors or covers shall be additionally secured by means of a hexagon socket set screw, or some equally effective method.~~ ~~Figure 22.~~
12.4	Delete sub-clause to align with US practice: ~~When cast iron is used, the material shall be not less than the quality 150 (ISO 185).~~

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
New 12.8	New sub-clause to align with US practice: **12.8** Flameproof enclosures, except for light transmitting parts, shall be made of iron, steel, copper, brass, bronze, or aluminium, or aluminium alloys containing a minimum of 80 percent aluminium, or shall be made of nonmetallic material which complies with the requirements in Clause 19. The enclosure of Group IIC apparatus shall not be constructed of copper or copper alloys unless coated with tin, nickel, or by other coatings, or by limiting the maximum copper content of the alloy to 30 percent. NOTE 1 The restriction of the use of copper in acetylene atmospheres is due to the potential formation of acetylides on the surface that can be ignited by friction or impact. NOTE 2 8.1.2 of ANSI/ISA-60079-0 limits the Mg content of the enclosure to 7.5% by mass.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
3	Modification of sub-clause as follows to align with the National Electrical Code: Threaded holes in enclosures to facilitate cable glands or conduit entries other than NPT, shall have the thread type and size identified, for example M25 ~~or 1/2NPT~~. This may be accomplished by – marking of the specific thread type and size adjacent to the hole per 20.3(a), Table 10, or – marking of the specific thread type and size on the nameplate per 20.3(a), Table 10, or – identification of the specific thread type and size as part of the installation instruction document, with a reference marking on the nameplate (via text or ISO symbol B.3.1 from ISO 3864) per 20.3(b), Table 10. The manufacturer shall state the following in the documents defining the electrical equipment: a) the places where entries can be fitted; and b) the maximum permitted number of these entries. Each entry shall have no more than one thread adapter when an adapter is used. A blanking element shall not be used with an adapter. NPT threaded entries shall not be smaller than trade size ½ nor larger than trade size 6. Where an integral bushing (historically known as a conduit stop) is not provided, the inner end of the entry shall be smooth and well-rounded. The dimensions of an integral bushing, if provided, shall be as shown in Table 11. NPS threaded entries may be provided for an enclosure for Group IIA or IIB locations and shall use a National Standard Pipe Straight (NPS) thread per ANSI/ASME B1.20.1 and shall include an integral bushing and shall provide for five full threads of engagement. The dimensions of the integral bushing shall be as shown in Table 11.

13 — Diameter of Integral Bushing

Trade Size of Conduit	Throat diameter of integral bushing (mm)	
	Minimum	Maximum
½	14	16
¾	19	21
1	24	27
1 ¼	32	35
1 ½	37	41
2	47	53
2 ½	56	63
3	70	78
3 ½	81	90
4	92	102
5	115	128
6	139	154

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
13.1	Modification of sub-clause as follows to align with National Electrical Code: **13.1 Cable glands** Cable glands, whether integral or separate, shall meet the requirements of ANSI/UL 2225~~this standard, the relevant requirements of Annex C~~ and create, on the enclosure, the joint widths and gaps prescribed in Clause 5. Where cable glands are integral with the enclosure or specific to the enclosure they shall be tested as part of the enclosure concerned. ~~Where cable glands are separate:~~ ~~– threaded Ex cable glands can be evaluated as equipment. Such cable glands do not have to be submitted to the tests of 15.1, nor to the routine test of Clause 16;~~ – other cable glands can only be evaluated as an Ex component.
13.2	Modification of sub-clause as follows to align with the National Electrical Code: **13.2 Conduit sealing devices** Conduit sealing devices, whether integral or separate, shall meet the requirements of this standard, the requirements of C.2.1.~~2~~3 and C.3.1.~~2~~3 ~~with "conduit sealing device" substituted for "cable gland"~~ and create, on the enclosure, the joint widths and gaps prescribed in Clause 5. ~~NOTE As such constructions preclude reuse, the requirement of C.2.1.2 that a conduit sealing device be capable of being fitted and removed without disturbing the compound seal after the specified curing period of the compound should not apply.~~ Where conduit sealing devices are integral with the enclosure or specific to the enclosure they shall be tested as part of the enclosure concerned. Where conduit sealing devices are separate: – threaded Ex conduit sealing devices can be evaluated as equipment. Such conduit sealing devices do not have to be submitted to the tests of 15.1, nor to the routine test of Clause 16; – other conduit sealing devices can only be evaluated as an Ex component.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
13.2.2	Modification of sub-clause as follows to align with the National Electrical Code: **13.2.2** A sealing device such as a stopping box with setting compound shall be provided, either as part of the flameproof enclosure or immediately at the entrance thereto. It shall satisfy the type test for sealing prescribed in Annex C. ~~An evaluated sealing device~~ <u>Alternatively, an explosionproof conduit seal</u> may be <u>permitted to be</u> applied by the installer or user of the equipment according to the <u>National Electrical Code®</u> ~~instructions provided by the manufacturer of the equipment.~~ NOTE A sealing device is considered as fitted immediately at the entrance of the flameproof enclosure when the device is fixed to the enclosure either directly or through an accessory necessary for coupling. ~~The sealing compound(s) and method(s) of application shall be specified in the certificate either of the stopping box or of the complete flameproof equipment. The part of the stopping box between the sealing compound and the flameproof enclosure shall be treated as a flameproof enclosure, i.e. the joints shall comply with Clause 5 and the assembly shall be submitted to the tests for non-transmission of 15.2.~~ ~~The distance from the face of the seal closest to the enclosure (or intended end-use enclosure), and the outside wall of the enclosure (or intended end-use enclosure) shall be as small as practical, but in no case more than the size of the conduit or 50 mm, whichever is the lesser.~~
13.3	Delete “cable couplers” as they are prohibited by the National Electrical Code.
13.3.4	Delete sub-clause to align with US practice: ~~**13.3.4** The requirements of 13.3.2 and 13.3.3 do not apply to plugs and sockets nor to cable couplers fixed together by means of special fasteners conforming to 11.1 and which bear a marking per 20.2(b), Table 9.~~

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
13.4	Modification of sub-clause as follows to align with US practice: **13.4 Bushings** Bushings, whether integral or separate, shall meet the requirements of this standard, the relevant requirements of Annex C and create, on the enclosure, the joint widths and gaps prescribed in Clause 5. Where bushings are integral with the enclosure or specific to the enclosure they shall be tested as part of the enclosure concerned. Where bushings are separate, they shall be evaluated as an Ex component: ~~– threaded Ex bushings can be evaluated as equipment. Such bushings do not have to be submitted to the tests of 15.1, nor to the routine test of Clause 16; and~~ ~~– other bushings can only be evaluated as an Ex component.~~
15	Modification of sub-clause as follows to align with the National Electrical Code: If the enclosure is designed so that it can be used in the absence of part of the enclosed equipment, the tests shall be made under the conditions considered to be the most severe. In both cases, the ~~certificate~~ installation instructions shall indicate the types of enclosed equipment permitted and their mounting arrangements.
15.1.1	Add new paragraph to sub-clause to align with US practice: For equipment marked with a nameplate circuit breaker interrupting rating greater than 10,000 r.m.s. symmetrical amperes, reference pressure tests are to be conducted under short-circuit conditions.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
15.1.2	Modification of text as follows to align with US practice Also shown is Corrigendum 1: *P* = 100[293 / (*T*a, min + 273)] kPa • For electrical equipment of Group IIA or IIB ~~other than rotating electrical machines (such as electric motors, generators and tachometers) that involve simple internal geometry (see Annex D) with an enclosure volume not exceeding 3 l, when empty, such that~~ where pressure-piling is not considered likely, the reference pressure shall be determined at normal ambient temperature using the defined test mixture(s), but is to be assumed to have a reference pressure increased by the factors given in the table below. ~~• For electrical equipment other than rotating electrical machines (such as electric motors, generators and tachometers) that involve simple internal geometry (see Annex D) with an enclosure volume not exceeding 10 l, when empty, such that pressure piling is not considered likely, the reference pressure shall be determined at normal ambient temperature using the defined test mixture(s), but is to be assumed to have a reference pressure increased by the factors given in the table below. Under this alternative, the test pressure for the overpressure type test in 15.1.3.1 shall be 4 times the increased reference pressure. The 1,5 times routine test is not permitted.~~
15.1.2.1	Modification of sub-clause as follows to align with US practice: The pressure developed during the explosion shall be determined and recorded during each test. The locations of the ignition sources as well as those of the pressure recording devices are varied as necessary ~~left to the discretion of the testing laboratory~~ to find the combination which produces the highest pressure. When detachable gaskets are provided by the manufacturer, these shall be fitted to the enclosure under test.
15.1.3	Modification of sub-clause as follows to align with US practice: **15.1.3 Overpressure test** This test shall be made using either of the following methods, which are considered as equivalent. ~~For electrical apparatus intended for use at an ambient temperature below -20°C, the overpressure test shall be conducted at a temperature not higher than the minimum ambient temperature. Where the tensile and yield strength properties of the material used is shown by material specifications to not decrease significantly at low temperature, the overpressure test may be conducted at normal room ambient.~~ NOTE – Some materials such as cast iron exhibit changes in material properties at extreme low temperatures such as a reduction in tensile strength.

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Clause	**US Differences to IEC 60079-1, 6th Ed., 2007-04**
15.2	Modification of sub-clause as follows to align with the National Electrical Code: ~~IEC 60079-14~~ The National Electrical Code® limits the installation of equipment employing type of protection "d" that incorporates flanged (flat) joints. Specifically, the flanged joints of such equipment are not permitted to be installed closer to solid objects, not part of the equipment, than the dimensions shown in Table 8, unless the equipment is so tested.
16.2	Modification of sub-clause as follows to align with US practice: Routine tests are not required for enclosures with a volume less than or equal to 10 cm^3. This exception also applies to enclosures with a volume greater than 10 cm^3 when the prescribed type test has been made at a static pressure of four times the reference pressure. ~~However, enclosures of welded construction shall in every case be submitted to the routine test.~~ For enclosures with a volume greater than 10 cm^3 where reference pressure measurement is impractical, exemption from routine pressure testing shall not apply.
19.3.2	Modification of sub-clause as follows to align with US practice: The test shall be carried out only for enclosures, or parts of enclosures, of plastic materials. The test shall be in accordance with IEC 60695-11-10 (Method V-2) or ANSI/UL 94 (Method V-2)
20.2(b)	Marking deleted as requirement has been removed.
20.3(d	Marking deleted as requirement has been removed.

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Clause	**US Differences to IEC 60079-1, 6th Ed., 2007-04**
C.1	Revise text and add note to explain deleted text of C.2.1, C.2.1.1, C.2.1.1.1, C.2.1.1.2, and C.2.1.2; as required by the National Electrical Code. This annex contains specific requirements which apply, in addition to those in I~~EC 60079-0~~ ANSI/ISA-60079-0, to the construction and testing of flameproof entry devices. Entry devices include ~~cable glands,~~ conduit sealing devices, Ex blanking elements, Ex thread adaptors, and bushings. NOTE The requirements for cable glands of type of protection “d” are located in ANSI/UL 2225.
C.2.1	Delete Clause
C.2.1.1	Delete Clause
C.2.1.1.1	Delete Clause
C.2.1.1.2	Delete Clause
C.2.1.2	Delete Clause
C.2.3.1	Modification of sub-clause as follows to align with US practice: **C.2.3.1** Ex blanking elements having male metric threads shall comply with one or more of the requirements of 11.9. Ex blanking elements having male NPT threads shall have an effective thread length not less than the “L2” dimension ~~be as type 22b (Figure 22) and with the external surface located at L1 (–0 +1/4)~~. ~~NOTE This requirement is intended to address concerns over entry into the enclosure by maintaining the outer surface of the blanking elements as close to the enclosure as possible.~~

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
C.3.1.1	Modification of sub-clause as follows to align with US practice. Also delete Figure C.1 **C.3.1.1 ~~Cable glands and~~ conduit sealing devices ~~with sealing ring~~** ~~These tests shall be carried out using, for each type of cable gland or conduit sealing device, one sealing ring from each of the different permitted sizes. In the case of elastomeric sealing rings, each ring is mounted on a clean, dry, polished mild steel cylindrical mandrel of diameter, equal to the smallest cable diameter permissible in the ring, as specified by the manufacturer of the cable gland or conduit sealing device.~~ ~~In the case of metallic or composite sealing rings, each ring is mounted on the metal sheath of a clean dry sample of cable, of diameter equal to the smallest diameter permissible in the ring, as specified by the manufacturer of the cable gland or conduit sealing device.~~ ~~In the case of sealing rings for non-circular cables, each ring is mounted on a clean dry sample of cable, of perimeter equal to the smallest value permitted in the ring, as specified by the manufacturer of the cable gland or conduit sealing device.~~ ~~The assembly is then fitted into the entry and a torque is applied to the screws (in the case of a flanged compression device) or to the nut (in the case of a screwed compression device) to obtain a seal under a hydraulic pressure of 2 000 kPa for group I and 3 000 kPa for group II.~~ ~~NOTE 1 The torque figures referred to in the preceding paragraph may be determined experimentally prior to the tests, or they may be supplied by the manufacturer of the cable gland or conduit sealing device.~~ The assembly is then mounted into a hydraulic testing device using coloured water or oil as the liquid, the principle of which is illustrated in Figure C.1. The hydraulic circuit is then purged. The hydraulic pressure is then gradually increased. The sealing is considered satisfactory if the blotting paper is free from any trace of leakage when the pressure has been maintained at 2 000 kPa for group I or 3 000 kPa for group II, for at least 10 s. ~~NOTE 2 It may be necessary to seal all the joints of the cable gland or conduit sealing device mounted in the test device, other than those associated with the sealing ring under test. When a sample of metal-sheathed cable is used, it may be necessary to avoid the application of pressure to the ends of the conductors or to the interior of the cable.~~
C.3.1.2	Delete sub-clause as it does not align with US practice.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
C.3.2	Delete Clause
C.3.2.1	Delete Clause
C.3.2.2	Delete Clause
C.3.2.3	Delete Clause
C.3.2.4	Delete Clause
C.3.3.1	Modification of sub-clause as follows to align with US practice. **C.3.3.1 Torque test** A sample Ex blanking element of each size shall be screwed into a test-block containing a threaded entry hole of size and form appropriate to the device under test. The sample shall be tightened to a torque at least equivalent to the appropriate torque given in column 2 of Table C.1, using a suitable tool. The test shall be deemed to be satisfactory if the correct thread engagement has been achieved and if, when dismantled, no damage is found~~, except of failure of the shearable neck of a type 22c plug which is required. Type 22b plugs shall be capable of being removed only by the appropriate tool~~. ~~Blanking elements of type 22b shall then be subjected to a further test at a torque at least equivalent to the appropriate torque given in column 3 of Table C.1, and shall be deemed to be satisfactory if the lip has not pulled fully into the thread.~~
D.1	Modification of sub-clause as follows to align with US practice. The purpose of an Ex component evaluation of ~~enclosure certificate for~~ empty enclosures is to enable a manufacturer of flameproof enclosures to obtain an evaluation ~~a certificate~~ without the internal equipment being defined, so as to enable the empty enclosure to be made available to third parties for incorporation into a full equipment evaluation ~~certificate~~ without the need for repetition of all the type tests. When an evaluation ~~a certificate~~ concerning the full equipment is required, an Ex component evaluation of ~~enclosure certificate for~~ the empty enclosure is not necessary.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
D.2	Modification of sub-clause as follows to align with US practice. The requirements for an Ex component evaluation of ~~enclosure certificate for~~ an empty enclosure are contained in this annex. This does not eliminate the need for a subsequent equipment evaluation ~~certificate~~, but it is intended to facilitate such an evaluation ~~a certificate~~. ~~The Ex component enclosure manufacturer shall be responsible for ensuring that each and every unit supplied~~ ~~a) is identical in construction with the original design as detailed in the documents mentioned in the Ex component enclosure certificate,~~ ~~b) has been subjected to such routine overpressure testing as is required, and~~ ~~c) meets the requirements of the applicable schedule of limitations imposed by the Ex component enclosure certificate.~~
D.3.5	Modification of sub-clause as follows to align with US practice. **D.3.5** No holes, whether for mechanical or electrical purposes, and whether blind or clear, shall be drilled in the Ex component enclosure other than those permitted by the installation instructions ~~Ex component enclosure certificate~~.
D.3.7	Modification of sub-clause as follows to align with US practice. Routine tests are not required for Ex Component enclosures when the prescribed type test has been made at a static pressure of four times the reference pressure. ~~However, enclosures of welded construction shall be in every case submitted to the routine test.~~

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
D.3.10	Modification of sub-clause as follows to align with US practice. **D.3.10** The following information shall be given in the installation instructions ~~Ex component enclosure certificate as part of the schedule of limitations as follows~~: – the maximum number of apertures, their maximum sizes and their positions shall be addressed through direct statement or reference to a drawing number; – rotating machines, or other devices which create turbulence, shall not be incorporated; – oil-filled circuit-breakers and contactors shall not be used; – (if other than -20 °C to +40 °C) the ambient range; – (for group I, IIA and IIB Ex component enclosures) the content of the Ex component enclosure equipment may be placed in any arrangement, provided that an area of at least 20 % of each cross-sectional area remains free to permit an unimpeded gas flow and, therefore, unrestricted development of an explosion. Separate relief areas may be aggregated provided that each area has a minimum dimension in any direction of 12,5 mm; – (for group IIC Ex component enclosures) the content of the Ex component enclosure equipment may be placed in any arrangement provided that an area of at least 40 % of each cross-sectional area remains free to permit unimpeded gas flow and, therefore, unrestricted development of an explosion. Separate relief areas may be aggregated provided that each area has a minimum dimension in any direction of 12,5 mm; and – any additional limitation required for the particular construction, e.g. maximum operating temperature of the window. – do not install switching devices with arcing contacts intended to interrupt a circuit with an available short circuit current of greater than 10,000 r.m.s. symmetrical amperes.

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Clause	US Differences to IEC 60079-1, 6th Ed., 2007-04
D.4.1	Modification of sub-clause as follows to align with US practice. Enclosures which have an Ex component enclosure certificate may be considered for incorporation in equipment evaluation in accordance ~~certificates~~ with ~~IEC 60079-0~~ ANSI/ISA- 60079-0 and this standard, normally without repetition of application of those requirements already applied to the Ex component enclosure, subject to compliance with the installation instructions ~~schedule of limitations detailed in D.3.10.~~ Documents shall be prepared for an equipment evaluation ~~certificate~~ depicting the specified equipment, any permitted substitutions or omissions, together with the mounting conditions within the Ex component enclosure, so that compliance can be verified with the installation instructions ~~schedule of limitations~~ of the Ex component ~~enclosure certificate~~. Any hole permitted in accordance with the Ex component enclosure evaluation ~~certificate~~ may be provided either by the Ex component enclosure manufacturer, or through agreement between the equipment manufacturer and the Ex component enclosure manufacturer.
D.4.2	Modification of sub-clause as follows to align with US practice. **D.4.2 Application of the installation instructions ~~schedule of limitations~~** In addition to compliance with the installation instructions ~~schedule of limitations~~, all application issues shall be considered and determined to comply with the applicable requirements of ~~IEC 60079-0~~ ANSI/ISA-60079-0 and this standard.

Section 3
National Differences

IEC Standard 60079-1: 5th Ed. 2003 Electrical apparatus for explosive gas atmospheres – Part 1: Flameproof enclosures 'd'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-5 2005
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-1:11
CH Switzerland	**NO**		EN 60079-1:2004
CN China*		**TBA**	
CZ Czech Republic	**NO**		ČSN EN 60079-1:2004
DE Germany	**NO**		EN 60079-1:2004
DK Denmark	**NO**		DS/EN 60079-1:2004
FI Finland	**NO**		EN 60079-1:2004
FR France	**NO**		EN 60079-1:2004
GB United Kingdom	**NO**		BS EN 60079-1:2004
HR Croatia	**N/A**		
HU Hungary	**NO**		MESZ EN 60079-1:2004
IN India		**N/R**	
IT Italy	**NO**		EN 60079-1:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-1:2005
NL The Netherlands	**NO**		EN 60079-1:2004
NO Norway	**NO**		EN 60079-1:2004
NZ New Zealand		**NO**	AS/NZS 60079-5 2005
PL Poland	**N/A**		
RO Romania	**NO**		EN 60079-1:2004
RU Russia*		**TBA**	
SE Sweden	**NO**		SS-EN 60079-1:2004 ed. 1
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-1:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-1 2005 ANSI/ISA 12.22.01 2005
ZA South Africa		**NO**	SANS 60079-1:2003

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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National Differences

Countries that have declared adoption of IEC 60079-1: 5th Ed. 2003 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-1: 5th Ed. 2003 With National/Group Differences- Differences follow

CA Canada

CAN/CSA-C22.2 No. 60079-1:11
Explosive atmospheres –
Part 1: Equipment protected by flameproof enclosures "d"

This is the second edition of CAN/CSA-C22.2 No. 60079-1, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-1 (Sixth edition, 2007-04). It replaces the CAN/CSA-C22.2 No.60079-1:07 (adopted CEI/IEC 60079-1: Fifth edition, 2003-11: Entitled *Electrical apparatus for explosive gas atmospheres – Part 1: Flameproof enclosures "d"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

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Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

See also CAN/CSA-C22.2 No. 60079-0:11 for further Canadian deviations.

5.1 General Requirements
[Delete the following paragraphs]

~~The dimensions given in 5.2 to 5.5 inclusive specify the minimum or maximum values that may be applied to the essential parameters of flamepaths. In instances where a dimension of a flameproof joint is other than the relevant minimum or maximum (i.e., in order to comply with the test for non-transmission of an internal ignition), the equipment shall be marked "X" according to 29.2 item i) of IEC60079-0 and the specific conditions of use on the certificate shall be in accordance with one of the following;~~
~~a) dimensions of the flameproof joints shall be detailed; or,~~
~~b) specific drawing referenced that details dimensions of the flameproof joints; or,~~
~~c) specific guidance noted to contact the original manufacturer for information on the dimensions of the flameproof joints.~~

5.3 Threaded joints

Table 3 – Cylindrical threaded joints

[Add the following entity (row) in the Table]

Threads engaged – IIC atmospheres ‖ ≥ 8

Table 4 – Taper threaded joints[a]

[Add the following entity (row) in the Table]

Threads engaged ‖ ≥ 5

11 Fasteners, associated holes and blanking elements

[Add a paragraph to the following effect]

A Canadian deviation has been stipulated in CAN/CSA-C22.2 No. 60079-0:11.

15.1.2 Determination of explosion pressure (reference pressure)

[Add the following paragraphs]

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For conduit connected enclosures having no normally arcing, sparking or excessive heat-generating parts, the test shall be conducted with lengths of conduit of the largest size permissible in the enclosure up to and including 1½ trade size.

The length of conduit shall be:
a) 2.5 m (10 ft), and also repeated with
b) 3.7 m (12 ft).

Where normally arcing, sparking or excessive heat-generating parts are present in the enclosure, the test shall be conducted with 450 mm (18 in) length of conduit of the largest size permissible in the enclosure up to and including 1½ trade size.

Ignition of the test mixture shall be at the far-end of the conduit. Pressure shall be determined at the ignition end, the opposite end, and at all points where higher pressures are likely to occur.

Testing without lengths of conduit is permitted, if and when the equipment is marked with the warning statement "A SEAL SHALL BE INSTALLED WITHIN 50mm OF THE ENCLOSURE" and "UN SCELLEMENT DOIT ÊTRE INSTALLÉ À MOINS DE 50 mm DU BOÎTIER", or wording with equal effect.

20 Marking
General

[Add a paragraph to the following effect]

A Canadian deviation has been stipulated in CAN/CSA-C22.2 No. 60079-0:11 that all warning and cautionary statement shall be in both English and French..

C.2.2 Threads

[Add the following "Note"]

For IIC equipment, the number of thread engagement is to be eight (8) fully engaged threads.

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US United States of America

Clause	US Differences to IEC 60079-1, 5th Ed, 2003
General	The U.S. version of IEC 60079-1 does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences. All references to Testing Station, Certificate, "X" marking and "U" marking have been removed.
1	**Modification of paragraph** **Scope** This ~~part of IEC 60079~~ standard contains specific requirements for the construction and testing of electrical apparatus with type of protection flameproof enclosure "d", intended for use in Class I, Zone 1 explosive gas atmospheres.
5.2.2	Modification of second and third paragraphs, deletion of fifth paragraph and Figure 1. The surfaces of joints shall be such that their average roughness R_a (derived from ISO 468) does not exceed 6.3 μm (250 microinches in accordance with ANSI/ASME B46.1).
5.2.8	Modification of fourth paragraph If the manufacturer's maximum constructional gap is different than that shown in Tables 1 or 2 for a flanged joint of the same length (determined by multiplying the pitch by the number of serrations), the maximum constructional gap shall be given in the ~~certificate~~ specific installation instructions in accordance with 29.2 i) of ANSI/ISA- 60079-0. ~~and the apparatus marked in accordance with 29.2 item i) of IEC 60079-0. In addition, this condition for safe use shall be specified in the manufacturer's instructions.~~

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<table>
<tr><th>Clause</th><th>US Differences to IEC 60079-1, 5th Ed, 2003</th></tr>
<tr><td>NEW 5.2.9</td><td>New Clause

Labyrinth joints
A labyrinth joint shall consist of not less than 3 adjacent segments where the path changes direction not less than two times. Labyrinth joints need not comply with the requirements of Tables 1 and 2 but shall satisfy the test requirements of 15.2, with the test length reduced to not more than 75% of the manufacturer’s specified design minimum length.</td></tr>
<tr><td>5.3</td><td>Modification of Table 4

4– Taper threaded joints [a]
<table>
<tr><td>Pitch</td><td>≥ 0.9 mm</td></tr>
<tr><td>Threads provided on each part
Threads engaged</td><td>≥ 5[b]
≥ 5[c]</td></tr>
<tr><td colspan="2">On male threaded fittings with a shoulder or interruption, a thread length not less than the L4 dimension defined by ANSI/ASME B1.20.1 shall be provided between the face of the shoulder and end of the fitting thread.</td></tr>
<tr><td colspan="2">[a] Internal and external thread shall have the same nominal size, cone angle, and thread form.
[b] Threads shall conform to the NPT requirements of ANSI/ASME B1.20.1, and shall be made up wrench tight.
[c] ~~Threads made in accordance with this Table will provide greater than 3.5 effective threads engaged.~~ Adjustment of gauging practices is required to achieve the required five engaged threads. See 13.2.2.1.</td></tr>
</table></td></tr>
<tr><td>8.1.2</td><td>Modification of Clause</td></tr>
</table>

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
	Labyrinth joints ~~Labyrinth joints which do not comply with the requirements of tables 1 and 2 may nevertheless be considered as complying with the requirements of this standard if the tests specified in clauses 14 through 16 are satisfied.~~ Labyrinth joints shall comply with 5.2.9.
9	Modification of Clause For light-transmitting parts ~~of luminaires and for inspection windows~~ of glass or plastic materials of flameproof enclosures, the requirements of ~~IEC 60079-0~~ ANSI/ISA-60079-0 apply.
10.9	Modification of Clause Breathing devices and draining devices when used as Ex components ~~Breathing and draining devices as Ex components are limited to application on flameproof enclosure volumes of 3 l (litres) or less.~~ ~~NOTE A breathing and draining device may be used as an integral part of flameproof enclosure volumes larger than 3 l provided it is tested with the specific enclosure in accordance with 15.4.~~
10.9.3	Modification of Clause **Marking** The marking of breathing and draining devices used as Ex components shall be as follows: ~~all breathing and draining devices shall be evaluated in the name of the original component manufacturer who shall ensure that all future component devices are made in accordance with the evaluated type;~~

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
10.9.4	Editorial correction of second paragraph In addition, the Ex component certificate shall require that each Ex component or package of Ex ~~package of~~ components be accompanied by a copy of the certificate, together with a manufacturer's declaration stating
11.1	Modification of Clause Fasteners accessible from the outside and necessary for the assembly of the parts of a flameproof enclosure shall comply with 11.2 through 11.8, as applicable. ~~for Group I, be special fasteners complying with the requirements of IEC 60079-0. When heads of fasteners are not protected, for example by counterbored holes, the electrical apparatus shall be marked "X";~~ ~~for Group II, be in conformity with 9.2 of IEC 60079-0 in respect of threads and heads.~~
11.3	Modification of Clause The lower yield stress of screws and nuts shall be at least 240 N/mm^2 according to ISO 6892. In carrying out the type tests specified in clause 15, ~~the testing station shall require the replacement of~~ all or some of the screws specified by the manufacturer shall be replaced, if these are of a higher yield stress than 240 N/mm^2, by screws of the lowest yield stress available, but with a minimum of 240 N/mm^2, unless a calculation based on a pressure of 1.5 times the reference pressure shows that a higher yield stress is necessary.

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
11.9	Modification of Clause If apertures provided in a flameproof enclosure (for example, for cable gland or conduit entry) are not used, they shall be closed so that the flameproof properties of the enclosure are maintained ~~(see figure 22 for examples)~~. The mechanically or frictionally locked blanking element ~~closing device~~ may be made so that it can be fitted or removed from either the outside or the inside of the wall of the flameproof enclosure. ~~The mechanically or frictionally locked blanking element shall comply with one or more of the requirements of 11.9.1 to 11.9.3.~~ ~~11.9.1 If the closing device is removable from the outside, this shall be possible only after disengagement of a retaining device inside the enclosure (see figure 22a).~~ ~~11.9.2 It may be so designed that it can be fitted or removed only by the use of a tool complying with the requirements of 9.2 of IEC 60079-0 (see figure 22b).~~ ~~11.9.3 It may be of a special construction in which insertion is done by a method other than that used for removal. Removal shall only be by one of the methods specified in 11.9.1 or 11.9.2 or by a special technique (see figure 22c).~~
11.10	Delete Clause, delete Figure 22. ~~11.10 Separate fastening arrangements, requiring the use of a tool, of the type required in 9.2 of IEC 60079-0 or some equally effective methods shall be provided to secure and release threaded doors or covers.~~ ~~Figure 22.~~
12.4	Delete Clause ~~When cast iron is used, the material shall be not less than the quality 150 (ISO 185).~~

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
New 12.7	New Clause The enclosure housing the electrical components, except those portions for viewing or light transmission, shall be made of iron, steel, copper, brass, bronze, or aluminum, or aluminum alloys containing a minimum of 80 percent aluminum, or shall be made of nonmetallic material which complies with the requirements in clause 19. A metal such as zinc or zinc alloy shall not be used. The enclosure of Group IIC apparatus shall not be constructed of copper or copper alloys unless coated with tin, nickel, or by other coatings, or by limiting the maximum copper content of the alloy to 30 percent. The restriction of the use of copper in acetylene atmospheres is due to the potential formation of acetylides on the surface that can be ignited by friction or impact. 8.2 of ANSI/ISA- 60079-0 limits the Mg content of the enclosure to 7.5% by mass.

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
13	Modification of Clause Threaded holes in enclosures to facilitate cable glands or conduit entries other than NPT, shall have the thread type and size identified, for example M25 ~~or 1/2NPT~~. This may be accomplished by marking of the specific thread type and size adjacent to the hole; identification of the specific thread type and size on the nameplate; or identification of the specific thread type and size as part of the installation instruction document, with a reference on the nameplate (via text or ISO symbol) to the installation instruction document. NPT threaded entries may be provided and shall use a modified National Standard Pipe Taper (NPT) thread with thread form per ANSI/ASME B1.20.1. Entries shall not be smaller than trade size ½ nor larger than trade size 6 and shall provide for five full threads of engagement with a conduit or fitting gauging at L1-1. Entries shall conform to ANSI/ASME B1.20.1 except that entries shall gauge from flush to +3-1/2 turns beyond the L-1 gauging notch in lieu of the -1 to +1 turns described in ANSI/ASME B1.20.1. Where an integral bushing (historically known as a conduit stop) is not provided, the inner end of the entry shall be smooth and well-rounded. The dimensions of an integral bushing, if provided, shall be as shown in Table 13. NPS threaded entries may be provided for an enclosure for Group IIA or IIB locations and shall use a National Standard Pipe Straight (NPS) thread per ANSI/ASME B1.20.1 and shall include an integral bushing and shall provide for five full threads of engagement. The dimensions of the integral bushing shall be as shown in Table 13. 13 — Diameter of Integral Bushing

Trade Size of Conduit	Throat diameter of integral bushing (mm)	
	Minimum	Maximum
½	14	16
¾	19	21
1	24	27
1 ¼	32	35
1 ½	37	41
2	47	53
2 ½	56	63
3	70	78
3 ½	81	90
4	92	102
5	115	128
6	139	154

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
13.1	Modification of Clause **Cable glands** Cable glands, whether integral or separate, shall meet the requirements of ~~this standard, the relevant requirements of annex C,~~ UL 2225 and create, on the enclosure, the joint widths and gaps prescribed in clause 5. Where cable glands are integral with the enclosure or specific to the enclosure they shall be tested as part of the enclosure concerned. ~~Where cable glands are separate~~ - ~~threaded Ex cable glands can be evaluated as apparatus. Such cable glands do not have to be submitted to the tests of 15.1 and the routine test of clause 16.~~ - ~~other cable glands can only be evaluated as an Ex component.~~

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
13.2.2	Modification of Clause A sealing device such as a stopping box with setting compound shall be provided, either as part of the flameproof enclosure or immediately at the entrance thereto. It shall satisfy the type test for sealing prescribed in annex C. ~~An evaluated sealing device~~ Alternatively, an explosionproof conduit seal may be permitted to be applied by the installer or user of the apparatus according to ~~instructions provided by the manufacturer of the apparatus~~ the National Electrical Code®. ~~NOTE A sealing device is considered as fitted immediately at the entrance of the flameproof enclosure when the device is fixed to the enclosure directly or through an accessory necessary for coupling.~~ ~~The sealing compound(s) and method(s) of application shall be specified in the certificate either of the stopping box or of the complete flameproof apparatus. The part of the stopping box between the setting compound and the flameproof enclosure shall be treated as a flameproof enclosure, i.e. the joints shall comply with clause 5 and the assembly shall be submitted to the tests for non-transmission of 15.2.~~ ~~The distance from the face of the seal closest to the enclosure (or intended end-use enclosure), and the outside wall of the enclosure (or intended end-use enclosure) shall be as small as practical, but in no case more than the size of the conduit or 50 mm, whichever is the lesser.~~
13.3	Modification of Heading Plugs and sockets ~~and cable couplers~~

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
13.3.2	Modification of Clause The widths and the gaps of the flameproof joints (see clause 5) of the flameproof enclosures of plugs and sockets ~~and cable couplers~~ shall be determined by the volume which exists at the moment of separation of the contacts other than those for earthing or bonding or those which are parts of circuits complying with ~~IEC 60079-11~~ ANSI/ISA- 60079-11.
13.3.3	Modification of Clause For plugs and sockets ~~and cable couplers~~, the flameproof properties of the enclosure shall be maintained in the event of an internal explosion, both when the plugs and sockets or cable couplers are connected together and at the moment of separation of the contacts other than those for earthing or bonding or those which are parts of circuits complying with ~~IEC 60079-11~~ ANSI/ISA- 60079-11.
13.3.4	Delete Clause ~~The requirements of 13.3.2 and 13.3.3 do not apply to plugs and sockets nor to cable couplers fixed together by means of special fasteners conforming to 11.1 and which bear a label with the warning~~ ~~"DO NOT SEPARATE WHEN ENERGIZED".~~
15	Modification of Clause Add requirements for installation instructions

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
15.1.2	Add two new paragraphs to Clause As a second alternative, apparatus rated for a lower ambient temperature of not less than -50ºC may be assumed to have a reference pressure of 150% of that determined at room temperature. For equipment marked with a nameplate circuit breaker interrupting rating greater than 10,000 r.m.s. symmetrical amperes, reference pressure tests are to be conducted under short-circuit conditions.
15.1.3	Modification of Clause **Overpressure test** This test shall be made using either of the following methods, which are considered as equivalent. ~~For electrical apparatus intended for use at an ambient temperature below -20°C, the overpressure test shall be conducted at a temperature not higher than the minimum ambient temperature. Where the tensile and yield strength properties of the material used is shown by material specifications to not decrease significantly at low temperature, the overpressure test may be conducted at normal room ambient.~~

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
15.2	Modification of Table 6 - added no joint reduction for NPT threads. Modification of Table 6 Note modified as follows: NOTE – For tapered threads the joint should be tested with the minimum handtight engagement permitted by the thread standard at the extremes of tolerances. Example of reduction of tapered threads: After marking the position of handtight engagement on the thread, the devices are removed and the length of engagement is reduced by ~~cutting the screw~~ removing 1/3 of the engaged male or female threads ~~or drilling out the hole~~. The parts are then reassembled to the marked position. Where the minimum distance of obstructions from flanged joints is intended to be less than the distance shown in Table 8, the apparatus may be appropriately tested and marked with the following statement or equivalent wording, on the product or in the installation instructions: “CAUTION – Do not install within ’X’ mm of any solid object that is not part of the equipment,” where “X” is determined by the proximity of the solid object during flame transmission testing, with the tested values less than those stated in Table 8. Addition of Table 8 as follows: **8 — Minimum distance of obstructions from flameproof “d” flange openings** Gas Group / Minimum Distance (mm): IIC – 40; IIB – 30; IIA – 10 NOTE The National Electrical Code®, ANSI/NFPA 70, limits the installation of apparatus employing type of protection “d” that incorporates flanged (flat) joints. The flanged joints of such apparatus are not permitted to be installed closer to solid objects, not part of the apparatus, than the dimensions shown in Table 8 unless the apparatus is marked to show a lesser dimension.

8 — Minimum distance of obstructions from flameproof “d” flange openings

Gas Group	Minimum Distance (mm)
IIC	40
IIB	30
IIA	10

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
16.2	Modification of Clause Routine tests are not required for enclosures with a volume less than or equal to 10 cm^3. This exception also applies to enclosures with a volume greater than 10 cm^3 when the prescribed type test has been made at a static pressure of four times the reference pressure. ~~However, enclosures of welded construction shall in every case be submitted to the routine test.~~ For enclosures with a volume greater than 10 cm^3 where reference pressure measurement is impractical, exemption from routine pressure testing shall not apply.
19	Modification of Clause **Non-metallic enclosures and non-metallic parts of enclosures** The following requirements apply to non-metallic enclosures and non-metallic parts of enclosures, except for - ~~sealing rings of cable glands or conduit sealing devices; and~~ - non-metallic parts on which the type of protection does not depend.
19.3.2.2	Delete Clause
19.3.2.2.1	Delete Clause
19.3.2.2.2	Modification of Clause **Test method** The test shall be carried out in accordance with IEC 60707 (Method FV 0: Flame – Vertical specimen) or ANSI/UL 94 (Method V-0), or IEC 60695-11-10 (Method V-0).

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
19.4	Delete Clause
A.1	Modification of Clause Crimped ribbon elements shall be constructed from cupro-nickel, stainless steel, or other metal investigated and found suitable for the purpose ~~a metal agreed between the manufacturer and the testing station~~. Aluminum, titanium, magnesium, and their alloys shall not be used.
B1.1. c)	Modification of Clause Other material investigated and found suitable for the purpose ~~a specific metal or specific alloy agreed between the manufacturer and the testing station~~. Aluminum, titanium, magnesium, and their alloys shall not be used.
B.2.1	Modification of Clause Pressed metal wire elements shall be constructed from stainless steel wire braid or other material investigated and found suitable for the purpose.~~another specified metal agreed between the manufacturer and the testing station.~~
C.2.1	Delete Clause
C.2.1.1	Delete Clause
C.2.1.1.1	Delete Clause
C.2.1.1.2	Delete Clause

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
C.2.1.2	Addition of Note: NOTE 13.2 applies to this subclause with “conduit sealing device” substituted for “cable gland.”
C.2.2	Delete Clause
C.3.1	Delete Clause
C.3.1.1	Delete Clause and Figure C.1
C.3.1.2	NOTE Modification of Clause: NOTE Cable glands sealed with setting compound NOTE NOTE 13.2 applies this subclause with “conduit sealing device” substituted for “cable gland”. For each size of cable, the test shall be carried out using metal mandrels, the number and diameter of which equate to the maximum diameter over cores with the maximum number of cores specified by the manufacturer in accordance with the requirements of C.2.1.2. The setting compound is prepared following the manufacturer's instructions and then filled ~~poured~~ into the appropriate volume. It is allowed to harden for the appropriate time. The tests prescribed in ~~23.4.7.3~~ 26.8 and ~~23.4.7.4~~ 26.9 of ANSI/ISA-60079-0 ~~IEC 60079-0~~ shall be applied. The assembly is then mounted into the hydraulic testing device~~, defined in C.3.1.1 above~~, and the same procedure is applied. The acceptance criteria are also the same.
C.3.2 through C3.2.4	Delete Clause

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Clause	US Differences to IEC 60079-1, 5th Ed, 2003
C.3.3.1	NOTE Modification of Clause: NOTE **Torque test** A sample Ex blanking element of each size shall be screwed into a steel block, containing a threaded entry hole of size and form appropriate to the device under test. The sample shall be tightened to a torque at least equivalent to the appropriate torque given in column 2 of Table C.1, using a suitable tool. The test shall be deemed to be satisfactory if the correct thread engagement has been achieved and if when dismantled no damage is found~~, except of failure of the shearable neck of a type 22c plug which is required. Type 22b plugs shall be capable of being removed only by the appropriate tool~~. ~~Blanking elements of type 22b shall then be subjected to a further test at a torque at least equivalent to the appropriate torque given in column 3 of table C.1, and shall be deemed to be satisfactory if the lip has not pulled fully into the thread.~~
D.3.7	Modification of Clause: Routine tests are not required for enclosures when the prescribed type test has been made at a static pressure of four times the reference pressure. ~~However, enclosures of welded construction shall be in every case submitted to the routine test.~~

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IEC Standard 60079-1 4th Edition 2001+ Corrigendum 1 Electrical apparatus for explosive gas atmospheres Part 1: Construction and verification test of flameproof enclosure of electrical apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-1 2002
BR Brazil		**N/R**	
CA Canada		**NO**	CAN/CSA-E60079-1-2001
CH Switzerland*	**TBA**		
CN China*		**TBA**	
CZ Czech Republic*	**TBA**		
DE Germany*	**TBA**		
DK Denmark*	**TBA**		
FI Finland*	**TBA**		
FR France*	**TBA**		
GB United Kingdom*	**TBA**		
HR Croatia*	**N/A**		
HU Hungary*	**TBA**		
IN India		**NO**	Adopted as IS 2148:2004/IEC 60079-1: 2001
IT Italy*	**TBA**		
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-1:2002 (ALREADY WITHDRAWN)
NL The Netherlands*	**TBA**		
NO Norway*	**TBA**		
NZ New Zealand		**NO**	AS/NZS 60079-1 2002
PL Poland*	**N/A**		
RO Romania*	**TBA**		
RU Russia		**YES**	GOST R 51330.1-99
SE Sweden*	**TBA**		
SG Singapore		**NO**	
SI Slovenia*	**TBA**		
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-1:2001

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-1: 4th Edition 2001+ Corrigendum 1 Without National Differences
AU Australia
CA Canada
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Section 3
National Differences

IEC Standard 60079-1 3rd Edition 1990 + Amendment 1 1993 and Amendment 2 1998 Electrical apparatus for explosive gas atmospheres Part1: Construction and verification test of flameproof enclosure of electrical apparatus.			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**Refer to IEC 60079-1:2001-02 Fourth Edition**	
CH Switzerland	**YES (Amd 1 only)**		EN 50018 1994
CN China		**YES**	GB3836.2 2000
CZ Czech Republic	**YES (Amd 1 only)**		EN 50018 1994
DE Germany	**YES (Amd 1 only)**		EN 50018 1994
DK Denmark	**YES (Amd 1 only)**		EN 50018 1994
FI Finland	**YES (Amd 1 only)**		EN 50018 1994
FR France	**YES (Amd 1 only)**		EN 50018 1994
GB United Kingdom	**YES (Amd 1 only)**		EN 50018 1994
HR Croatia	**N/A**		
HU Hungary	**YES (Amd 1 only)**		EN 50018 1994
IN India		**N/R**	
IT Italy	**YES (Amd 1 only)**		EN 50018 1994
JP Japan		**NO**	
KR Republic of Korea		**YES (Amd 1 only)**	Labor Ministry Notice1998-65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES (Amd 1 only)**		EN 50018 1994
NO Norway	**YES (Amd 1 only)**		EN 50018 1994
NZ New Zealand		**NO**	
PL Poland*	**N/A**		
RO Romania	**YES (Amd 1 only)**		EN 50018 1994
RU Russia		**N/R**	
SE Sweden	**YES (Amd 1 only)**		EN 50018 1994
SG Singapore		**N/R**	
SI Slovenia	**YES (Amd 1 only)**		EN 50018 1994
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed. Please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Section 3
National Differences

Countries that have declared adoption of IEC 60079-1: 3rd Edition 1990 + Amendment 1 1993 and Amendment 2 1998 Without National Differences
AU Australia
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 60079-1: 3rd Edition 1990 + Amendment 1 1993 and Amendment 2 1998 With National/Group Differences- Differences follow

CN China

IEC Clause or Annex Number Details of differences to IEC 60079-1 1990 (edition 3) + A1: 1993

4.3.1 Add the following extra requirements to this Clause:
The threaded joints must have measures to avoid loosing.

10.1 Add the following extra requirement to this Clause:
For materials of flameproof enclosure Group I: Enclosure of electrical apparatus for exploit mine (including electrical apparatus on winning Machine, loading machine and transfer machine) shall be made of steel plate or steel casting.
Other enclosure, which will not be submitted to the impact and enclosure with volume no more than 2000 cm^3
may be made of gray cast iron with sort no less the HT250. Stator enclosure of electrical motor shall be made
of steel plate or steel casting, other component may be made of gray cast iron with sort no HT250.
The enclosure of electrical apparatus which is not located in exploit mine, may be made of gray cast iron with sort no less than HT250.
Enclosure of electrical apparatus with volume no more than 2000 cm^3 may be made of plastic, and threads shall not be tapped into plastic material.
Except cable entry, holes for the threaded fasteners could not be drilled on the non-mental enclosures.

11.1 Add the following extra requirement to this Clause:
For direct entries for flameproof enclosure of electrical apparatus Group I: Direct entries may be used only if electrical apparatus doesn’t produce arcs, sparks or hazardous hot in normal service and has rated power no more than 250 W and current no more than 5 A. The clearances and creepage distances between terminals in connection compartment and for direct entry shall meet with requirements in accordance with IEC 60079-7 “e”.

Annex Add Annex C (normative) Additional requirements for electrical apparatus Gruppe I. The contents of Annex C are described as above.

Add Annex D (informative) Additional requirements for flameproof cable entries.
The contents of Annex D are described as Annex C of IEC 60079-1: 2001.

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Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Sub-clauses, clauses or annex in EN 50018:1994, which contain additional requirements and which do not correspond directly to any sub-clause or annex in the IEC Publication, have been specified as ”Additional sub-clauses”, ”Additional clauses” or ”Additional annex”.

Nearly the whole IEC standard is replaced by text according to EN 50018:1994 due to differences in requirements or editorial presentation. A lot of the figures and requirements are however technically equal for the standards. Sub-clauses in the IEC standard with text which remain unchanged (possibly except headings and references to other sub-clauses etc), in relation to corresponding sub-clauses in EN 50018:1994, include: 4.5.2, A.1, A.3.2.1, A.3.2.2, A.3.2.4, A.3.3.1, A.3.3.2, A.3.3.3.1, B.4, B.5, B.6, B.7 (first two paragraphs), B.7.2.1.3, B.7.2.1.4, B.7.2.2.4, B.7.2.2.5, B.7.2.3.2, B.7.2.3.3, B.7.2.3.4, B.7.2.3.5, B.8, B.10 and B.11 (first two paragraphs).

IEC Clause or Annex Number **Details of Differences**

Replace the lists of publications headed ”The following IEC publications are quoted in this standard:” and ”Other publications quoted:”, under the clause ”PREFACE”, with lists according clause 2 below.

Section one Amend the heading of the section to read ”GENERAL”

1.1 Replace the sub-clause with the following:

This European Standard contains the specific requirements for the construction and testing of electrical apparatus with type of protection flameproof enclosure "d", intended for use in potentially explosive atmospheres.

Replace the sub-clause with the following:

This European Standard supplements European Standard EN 50014, the requirements of which apply to electrical apparatus with flameproof enclosure.

Delete the sub-clause.

Renumber the clause to ”3” (all sub-clauses will commence with ”3” instead of ”2” accordingly). Insert a new clause 2 as following:

2 Publications

2.1 IEC PUBLICATIONS REFERRED TO IN EUROPEAN STANDARD EN 50018

IEC 61 Lamp caps and holders together with gauges for the control of interchangeability and safety.

IEC 61-1 Part 1: Lamp caps
- supplement K (1983) (HD 65.1 S1(1978))

IEC 61-2 Part 2: Lampholders
- supplement G (1983) (HD 65.2 S1 (1978))

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IEC 79-1A (1975) Electrical apparatus for explosive gas atmospheres. Part 1: Construction and test of flameproof enclosures of electrical apparatus. First supplement: Appendix D:

Method of test for ascertainment of maximum experimental safe gap.

IEC 82 (1984) Ballasts for tubular fluorescent lamps.

IEC 112 (1979) Recommended method for determining the comparative tracking index of solid insulating materials under moist conditions (HD 214 S2 (1980)).

IEC 529 (1989) Degrees of protection provided by enclosures (IP Code). (EN 60529(1991))
IEC 707 (1981) Methods of test for the determination of the flammability of solid electrical insulating materials when exposed to an igniting source. (HD 441 S1(1983))

2.2 ISO STANDARDS REFERRED TO IN EUROPEAN STANDARD EN 50018

ISO 31-0 (1992) Quantities and units; part 0: General principles

ISO 185 (1988) Grey cast iron; classification

ISO 468 (1982) Surface roughness; Parameters, their values and general rules for specifying requirements.

ISO 965-1 (1980) ISO general purpose metric screw threads; Tolerances; Part 1: Principles and basic data.

ISO 965-3 (1980) ISO general purpose metric screw threads; Tolerances; Part 3: Deviations for constructional threads.

ISO 1210 (1992) Plastics; Determination of burning behaviour of horizontal and vertical specimens in contact with a small-flame ignition source

ISO 2738 (1987) Permeable sintered metal materials; Determination of density, oil content, and open porosity.

ISO 4003 (1977) Permeable sintered metal materials; Determination of bubble test pore size.

ISO 4022 (1987) Permeable sintered metal materials; Determination of fluid permeability.
ISO 6892 (1984) Metallic materials; Tensile testing.
(EN 10002-1 (1990) and EN 10002-1 AC1 (1990))

2.2 EUROPEAN STANDARDS REFERRED TO IN EUROPEAN STANDARD EN 50018

EN 50014 (1992) Electrical apparatus for potentially explosive atmospheres
General requirements

EN 50019 (1994) Electrical apparatus for potentially explosive atmospheres.
Increased safety "e"

EN 50020 (1994) Electrical apparatus for potentially explosive atmospheres.
Intrinsic safety "i"

3 Replace the clause with the following:
(renumbered clause 2)

3 Definitions

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The following definitions specific to type of protection flameproof enclosure "d"are applicable in this European Standard; they supplement the definitions which are given in European Standard EN 50014.

3.1 Flameproof enclosure "d"

A type of protection in which the parts which can ignite an explosive atmosphere are placed in an enclosure which can withstand the pressure developed during an internal explosion of an explosive mixture and which prevents the transmission of the explosion to the explosive atmosphere surrounding the enclosure.

3.2 Volume

The total internal volume of the enclosure. However, for enclosures in which the contents are essential in service, the volume to be considered is the remaining free volume.
Note: For luminaires, the volume is determined without lamps fitted.

3.3 Flameproof joint

The place where corresponding surfaces of two parts of an enclosure come together, or the conjunction of enclosures, and prevent the transmission of an internal explosion to the explosive atmosphere surrounding the enclosure.

3.4 Width of flameproof joint (*L*)

The shortest path through a flameproof joint from the inside to the outside of an enclosure.

3.5 Distance (*l*)

The shortest path through a flameproof joint, when the width of the joint *L* is interrupted by holes intended for the passage of fasteners for assembling the parts of the flameproof enclosure.

3.6 Gap of flameproof joint (*i*)

The distance between the corresponding surfaces of a flameproof joint when the electrical apparatus enclosure has been assembled. For cylindrical surfaces, forming cylindrical joints, the gap is the difference between the diameters of the bore and the cylindrical component.

3.7 Maximum experimental safe gap (MESG) (for an explosive mixture)

The maximum gap of a joint of 25 mm width which prevents any transmission of an explosion in 10 tests made under the conditions specified in IEC 79-1A.

3.8 Shaft

A part of circular cross section used for the transmission of rotary movement.

3.9 Operating rod

A part used for the transmission of control movements which may be rotary or linear or a combination of the two.

3.10 Pressure-piling

The results of an ignition, in a compartment or subdivision of an enclosure, of a gas mixture pre-compressed for example due to a primary ignition in another compartment or subdivision.

3.11 Quick-acting door or cover

A door or cover provided with a device which permits opening or closing by a simple operation, such as the movement of a lever or the rotation of a wheel. The device is arranged so that the operation has two stages:
- one for locking or unlocking

another for opening or closing

3.12 Door or cover fixed by threaded fasteners

A door or cover the opening or closing of which requires the manipulation of one or more threaded fasteners (screws, studs, bolts or nuts).

3.13 Threaded door or cover

A door or cover which is assembled to a flameproof enclosure by a threaded flameproof joint.

3.14 Breathing device

An integral or separable part of a flameproof enclosure designed to permit exchange between the atmosphere inside the enclosure and the surrounding atmosphere.

3.15 Draining device

An integral or separable part of a flameproof enclosure designed to permit water formed by condensation to escape from the enclosure.

3 Renumber the clause to ”4” and replace the clause with the following:

4 Apparatus grouping and temperature classification

The apparatus grouping and temperature classification defined in EN 50014 for

the use of electrical apparatus in potentially explosive atmospheres apply to flameproof enclosures. The subdivisions A, B, C for electrical apparatus of Group II also apply.

Section two Amend the heading of the section to read ”SPECIFIC CONSTRUCTIONAL REQUIREMENT”

4 Renumber the clause to ”5” and delete ”(joints)” in the heading.

4.1 Renumber the sub-clause to ”4.1” and replace the sub-clause with the following:

5.1 General requirements

All flameproof joints, whether permanently closed or designed to be opened from time to time, shall comply, in the absence of pressure, with the appropriate requirements of clause 5.

The design of joints shall be appropriate to the mechanical constraints applied to them.

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Note: The values given in clause 5 constitute the necessary conditions. Additional measures may be necessary in order to pass the non transmission test of 15.2.

The surface of joints may be protected against corrosion.

Note: Coating with paint is not permitted. Other coating material may be used if the material and application procedure have been shown not to adversely affect the flameproof properties of the joint.

Figure 1 Renumber the figure to ”19” and replace the text in the heading of the figure, with the following:

Figure 19: Example of labyrinth joint for shaft of rotating electrical machine

Figure 2 Delete this figure.

4.2 Renumber the sub-clause to ”5.2”.

4.2.1 Renumber the sub-clause to ”5.2.1” and replace the sub-clause with the following:

5.2.1 Width of joints (*L*)

The width of joints shall not be less than the minimum values given in tables 1 and 2. The width of joint for cylindrical metallic parts press-fitted into the walls of a metallic flameproof enclosure of volume not greater than 2 000 cm^3 may be reduced to 5 mm, if:

- the design does not rely only on an interference fit to prevent the part being displaced during the type tests of clause 15, and

- the assembly meets the impact test requirements of EN 50014, taking the worst case interference fit tolerances into account and,

- the external diameter of the press-fitted part, where the width of joint is measured, does not exceed 60 mm.

4.2.2-4.2.3 Replace these two sub-clauses with the following sub-clause numbered ”5.2.2”:

5.2.2 Gap (i)

The gap, if one exists, between the surfaces of a joint shall nowhere exceed the maximum values given in tables 1 and 2.

The surfaces of joints shall be such that their average roughness R_a (ISO 468) does not exceed 6,3 µm.

For flanged joints there shall be no intentional gap between the surfaces, except for quick acting doors or covers.

For electrical apparatus of Group I, it shall be possible to check, directly or indirectly, the gaps of flanged joints of covers and doors designed to be opened from time to time. Figure 1 shows an example of construction for indirect checking of a flameproof joint.

Tables I, IIA,
IIB and IIC Replace these tables with the enclosed tables 1 and 2.

Figure 3 Renumber the figure to ”1” and replace the text in the heading of the figure, with the following:

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Figure 1: Example of construction for indirect checking of a flanged Group I flameproof joint

4.2.4 Renumber the sub-clause to "5.2.3" and replace the sub-clause with the following:

5.2.3 Spigot joints

For the determination of the width L of spigot joints the following shall be taken into account:

-either the cylindrical part and the plane part (see figure 2).

The gap if one exists, between the surfaces of the joint shall nowhere exceed the maximum values given in tables 1 and 2.

-or the cylindrical part only (see figure 3).

In this case the plane part need not comply with the requirements of tables 1 and 2.

Note: For gaskets see also 5.4

Figure 4 and 5 Renumber Figure 4 to "2" and Figure 5 to "3". Delete the text (commencing "L=c+d...") in Figure 4. Introduce the following common text for the figures:

L = c + d (I, IIA, IIB, IIC)
$c \geq$ 6,0 mm (IIC)
$\geq$ 3,0 mm (I, IIA, IIB)
$d \geq$ 0,50 L (IIC)
$f \leq$ 1,0 mm (I, IIA, IIB, IIC)

1 Interior of enclosure

Figures 2 and 3: Spigot joints

Figure 6 and 7 Delete these figures.

Delete this sub-clause.

4.2.6 Renumber the sub-clause to "5.2.4" and replace the sub-clause with the following:

5.2.4 Holes in joint surfaces

Where a plane joint or the plane part or partial cylindrical surface (see 5.2.6) of a joint is interrupted by holes intended for the passage of threaded fasteners for assembling the parts of a flameproof enclosure, the distance *l* to the edge of the hole shall be equal to or greater than:

- 6 mm when the width of joint *L* is less than 12,5 mm
- 8 mm when the width of joint *L* is equal to or greater than 12,5 mm but less than 25 mm;
- 9 mm when the width of joint *L* is equal to or greater than 25 mm.

The distance / is determined as follows:

5.2.4.1 Flanged joints with holes outside the enclosure (see figures 4 and 6)

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The distance *l* is measured between each hole and the inside of the enclosure.

5.2.4.2 Flanged joints with holes inside the enclosure (see figure 5)

The distance *l* is measured between each hole and the outside of the enclosure.

5.2.4.3 Spigot joints where, to the edges of the holes, the joint consists of a cylindrical part and a plane part (see figure 7)

The distance *l* is:
-the sum of the width *a* of the cylindrical part and the width *b* of the plane part, if f is less than or equal to 1 mm and if the gap of the cylindrical part is less than or equal to 0,2 mm for electrical apparatus of Groups I and IIA, 0,15 mm for electrical apparatus of Group IIB, or 0,1 mm for electrical apparatus of Group IIC (reduced gap);

- the width *b* of the plane part alone, if either of the above-mentioned conditions is not met.

5.2.4.4 Spigot joints where, to the edges of the holes, the joint consists only of the plane part (see figures 8 and

9), in so far as plane joints are permitted (see 5.2.7)

The distance *l* is the width of the plane part between the inside of the enclosure and a hole, where the hole is outside the enclosure (see figure 8), or between a hole and the outside of the enclosure where the hole is inside the enclosure (see figure 9).

Figure 8, 9, 10 and 11 — Renumber Figure 8 to "4", Figure 9 to "5", Figure 10 to "6" and replace the common heading of the figures with the following:

Figures 4, 5, 6: Holes in surfaces of flanged joints

Replace Figure 11 with the following Figure 7 and add the following Figure 8 and Figure 9:

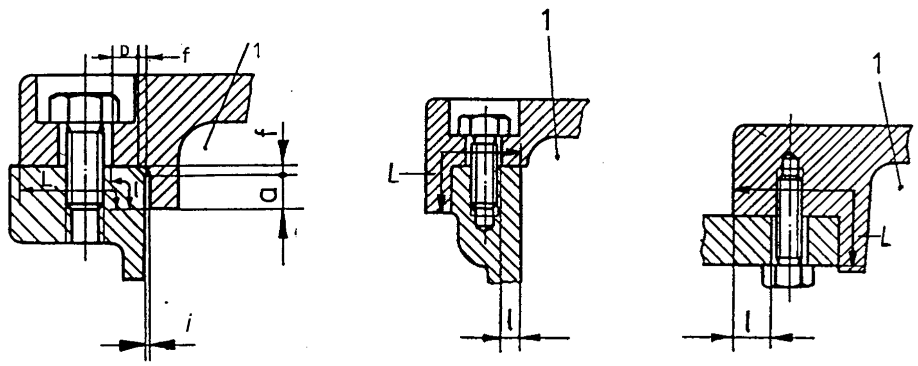

Figure 7 Figure 8 Figure 9

1 interior of enclosure

Figures 7, 8, 9: Holes in surfaces of spigot joints

Additional sub-clauses — Add the following sub-clauses:

5.2.5 Conical joints

Where joints include conical surfaces, the width of joint, and the gap normal to the joint surfaces shall comply with the relevant values in tables 1 and 2. The gap shall be uniform through the conical part. For electrical apparatus of group IIC, the cone angle shall not exceed 5°.

5.2.6 Joints with partial cylindrical surfaces (not permitted for Group IIC)

There shall by no intentional gap between the two parts (see figure 10).

The width of the joint shall comply with the requirements of table 1.

The diameters of the cylindrical surfaces of the two parts forming the flameproof joint, and their tolerances, shall ensure compliance with the relevant requirements for the gap of a cylindrical joint as given in table 1.

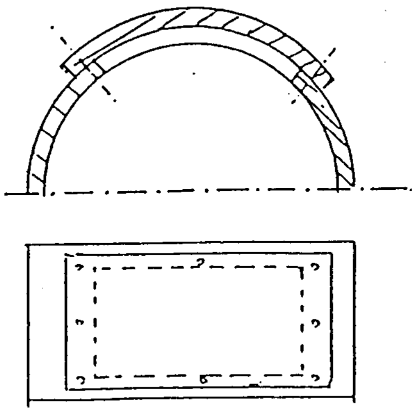

Figure 10: Example for a joint with partial cylindrical surfaces

Additional requirements for joints of electrical apparatus of Group IIC

Flanged joints are not permitted for electrical apparatus of Group IIC intended for use in potentially explosive atmospheres containing acetylene; they are permitted for potentially explosive atmospheres containing no acetylene if the volume of the enclosure does not exceed 500 cm^3.

4.3, Table III Renumber the sub-clause to "5.3" and replace the sub-clause including Table III with the following:

5.3 Threaded joints

Threaded joints shall comply with the requirements in tables 3 or 4.

Table 3: Cylindrical threaded joints

Pitch:	≥ 0,7 mm 1)
Thread form and quality of fit:	medium or fine tolerance quality according to ISO 965/1 and ISO 965/3 2)
Threads engaged:	≥ 5
Depth of engagement:	
Volume ≤ 100 cm^3	≥ 5 mm
Volume > 100 cm^3	≥ 8 mm

1) Where the pitch exceeds 2 mm, special manufacturing precautions may be necessary (e. g. more threads engaged) to ensure that the electrical apparatus can pass the test for non-transmission of an internal ignition which is prescribed in 15.2.

2) Cylindrical threaded joints which do not conform with the ISO standard, in respect of thread form or quality of fit, are permitted if the test for non-transmission of an internal ignition that is prescribed in 15.2 is passed when the width of threaded joint specified the manufacturer is reduced by the amount specified in table 6.

Table 4: Taper threaded joints

Pitch	≥ 0,9 mm
Threads provided on each part	≥ 6

The internal and external thread shall have the same cone angle and thread form, which shall be defined.

Taking the maximum permissible tolerances, the actual number of engaged threads may be less than 5.

4.4 Renumber the sub-clause to ”5.4” and replace the sub-clause with the following:

5.4 Gaskets (including O-rings)

If a gasket of compressible or elastic material is used, for example to protect against the ingress of moisture or dust or against leakage of a liquid, it shall be applied as a supplement, that is to say neither be taken into account in the determination of the width of the flameproof joint nor interrupt it.

The gasket shall then be mounted so that

- the permissible gap and width of flanged joints or the plane part of a spigot joint are maintained

- the minimum width of joint of a cylindrical joint or the cylindrical part of a spigot joint are maintained before and after compression.

These requirements do not apply to cable entries (see 13.1) or to joints which contain a sealing gasket of metal or of a non-flammable compressible material with a metallic sheath. Such a sealing gasket contributes to the explosion protection, and in this case the gap between each surface of the plane part shall be measured after compression. The minimum width of the cylindrical part shall be maintained before and after compression.

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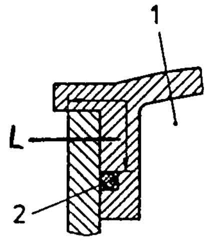
Figure 12, 13,14 and 15 Replace Figure 12-15 with the following technically identical Figure 11-14 and add the following Figure 15, 16 and 17:



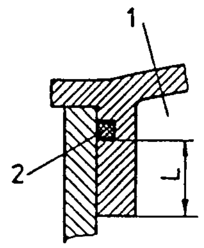
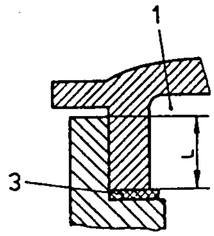


Figure 11 Figure 12



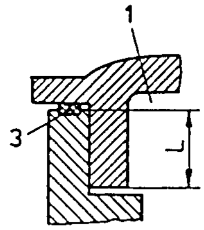
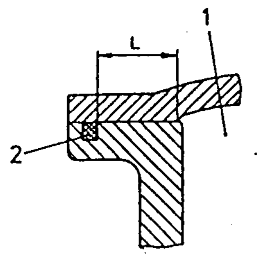


Figure 13 Figure 14



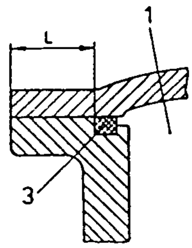


Figure 15 Figure 16

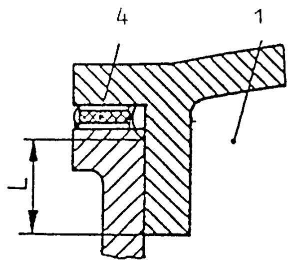


Figure 17

Figures 11 to 17: Illustration of the requirements concerning gaskets

1 Interior of enclosure
2 O-ring
3 Gasket
4 Metallic or metal sheath gasket

4.5 Renumber the sub-clause to ”6”.
Renumber sub-clause 4.5.2 to ”6.3” with the heading ”6.3 Width of cemented joints” and replace sub-clause 4.5.1 and 4.5.3 with the following sub-clauses:

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6.1 General

Parts of a flameproof enclosure may be cemented either directly into the wall of the enclosure so as to form with the latter an inseparable assembly, or into a metallic frame such that the assembly can be replaced as a unit without damaging the cement.

If a joint which is cemented does not fulfill the requirements of clause 5 in the absence of the cement it shall be subjected to 23.4.7.3 and 23.4.7.4 in EN 50014:1992.

6.2 Mechanical strength

Cemented joints are only permitted to ensure the sealing of the flameproof enclosure of which they form a part. Arrangements shall be made in the construction so that the mechanical strength of the assembly does not depend upon the adhesion of the cement alone. Cemented joints shall comply with a test based on 15.3 with the relevant overpressure value given in 15.1.3.

5 Renumber the clause to ”7” and replace it with the following clause:

7 Operating rods

Where an operating rod passes through the wall of a flameproof enclosure, the following requirements shall be met:

7.1 If the diameter of the operating rod exceeds the minimum width of joint specified in tables 1 and 2, the width of joint shall be at least equal to this diameter but without, however, having to exceed 25 mm.

7.2 If the diametral clearance is liable to be enlarged as a result of wear in normal service, appropriate arrangements shall be made to facilitate a return to the original state, e. g. by means of a replaceable bush. Alternatively, gap enlargement due to wear may be prevented by use of bearings complying with clause 8.

6 Renumber the clause to ”8” with the heading ”8 Supplementary requirements for shafts and bearings”.

Replace the first 4 paragraphs of clause 6 (”A flameproof gland...for Group IIC”) with the following sub-clauses:

8.1 Joints of shafts

Flameproof joints of shafts of rotating electrical machines shall be arranged so as not to be subject to wear in normal service.

The flameproof joint may be:

-a cylindrical joint (see figures 18 and 21), or
-a labyrinth joint (see figure 19 and 21), or
-a joint with a floating gland (see figure 20).

8.1.1 Cylindrical joints

Where a cylindrical joint contains grooves for the retention of grease, the region containing the grooves shall neither be taken into account when determining the width of a flameproof joint nor interrupt it (see figure 18).

The minimum radial clearance *k* (see figure 21) of shafts of rotating electrical machines shall not be less than 0,05 mm.

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8.1.2 Labyrinth joints

Labyrinth joints which do not comply with the requirements of tables 1 and 2 may nevertheless be considered as complying with the requirements of this European Standard if the tests specified in section "Verifications and tests" are satisfied.

The minimum radial clearance *k* (see figure 21) of shafts of rotating electrical machines shall not be less than 0,05 mm.

8.1.3 Joints with floating glands

The determination of the maximum degree of float of the gland shall take account of the clearance in the bearing and the permissible wear of the bearing as specified by the manufacturer. The gland may move freely radially with the shaft and axially on the shaft but it shall remain concentric with it. A device shall prevent rotation of the gland (see figure 20).

Floating glands are not permitted for electrical apparatus of Group IIC.

Figure 16
and 17 Replace Figure 16 with the following Figure 18 and replace Figure 17 with the following Figure 20:

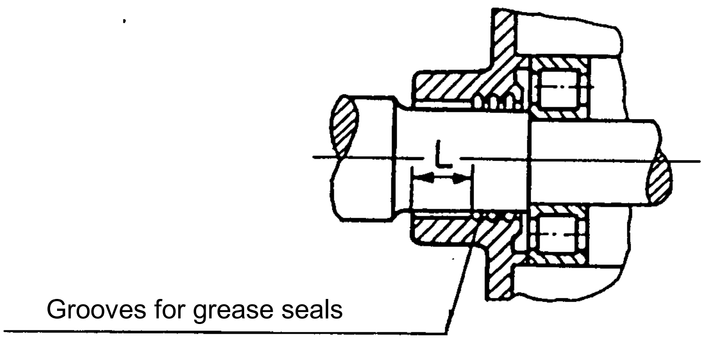
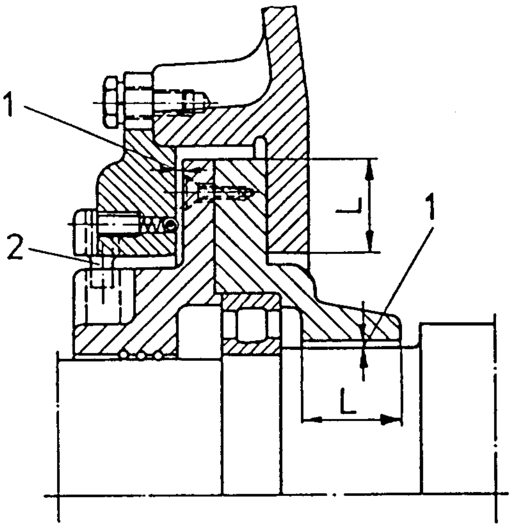


Figure 18: Example of cylindrical joint for shaft of rotating electrical machine



1 Gap
2 Stop to prevent rotation of gland

Figure 20: Example of joint with floating gland for shaft of rotating electrical machine

6.1-6.3 Delete sub-clause 6.3, renumber sub-clause 6.1 to ”8.2.1” and renumber sub-clause 6.2 to ”8.2.2”. Introduce a common heading as follows and replace the sub-clauses with the following (last sentence in 6.1 remaining identical):

8.2 Bearings

8.2.1 Sleeve bearings

A flameproof joint of a shaft gland associated with a sleeve bearing shall be provided in addition to the joint of the sleeve bearing itself and shall have a width of joint at least equal to the diameter of the shaft but without having to exceed 25 mm.

If a cylindrical or labyrinth flameproof joint is used in a rotating electrical machine with sleeve bearings, at least one face of the joint shall be of non-sparking metal (e. g. leaded brass) whenever the air gap between stator and rotor is greater than the minimum radial clearance *k* (see figure 21) specified by the manufacturer. The minimum thickness of the non-sparking metal shall be greater than the air gap.

Sleeve bearings are not permitted for rotating electrical machines of Group IIC.

8.2.2 Rolling-element bearings

In shaft glands equipped with rolling-element bearings, the maximum radial clearance *m* (see figure 21) shall not exceed two-thirds of the maximum gap permitted for such glands in tables 1 and 2.

Figure 18 Renumber the figure to ”21” and replace ”m = maximum radial clearance” with:
”m = maximal radial clearance taking *k* into account”

Figure 19
and 20 Delete the figures.

7 Renumber the clause to ”9” and replace it with the following:

9 Light-transmitting parts

For light-transmitting parts of luminaires and for inspection windows of glass or plastics materials of flameproof enclosures, the requirements of EN 50014 apply.

Note: Precautions should be taken so that the mountings of light-transmitting parts do not produce internal mechanical stress in those parts.

8 Renumber the clause to ”10” and amend the heading to read ”10 Breathing and draining devices which form part of a flameproof enclosure”.

8.1-8.3 Replace the sub-clauses with the following (which including sub-clause 10.1-10.8):

Breathing and draining devices shall incorporate permeable elements which can withstand the pressure created by an internal explosion in the enclosure to which they are fitted, and which shall prevent the transmission of the explosion to the explosive atmosphere surrounding the enclosure.

They shall also withstand the dynamic effects of explosions within the flameproof enclosure without permanent distortion or damage which would impair their flame arresting properties. They are not intended to withstand continuous burning on their surfaces.

These requirements apply equally to devices for the transmission of sound but do not cover devices for

-relief of pressure in the event of internal explosion or
-use with pressure lines containing gas which is capable of forming an explosive mixture with air and is at a pressure in excess of 1,1 times atmospheric pressure.

10.1 Openings for breathing or draining

The openings for breathing or draining shall not be produced by deliberate enlargement of gaps of flanged joints.

Note: If for technical reasons breathing or draining devices have to be provided, they should be so constructed that they are not liable to become inoperative in service (e. g. because of the accumulation of dust or paint).

10.2 Composition limits

The composition limits of the materials used in the device shall be specified either directly or by reference to an existing applicable specification.

The elements of breathing or draining devices for use in potentially explosive atmosphere containing acetylene shall comprise not more than 60 % of copper by mass to limit acetylide formation.

10.3 Dimensions

The dimensions of the breathing and draining devices and their component parts shall be specified.

10.4 Elements with measurable paths

Interstices and measurable lengths of path need not comply with the values given in tables 1 and 2 provided that the elements pass the tests of section "Verifications and tests".

Additional requirements for crimped ribbon elements are given in Annex A.

10.5 Elements with non-measurable paths

Where the paths through the elements are not measurable (for example, sintered metal elements), the element shall comply with the relevant requirements of Annex B.

The elements are classified according to their density as well as their pore size in accordance with the standard methods for the particular material and the particular manufacturing methods (see Annex B).

Note: For functional reasons, it may also be necessary to state the fluid permeability and the open porosity specified in accordance with the standard methods for the particular material and the particular manufacturing methods (see Annex B).

10.6 Removable devices

If a device can be dismantled, it shall be designed to avoid reduction or enlargement of the openings during re-assembly.

10.7 Mounting arrangements of the elements

The breathing and draining elements shall be sintered, or fixed by other suitable methods:

-either directly into the enclosure to form an integral part of the enclosure or

-in a suitable mounting component, which is clamped or screwed into the enclosure so that it is replaceable as a unit.

Alternatively, the element can be mounted, for example press fitted in accordance with 5.2.1, so as to form a flameproof joint. In this case, the appropriate requirements of clause 5 are to be applied, with the exception that the surface roughness of the element need not comply with 5.2.2, if the element arrangement passes the type test in section "Verifications and tests".

If necessary, a clamping ring or similar means can be used to maintain the integrity of the enclosure. The breathing or draining element can be mounted:

-either from within, in which case the accessibility of screws and clamping ring shall be possible only from the inside or

-from outside the enclosure, in which case the fasteners shall comply with clause 11.

10.8 Mechanical strength

The device and its guard, if any, shall, when mounted normally, pass the test for resistance to impact in 23.4.7.7 of EN 50014:1992.

(Note provided by comparison compiler: Parts of clause 10 above in EN 50018 deal with the same subject – some of them with equal requirements - as parts of clauses/sub-clauses they replace and parts of the

normative Annex B in IEC 60079-1 + A1, according to the following:

- *The first two paragraphs in clause 10 (”Breathing and draining…their surfaces.”): the first two paragraphs in sub-clause B.2 (”Breathing or draining...their surfaces.”)*

- *The 3rd paragraph in clause 10 (”These requirements…atmospheric pressure.”): the 2nd paragraph in sub-clause B.1 (”The following requirements... atmospheric pressure.”)*

- *Sub-clause 10.1: 3rd and 4th paragraph in sub-clause B.2 (”The openings…dust or paint).”), sub-clause 8.1*

- *Sub-clause 10.2: sub-clause B.3*

- *Sub-clause 10.3: sub-clause B.4 (identical except the heading)*

- *Sub-clause 10.4: sub-clause B.5 (identical except references to tables and a sub-clause)*

- *Sub-clause 10.5: the first two paragraphs in sub-clause B.7*

- *Sub-clause 10.6: sub-clause B.8 (identical except the heading), sub-clause 8.3*

- *Sub-clause 10.7: sub-clause B.9*

- *Sub-clause 10.8: sub-clause B.10 (identical except a reference to a standard))*

9 Renumber the clause to ”11” and amend the heading to read ”Fasteners, associated holes and closing devices”.

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9.1-9.6 Replace the sub-clauses with the following sub-clauses 11.1-11.10:

11.1 Fasteners accessible from the outside and necessary for the assembly of the parts of a flameproof enclosure shall:

- For Group I, be special fasteners complying with the requirements of EN 50014.

When heads of fasteners are not protected, for example by counterbored holes, the electrical apparatus shall be marked "X"

- For Group II, be in conformity with 9.2 of EN 50014:1992 in respect of threads and heads.

11.2 Fasteners of plastic material or light alloys are not permitted.

11.3 The lower yield stress of screws and nuts shall be at least 240 N/mm^2 according to ISO 6892.

In carrying out the type tests specified in clause 15, the testing station shall require the replacement of all or some of the screws specified by the manufacturer, if these are of a higher yield stress than 240 N/mm^2, by screws of the lowest yield stress available, but with a minimum of 240 N/mm^2, unless a calculation based on a pressure of 1,5 times the reference pressure shows that a higher yield stress is necessary.

If a yield stress higher than 240 N/mm^2 is necessary, the required yield stress shall be:
- either marked on the apparatus or

- specified in the relevant certificate, in which case the apparatus shall be marked with an "X".

The type test is then carried out with the screws and nuts specified by the manufacturer.

11.4 Studs shall be securely fixed i. e. they shall be welded or riveted or permanently attached to the enclosure by another equally effective method.

If a yield stress higher than 240 N/mm^2 is necessary, the required yield stress shall be:
- either marked on the apparatus or

- specified in the relevant certificate, in which case the apparatus shall be marked "X".

The type test is then carried out with the studs specified by the manufacturer.

11.5 Fasteners shall not pass through the walls of a flameproof enclosure unless they form a flameproof joint with the wall and they are non-detachable from the enclosure e. g. by welding, riveting or an equally effective method.

11.6 In the case of holes for screws or studs which do not pass through the walls of flameproof enclosures, the remaining thickness of the wall of the flameproof enclosure shall be at least one third of the nominal diameter of the screw or stud with a minimum of 3 mm.

11.7 When screws are fully tightened into blind holes in enclosure walls, with no washer fitted, at least one full thread shall remain free at the base of the hole.

11.8 If, for ease of manufacture, a wall of a flameproof enclosure has to be drilled through, the resulting hole shall subsequently be closed by a device so that the flameproof properties of the enclosure are maintained. This device shall be securely fixed in accordance with the requirements of 11.4 for studs.

If apertures provided in a flameproof enclosure (e. g. for cable or conduit entry) are not used, they shall be closed so that the flameproof properties of the enclosure are maintained (see Figure 22 for examples).

The closing device may be made so that it can be fitted or removed from either the outside or the inside of the wall of the flameproof enclosure.

The mechanically or frictionally locked closing device shall comply with one or more of the requirements of 11.9.1 to 11.9.3.

11.9.1 If the closing device is removable from the outside, this shall be possible only after disengagement of a retaining device inside the enclosure (see Figure 22a).

11.9.2 It may be so designed that it can be fitted or removed only by the use of a tool complying with the requirements of 9.2 of EN 50014:1992 (see Figure 22b).

11.9.3 It may be of a special construction in which insertion is done by a method other than that used for removal. Removal shall only be by one of the methods specified in 11.9.1 or 11.9.2 or by a special technique (see Figure 22c).

Separate fastening arrangements, requiring the use of a tool, of the type required in 9.2 of EN 50014:1992 or some equally effective methods shall be provided to secure and release threaded doors or covers.

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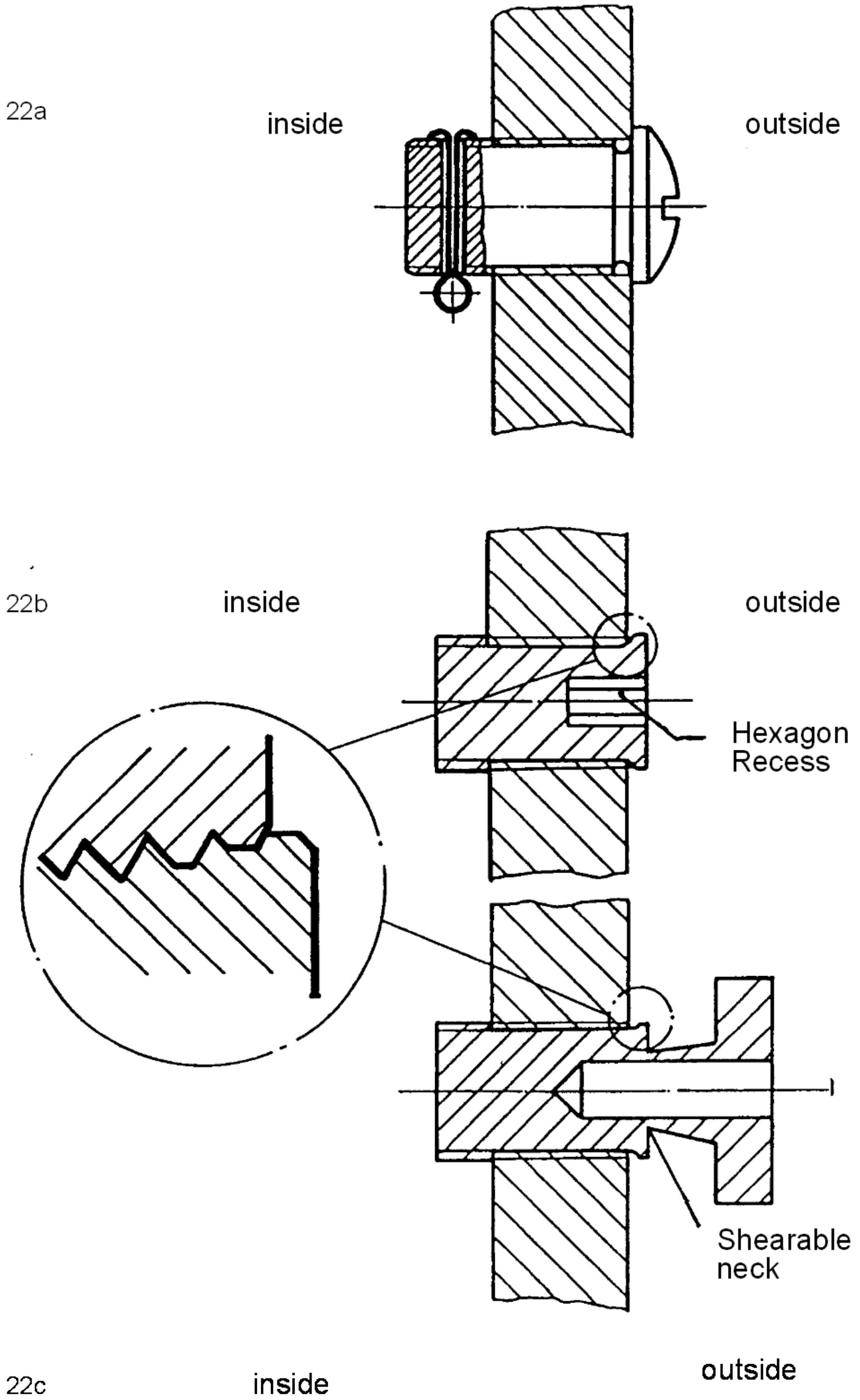

Figure 22: Examples of closing devices for unused apertures

10 Renumber the clause to ”12” and amend the heading to read ” Materials and mechanical strength of enclosures; materials inside the enclosures”.

10.1-10.4 Replace the sub-clauses with the following sub-clauses 12.1-12.6:

12.1 Flameproof enclosures shall withstand the relevant tests prescribed in Section "Verifications and tests".

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12.2 When several flameproof enclosures are assembled together, the requirements of this European Standard apply to each of them separately, and in particular to the partitions separating them and to all the bushings and operating rods which pass through the partitions.

12.3 When an enclosure contains several intercommunicating compartments or when it is subdivided because of the disposition of the internal parts, pressures and rates of rise of pressure greater than normal may be produced.

Such phenomena shall be precluded as far as possible by the construction. If it is impossible to avoid these phenomena, the resulting higher stresses shall be taken into account in the construction of the enclosure.

12.4 When cast iron is used, the material shall be not less than the quality 150 (ISO 185).

12.5 Liquids shall not be used in flameproof enclosures when there is a risk of producing an explosive mixture, more hazardous than that for which the enclosure was designed, by the decomposition of these liquids. They may, however, be used if the enclosure passes the tests prescribed in section "Verifications and tests" for the type of explosive mixture produced; however, the surrounding explosive atmosphere shall be appropriate to the Group for which the electrical apparatus is constructed.

12.6 In flameproof enclosures of Group I, insulating materials subjected to electrical stresses capable of causing arcs in air and which result from rated currents of more than 16 A (in switching apparatus such as circuit-breakers, contactors, isolators) shall have a comparative tracking index equal to or greater than CTI 400 M, according to IEC 112.

However, if the above-mentioned insulating materials do not pass this test they may be used if their volume is limited to 1 % of the total volume of the empty enclosure or if a suitable detection device enables the power supply to the enclosure to be disconnected, on the supply side, before possible decomposition of the insulating material leads to dangerous conditions. The presence and effectiveness of such a device shall be verified by the testing station.

(Note provided by comparison compiler: 1st paragraph in sub-clause 12.6 above in EN 50018 is identical with the 2nd paragraph in sub-clause A.2.1 in the normative Annex A in IEC 60079-1 + A1, except that ”However, for” should read ”In”)

11 Renumber the clause to ”13” and amend the heading to read ” Entries for flameproof enclosures”.

11.1-11.4 Replace the sub-clauses with the following:

The flameproof properties of the enclosure are not altered if all entries meet the relevant requirements given in this clause.

The following different means can be used to provide the connection of electrical apparatus within a flameproof enclosure to external circuits or to other electrical apparatus; nevertheless, the manufacturer shall state, in the documents defining the electrical apparatus, those means which are explicitly intended to be used for this purpose, the places where they can be mounted and the maximum permitted number of these means.

13.1 Cable entries

Cable entries, whether integral or separate, shall meet the requirements of this standard, the relevant requirements of Annex C "Flameproof cable entries" and create, on the enclosure the joint widths and gaps prescribed in clause 5.

Where cable entries are integral with the enclosure or specific to the enclosure they will be tested as part of the enclosure concerned.

Where cable entries are separate

- threaded Ex cable entries can be certified as apparatus. Such cable entries do not have to be submitted to the tests of 15.1 and the routine test of clause 16.

Threaded cable entries and their associated holes with threads not complying with ISO standards shall be marked in such a manner that any confusion can be precluded

- other cable entries can only be certified as an Ex component

13.2 Conduit entries

13.2.1 Conduit entries are permitted only for electrical apparatus of Group II.

13.2.2 Conduit entries shall create on the enclosure the joint widths and gaps prescribed in clause 5.

13.2.3 In addition, a sealing device such as a stopping box with setting compound shall be provided, either in the flameproof enclosure or immediately at the entrance thereto. A sealing device is considered as fitted immediately at the entrance of the flameproof enclosure when the device is fixed to the enclosure either directly or through an accessory necessary for coupling (such as a nipple or a three-piece union); it shall satisfy the type test for sealing prescribed in 15.3. The setting compound shall be specified in the certificate either of the stopping box or of the complete electrical apparatus having the flameproof enclosure. The part of the stopping box between the setting compound and the flameproof enclosure shall be treated as a flameproof enclosure, i. e. the joints shall comply with clause 5 and the assembly shall be submitted to the tests for non-transmission of 15.2.
Note: The sealing device may be applied by the installer or user of the electrical apparatus according to instructions provided by the manufacturer.

13.3 Plugs and sockets and cable couplers

13.3.1 Plugs and sockets shall be constructed and mounted so that they do not alter the flameproof properties of the enclosure on which they are mounted, even when the two parts of the plugs and sockets are separated.

13.3.2 The widths and the gaps of the flameproof joints (see clause 5) of the flameproof enclosures of plugs and sockets and cable couplers shall be determined by the volume which exists at the moment of separation of the contacts other than those for earthing or bonding or those which are parts of circuits complying with EN 50020 and EN 50039.

13.3.3 For plugs and sockets and cable couplers the flameproof properties of the enclosure shall be maintained in the event of an internal explosion both when the plugs and sockets or cable couplers are connected together and at the moment of separation of the contacts other than those for earthing or bonding or those which are parts of circuits complying with EN 50020 and EN 50039.

13.3.4 The requirements of 13.3.2 and 13.3.3 do not apply to plugs and sockets nor to cable couplers fixed together by means of special fasteners conforming to 11.1 and which bear a label with the warning

"DO NOT SEPARATE WHEN ENERGIZED".

13.4 Bushings

13.4.1 Bushings may contain one or more conductors. When they are correctly assembled and mounted in the walls of the enclosure, all joint widths, gaps or cemented joints shall conform with the relevant requirements of clauses 5 and 6.

When the bushing is formed by moulding insulation on metallic parts the requirements of 5.2, 5.3 and 5.4 do not apply, but clause 6 is applicable. The insulation material itself can contribute to the mechanical strength of the enclosure.

When the bushing includes parts assembled with adhesive, this is considered as a cement if it complies with the requirements of clause 6. Should this not be the case, the requirements of 5.2, 5.3 and 5.4 are applicable.

13.4.2 The parts of bushings outside the flameproof enclosure shall be protected in accordance with one of the types of protection listed in EN 50014.

13.4.3 Bushings specific to a flameproof enclosure shall satisfy the type tests and routine tests for that enclosure.

13.4.4 Bushings not specific to one flameproof enclosure shall be submitted to a type test for resistance to pressure carried out by means of a static pressure test as specified in 15.1.3.1 at the following values:

- 20 bar for electrical apparatus of Group I
- 30 bar for electrical apparatus of Group II

These bushings shall be subject to a routine pressure test as specified in 16.1, except where the assembly procedure used is described in the manufacturer's documentation and is such as to ensure consistency in the manufactured products.

(Note provided by comparison compiler: Parts of clause 13 above in EN 50018 deals with the same subject as clauses/sub-clauses they replaces, according to the following:

- *Sub-clause 13.2.3: sub-clause 11.4*
- *Sub-clause 13.3.1: sub-clause 11.3.3*
- *Sub-clause 13.3.2: sub-clause 11.3.3.1*
- Sub-clause 13.3.3: sub-clause 11.3.3.2
- *Sub-clause 13.3.4: sub-clause 11.3.3.3*
- *Sub-clause 13.4.1: sub-clause 11.3.1*
- *Sub-clause 13.4.2: 1st paragraph in 11.3*

Figure 21, 22 and 23 — Delete the figures.

12 Delete the clause.

Section three Amend the heading of the section to read ”VERIFICATIONS AND TESTS”

13 Replace the clause with the following:

The requirements of EN 50014 concerning verifications and tests are, for type of protection flameproof enclosure "d", supplemented by the following requirements.

The determination of the maximum surface temperature specified in 23.4.6.1 of

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EN 50014:1992 shall be made under the conditions defined in Table 5 of this standard.

Table 5: Conditions for the determination of maximum surface temperature

Type of electrical apparatus	Test voltage	Overload or fault conditions
Luminaires (without ballast)	U_n + 10 %	None
Ballast	U_n + 10 %	U_n + 10 % Rectifier effect simulated by diode 1)
Motors	$U_n \pm 5$ %	None
Resistors	U_n + 10 %	None
Electromagnets	U_n + 10 %	U_n and worst case air-gap
Other apparatus	$U_n \pm 10$ %	2)
Note: U_n is the rated voltage of the apparatus. 1) The rectifier effect is only to be simulated in the case of ballasts for tubular fluorescent lamps; it is therefore done according to IEC 82. 2) To be agreed between manufacturer and testing station, depending on the type of apparatus.		

14 Renumber the clause to ”15” and replace the first paragraph (”Normally...given”) with the following:

The type tests shall be carried out in the following sequence on one of the samples which has been subjected to the mechanical tests in accordance with 23.4.3 of EN 50014:1992.

1) Determination of the explosion pressure (reference pressure) in accordance with 15.1.2

2) Overpressure test in accordance with 15.1.3

3) Test for non-transmission of an internal ignition in accordance with 15.2

Testing stations may deviate from this test sequence in that the static or dynamic overpressure test may be carried out either after the test for non-transmission of an internal ignition or on another sample which has also been subjected to those other tests affecting mechanical strength already applied to the first sample; in no case, after the overpressure test, shall the joints of the enclosure have suffered a permanent deformation nor shall the enclosure have suffered any damage affecting the type of protection.

The enclosure shall, in general, be tested with all the enclosed apparatus in place. However, this may with the agreement of the testing station, be replaced by equivalent models.

If an enclosure is designed to take different types of apparatus and components, declared with the detailed mounting arrangements by the manufacturer, the enclosure may be tested empty provided that this is the most severe condition for explosion pressure development, and compliance with the other safety requirements of EN 50014 can be confirmed.

If the enclosure is designed so that it can be used in the absence of part of the enclosed apparatus the tests shall be made under the conditions considered by the testing station to be the most severe. In both cases the testing station shall then indicate in the certificate, on the basis of the proposals made by the manufacturer, the kinds of enclosed apparatus permitted and their mounting arrangements.

Joints of removable parts of flameproof enclosures shall be tested in the worst assembly conditions.

14.1 Renumber the sub-clause to "15.1" and replace it with the following:

15.1 Tests of ability of the enclosure to withstand pressure
15.1.1 General
The object of these tests is to verify that the enclosure can withstand the pressure of an internal explosion.

The enclosure shall be subjected to tests in accordance with 15.1.2 and 15.1.3.

The tests are considered satisfactory if the enclosure suffers no permanent deformation or damage, affecting the type of protection. In addition, the joints shall in no place have been permanently enlarged.

15.1.2 Determination of explosion pressure (reference pressure)

The reference pressure is the highest value of the maximum smoothed pressure, relative to atmospheric pressure, observed during these tests. For smoothing, a frequency limit of 5 kHz ± 10 % shall be adopted.

15.1.2.1 Each test consists of igniting an explosive mixture inside the enclosure and of measuring the pressure developed by the explosion.

The mixture shall be ignited by one or more sparking plugs or another low energy source. However, when the enclosure contains a device which produces sparks capable of igniting the explosive mixture, this device may be used to produce the explosion. (It is nevertheless not necessary to produce the maximum power for which the device is designed.)

The pressure developed during the explosion shall be determined and recorded during each test. The locations of the sparking plug(s) as well as those of the pressure gauge(s) are left to the discretion of the testing station, to find the combination which produces the highest pressure. When detachable gaskets are provided by the manufacturer, these shall be fitted to the enclosure under test.

The number of tests to be made and the explosive mixture to be used, in volumetric ratio with air and at atmospheric pressure, are as follows:

- Electrical apparatus of Group I:
3 tests with (9,8 ± 0,5) % methane

- Electrical apparatus of Group IIA:
3 tests with (4,6 ± 0,3) % propane

- Electrical apparatus of Group IIB:
3 tests with (8 ± 0,5) % ethylene

- Electrical apparatus of Group IIC:
3 tests with (14 ± 1) % acetylene and 3 tests with (31 ± 1) % hydrogen

15.1.2.2 Rotating electrical machines shall be tested at rest and, when the testing station considers it necessary, when running. When they are tested running, they may be driven either by their own source of power or by an auxiliary motor. The speed shall be between 90 % and 100 % of the rated speed of the machine.

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The pressures shall be determined at the ignition end, at the opposite end and at all points where higher pressures are likely to occur.

15.1.2.3 In cases where pressure-piling may occur during the test of flameproof enclosures, the tests shall be made with each gas at least five times. For Group IIB they shall afterwards be repeated at least five times with a mixture of (24 ± 1) % hydrogen/methane (85/15).

Note: There is presumption of pressure-piling when
either the pressure values obtained during a series of tests, deviate from one to another by a factor of ≥ 1,5 or

- the pressure rise time is less than 5 ms.

15.1.2.4 Electrical apparatus intended to be used in a single specified gas may be tested with the mixture of that gas with air at atmospheric pressure that gives the highest explosion pressure. Such electrical apparatus shall then be certified not for the corresponding Group but only for the gas considered. The restriction of use shall be indicated accordingly as specified in 27.2 Point 5 EN 50014:1992.

Where exclusion of a specific gas or gases is required, the apparatus shall be marked "X".

Double marking can be applied for a specific gas and for the next lower Group than the Group of this gas (e. g. IIB + H_2), if the enclosure has been submitted not only to the tests for the specific gas, but also to those necessary for the lower Group.

15.1.3 Overpressure test

This test shall be made by one of the following methods, which are considered as equivalent:

15.1.3.1 Overpressure test: first method (static)

The relative pressure applied shall be
- 1,5 times the reference pressure, with a minimum of 3,5 bar, or

-4 times the reference pressure for enclosures not subject to routine overpressure testing, or

-the following pressures, when reference pressure determination has been impracticable.

Volume (cm^3)	Group	Pressure (bar)
≤ 10	I, IIA, IIB, IIC	10
> 10	I	10
> 10	IIA, IIB	15
> 10	IIC	20

The period of application of the pressure shall be at least 10 s but need not exceed 60 s.

The test is made once.

The overpressure test shall be considered satisfactory if the test result is in compliance with 15.1.1 and if there is no leakage through the walls of the enclosure.

15.1.3.2 Overpressure test: second method (dynamic)

The dynamic tests shall be so carried out that the maximum pressure to which the enclosure is subjected is

1,5 times the reference pressure, but with a minimum of 3,5 bar.

When the test is carried out with mixtures specified in 15.1.2.1 these may by precompressed to produce an explosion-pressure of 1,5 times the reference pressure.

The test shall be made once only except for electrical apparatus of Group IIC for which each test shall be made three times with each gas.

The overpressure test shall be considered satisfactory if the test result is in compliance with 15.1.1.

Table IV Delete the table.

14.2 Renumber the sub-clause to ”15.2” and replace it with the following:

15.2 Test for non-transmission of an internal ignition

Gaskets, (see 5.4) are to be removed. The enclosure is placed in a test-chamber. The same explosive mixture is introduced into the enclosure and the test-chamber, at atmospheric pressure.

The length of threaded joints shall be reduced according to table 6.

Flanged gaps of spigot joints, where the width of the joint *L* consists only of a cylindrical part, (see figure 3) are to be enlarged to values of 1 mm for Group I and IIA, 0,5 mm for Group IIB and 0,3 mm for Group IIC.

Table 6: Reduction in length of a threaded joint for non-transmission test

Type of threaded joint	Reduction in length by			
	Groups I,IIA and IIB (15.2.1)		Group IIC (15.2.2)	
	15.2.1.1	15.2.1.2	15.2.2.1	15.2.2.2
Cylindrical, complying with ISO 965 fit medium or better	no reduction	1/3	1/3	no reduction
Cylindrical, with larger tolerances than permitted above	1/3	1/2	1/2	1/3
taper	no reduction	1/3	1/3	no reduction

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For tapered threads the joint shall be tested with the minimum hand tight engagement permitted by the thread standard at the extremes of tolerances.

Example of reduction of tapered threads:

After marking the position of hand tight engagement on the thread, the devices are removed and the length of engagement is reduced by cutting the screw or drilling out the hole. The parts are then reassembled to the marked position.

15.2.1 Electrical apparatus of Groups I, IIA and IIB

15.2.1.1 The gaps i_E of the enclosure shall be at least equal to 90 % of the maximum constructional gap i_C as specified in the manufacturer's drawings ($0,9\ i_C \leq i_E \leq i_C$).

The explosive mixtures to be used, in volumetric ratio with air and at atmospheric pressure, are as follows:

-electrical apparatus of Group I:
(12,5 ± 0,5) % methane-hydrogen [(58 ± 1) % methane and (42 ± 1) % hydrogen]
(MESG = 0,8 mm)

-electrical apparatus of Group IIA:
(55 ± 0,5) % hydrogen (MESG = 0,65 mm)

-electrical apparatus of Group IIB:
(37 ± 0,5) % hydrogen (MESG = 0,35 mm)

Note: The explosive mixtures chosen for this test ensure that the joints prevent the transmission of an internal ignition, with a known margin of safety. This margin of safety, *K*, is the ratio of the maximum experimental safe gap of the representative gas of the Group concerned to the maximum experimental safe gap of the chosen test gas:

-electrical apparatus of Group I: K = 1,14/0,8 = 1,42 (methane),

-electrical apparatus of Group IIA: K = 0,92/0,65 = 1,42 (propane),

-electrical apparatus of Group IIB: K = 0,65/0,35 = 1,85 (ethylene).

Alternatively, by agreement between testing station and the manufacturer, if the gaps of a test specimen do not fulfill the above condition one of the following methods may be used for the type test for non-transmission of an internal ignition:

- a gas/air mixture with a smaller MESG value:

	i_E/i_C	mixture
Group I	$\geq 0{,}75$	55 % H_2 ± 0,5
	$\geq 0{,}6$	50 % H_2 ± 0,5
Group IIA	$\geq 0{,}75$	50 % H_2 ± 0,5
	$\geq 0{,}6$	45 % H_2 ± 0,5
Group IIB	$\geq 0{,}75$	28 % H_2 ± 1

$\geq$ 0,6	28 % H_2 ± 1 at 1,4 bar

-Precompression of the normal test mixtures according to the following formula

P_k = 0,9 x i_C/i_E

P_k = Precompression factor

15.2.1.2 If enclosures of Groups IIA and IIB could be destroyed or damaged by the test in 15.2.1.1, it is permitted that the test be made by increasing the gaps above the maximum values specified by the manufacturer. The enlargement factor of the gap is 1,42 for Group IIA electrical apparatus and 1,85 for Group IIB electrical apparatus. The explosive mixtures to be used in the enclosure and in the test chamber, in volumetric ratio with air and at atmospheric pressure, are as follows:

-electrical apparatus of Group IIA: (4,2 ± 0,1) % propane

- electrical apparatus of Group IIB: (6,5 ± 0,5) % ethylene

15.2.1.3 The test in 15.2.1.1 or 15.2.1.2 shall be made five times. The test result is considered satisfactory if the ignition is not transmitted to the test chamber.

15.2.2 Electrical apparatus of Group IIC

The following methods can be used for this test.

15.2.2.1 First method

All gaps of joints other than threaded joints shall be increased to the value

i_E = 1,5 x i_C
with a minimum of 0,1 mm for flanged joints

i_E = the test gap

i_C = the maximum constructional gap, as specified on the manufacturer's drawings.

The following explosive mixtures, in volumetric ratio with air and at atmospheric pressure, are to be used in the enclosure and in the test-chamber:

- (28 ± 1) % hydrogen, and
- (7,5 ± 1) % acetylene.

Five tests shall be made with each mixture. If the apparatus is intended for use solely with hydrogen or solely with acetylene, the tests shall be made only with the corresponding gas mixture.

15.2.2.2 Second method

The enclosure shall be tested with a test gap i_E according to the following formula

$$0{,}9\, i_C \leq i_E \leq i_C$$

The enclosure and the test chamber are filled with one of the gas mixtures specified for the first method at a pressure equal to 1,5 times atmospheric pressure.

The test shall be made 5 times with each explosive mixture.

Alternatively if the gaps of a test specimen do not fulfill the above condition, by agreement between testing station and manufacturer, the following method may by used.

Precompression of the normal test mixtures according to the following formula:

$$P_k = 1{,}35 \times i_C/i_E$$

P_k = Precompression factor

15.2.2.3 Electrical apparatus which are single constructions shall be tested five times with unaltered gaps and with each of the explosive mixtures specified in 15.2.2.1 at atmospheric pressure.

Additional sub-clauses Add the following sub-clause 15.3 and 15.4:

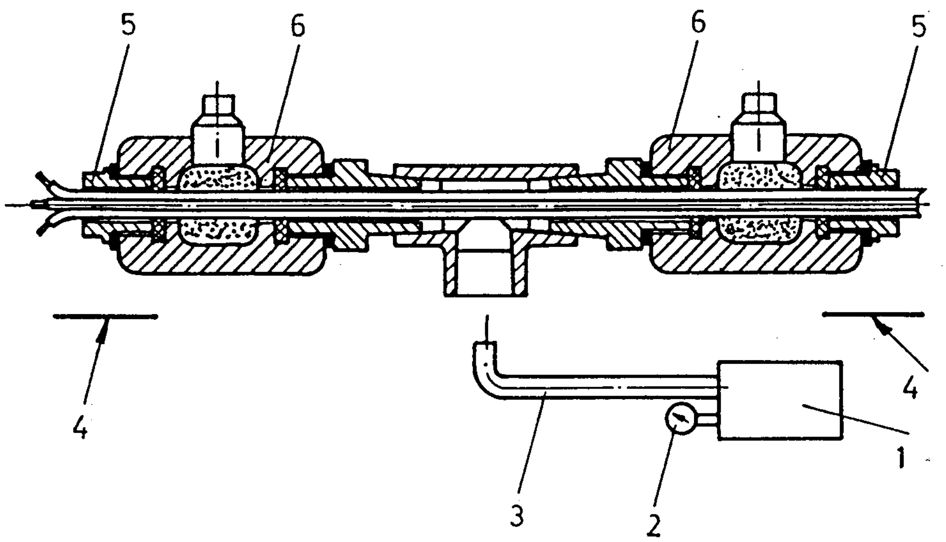
15.3 Test for sealing of stopping boxes with setting compound

A hydraulic testing device is to be used which avoids the application of pressure to the ends of the conductors; figure 23 shows an example of a device fulfilling this condition and allowing a simultaneous test on two stopping boxes.

The combination of conductors or cables to be placed in the conduit is chosen by the testing station so as to obtain the most unfavourable conditions.

Filling of the stopping box submitted to the sealing test is carried out in accordance with the instructions of the manufacturer of the stopping box.

A piece of clean white blotting paper is placed under the stopping boxes being tested so as to detect any leakage of liquid. The liquid to be used is coloured water. The hydraulic circuit is to be purged.



1 Hydraulic pump
2 Pressure gauge
3 Hose
4 Blotting paper
5 Packing gland to be removed for the test
6 Stopping box

Figure 23: Sealing test for stopping boxes with setting compound

The hydraulic pressure is then progressively raised to a value of 20 bar for Group I and 30 bar for Group II, which shall be reached in a maximum of 1 min. The pressure shown by the gauge is observed for 2 min and no reduction of pressure shall be observed.

At the end of the test, the blotting paper shall be free from any trace of leakage.

Note: It may be necessary to seal all joints of the test device other than those of the parts under test.

15.4 Tests of flameproof enclosures with breathing and draining devices

The tests in accordance with 15.4.1 to 15.4.3 shall be carried out in the following order on a sample after the impact strength test of 10.8.

For devices with non-measurable paths, the pore size of the sample shall not be less than 85 % of the specified maximum pore size.

15.4.1 Tests of ability of the enclosure to withstand pressure.

The tests shall be made in accordance with 15.1 with the following additions and modifications.

15.4.1.1 For the determination of the explosion pressure in accordance with 15.1.2, breathing and draining devices shall be replaced by solid plugs.

15.4.1.2 For the overpressure test in accordance with 15.1.3 a thin flexible membrane (e. g. a thin plastic sheet) shall be fitted to the inner surfaces of the breathing and draining devices. After the overpressure test, the device shall show no permanent deformation or damage, affecting the type of protection.

15.4.2 Thermal tests

15.4.2.1 Test procedure

The enclosure with the device(s) fitted shall be tested in accordance with the method 15.4.3.1 but with the ignition source only in the position giving the most unfavourable thermal results.

The temperature of the external surface of the device(s) shall be monitored during the test. The test shall be made five times. The test mixture to be used shall be (4,2 ± 0,1) % propane in volumetric ratio with air and at atmospheric pressure. Additionally, for devices intended for use in acetylene, (7,5 ± 1) % acetylene in volumetric ratio with air and at atmospheric pressure shall be used.

In an enclosure where there is the possibility of a forced or induced flow of a potentially dangerous gas, the enclosure shall be arranged during the tests so that the gas can flow through the device(s) and the enclosure.

Any ventilation or sampling system shall be operated as specified in the manufacturer's documentation. After each of the five tests the external explosive mixture shall be maintained for a sufficient time to allow any continuous burning on the face of the device to become evident (e. g. for at least 10 min so as to increase the temperature of the external surface of the device or to make heat transfer to the outer face possible.)

15.4.2.2 Acceptance criterion

No continuous burning shall be observed. No flame transmission shall occur. The measured external surface temperature rise of the device shall be multiplied by a safety factor of 1,2 for the determination of the temperature class of the electrical apparatus.

15.4.3 Test for non-transmission of an internal ignition

This test shall be made in accordance with 15.2 with the following additions and modifications.

15.4.3.1 Test procedure

An ignition source shall be placed first close to the inner surface of the breathing and draining device and subsequently in one or more places if a high peak explosion pressure and rate of rise of pressure at the face of the device is likely to occur. Where the enclosure has more than one identical device, the device to be tested shall be that which gives the most unfavourable results. The test mixture within the enclosure shall be ignited. The test shall be made five times for each position of the ignition source.

15.4.3.2 Non-transmission test for breathing and draining devices
For breathing and draining devices of Groups I, IIA and IIB the non-transmission test of 15.2.1 shall be applied.

For breathing and draining devices of Group IIC with measurable paths, 15.2.2 and either 15.4.3.2.1 or 15.4.3.2.2 is to be applied. For breathing and draining devices of Group IIC with non-measurable paths, 15.4.3.2.1 or 15.4.3.2.2 is to be applied.

15.4.3.2.1 Method A
For devices intended for use only in hydrogen, only the test with the hydrogen/air mixture is required. The tests are made five times with each test mixture. The tests are made according to 15.2.2.2 and 15.4.3.1.

15.4.3.2.2 Method B
The use of this method involves limitation of the range of Group IIC gases covered. The restriction of use shall be indicated accordingly as specified in 27.2. Point 5 of EN 50014:1992**.**

Where exclusion of a specific gas or gases is required, the apparatus shall be marked "X".

Carbon disulphide is excluded for enclosures with a volume greater than 100 cm^3.

The test mixtures to be used consist of the following, in volumetric ratio and at atmospheric pressure:

a) (40 $\pm$ 1) % hydrogen, (20 $\pm$ 1) % oxygen and the rest nitrogen
b) (10 $\pm$ 1) % acetylene, (24 $\pm$ 1) % oxygen and the rest nitrogen

The tests shall be made 5 times with each test mixture, in accordance with 15.4.3.1.

For devices intended for use only in hydrogen, only test mixture
a) is to be used.

15.4.3.3 Acceptance criterion

The test result is considered satisfactory if no ignition is transmitted to the test-chamber.

(Note provided by comparison compiler: Parts of sub-clause 15.4 above in EN 50018 deal with the same subject – some of them with equal requirements – as parts of the normative Annex B in IEC 60079-1 + A1, according to the following:

- *The first two paragraphs in sub-clause 15.4 (”The tests…pore size.”): the first two paragraphs in sub-clause B.11 (identical except references to sub-clauses)*

- *Sub-clause 15.4.1: sub-clause B.11.1*

- *Sub-clause 15.4.2: sub-clause B.11.2*

- *Sub-clause 15.4.3: sub-clause B.11.3*

15 Renumber the clause to ”16” and replace it with the following:

16 Routine tests

16.1 The following routine tests are intended to ensure that the enclosure withstands the pressure and also that it contains no holes or cracks connecting to the exterior.

The routine tests include an overpressure test made according to one of the methods described for the type tests in 15.1.3. The method is to be agreed between the manufacturer and the testing station.

16.1.1 The routine overpressure test may be made by the first method even when the overpressure type test has been made by the second method.

When the determination of the reference pressure has been impracticable (the pressure appearing abnormal) and when a dynamic test involves a risk to the enclosed apparatus (windings, etc.), the static pressures to be applied are as follows:

Volume (cm³)	Group	Pressure (bar)
≤ 10	I, IIA, IIB, IIC	10
> 10	I	10
> 10	IIA, IIB	15
> 10	IIC	20

16.1.2 When the second method is chosen, the routine test consists of

-either an explosion test with, inside and outside the enclosure, the appropriate explosive mixture specified in 15.1.2 (for the determination
-
-of explosion pressure) at 1,5 times atmospheric pressure;

-or a dynamic overpressure test as described in 15.1.3.2 for type tests, followed by a non- transmission test with explosive mixtures as specified in 15.2.1.2 or 15.2.2.1 (test for non-transmission of an internal ignition, with enlarged gaps) inside and outside the enclosure at atmospheric pressure;

-or a dynamic overpressure test as described in 15.1.3.2 for type tests, followed by a static test at a pressure of at least 2 bar.

16.1.3 For the routine test, it is sufficient to test the enclosure empty. However, if the routine test is dynamic and the enclosed apparatus influences the pressure rise during an internal explosion, the test conditions shall be decided by agreement between the manufacturer and the testing station.

The individual parts of a flameproof enclosure (e. g. cover and base) can be tested separately. The test conditions shall be such that the stresses are comparable to those to which these parts are exposed in the complete enclosure.

16.2 Routine tests are not required for enclosures with a volume less than or equal to 10 cm^3. This exception also applies to enclosures with a volume greater than 10 cm^3 when the prescribed type test has been made at a static pressure of four times the reference pressure. However, enclosures of welded construction shall in every case be submitted to the routine test.

For enclosures where reference pressure measurement is impractical, exemption from routine pressure testing shall not apply.

Routine tests are not required for bushings not specific to one flameproof enclosure, if the assembly procedure is sufficiently documented (see 13.4.4).

16.3 The routine tests are considered satisfactory if:

-the enclosure withstands the pressure without suffering permanent deformation of the joints or damage to the enclosure, and

-when the test has been made by the dynamic followed by static tests of 16.1.2 there is no leakage through the walls of the enclosure or if made dynamically there is no transmission of an internal ignition.

Additional clauses Add the following additional clauses under a common heading "OTHER REQUIREMENTS":

17 Switchgear

Group I flameproof enclosures which are to be opened from time to time on site, e. g. for adjustment purposes or for resetting of protection relays and which contain remotely operated switching devices in which circuits can be made or broken by a separate influence (which may be mechanical, electrical, electro-optical, pneumatic, acoustic, magnetic or thermal) when this influence is not applied manually to the apparatus itself, which produce, in service arcs or sparks capable of ignition an explosive mixture shall comply with the following requirements.

17.1 Means of isolation

All accessible conductors, except those of intrinsically safe circuits complying with EN 50020 or EN 50039 and those for bonding or earthing, shall be capable of being isolated from the supply before the opening of the flameproof enclosure.

The means of isolation of these flameproof enclosures shall be either

17.1.1 fitted inside the flameproof enclosure, in which case the parts which remain energised after the means of isolation has been opened shall either

-be protected by one of the standard types of protection listed in EN 50014 or

-shall have clearances and creepage distances between phases and to earth in accordance with the requirements of EN 50019, and be protected by an enclosure that provides a degree of protection of at least IP20 according to IEC 529 arranged so that a tool cannot contact the energized parts through any openings. This does not apply to parts of intrinsically safe circuits complying with EN 50020 or EN 50039 which remain energized.

In either case, a warning label "DO NOT OPEN WHEN ENERGIZED" shall be provided on the cover protecting the parts which remain energised.

or

17.1.2 fitted inside another enclosure complying with one of the standard types of protection listed in EN 50014.

or

17.1.3 a plug and socket or a cable coupler complying with the requirements of 13.3.

17.2 Doors or covers

17.2.1 Quick-acting doors or covers

These doors or covers shall be mechanically interlocked with an isolator so that

17.2.1.1 the enclosure retains the properties of the flameproof enclosure, type of protection "d", as long as the isolator is closed and

17.2.1.2 the isolator can only be closed when these doors or covers ensure the properties of the flameproof enclosure, type of protection "d".

17.2.2 Doors or covers fixed by screws

These doors or covers shall bear a label 'DO NOT OPEN WHEN ENERGIZED'.

17.2.3 Threaded doors or covers

These doors or covers shall bear a label 'DO NOT OPEN WHEN ENERGIZED'.

18 Lampholders and lampcaps

The following requirements apply to lampholders and lampcaps which together have to form a flameproof enclosure, type of protection "d", so that they may be used in luminaires of increased safety, type of protection "e".

18.1 Device preventing lamps working loose

The device which prevents lamps working loose, required in B.1 of EN 50019:1992, increased safe- ty "e", may be omitted for threaded lampholders provided with a quick-acting switch in a flameproof enclosure, type of protection "d", which breaks all poles of the lamp circuit before contact separation.

18.2 Holders and caps for lamps with cylindrical caps

18.2.1 Holders and caps for tubular fluorescent lamps shall comply with the dimensional requirements of data sheets Fa 6 of IEC 61.

18.2.2 For other holders, the requirements of clause 5 shall apply, but the width of the flameproof joint between the holder and the cap shall be at least 10 mm at the moment of contact separation.

18.3 Holders for lamps with threaded caps

18.3.1 The threaded part of the holder shall be of a material which is resistant to corrosion under the likely conditions of service.

18.3.2 At the moment of contact separation during unscrewing of the lamp, at least two complete turns of the thread shall be engaged.

18.3.3 For threaded lampholders E 27 and E 40, electrical contact shall be established by spring- loaded contact-elements. In addition, for electrical apparatus of Group IIB or IIC, the making and breaking of contact during insertion and removal of the lamp, shall take place within a flameproof enclosure, type of protection "d", of Group IIB or IIC respectively.

Note: For threaded lampholders E 10 and E 14, the requirements of 18.3.3 are not necessary.

19 Non-metallic enclosures and non-metallic parts of enclosures

The following requirements apply to non-metallic enclosures and non-metallic parts of enclosures, except for

-sealing rings of cable entries and

-non-metallic parts on which the type of protection does not depend.

19.1 Permitted non-metallic enclosures

Non-metallic enclosures are permitted
- if their free volume is $\leq$ 3 000 cm^3

-without limitation of volume if the enclosure is partly made of non-metallic material and if the surface area of each of the parts of nonmetallic material does not exceed 500 cm^2; however, the light transmitting part

of a luminaire may have a surface area not exceeding 8 000 cm^2.

19.2 Special constructional requirements

19.2.1 Resistance to tracking and creepage distances on internal surfaces of the enclosure walls

When an enclosure or a part of an enclosure of non-metallic material serves directly to support live bare parts, the resistance to tracking and the creepage distances on the internal surfaces of the walls of the enclosure shall comply with the requirements of EN 50019:1992.

However, for enclosures of electrical apparatus of Group I which may be subjected to electrical stresses capable of producing arcs in air and which result from rated currents of more than 16 A, the requirements stated in 12.6 are to be observed.

19.3 Supplementary requirements for type tests

The type tests according to 23.4 of EN 50014: 1992 shall be supplemented by the tests which are indicated in 19.3.1 and 19.3.2.

19.3.1 Tests for flameproofness

19.3.1.1 Test procedure

The tests for flameproofness shall be made in the following order on the enclosures which have been previously subjected, as far as these tests are applicable, to the tests of 23.4.7 of the European standard EN 50014:1992.

19.3.1.2 Tests of ability of the enclosure to withstand pressure

These tests shall be made as specified in 15.1.

19.3.1.3 Test of erosion by flame.

This test shall be made only on enclosures of volume greater than 100 cm^3 and of which the flameproof joints have at least one face of plastics material. For this
Test

-static gaps of flanged joints and plane parts of spigot joints of the enclosure shall be set to a value between 0,1 mm and 0,15 mm; however, if the maximum permitted static gap for the Group under consideration is less than 0,15 mm, the gaps shall be set to the maximum permitted value;

-cylindrical joints and cylindrical parts of spigot joints, as well as threaded joints, shall not be modified;
-for bushings which are common to two adjacent flameproof enclosures, the test shall be carried out in the enclosure giving the worst conditions.

The test consists of 50 ignitions of the explosive mixture specified in 15.1.2.1 for the corresponding Group. In the case of electrical apparatus of Group IIC, 25 ignitions shall be made with each of the two explosive mixtures specified in 15.1.2.1.

The test is judged satisfactory if the following test for non-transmission is satisfactory.

19.3.1.4 Test for non-transmission of an internal ignition

This test shall be made as specified in 15.2.

19.3.2 Flammability

This test shall be carried out only for enclosures or parts of enclosures of plastic materials.

19.3.2.1 The test shall be carried out in accordance with ISO 1210.

The test pieces shall

-be cut from the enclosure of the electrical apparatus or
-be moulded as individual pieces or
-be cut from plates prepared for this purpose.

The test pieces moulded as individual pieces or the plates from which the test pieces are cut shall be produced under conditions as close as possible to those used to produce the enclosures of the electrical apparatus. These conditions shall be recorded in the manufacturer's documentation.

Note: If the conditions under which the enclosures are produced are critical, they should be recorded in the certification documents.

The time during which any test piece continues to burn after removal of the flame shall be less than 15 s. During this time, the test piece shall not be burnt completely (ISO 1210).

19.3.2.2 If the test in 19.3.2.1 is not applicable due to distortion of the test piece out of flame one of the following tests shall be applied.

19.3.2.2.1 First alternative test method

The burning test is to be conducted in a chamber, enclosure or laboratory hood that is free from draughts. Each specimen is to be supported from the upper (6 mm) end of the specimen, with the longitudinal axis vertical, by the clamp on the ring stand so that the lower end of the specimen is 10 mm above the top of the burner tube and 300 mm above the horizontal layer of dry absorbent surgical cotton (50 mm x 50 mm swatch thinned

to a maximum free standing thickness of 6 mm).

The bunsen burner shall have a tube with a length of 100 mm and an inside diameter of (9,5 ± 0,5) mm. The tube shall not be equipped with end attachments such as a stabilizer.

The gas should be technical grade methane gas with suitable regulator and meter for uniform gas flow. (Natural gas having a heat content of approximately 37 MJ per cubic meter has been found to provide similar results).

The test specimens shall be (125 ± 5) mm in length, (13 ± 0,3) mm in width and (4 ± 0,2) mm in thickness.

When necessary, the specimens shall be preconditioned (see 5.2 of ISO 1210). The burner is to be placed remote from the specimen, ignited, and adjusted to produce a blue flame 20 mm high. The flame should be obtained by adjusting the gas supply and the air ports of the burner untill a 20 mm yellow tipped blue flame is produced and then an increase in the air supply is to be made until the yellow-tip disappears. The height of the flame is to measured again and corrected, if necessary.

The test flame is to be placed centrally under the lower end of test specimen and allowed to remain for 10 seconds. The test flame is then to be withdrawn at least 150 mm away and the duration of flaming of the specimen noted. When flaming of the specimen ceases; the test flame is to be immediately placed again under the specimen.

After 10 seconds, the test flame is again to be withdrawn, and the duration of flaming and glowing is to be noted.

The flammability properties of the tested material are acceptable when:

-no specimen burns with flaming combustion for more than 10 seconds after each application of the test flame.

-the total flaming combustion time does not exceed 50 seconds for the 10 flame applications for each set of 5 specimens.

-no specimen burns with flaming or glowing combustion up to the holding clamp.

-no specimen drips flaming particles that ignite the dry absorbent surgical cotton located 305 mm below the test specimen.

- no specimen burns with glowing combustion which persists beyond 30 seconds after the second removal of the test flame.

19.3.2.2.2 Second alternative test method

The test shall be carried out in accordance with IEC 707 (Method FV: Flame - Vertical specimen)

The test pieces shall

-be cut from the enclosure of the electrical apparatus or

-be moulded as individual pieces or
-be cut from plates prepared for this purpose.

The test pieces moulded as individual pieces or the plates from which the test pieces are cut shall be produced under conditions as close as possible to those used to produce the enclosures of the electrical apparatus. These conditions
shall be recorded in the manufacturer's documentation.

19.3.2.2.3 In these cases 50 explosions according to 19.3.1.3 shall be done inside the enclosure as a type test before applying the tests according 19.3.1.2 and 19.3.1.4, except if the test of erosion by flame has already been carried out successfully.

19.4 Test report

The test report shall include:

-the complete reference of the electrical apparatus;

-the complete reference of the non-metallic material used for the manufacture of the enclosure or parts of the enclosure;

-the result obtained in each of the tests specified;

-the description of tests which have not been made according to the specified requirements and the reasons for the deviations.

20 Pressure transducers using capillaries

The capillaries shall either comply with the gap dimensions given in table 1 or table 2 for cylindrical joints using 0 as the diameter of the inner part, or when the capillaries do not conform to the gaps given in these tables, the pressure transducers shall be certified if they pass the test for non-transmission of an internal ignition given in 15.2.

(Note provided by comparison compiler: Parts of clause 19 deal with the same subject – some of them with equal requirements - as parts of the normative Annex A in IEC 60079-1 + A1, according to the following:

- *1st paragraph in clause 19 ("The following…depend."): clause A.1 (identical except a reference)*

- *Sub-clause 19.2.1 (including reference to EN 50019:1992):*
  sub-clause A.2.1 (including reference to IEC 79-7)

- 1st paragraph in sub-clause 19.3 (including reference to EN 50014:1992):
  sub-clause A.3.1

- *Sub-clause 19.3.1.1: sub-clause A.3.2 1 (identical except reference to sub-clause and heading)*

- *Sub-clause 19.3.1.2: sub-clause A.3.2.2 (identical except reference to sub-clause)*

- *Sub-clause 19.3.1.3: sub-clause A.3.2.3*

- *Sub-clause 19.3.1.4: sub-clause A.3.2.4 (identical except reference to sub-clause)*

- *Sub-clause 19.3.2.1: sub-clause A.3.3.1 (identical)*

- *First paragraph in sub-clause 19.3.2.2: sub-clause A.3.3.2 (identical except reference to sub-clause)*

- *Sub-clause 19.3.2.2.2: sub-clause A.3.3.3.1 (identical except the heading)*

- *Sub-clause 19.3.2.2.1: sub-clause A.3.3.3.2*

EN 50018 refers in sub-clause 19.3 to requirements according to EN 50014:1992, regarding subjects dealt with in sub-clause A.3.1 in Annex A of IEC 60079-1 + A1.)

Annex A Delete the annex.

(Note provided by comparison compiler: Parts of EN 50018 (see clause 19) and the associated EN 50014, deal with the same subject – some of them with equal requirements - as parts of this normative annex in IEC 60079-1 + A1)

Annex B Replace the annex with the following normative Annex A and B:

ANNEX A (normative)

Additional requirements for crimped ribbon elements of breathing and draining devices

A.1 Crimped ribbon elements shall be constructed from cupro-nickel, stainless steel or a metal agreed between the manufacturer and the testing station. Aluminium, titanium, magnesium and their alloys shall not be used.

A.2 Where the paths through the device can be specified in the drawings and can be measured in the complete device, an upper and lower tolerance limit for the path dimensions shall be specified and shall be monitored in production.

A.3 Where A.2 does not apply, the relevant requirements of Annex B shall apply.

A.4 The type tests of 15.4.3 shall be carried out with samples manufactured with the largest permitted gap dimensions.

(Note provided by comparison compiler: the text of Annex A above in EN 50018 is identical with the text of sub-clause B.6 in the normative Annex B in IEC 60079-1 + A1, except references to sub-clauses.)

ANNEX B (normative)

Additional requirements for elements, with non-measurable paths, of breathing and draining devices

B.1 Sintered metal elements

B.1.1 Sintered metal elements shall be constructed from one of the following:

-stainless steel

-90/10 copper-tin bronze (see however 10.2)

-a specific metal or specific alloy agreed between the manufacturer and the testing station. Aluminium, titanium, magnesium and their alloys shall not be used.

B.1.2 The equivalent bubble test pore size shall be determined by the method specified in ISO 4003.

B.1.3 The density of the sintered metal element shall be determined in accordance with ISO 2738.

B.1.4 Where determination of open porosity and/or fluid permeability of elements is required in connection with functional aspects of devices, measurements shall be made in accordance with ISO 2738 and ISO 4022.

B.1.5 Sintered metal elements shall be clearly identified in the documentation by declaring:

-the material in accordance with 10.2 and B.1.1
-the maximum bubble test pore size in μm in accordance with B.1.2
-the minimum density in accordance with B.1.3
-the minimum thickness
-where appropriate the fluid permeability and open porosity in accordance with B.1.4.

B.2 Pressed metal wire elements

B.2.1 Metal wire elements shall be constructed from stainless steel wire braid or another specified metal agreed between the manufacturer and the testing station. Aluminium, titanium, magnesium and their alloys shall not be used. Manufacture shall start from a wire braid which is compressed in a die to form a homogeneous matrix.

B.2.2 In order to evaluate the density, the wire diameter shall be specified. Information shall also be given on the mass, length of wire braid, thickness of the element and mesh size. The ratio between the mass of the element and the mass of an identical volume of the same solid metal shall be between 0,4 and 0,6.

B.2.3 The equivalent bubble test pore size shall be determined by the method specified in ISO 4003.

B 2.4 The density of the element shall be determined in accordance with ISO 2738.

B 2.5 Where determination of open porosity and/or fluid permeability is required in connection with functional aspects of elements, measurements shall be made in accordance with ISO 2738 and ISO 4022.

B.2.6 Metal wire elements shall be clearly identified in the documentation by declaring:

-the material in accordance with 10.2 and B.2.1
-the maximum bubble test pore size in μm in accordance with B.2.3
-the minimum density in accordance with B.2.4
-the dimensions, including tolerances
-the original wire diameter
-where appropriate the fluid permeability and open porosity in accordance with B.2.5.

B.3 Metal foam elements

B.3.1 Metal foam elements shall be produced by coating a reticulated polyurethane foam with nickel, removing the polyurethane by thermal decomposition, converting the nickel into a nickel-chrome alloy e.g. by gaseous diffusion, and compressing the material as necessary.

B.3.2 Metal foam elements shall contain at least 15 % chromium by mass.

B.3.3 The equivalent bubble test pore size shall be determined by the method specified in ISO 4003.

B.3.4 The density of the element shall be determined in accordance with ISO 2738.

B.3.5 Where determination of open porosity and/or fluid permeability is required in connection with functional aspects of elements, measurements shall be made in accordance with ISO 2738 and ISO 4022.

B.3.6 Metal foam elements shall be clearly defined in the documentation by declaring:

-the material, in accordance with 10.2, B.3.1 and B.3.2
-the maximum bubble test pore size in µm in accordance with B.3.3
-the minimum thickness
-the minimum density
-where appropriate, the open porosity and fluid permeability in accordance with B.3.5.

(Note provided by comparison compiler: Parts of Annex B above in EN 50018 deal with the same subject – some of them with equal requirements - as parts of the normative Annex B in IEC 60079-1 + A1, according to the following:

- *Sub-clause B.1.1: sub-clause B.7.2.1.1*
- *Sub-clause B.1.2: sub-clause B.7.1.2*
- *Sub-clause B.1.3: sub-clause B.7.2.1.3 (identical)*
- *Sub-clause B.1.4: sub-clause B.7.2.1.4 (identical)*
- *Sub-clause B.1.5: sub-clause B.7.2.1.5*
- *Sub-clause B.2.1: sub-clause B.7.2.2.1*
- *Sub-clause B.2.2: sub-clause B.7.2.2.2*
- *Sub-clause B.2.3: sub-clause B.7.2.2.3*
- *Sub-clause B.2.4: sub-clause B.7.2.2.4 (identical)*
- *Sub-clause B.2.5: sub-clause B.7.2.2.5 (identical)*
- *Sub-clause B.2.6: sub-clause B.7.2.2.6*
- *Sub-clause B.3.1: sub-clause B.7.2.3.1*
- *Sub-clause B.3.2: sub-clause B.7.2.3.2 (identical)*
- *Sub-clause B.3.3: sub-clause B.7.2.3.3 (identical)*
- *Sub-clause B.3.4: sub-clause B.7.2.3.4 (identical)*
- *Sub-clause B.3.5: sub-clause B.7.2.3.5 (identical)*
- *Sub-clause B.3.6: sub-clause B.7.2.3.6*

Additional annex   Add the following normative Annex C:

ANNEX C (normative)

Additional requirements for flameproof cable entries

Section 3

National Differences

C.1 This annex contains specific requirements which apply, in addition to those in the European standard EN 50014: "General requirements" to the construction and testing of flameproof cable entries.

C.2 Constructional requirements

C.2.1 Sealing methods

C.2.1.1 Cable entries with elastomeric sealing rings

C.2.1.1.1 If a cable entry can accept any sealing ring with the same outside diameter but with different internal dimensions, the ring shall have a minimum uncompressed axial height of:

-20 mm, for circular cables of diameter not greater than 20 mm and for non-circular cables of perimeter not greater than 60 mm.

-25 mm, for circular cables of diameter greater than 20 mm and for non-circular cables of perimeter greater than 60 mm.

C.2.1.1.2 If a cable entry can accept only one specific elastomeric sealing ring, this ring shall have a minimum uncompressed axial height of 5 mm. In this case the cable entry shall be marked with an "X". However, for flameproof enclosures of Group I and IIC with a volume greater than 2 000 cm^3, the minimum axial heights are those required in C.2.1.1.1.

C.2.1.2 Cable entries sealed with setting compound

The minimum length of the compound shall be 20 mm when installed.

The manufacturer shall specify:

-the maximum diameter over cores of the cable that the entry is intended to accept.
-the maximum numbers of cores that can pass through the compound.

These specified values shall ensure that at all points along the 20 mm seal length at least 20 % of the cross section area is filled with compound.

The cable entry shall be capable of being fitted and removed from electrical apparatus without disturbing the compound seal after the specified curing period of the compound.

The filling compound and appropriate installation instructions are to be provided with the cable entry to the user by the manufacturer. These instructions are to form part of the descriptive documents.

C.2.2 Threaded cable entries

Threads forming a flameproof joint shall comply with the relevant requirements of 5.3.

For cylindrical threads the threaded part shall be at least 8 mm in length and comprise at least 6 full threads. If the thread is provided with an undercut, then a non-detachable and non compressible washer or equivalent device shall be fitted to ensure the required length of thread engagement.

Note: The requirement for 6 threads is to ensure that at least 5 full threads will be in engagement when the cable entry is assembled onto the flameproof enclosure.

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C.3 Type tests

C.3.1 Sealing test

C.3.1.1 Cable entries with sealing ring

These tests shall be carried out using, for each type of cable entry, one sealing ring of each of the different permitted sizes. In the case of elastomeric sealing rings, each ring is mounted on a clean, dry, polished mild steel cylindrical mandrel of diameter equal to the smallest cable diameter permissible in the ring as specified by the manufacturer of the cable entry.

In the case of metallic or composite sealing rings, each ring is mounted on the metal sheath of a clean dry sample of cable of diameter equal to the smallest diameter permissible in the ring, as specified by the manufacturer of the cable entry.

In the case of sealing rings for non-circular cables, each ring is mounted on a clean dry sample of cable of perimeter equal to the smallest value permitted in the ring, as specified by the manufacturer of the cable entry.

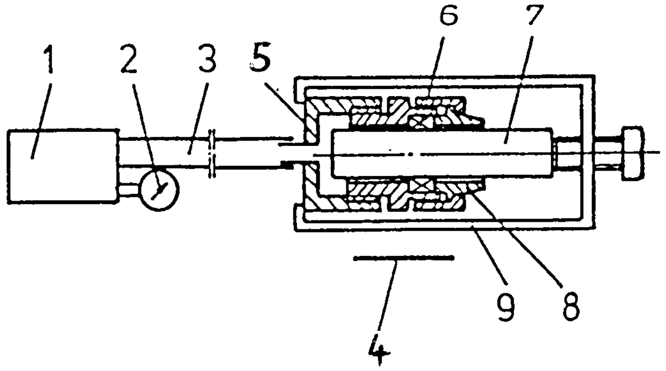
The assembly is then fitted into the entry and a torque is applied to the screws (in the case of a flanged compression device) or to the nut (in the case of a screwed compression device) to obtain a seal under a hydraulic pressure of 20 bar for Group I and 30 bar for Group II.

Note: The torque figures referred to in the preceding paragraph may be determined experimentally prior to the tests, or they may be supplied by the manufacturer of the cable entry.

The assembly is then mounted into a hydraulic testing device using coloured water or oil as the liquid, as shown in principle in figure 24. The hydraulic circuit is then purged. The hydraulic pressure is then gradually increased.

The sealing is considered satisfactory if the blotting paper is free from any trace of leakage when the pressure has been maintained at 20 bar for Group I or 30 bar for Group II, for two minutes.

Note: It may be necessary to seal all the joints of the cable entry mounted in the test device, other than those associated with the sealing ring under test. When a sample of metal-sheathed cable is used, it may be necessary to avoid the application of pressure to the ends of the conductors or to the interior of the cable.



1	Hydraulic pump	6	Sealing ring
2	Pressure gauge	7	Mandrel/Metal-sheathed cable
3	Hose	8	Compression component
4	Blotting paper	9	Retaining clamp
5	Adaptor		

Figure 24: Device for the sealing tests for cable entries

C.3.1.2 Cable entries sealed with setting compound

The test shall be carried out using, for each size of cable entry, metal mandrels, the number and diameter, of which result in the least practical amount of compound as specified in C.2.1.2 paragraph 2 through any cross section of the entry.

The setting compound is prepared following the manufacturer's instructions and then filled into the appropriate volume. It is allowed to harden for the appropriate time. The test prescribed in 23.4.7.3 and 23.4.7.4 of EN 50014:1992 shall be applied.

The assembly is then mounted into the hydraulic testing device defined in C.3.1.1 above and the same procedure is applied. The acceptance criteria are also the same.

C.3.2 Test of mechanical strength

C.3.2.1 Cable entries with a screwed compression element

A torque of twice that required in the sealing test shall be applied to the compression element; however, the value of this torque expressed in Nm shall always be at least 3 times the value in millimeters of the maximum permissible cable diameter when the cable entry is designed for circular cables or equal to the value in millimeters of the maximum permissible cable perimeter when the cable entry is designed for non circular cables.

The cable entry is then dismantled and its parts are examined.

C.3.2.2 Cable entries with a compression element fixed by screws

A torque of twice that required in the sealing test shall applied to the compression element screws; however, the value of this torque shall always be at least equal to the following values:

M6:10 Nm	M 12 : 60 Nm
M8:20 Nm	M 14 :100 Nm
M10:40 Nm	M 16 :150 Nm

The cable entry is then dismantled and its parts are examined.

C.3.2.3 Cable entries sealed with setting compound

In the case of threaded entries, a torque in Nm equal to the minimum value specified in C.3.2.1 shall applied to the entry when screwed into a steel test block having a suitable threaded hole.

The cable entry is then dismantled and its parts are examined.

C.3.2.4 Acceptance criteria

The tests C.3.2.1 to C.3.2.3 shall be considered to be satisfactory if no damage is found to any of the parts of the cable entry.

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Note: Any damage to the sealing ring may be disregarded, as the test is intended to show that the mechanical strength of the cable entry is sufficient to withstand the conditions of use.

(Note provided by comparison compiler: Parts of Annex C above in EN 50018 deal with the same subject as parts of clause 11 in IEC 60079-1 + A1, according to the following:

- *Sub-clause C.2.1.1: sub-clause 11.4 (2nd paragraph)*
- *1st paragraph in sub-clause C.2.1.2: sub-clause 11.4 (2nd paragraph)*
- *Sub-clause C.2.2: sub-clause 11.2*

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Table 1: Minimum width of joint and maximum gap for enclosures of Groups I, IIA and IIB

Type of joint		Minimum width of joint L mm	Maximum gap in mm for Volume V (cm^3)											
			$V \leq 100$			$100 < V \leq 500$			$500 < V \leq 2\ 000$			$V > 2\ 000$		
			I	IIA	IIB	I	IIA	IIB	I	IIA	IIB	I	IIA	IIB
Flanged, cylindrical or spigot joints		6	0,30	0,30	0,20	-	-	-	-	-	-	-	-	-
		9,5	0,35	0,30	0,20	0,35	0,30	0,20	-	-	-	-	-	-
		12,5	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,20	0,15
		25	0,50	0,40	0,20	0,50	0,40	0,20	0,50	0,40	0,20	0,50	0,40	0,20
Cylindrical joints for shaft glands of rotating electrical machines with	Sleeve bearings	6	0,30	0,30	0,20	-	-	-	-	-	-	-	-	-
		9,5	0,35	0,30	0,20	0,35	0,30	0,20	-	-	-	-	-	-
		12,5	0,40	0,35	0,25	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,20	-
		25	0,50	0,40	0,30	0,50	0,40	0,25	0,50	0,40	0,25	0,50	0,40	0,20
		40	0,60	0,50	0,40	0,60	0,50	0,30	0,60	0,50	0,30	0,60	0,50	0,25
	Rolling-element bearings	6	0,45	0,45	0,30	-	-	-	-	-	-	-	-	-
		9,5	0,50	0,45	0,35	0,50	0,40	0,25	-	-	-	-	-	-
		12,5	0,60	0,50	0,40	0,60	0,45	0,30	0,60	0,45	0,30	0,60	0,30	0,20
		25	0,75	0,60	0,45	0,75	0,60	0,40	0,75	0,60	0,40	0,75	0,60	0,30
		40	0,80	0,75	0,60	0,80	0,75	0,45	0,80	0,75	0,45	0,80	0,75	0,40

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National Differences

Table 2: Minimum width of joint and maximum gap for

Type of joint	Minimumwidth of joint *L* mm	Maximum gap in mm for Volume *V* (cm^3) *V* < 100	100 < *V* ≤ 500	500 < *V* ≤ 2 000	*V* > 2 000
Flanged joints 1)	6	0,10	-	-	-
	9,5	0,10	0,10	-	-
Spigot joints *c* ³ 6 mm (figure 2)	12,5	0,15	0,15	0,15	
d_{mi}	25	0,18 2)	0,18 2)	0,18 2)	0,18 2)
n = 0,5 *L*					
L = *c* + *d*	40	0,20 3)	0,20 3)	0,20 3)	0,20 3)
f □ 1 mm					
Cylindrical joints Spigot joints (figure 3)	6	0,10	-	-	-
	9,5	0,10	0,10	-	-
	12,5	0,15	0,15	0,1□	-
	25	0,15	0,15	0,15	0,15
	40	0,20	0,20	0,20	0,20
Cylindrical □oints for shaft glands of rotating electrical machines	6	0,15	-	-	-
	9,5	0,15	0,15	-	-
with rolling-element bearings	12,5	0,25	0,25	0,25	-
	25	0,25	0,25	0,25	0,25
	40	0,30	0,30	0,30	0,30

1) Flanged joints are not permitted for explosive mixtures of acetylene/air.

2) i_T of cylindrical part increased to 0,20 if *f* ≤ 0,5.

3) i_T of cylindrical part increased to 0,25 if *f* ≤ 0,5. Note: The constructional values rounded according to ISO 31/0 should be taken when determining the maximum gap.

Note: **The constructional values rounded according to ISO 31/0 should be taken when determining the maximum gap**.

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KR Korea

IEC Clause or Annex Number
Details of differences to IEC 60079-1 1990 (edition 3) + A1: 1993

The requirements of Group I do not apply

The requirement of Note 1 does not apply.

4.2.1 The following requirement does not apply
- meets the impact test requirements of IEC 79-0, as required below by clause 13 taking worst interference fit tolerances into account and
- the diameter of the press-fitted part does not exceed 60mm

- Where joints include conical surfaces, the width of joint and the gap normal to the joint surfaces shall comply with the relevant dimensions given in Table I, IIA, IIB and IIC. The gap shall be uniform through out the conical part. For Group IIC enclosures, the cone angle shall not exceed
5 degree

6 Replace requirements of the minimum radial clearance k with the following. Only for a plain fixed gland, the minimum radial clearance k of shafts of rotating machines shall be not less than 0.075mm for Group IIA and IIB and not less than 0.05mm for Group IIC.

6.2 Replace the Clause with the following.

Shaft glands equipped with rolling-element bearings, shall have a maximum
calculated radial clearance “m” not greater than the maximum gap permitted for the flameproof gland associated with a sleeve bearing.

6.3 Replace requirements of the diametrical clearance with the following.
The diametrical clearance shall not exceed the appropriate value given in Table I, IIA, IIB or IIC.

11.4 Replace the requirement of the cable sealed into the main enclosure with the following. A length of at least 3m shall be attached to the enclosure.

14.1.2.1 Replace the period of application of the pressure with the following.
Between 10 and 50s

14.2.1.1The equation form does not apply

14.2.2.1Replace the gap with the following
iE = iC+1/2
The requirement of joints where the fit is worse than ISO does not apply.
The requirements of Note 1 and 2 do not apply.

Table IIC Replace the 2) of table IIC with the following.
"Flanged joints are not permitted explosive mixtures of acetylene without exception."

Appendix A

A.3.1.5	This requirement does not apply
A.3.1.6	This requirement does not apply
A.3.2.3	This requirement does not apply

Appendix B

B.6.3	This requirement does not apply
B.7	This requirement does not apply
B.11	The requirement of the second paragraph

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IECEx Bulletin –
Section 3: National Differences for 60079-2

IECEx Bulletin:2011(E)

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-2

INTERNATIONAL
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Section 3
National Differences

IEC Technical Report: IEC60079-2 5th Ed 2007 Electrical apparatus for explosive gas atmospheres Part 2: Electrical apparatus-Type of Protection "p"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-2:11
CH Switzerland	**NO**		EN 60079-2:2007
CN China		**YES**	GB3836.5-2004 modified IEC60079-2 Ed4.0
CZ Czech Republic	**NO**		EN 60079-2:2007
DE Germany	**NO**		EN 60079-2:2007
DK Denmark	**NO**		EN 60079-2:2007
FI Finland	**NO**		EN 60079-2:2007
FR France	**NO**		EN 60079-2:2007
GB United Kingdom	**NO**		EN 60079-2:2007
HR Croatia	**NO**		EN 60079-2:2007
HU Hungary	**NO**		EN 60079-2:2007
IN India		**NO**	Adopted as IS /IEC 60079-2: 2007
IT Italy	**NO**		EN 60079-2:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia	**NO**	**NO**	MS IEC 60079-2:2007
NL The Netherlands	**NO**		EN 60079-2:2007
NO Norway	**NO**		EN 60079-2:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-2:2007
RO Romania	**NO**		EN 60079-2:2007
RU Russia		**NO**	GOST R 52350.2-2006
SE Sweden	**NO**		EN 60079-2:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-2:2007
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-2:2007

(N/R Not recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section 1 for contact details

Section 3
National Differences

Countries that have declared adoption of IEC 60079-2 5th Ed 2007 Without National Differences
AU Australia
BR Brazil
***Group:** CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia*
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-2 5th Ed 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-2:11
Explosive atmospheres –
Part 2: Equipment protected by pressurized enclosure "p"

This is the third edition of CAN/CSA-C22.2 No. 60079-2, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-2 (Fifth edition, 2007-02). It replaces the CAN/CSA-C22.2 No.60079-2:02 (adopted CEI/IEC 60079-0: Fourth edition, 2001-02: Entitled *Electrical apparatus for explosive gas atmospheres – Part 2: Pressurized enclosures "p"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

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National Differences

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Electrical equipment of **Group I**, (intended for use in mines susceptible to firedamp), is under the jurisdiction of the Canadian Federal Government. Such types of equipment, (for example, reducing an EPL Mb to non-hazardous), shall be referred to the respective AHJ (authority having jurisdiction).

NOTE: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)
C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

5.8 Spark and particle barriers

[Delete the last paragraph. That is, the spark and particle barriers must be provided by the manufacturer.]

7.3 Provider of safety devices

[Replace this clause with the following paragraph]

All safety devices, that are required for the proper functioning of the pressurizing system, shall be provided by the manufacturer of the equipment.

12.3 Containment system with a limited release

[Delete the fourth paragraph, that is paragraph stating: "If the flow limiting device is not included ……."]

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14 Ignition-capable apparatus

[Add a NOTE to the following effect]

NOTE 3: Notwithstanding NOTE 2, a flame arrester is required when the containment system supports a burning flame.

18.2 Warnings

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

CN China

IEC Standard Number IEC 60079-2:2007 Ed5.0

IEC Clause or Annex Number	Details of Differences

The details of national differences are differences between GB3836.5-2004 and IEC60079-2 Ed4.0 plus the technical changes from IEC60079-2 Ed4.0 to IEC60079-2 Ed5.0.

Section 3
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IEC Technical Report: IEC60079-2 4th Edition 2001 Electrical apparatus for explosive gas atmospheres Part 2: Electrical apparatus-Type of Protection "p"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-2 2002
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA E60079-2-2001
CH Switzerland	**NO**		EN 60079-2:2004
CN China*		**TBA**	
CZ Czech Republic	**NO**		ČSN EN 60079-2:2004
DE Germany	**NO**		EN 60079-2:2004
DK Denmark	**NO**		DS/EN 60079-2:2004
FI Finland	**NO**		EN 60079-2:2004
FR France	**NO**		EN 60079-2:2004
GB United Kingdom	**NO**		BS EN 60079-2:2004
HR Croatia	**NO**		
HU Hungary	**NO**		MESZ EN 60079-2:2004
IN India		**NO**	Adopted as IS 7389:2004/IEC 60079-2: 2001
IT Italy	**NO**		EN 60079-2:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-2:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**NO**		EN 60079-2:2004
NO Norway	**NO**		EN 60079-2:2004
NZ New Zealand		**NO**	AS/NZS 60079-2 2002
PL Poland	**NO**		
RO Romania	**NO**		EN 60079-2:2004
RU Russia		**YES**	GOST R 51330.3-99
SE Sweden	**NO**		SS-EN 60079-2:2004 ed. 1
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-2:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-2:2003
ZA South Africa		**NO**	SANS 60079-2:2001

(N/R Not recognised)

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Section 3
National Differences

Countries that have declared adoption of IEC 60079-2 4th Edition 2001 Without National Differences
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ZA South Africa

Countries that have declared adoption of IEC 60079-2 4th Edition 2001 With National/Group Differences- Differences follow.

CA Canada

Clause 5.8
[Delete the following paragraph]

If the manufacturer does not provide the spark and particle barriers, the apparatus shall be marked in accordance with 27.2 i) of IEC 60079-0.

Clause 7.3
[Change to read as follow]

The safety devices shall be provided by the manufacturer of the apparatus.

Clause 12.3
[Delete the following paragraph]

If the flow limiting device is not included as part of the apparatus, the pressurized enclosure shall be marked “X”. The special condition for safe use shall specify the maximum pressure and flow of the flammable substance into the containment system.

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US United States of America

Clause	US Differences to IEC 60079-2, 4th Ed, 2001
General	The U.S. version of IEC 60079-2 (ANSI/ISA 60079-2:2003 does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
General	Where appropriate, the US customary units have been added to the metric units of measurement.
2	Additional reference documents: ANSI/NFPA 70, National Electrical Code® (NEC®). ANSI/ISA-12.13.01:2003, Performance requirements for combustible gas detectors
3.25	Addition of 3.25: **ventilated equipment** equipment, such as motors, that requires airflow for heat dissipation as well as pressurization to prevent entrance of flammable gases, vapours, or dusts.
3.26	Addition of 3.26:

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Clause	US Differences to IEC 60079-2, 4th Ed, 2001
	hazardous area an area in which an flammable gas atmosphere is or may be expected to be present in quantities such as to require special precautions for the construction, installation and use of electrical apparatus.(IEC 60050-426-03-01) NOTE — The NEC uses “Hazardous (Classified) Location” to describe these areas. The definition for hazardous location is “A location in which fire or explosion hazards may exist due to an flammable atmosphere of flammable gases or vapours, flammable liquids, combustible dust, or easily ignitible fibers or flyings.”
3.27	Addition of 3.27: **non-hazardous area** an area in which an explosive gas atmosphere is not expected to be present in quantities such as torequire special precautions for the construction, installations and use of electrical apparatus. (IEC60050-426-03-02) NOTE — The NEC uses “unclassified” to describe these areas.
3.28	Addition of 3.28: **autoignition temperature** the minimum temperature required to initiate or cause self-sustained combustion of a solid, liquid, or gas independently of the heating or heated element.

Section 3
National Differences

Clause	US Differences to IEC 60079-2, 4th Ed, 2001
3.29	Addition of 3.29: **protected equipment** The electrical equipment internal to the protected enclosure.
5.3.3	Modification of text as follows: ~~For Group II pressurized enclosures, The requirements for fasteners in 9.2 of IEC 60079-0 do not apply.~~ For Type px, Doors and covers, except for those which can be opened only by the use of a tool or key, shall be interlocked so that the electrical supply to electrical apparatus not protected by ………………..

Section 3
National Differences

Clause	US Differences to IEC 60079-2, 4th Ed, 2001
5.4	Modification of text as follows: **Mechanical strength** The pressurized enclosure ducts, if any, and their connecting parts shall withstand a pressure equal to 1.5 times the maximum overpressure specified by the manufacturer for normal service with all outlets closed with a minimum of 200 Pa (8 in of water). If a pressure can occur in service that can cause a deformation of the enclosure ducts, if any, or connecting parts, a safety device shall be fitted to limit the maximum internal overpressure to a level below that which could adversely affect the type of protection. If the manufacturer does not provide the safety device, the apparatus shall be marked ~~"X"~~ according to 27.2 (i) to ANSI/ISA-60079-0:2002 or ANSI/UL 60079-0:2002 and the description documents shall contain all necessary information required by the user to ensure conformity with the requirements of this standard.
7.3	Modification of text as follows: **7.3** The safety devices shall be provided by the manufacturer of the apparatus or by the user. If the manufacturer does not provide the safety devices, the apparatus shall be marked ~~"X"~~ according to 27.2 (i) of ANSI/ISA-60079-0:2002 or ANSI/UL 60079-0:2002 and the description documents shall contain all necessary information required by the user to ensure conformity with the requirements of this standard.

Section 3
National Differences

Clause	US Differences to IEC 60079-2, 4th Ed, 2001
7.9	Modification of 7.9 d) 1) as follows: the protective gas supply shall be equipped with an alarm <u>located at a constantly attended location</u> to indicate failure of the protective gas supply to maintain the minimum pressurized enclosure pressure;
8.1	Revise text as follows: ~~All safety devices used to prevent electrical apparatus protected by static pressurization causing an explosion shall themselves not be capable of causing an explosion and if the safety device is electrically operated, it shall be protected by one of the types of protection recognized in IEC 60079-0 or shall be mounted outside the hazardous area.~~ <u>All pressurizing system components that may be electrically energized in the absence of the protective gas shall be suitable for the classified location in which they are installed.</u>
9.3	Add new sub-clause 9.3: <u>**9.3 Double pressurization**</u> <u>When "double pressurization" is used, the protective gas supplies shall be independent.</u>

Section 3
National Differences

Clause	US Differences to IEC 60079-2, 4th Ed, 2001
11.1.2 b)	Revise text of second paragraph as follows: The conditions to be met in this sub-clause require the apparatus to be marked according to 27.2 (i) of ANSI/ISA-60079-0:2002 or ANSI/UL 60079-0:2002 ~~"X" in accordance with 27.2 i) of IEC 60079-0~~.
12.3	Revise text of fourth paragraph as follows: If the flow limiting device is not included as part of the apparatus, the pressurized enclosure shall be marked ~~"X"~~ according to 27.2 (i) of ANSI/ISA-60079-0:2002 or ANSI/UL 60079-0:2002. The special condition for safe use shall specify the maximum pressure and flow of the flammable substance into the containment system.

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National Differences

IEC Technical Report: IEC60079-2 3rd Edition:1983 Electrical apparatus for explosive gas atmospheres Part 2: Electrical apparatus-Type of Protection "p"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**Refer to IEC 60079-2:2001-02 Fourth Edition**	
CH Switzerland	**YES**		EN 50016 1995
CN China*		**TBA**	
CZ Czech Republic	**YES**		
DE Germany	**YES**		EN 50016 1995
DK Denmark	**YES**		EN 50016 1995
FI Finland	**YES**		EN 50016 1995
FR France	**YES**		EN 50016 1995
GB United Kingdom	**YES**		EN 50016 1995
HR Croatia	**YES**		
HU Hungary	**YES**		EN 50016 1995
IN India		**N/R**	
IT Italy	**YES**		
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998-65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 50016 1995
NO Norway	**YES**		EN 50016 1995
NZ New Zealand		**NO**	
PL Poland	**YES**		
RO Romania	**YES**		EN 50016 1995
RU Russia		**N/R**	
SE Sweden	**YES**		EN 50016 1995
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 50016 1995
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-2:1983

(N/R Not recognised)

NOTE: For information relating to National Differences not listed please contact the IECEx Member Body of that country. Refer to Section1 for contact details

.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-2 3rd Edition 1983 Without National Differences
AU Australia
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-2 3rd Edition 1983 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number	*Details of Differences - EN 50016:1995 Clause*
1	**Comparable clause 1** More descriptive 1. SCOPE 1.1 This European Standard contains the specific requirements for the construction and testing of electrical apparatus with type of protection pressurisation 'p', intended for use in potentially explosive atmospheres. This European Standard supplements European Standard EN 50014, the requirements of which apply to electrical apparatus with type of protection 'p'. 1.2 This European Standard includes the requirements for the construction of the enclosure and its associated components, including, if any, the inlet and outlet ducts for the protective gas, and for the safety provisions and devices necessary for the type of protection pressurisation 'p'. 1.3 This European Standard specifies the requirements for pressurised enclosures with or without an internal source of release, with the exceptions given in 1.5 and 1.6. 1.4 This European Standard specifies requirement for pressurised enclosures containing an unlimited source of release of flammable gas or vapour only where the unlimited source of release is from the surface of a liquid. 1.5 This European Standard does not contain requirements for pressurised rooms or analyser houses. 1.6 This European Standard does not contain the requirements for pressurised enclosures where, in a containment system with limited

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	or unlimited release there is: a) air with an oxygen content greater than normal b) Or oxygen in combination with inert gas in a proportion greater than 21%. 1.7 Due to the safety factors incorporated in the type of protection the uncertainty of measurement inherent in good quality, regularly calibrated measurement equipment is considered to have no significant detrimental effect and need not be taken into account when making the measurements necessary to verify compliance of the apparatus with the requirements of this standard.
2	Comparable clause 2, Definitions
2.1	Comparable clause 2.1
2.2	No direct comparison
2.3	Comparable clause 2.3
2.4	No direct comparison
2.5	Comparable clause 2.2
2.6	Comparable clause 2.6
2.7	Comparable clause 2.7
2.8	Comparable clause 2.11
2.9	Comparable clause 2.4
2.10	Comparable clause 2.12
2.11	No direct comparison
2.12	Comparable clause 2.8
	Additional definitions covered (2.5, 2.9, 2.10, 2.13, 2.14, 2.15, 2.16, 2.17, 2.18) 2.5 **Static pressurisation** The maintenance of an overpressure within a pressurized enclosure without the addition of protective gas in a hazardous area. 2.9 **Containment system** The part of the apparatus containing the flammable gas, vapour or liquid that may constitute an internal source of release. 2.10 **Infallible containment system** The construction of an infallible containment system is of such integrity that the possibility of a leak is so remote that it can be ignored. 2.11 **Dilution** The continuous supply of a protective gas, after purging, at such a rate that

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	the concentration of a flammable mixture inside the pressurised enclosure is maintained at a value outside the explosive limits except in a dilution area. Note: Dilution of oxygen by inert gas may result in a concentration of flammable gas above the UEL. 2.12 **Ignition capable apparatus** Apparatus which in normal operation constitutes a source of ignition for a specified explosive atmosphere. This includes electrical apparatus not protected by a type of protection listed in 1.8.1 of this standard. 2.13 **Dilution area** An area in the vicinity of a source of release where the concentration of flammable gas or vapour is not diluted to a safe concentration. 2.14 **Limited release** A release of flammable gas or vapour the maximum flow rate of which can be predicted. 2.15 **Unlimited release** A release of flammable gas or vapour the maximum flow rate of which cannot be predicted. Note: This refers to liquids which can evolve flammable gas or vapour where the rate of release cannot be predicted 2.16 **Lower explosive limit (LEL)** Volume ratio of flammable gas or vapour in air below which an explosive gas atmosphere will not be formed. 2.17 **Upper explosive limit (UEL)** Volume ratio of flammable gas or vapour in air above which an explosive gas atmosphere will not be formed. 2.18 **Volume ratio (v/v)** Ratio of the volume of a component to the volume of the gas mixture under specified conditions of temperature and pressure.
3	No direct comparison, reference EN 60079-1997 or BS 5345:Part 5: 1983**.**
4	Comparable clause 3, Enclosures and Ducting.
4.1	IEC 79-2 More descriptive.
4.2	Comparable clause 3.1, General.
4.3	Comparable clause 3.3, Mechanical strength.
4.4	IEC 79-2 More descriptive, Comparable clause 3.4 Apertures.

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4.5	No direct comparison can be made but the secondary purge system is mentioned as a method of protection for hot internal components in clause 4.2.
4.6	No direct comparison in the main text but Annex A corresponds to IEC Annex A, reference EN 60079-14:1997 or BS 5345: Part 5: 1983
4.7	No direct comparison, reference EN 50014:1992.
4.8	Comparable clauses 3.6.1 and 3.6.2 Apertures, which are more descriptive for both group II apparatus and I.
4.9	No direct comparison, reference EN 50014:1992
4.10	Batteries are not yet covered but amendments have been suggested, which will cover the usage of such devices. These amendments are more detailed.
4.11	Comparable clause 4.1.
4.12	No direct comparison but implied and clause 5.1 can be referenced.
5	Comparable clause 4, Temperature Limits.
6	Comparable clause 7, Supply of Protective Gas.
6.1	Comparable clause 7.1.
6.2	Comparable clause 7.1.
	Additional requirements of 7.2 7.2 The manufacturer shall specify the density limits of the range of flammable gases for which the pressurised enclosure is designed except where the pressurised enclosure is designed for all flammable gases or for a specific flammable gas.
7	Comparable clause 16, Marking The main clauses 1 to 5 are covered by clause 16.1; Pressurised enclosures other than those protected by static pressurisation. Other considerations listed are as follows: e) The maximum leakage rate from the pressurised enclosure. f) A special temperature or range of temperatures for the protective gas at the inlet to the pressurised enclosure when specified by the manufacturer.
8	14 Type Verification and Type Tests
8.1	No direct comparison
8.2	The general clause 14 (purge tests) and the relevant purge test from clauses 14.3 and 14.4 cover item 1.

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	The general clause 14 and the specific clause 14.5 cover item 2 for minimum overpressure. Item 3 is covered by the general clause 14 (reference EN 50014:1992). Item 4 is covered by the general clause 14 (reference EN 50014:1992). Item 5 is covered by the general clause 14 (reference EN 50014:1992). Item 6 is covered by the general clause 14 and the requirements in clause 3.2, Spark and particle barriers.
Section 2	No direct comparable section all relevant information is split into separate areas.
9	Comparable clauses 9, Release Condition and 11, Protective Gas and Pressurised Technique.
10	No direct comparison, Table I not detailed (but inferred in overall text and reference EN 60079-14:1997 and BS 5345: Part 5:1983).
11	
11.1	No direct comparison
11.2	Warning statement or additional requirements covered by EN 50014:1992
11.3	Item a) comparable clause 3.6. Item b) comparable clause 5.10.
12	Comparable clause 5.9
Section 3	No direct comparable section all relevant information is split into separate areas.
13	Comparable clause 9, Release Conditions and 11, Protective Gas and Pressurised Technique.
14	No direct comparison inferred.
15	Comparable clause 9. Release Conditions.
15.1	Different explanations of release condition but essentially the same. Reference EN 60079:1997 and BS 5345: Part 5: 1983.
15.2	No direct comparison, Table II also no direct comparison.
16	Comparable clause 5.9 and 10.2.
17	Comparable clause 3, Enclosures and ducting.
17.1	Comparable clause 3.4, Apertures
17.2	Comparable clause 3.4, Apertures
18	No direct comparison. Reference EN 60079-14:1997 and BS 5345: Part 5: 1983.

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18.1	No direct comparison.
18.2	No direct comparison.
18.3	Item a) no direct comparison but clause 3.6 covers some requirements. Item b) no direct comparison
19	
19.1	Comparable clause 11.
19.2	Comparable clause 11.
19.3	No direct comparison.

Note 79-2 combine's code of practice, general and EN 50016 requirements

KR Korea

IEC Clause or
Annex
Number
Details of Differences to IEC 60079-2: 1983(edition 3)

Section One
Replace this clause with the following.

Doors and covers which can be opened without the use of tools or keys should have warning label
Affixed stating the following or equivalent.
"Warning! Do not open the door or cover while energized"

Section Two
These requirements of the second sentence and note do not apply.

Section Three
These requirements do not apply

Appendix A These requirements do not apply

Appendix B These requirements do not apply

Appendix C These requirements do not apply

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Section 3: National Differences for 60079-5

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 60079-5 3rd Edition 2007 Electrical apparatus for explosive gas atmospheres - Part 5: Powder filling "q"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-5:11
CH Switzerland	**NO**		EN 60079-5:2007
CN China		**YES**	GB38367-2004 identified IEC 60079-5:1995 Ed2.0
CZ Czech Republic	**NO**		EN 60079-5:2007
DE Germany	**NO**		EN 60079-5:2007
DK Denmark	**NO**		EN 60079-5:2007
FI Finland	**NO**		EN 60079-5:2007
FR France	**NO**		EN 60079-5:2007
GB United Kingdom	**NO**		EN 60079-5:2007
HR Croatia	**NO**		EN 60079-5:2007
HU Hungary	**NO**		EN 60079-5:2007
IN India		**NO**	Adopted as IS /IEC 60079-5: 2007
IT Italy	**NO**		EN 60079-5:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-5:2007
NL The Netherlands	**NO**		EN 60079-5:2007
NO Norway	**NO**		EN 60079-5:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-5:2007
RO Romania	**NO**		EN 60079-5:2007
RU Russia		**NO**	GOST R 52350.5-2006
SE Sweden	**NO**		EN 60079-5:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-5:2007
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-5 2009 ANSI/UL 60079-5 2009
ZA South Africa		**NO**	SANS 60079-5:2007

(N/R Not recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-5 3rd Edition 2007 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-5 3rd Edition 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-5:11
Explosive atmospheres –
Part 5: Equipment protection by powder filling "q"

This is the second edition of CAN/CSA-C22.2 No. 60079-5, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-5 (Third edition, 2007-03). It replaces the CAN/CSA-C22.2 No. 60079-5:02 (adopted CEI.IEC 60079-5: Second edition, 1997-07: Entitled *Electrical apparatus for explosive gas atmospheres – Part 5: Powder filling "q").*

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

[Add a NOTE to the following effect]

NOTE: See also CAN/CSA-C22.2 No. 60079-0:11 for further Canadian deviations.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety standard for electrical installations

4.9 Temperature limitations under fault conditions

[Add the following NOTE beneath the 2nd last paragraph]

NOTE: Notwithstanding the "X" marking, if and when the fuse is necessary for the safe operation of the equipment, it shall be provided as part of the package of the equipment for installation by the end-user.

4.9.1 Fault exclusions

[Add a NOTE to the following effect; before the 2nd last paragraph]

NOTE: A soldering mask would cover the traces of the circuit board but would not cover the soldering pad or the conductive end of the component. Therefore, a soldering mask cannot be considered as one of the two overall coatings.

4.9.2 Power supply prospective short-circuit current

[Add the following NOTE beneath the last paragraph]

NOTE: Notwithstanding the "X" marking, if and when the short-circuit device is necessary for the safe operation of the equipment, it shall be provided as part of the package of the equipment for installation by the end-user.

6 Marking

[Add a NOTE to the following effect]

NOTE: Any cautionary or warning statement, [e.g.: Items (a) and (b)], that relates to the safety of use of the equipment shall be in both English and French.

7 Instructions

[Add a NOTE to the following effect]

NOTE: Any instruction that provides a safe installation, operation and maintenance of the equipment shall be in both English and French.

CN China

IEC Standard Number IEC 60079-5:2007 Ed3.0…..

IEC Clause or Annex Number	Details of Differences

The details of national differences are the technical changes from IEC60079-5 Ed2.0 to IEC60079-5 Ed3.0.

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National Differences

US United States of America

Clause	US Differences to IEC 60079-5, 3rd Ed, 2007-03
General	The U.S. version of IEC 60079-5 (ANSI/ISA 60079-5:2009 and ANSI/UL 60079-5:2009) does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
2	Additional reference to align with installation requirements of the National Electrical Code: ANSI/UL 2225, Cables and Cable-Fittings For Use In Hazardous (Classified) Locations
4.1.3	Modification of text as follows to indicate Specific Conditions of Use aligning with ANSI/ISA 60079-0: The ingress protection of enclosures or parts of electrical equipment protected by powder filling “q”, intended for use only in clean, dry rooms, may be reduced to degree of protection IP43. These enclosures shall be ~~marked with the symbol "X"~~ in accordance with ANSI/ISA-60079-0 ~~29.2 i) of IEC 60079-0~~ to indicate that there are specific conditions of use and the specific conditions for use shall detail the restrictions of use.
4.5	Modification of text to address the installations necessary with wiring methods mandated by the National Electrical Code: **4.5 External connections** Terminations ~~or cables~~ used for the entry of electrical conductors into a powder filled “q” enclosure shall be protected using a suitable type of protection such as “d”, “e” or “i”. The connection between the terminations. The connection between the terminations and the powder filled “q” enclosure shall be ~~and~~ sealed as specified in 4.1.1. When a cable provides the direct entry of conductors into powder filled “q” equipment or Ex component, the cable gland ~~clamping means~~ shall comply with ANSI/UL 2225 ~~the cable gland requirements of IEC 60079-0~~ and shall not be capable of being removed without obvious damage to the powder filled “q” enclosure. NOTE A flameproof “d” cable gland in accordance with ANSI/UL 2225 may be required as the increased safety “e” cable glands in accordance with ANSI/UL 2225 may not provide adequate pressure sealing (see 5.1.1) of the powder filled “q” enclosure. This construction

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Clause	US Differences to IEC 60079-5, 3rd Ed, 2007-03
	<u>is generally only practical with a factory-installed cable and cable gland.</u>
4.9	Deletion of text to acknowledge that ordinary location requirements always apply in the US. ~~Where there is no product standard, the overloads to be considered are those specified by the manufacturer.~~
4.9	Modification of text as follows to indicate Specific Conditions of Use: aligning with ANSI/ISA 60079-0 When the fuse is not integral to the electrical equipment or parts of electrical equipment, the <u>equipment shall be marked</u> ~~marking shall include the symbol "X"~~ in accordance with ~~29.2 i) of~~ <u>ANSI/ISA-60079-0</u> ~~IEC 60079-0~~ to indicate that there are specific conditions of use and the specific conditions for use shall detail the required fuse. When the fuse is not integral to an Ex component, the ~~marking shall include the symbol "U" in accordance with 29.5 g) of IEC 60079-0 and the schedule of limitations~~ <u>instructions</u> shall detail the required fuse.
4.9.3	Modification of text as follows to indicate Specific Conditions of Use aligning with ANSI/ISA 60079-0: If the manufacturer does not provide a required short-circuit protective device, the electrical equipment or parts of electrical equipment shall be ~~marked with the symbol "X"~~ in accordance with ~~29.2 i) of~~ <u>ANSI/ISA-60079-0</u> ~~IEC 60079-0~~ <u>to indicate that there are specific conditions of use</u> and the specific conditions of use shall detail the short-circuit protective devices required.
5.1.2	Modification of text as follows to clarify that only the degree of protection required by 4.1.3 needs to be verified. The degree of protection of the enclosure <u>required by 4.1.3</u> shall be verified in accordance with the method specified in IEC 60529. Any breathing devices shall be in place. This test shall be carried out after the pressure type test in 5.1.1.
5.1.4	Modification of text as follows to clarify that re-testing of the same sample is not permitted. The filling material complies with the requirements, if the leakage current does not exceed 10^{-6} A. If the material fails to comply, further conditioning and retesting <u>of the same sample</u> are not permitted.
5.1.5	Note 2 added to clarify text. The added note is consistent with IECEx ExTAG Decision Sheet DS2008/001.

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Clause	US Differences to IEC 60079-5, 3rd Ed, 2007-03
	NOTE 2 The term “overload” should be understood to apply to a situation where the Ex q equipment (or component – such as a solid state relay or a luminaire ballast) controls rather than consumes power and the temperature rise is, at least partly, related to the external load. In such cases, where the equipment is protected by a fuse rated at not more than 170 % of the maximum normal current, the external load should be adjusted to achieve the maximum current through the fused circuit, but no more than 1.7 times the fuse rating. Internal faults should not be applied and are not considered to result in an “overload”.
6	Text clarified as follows to highlight that the specified text was an example, not a specific required marking. c) each connection facility for external connection shall be marked with an identification of rated voltage and rated current (for example: “24 V d.c., 200 mA”, “230 V, 100 mA”); d) external fuse data if the type of protection depends upon such a fuse, for example: “Required external fuse: 315 mA”;

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National Differences

IEC Standard 60079-5:2nd Edition 1997 Amd 1 2003 Electrical apparatus for explosive gas atmospheres - Part 5: Powder filling "q"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-5 2000
BR Brazil		**N/R**	
CA Canada		**NO**	CAN/CSA E60079-5-2001
CH Switzerland	**YES without Amd**		EN 50017 1994
CN China		**NO**	GB3836.7 2nd[t] ed 2004
CZ Czech Republic	**YES without Amd**		
DE Germany	**YES without Amd**		EN 50017 1994
DK Denmark	**YES without Amd**		DS/EN 50017 1994
FI Finland	**YES without Amd**		EN 50017 1994
FR France	**YES without Amd**		EN 50017 1994
GB United Kingdom	**YES without Amd**		EN 50017 1994
HR Croatia	**N/A**		
HU Hungary	**YES without Amd**		EN 50017 1994
IN India		**NO**	Adopted as IS 7724:2004/IEC 60079-5: 1997
IT Italy	**YES without Amd**		EN 50017 1994
JP Japan		**NO**	
KR Republic of Korea		**NO**	Labor Ministry Notice1998-65
MY Malaysia		**NO**	MS IEC 60079-5:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**YES without Amd**		EN 50017 1994
NO Norway	**YES without Amd**		EN 50017 1994
NZ New Zealand		**NO**	AS/NZS 60079-5 2000
PL Poland	**N/A**		
RO Romania	**YES without Amd**		EN 50017 1994
RU Russia		**YES**	GOST R 51330.6-99
SE Sweden	**YES without Amd**		
SG Singapore		**NO**	
SI Slovenia	**YES without Amd**		EN 50017 1994
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-5 2002 ANSI/ISA 12.25.01 2002
ZA South Africa		**NO**	SANS 60079-5:1997

(N/R Not recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-5 2nd Edition 1997 Amd 1 2003 Without National Differences
AU Australia
CA Canada
CN China
IN India
JP Japan
KR Korea
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-5 2nd Edition 1997 Amd 1 2003 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number Details of Differences to IEC 60079-5:1997 Edition 2
Details of Differences to IEC 60079-5:1997 Edition 2

IEC Clause	Detail of difference	Remarks
Foreword	Second edition of IEC 60079-5 replaces the first edition published in 1967. EN50017 1994 replaces the first edition published in 1977	Editorial
1	Delete subclause number 1.1 & 1.2,1.3	Editorial
2.1	Delete subclause number 2.1 ,2.2 & 2.3 Replace EN50014 1992 with IEC60079-0:1983 Replace EN50019 1994 with IEC60079-7 1990 Replace EN50020 1994 with IEC60079-11 1991 Replace EB60529 1991 with IEC 60529:1989 Add IEC60079-1:1990	Editorial
3	Add the following additional definitions: Externally applied maximum voltage, U_m; Creepage distance under coating; Distance through filling material; Fuse rating, I_n	Technical
4.1	Replace clause number 4 in EN 50017	Editorial
4.1.1	Replace clause number 4.1 in EN 50017	Editorial
4.1.2	Replace clause number 4.2 in EN 50017	Editorial
4.1.3	Replace clause number 4.3 in EN 50017	Editorial

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National Differences

IEC Clause	Detail of difference	Remarks
4.1.4	Replace clause number 4.4 in EN 50017	Editorial
4.2	Replace clause number 5 in EN 50017 named ”Filling material”	Editorial
4.2.1	Replace clause number 5.1 in EN 50017 with the new name ”Documentation”	Editorial
4.2.2	Replace clause number 5.1.2 in EN 50017 with the new name ”Requirements”	Editorial
4.2.3	Replace clause number 5.2 in EN 50017 with the new name ”Testing”	Editorial
4.3	Replace clause number 6 in EN 50017. Add note ”While this standard is applicable to apparatus with a rated supply not exceeding 1000 V, table takes into account voltage greater than 1000 V which may be developed or created within the equipment.	Technical
4.4	Replace clause number 7 in EN 50017	
4.5	Replace clause number 8 in EN 50017	Editorial
4.6	Replace clause number 9 in EN 50017	Editorial
4.7	Replace clause number 10 in EN 50017	Editorial
4.8	Replace clause number 11 in EN 50017	Editorial
4.8.1	Replace clause number 11.1 in EN 50017. Add the following extra requirement to this clause: ”Conductive parts protruding from the insulation (including soldered component pins) shall not be considered as coated unless special measures have been applied to obtain an effective unbroken seal"	Editorial and technical
4.8.2	Replace clause number 11.2 in EN 50017	Editorial
4.8.3	Replace clause number 11.3 in EN50017	Editorial
5.1.1	Replace clause number 12.1 in EN 50017	Editorial
5.1.2	Replace clause number 12.2 in EN 50017	Editorial
5.1.3	Replace clause number 12.3 in EN 50017. Replace the whole requirements with: ”The flammability requirements of IEC 60079-1 shall be applied. (See A.3.3 of Amendment 1 to IEC 60079-1).”	Editorial and technical
5.1.4	Replace clause number 12.4 in EN 50017	Editorial
5.1.5	Replace clause number 12.5 in EN 50017	Editorial
5.2	Replace clause number 13 in EN 50017	Editorial
5.2.1	Replace clause number 13.1 in EN 50017	Editorial

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National Differences

IEC Clause	Detail of difference	Remarks
5.2.2	Replace clause number 13.2 in EN 50017	Editorial
6	Replace clause number 14 in EN 50017	Editorial
Figure 1	Replace clause "Annex A" in EN 50017	Editorial

EN Clause or Annex Number details of differences

1 Refers to EN 50014

2 List of standards refers to EN 50019, 50020 and CEI 1210

3 No direct reference to definitions 3.3, 3.6, 3.7, and 3.8 of relevant CEI

4 No significant differences editorial

5 §5.2 of EN specifies that filling material shall be subjected to measurement of dielectric strength similar to 5.1.3 of relevant CEI

6 Identical to 4.3 of CEI

7 No significant differences with §4.4 of CEI

8 No significant differences with §4.5 of CEI

9 §4.6 of CEI adds that « Cells and batteries which might impair the type of protection are not permitted. Note of EN is more developed.

10 No differences with CEI

11 EN does not specify that the voltage Uum shall be assumed to be applied to the supply terminals when considering fault conditions and fault exclusions » as in § 4.8 of CEI

11.1 Does not specify that « Conductive parts protruding from the insulation (including soldered components pins) shall not be considered as coated unless special measures have been applied to obtain an effective unbroken seal » as in 4.8.1 of CEI

11.2 No differences with CEI 4.8.2

11.3 No differences with CEI 4.8.3

12.1 No differences with CEI 5.1.1

12.2 No differences with CEI 5.1.2

12.3 EN refers directly to ISO 1210 and proposes two tests variant, one being based on CEI 707

12.4 EN gives a test voltage of 1000V when CEI 5.1.4 gives a test voltage of 1000V+5%-0 and specifies that « if the material fails to comply, further conditioning and retesting is not permitted »

12.5 No differences with CEI 5.1.5

13.1 No differences with CEI 5.2.1

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National Differences

13.2 No differences with CEI 5.2.2

14 No differences excepted for the reference made to EN 50014
Annex A does not provide the tolerance for the dimension where CEI figure 1 gives +-1.0

Section 3
National Differences

US United States of America National differences to IEC 60079-5 (1997)		
	Clause	**Difference**
IEC 60079-5 (1997)	General	The U.S. version of IEC 60079-5 does not include requirements for Group I electrical apparatus.
IEC 60079-5 (1997)	General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 60079-5 (1997)	2	ANSI/UL 50 Enclosures for Electrical Equipment ANSI/UL 248-1: 1994 Standard for Low Voltage Fuses – General Requirements
IEC 60079-5 (1997)	4.1.1	**4.1.1 Mechanical strength** (second paragraph, last sentence) This apparatus shall be marked with specific installation instructions or reference to a specific installation document when necessary to indicate special conditions for safe use ~~the symbol "X" according to IEC 60079-0~~ if not an Ex component.
IEC 60079-5 (1997)	4.1.2	**4.1.2 Degree of protection of the enclosure** (second paragraph, last sentence) These enclosures shall be marked with specific installation instructions or reference to a specific installation document when necessary to indicate special conditions for safe use ~~the symbol "X"~~.
IEC 60079-5 (1997)	4.2.1	**4.2.1** (first paragraph) The ~~documents presented by the manufacturer and verified by the testing station in accordance with IEC 60079-0 (see type verification of documents) shall describe precisely the~~ filling material as well as the filling process and the measures taken to ensure proper filling shall be precisely described by the manufacturer.
IEC 60079-5 (1997)	4.2.2	**4.2.2** (last paragraph) ~~The testing station is not required to verify compliance of the filling material with 4.2.1 and 4.2.2.~~
IEC 60079-5 (1997)	4.8	**Fault Conditions** (first paragraph) The type of protection "q" shall be maintained even in the case of overloads prescribed in any relevant product standard ~~specified by the manufacturer~~ and any single internal electrical fault which may cause either an overvoltage or overcurrent, for example:
IEC 60079-5 (1997)	4.8	**4.8 Fault Conditions** (penultimate paragraph) ~~Where there is no product standard, the overloads shall be those specified by the manufacturer.~~

Section 3
National Differences

US United States of America National differences to IEC 60079-5 (1997)		
	Clause	**Difference**
IEC 60079-5 (1997)	4.8.2	**4.8.2 Protective devices for temperature limitation** (second paragraph) Where fuses are used as protective devices, ~~the fusing element~~ they shall be of the enclosed type, for example, in glass or ceramic. ~~For voltages above 60 V,~~ Fuses shall comply with ANSI/UL 248-1 and shall have a voltage rating not less than the supply circuit and shall have a breaking capacity not less than the available short circuit current of the supply ~~in accordance with IEC 127-1 or IEC 269-1~~.
IEC 60079-5 (1997)	5.1.1	**5.1.1 Pressure type test of enclosure** (second paragraph) For powder-filled enclosures, without breathing or degassing openings, which contain capacitors other than plastic foil, paper or ceramic type and where the volume of the filling material is lower than eight times the volume of the capacitors, a pressure type test with an overpressure of 15 bar (1,5 Mpa) without the occurrence of permanent deformation exceeding 0.5 mm in any of its dimensions shall be performed with an application time of 60 + 5 / 0 s.

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IECEx Bulletin

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IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-6

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-6

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Section 3
National Differences

IEC Standard 60079-6 3rd Edition 2007 Electrical apparatus for explosive gas atmospheres Part 6: Oil-immersion "o"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-6:11
CH Switzerland	**NO**		EN 60079-6:2007
CN China		**YES**	GB3836.6-2004 identified IEC 60079-6:1995 Ed2.0
CZ Czech Republic	**NO**		EN 60079-6:2007
DE Germany	**NO**		EN 60079-6:2007
DK Denmark	**NO**		EN 60079-6:2007
FI Finland	**NO**		EN 60079-6:2007
FR France	**NO**		EN 60079-6:2007
GB United Kingdom	**NO**		EN 60079-6:2007
HR Croatia	**NO**		EN 60079-6:2007
HU Hungary	**NO**		EN 60079-6:2007
IN India		**NO**	Adopted as IS /IEC 60079-6: 2007
IT Italy	**NO**		EN 60079-6:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-6:2007
NL The Netherlands	**NO**		EN 60079-6:2007
NO Norway	**NO**		EN 60079-6:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-6:2007
RO Romania	**NO**		EN 60079-6:2007
RU Russia*		**TBA**	In the national standardization program for 2010
SE Sweden	**NO**		EN 60079-6:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-6:2007
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-6 2009 ANSI/UL 60079-6 2009
ZA South Africa		**NO**	SANS 60079-6:2007

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-6 3rd Edition 2007 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-6 3rd Edition 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-6:11
Explosive atmospheres –
Part 6: Equipment protection by oil immersion "o"

This is the second edition of CAN/CSA-C22.2 No. 60079-6, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-6 (Third edition, 2007-03). It replaces the CAN/CSA-C22.2 No.60079-6:02 (adopted CEI/IEC 60079-6: Second edition, 1995-01: Entitled *Electrical apparatus for explosive gas atmospheres – Part 6: Oil immersion "o"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Section 3
National Differences

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Electrical equipment of **Group I**, (intended for use in mines susceptible to firedamp), is under the jurisdiction of the Canadian Federal Government. Such types of equipment shall be referred to the respective AHJ (authority having jurisdiction.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety standard for electrical installations

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

6 Marking

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

7 Instructions

[Add a paragraph to the following effect]

Instructions, relevant to the safe installation, usage and operation of the equipment, shall be in both English and French.

CN China

IEC Standard Number IEC 60079-6:2007 Ed3.0

IEC Clause or Annex Number	Details of Differences

The details of national differences are the technical changes from IEC60079-6 Ed2.0 to IEC60079-6 Ed3.0.

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National Differences

US United States of America

Clause	US Differences to IEC 60079-6, 3rd Ed, 2007-03
General	The U.S. version of IEC 60079-6 (ANSI/ISA 60079-6:2009 and ANSI/UL 60079-6:2009) does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
2	Additional references to align with US practice and the National Electrical Code: ASTM D3487, Mineral Insulating Oil Used in Electrical Apparatus ANSI/UL 2225, Cables and Cable-Fittings For Use In Hazardous (Classified) Locations
3.2	Modification of text as follows to align with US practice: mineral oil conforming to IEC 60296, ASTM D3487, or an alternative liquid meeting the requirements of 4.2
4.2	Modification of text as follows to align with US practice: Protective liquid other than mineral oil conforming to IEC 60296 or ASTM D3487 shall comply with all of the following requirements:
4.8	Modification of text to reflect that the EPL concept is not currently recognized in the US. Equipment, components and conductors not complying with the above requirement shall be suitable for Class I, Zone 0 or Zone 1 with an appropriate apparatus group and temperature class. ~~have an equipment protection level of:~~ ~~• Ma or Mb for group I equipment, or~~ ~~• Ga or Gb for group II equipment.~~
4.13	Modification of text to address the installations necessary with wiring methods mandated by the National Electrical Code:

Section 3
National Differences

Clause	US Differences to IEC 60079-6, 3rd Ed, 2007-03
	4.13 External connections Terminations ~~or cables~~ used for the entry of electrical conductors into an oil immersion “o” enclosure shall be protected using a suitable type of protection such as “d”, “e” or “i”. The connection between the terminations and the oil immersion “o” enclosure shall be an integral part of the enclosure. When a cable provides the direct entry of conductors into oil immersion “o” equipment or an Ex component, the cable gland ~~clamping means~~ shall comply with ANSI/UL 2225 ~~the cable gland requirements of IEC 60079-0~~ and shall not be capable of being removed without obvious damage to the oil immersion “o” enclosure. NOTE A flameproof “d” cable gland in accordance with ANSI/UL 2225 may be required as the increased safety “e” cable glands in accordance with ANSI/UL 2225 may not provide adequate pressure sealing (see 5.1.1) of the oil immersion “o” enclosure. This construction is generally only practical with a factory-installed cable and cable gland.

Section 3
National Differences

IEC Standard 60079-6 2nd Edition 1995 Electrical apparatus for explosive gas atmospheres Part 6: Oil-immersion "o"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-6 2000
BR Brazil		**N/R**	
CA Canada		**NO**	CAN/CSA-E60079-6-2001
CH Switzerland	**YES**		EN 50015 1994
CN China		**NO**	GB3836.6-2004 identified IEC 60079-6:1995 Ed2.0
CZ Czech Republic	**YES**		EN 50015 1994
DE Germany	**YES**		EN 50015 1994
DK Denmark	**YES**		EN 50015 1994
FI Finland	**YES**		EN 50015 1994
FR France	**YES**		EN 50015 1994
GB United Kingdom	**YES**		EN 50015 1994
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 50015 1994
IN India		**NO**	Adopted as IS 7693:2004/IEC 60079-6:1995
IT Italy	**YES**		EN 50015 1994
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998-65
MY Malaysia		**NO**	MS IEC 60079-6:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**YES**		EN 50015 1994
NO Norway	**YES**		EN 50015 1994
NZ New Zealand		**NO**	AS/NZS 60079-6 2000
PL Poland	**N/A**		
RO Romania	**YES**		EN 50015 1994
RU Russia		**NO**	GOST R 52350.6-2006
SE Sweden	**YES**		EN 50015 1994
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 50015 1994
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-6 2002 ANSI/ISA 12.26.01 2002
ZA South Africa		**NO**	SANS 60079-6:1995

(N/R Not Recognised)

***NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details*.**

Section 3
National Differences

Countries that have declared adoption of IEC 60079-6 2nd Edition 1995 Without National Differences
AU Australia
CA Canada
CN China
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-6 2nd Edition 1995 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

The following countries Switzerland **(CH)** Czech Republic **(CZ)** Germany (DE) Denmark **(DK)** Finland **(FI)** France **(FR)** United Kingdom **(GB)** Hungary **(HU)** Italy **(IT)** the Netherlands **(NL)** Norway **(NO)** Romania **(RO)** Sweden **(SE)** and Slovenia **(SI)** have declared the following identical differences to IEC 60079-6 1995 (edition 2)

Foreword
Note the penultimate paragraph of the Foreword to IEC 79-6 1995 states:

" Apart from editorial changes, this standard is identical to the European Standard EN 50015, second edition, from CENELEC."

IEC Clause or Annex Number	Detail of Differences
All	References to standards are to the equivalent EN rather than the IEC where an EN exists:
IEC 79-0	EN50014
IEC 79-7	EN50019
IEC 79-11	EN50020
IEC 529	EN60529
5	Editorial change required such that 5.1.1 becomes 5.1, 5.1.2 becomes 5.2 and 5.1.3 becomes 5.3; also 5.2 become 6. Internal cross references to be amended accordingly.
6	Change clause number to 7.

KR Korea

IEC Clause or Annex Number	Details of Differences IEC 60079-6 1995 (edition 2)
	The requirements for Group I do not apply
4.2	This requirement does not applied
5.1.1	This requirement does not applied

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National Differences

US United States of America

US National differences to IEC 60079-6 (1995)		
IEC Standard	**Clause**	**Difference**
IEC 60079-6 (1995)	General	The U.S. version of IEC 60079-6 does not include requirements for Group I electrical apparatus.
IEC 60079-6 (1995)	General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 60079-6 (1995)	2	ASTM D3487:1993 Mineral Insulating Oil Used in Electrical Apparatus ASTM D5222:1992 High Fire-Point Electrical Insulating Oils of Petroleum Origin ~~IEC 588-2: 1978, Askarels for transformers and capacitors – Part 2: Test methods~~
IEC 60079-6 (1995)	3.2	**3.2 Protective liquid:** Mineral oil conforming to IEC 296, ASTM D3487 or an alternative liquid meeting the requirements of 4.1.
IEC 60079-6 (1995)	4.1	**4.1** Protective liquid other than mineral oil conforming to IEC 296 or ASTM D3487 shall comply with the following specific requirements
IEC 60079-6 (1995)	4.1 g)	~~g) the acidity (neutralization value) shall be 0,03 mg KOH/g (maximum) determined in accordance with IEC 588-2.~~
IEC 60079-6 (1995)	4.3.2	**4.3.2** (last sentence) ~~The testing station is not required to verify the suitability of the drying agent nor its maintenance.~~

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Section 3
National Differences

IEC Standard 60079-7 4th Edition 2006 Electrical apparatus for explosive gas atmospheres - Part 7: Increased safety "e"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-7:11
CH Switzerland	**NO**		EN 60079-7:2007
CN China		**NO**	GB 3836.3-2010 Explosive atmospheres Part 3: Equipment protection by increased safety "e" （IEC 60079 -7:2006, IDT）
CZ Czech Republic	**NO**		EN 60079-7:2007
DE Germany	**NO**		EN 60079-7:2007
DK Denmark	**NO**		EN 60079-7:2007
FI Finland	**NO**		EN 60079-7:2007
FR France	**NO**		EN 60079-7:2007
GB United Kingdom	**NO**		EN 60079-7:2007
HR Croatia	**NO**		EN 60079-7:2007
HU Hungary	**NO**		EN 60079-7:2007
IN India		**NO**	Adopted as IS /IEC 60079-7: 2006
IT Italy	**NO**		EN 60079-7:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-7:2007
NL The Netherlands	**NO**		EN 60079-7:2007
NO Norway	**NO**		EN 60079-7:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-7:2007
RO Romania	**NO**		EN 60079-7:2007
RU Russia		**NO**	GOST R 52350.7-2005
SE Sweden	**NO**		EN 60079-7:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-7:2007
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**NO**	SANS 60079-7:2006

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-7 4th Edition 2006 Without National Differences
AU Australia
BR Brazil
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-7 4th Edition 2006 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-7:11
Explosive atmospheres –
Part 7: Equipment protection by increased safety "e"

This is the third edition of CAN/CSA-C22.2 No. 60079-7, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-7 (Fourth edition, 2006-07). It replaces the CAN/CSA-C22.2 No.60079-7:03 (adopted CEI/IEC 60079-7: Third edition, 2001-11: Entitled *Electrical apparatus for explosive gas atmospheres – Part 7: Increased safety"e"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Section 3
National Differences
Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Electrical equipment of **Group I**, (intended for use in mines susceptible to firedamp), is under the jurisdiction of the Canadian Federal Government. Such types of equipment, (for example, caplights), shall be referred to the respective AHJ (authority having jurisdiction).

NOTE: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11

2 Normative references

[Add the following]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

9.3 Warning markings

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

Section 3
National Differences

IEC Standard: 60079-7:3rd Edition 2001 Electrical apparatus for explosive gas atmospheres - Part 7: Increased safety "e"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-7 2002
BR Brazil		**N/R**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-7:2003
CN China*		**TBA**	
CZ Czech Republic	**NO**		ČSN EN 60079-7:2004
DE Germany	**NO**		EN 60079-7:2003
DK Denmark	**NO**		DS/EN 60079-7:2003
FI Finland	**NO**		EN 60079-7:2003
FR France	**NO**		EN 60079-7:2003
GB United Kingdom	**NO**		BS EN 60079-7:2003
HR Croatia	**N/A**		
HU Hungary	**NO**		MESZ EN 60079-7:2003
IN India		**NO**	Adopted as IS 6381: 2001/IEC 60079-7:2001
IT Italy	**NO**		EN 60079-7:2003
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-7:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**NO**		EN 60079-7:2003
NO Norway	**NO**		EN 60079-7:2003
NZ New Zealand		**NO**	AS/NZS 60079-7 2002
PL Poland	**N/A**		
RO Romania	**NO**		EN 60079-7:2003
RU Russia		**YES**	GOST R 51330.8-99
SE Sweden	**NO**		SS-EN 60079-7:2003 ed.1
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-7:2003
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-7 2002 ANSI/ISA 12.16.01 2002
ZA South Africa		**NO**	SANS 60079-7:2001

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-7 3rd Edition 2001 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-7 3rd Edition 2001 With National/Group Differences- Differences follow.

RU Russia

Group I electrical equipment with types of protection "e" and "m"
(standards GOST R 51330.8-99 (IEC 60079-7:2001) and GOST R 51330.8-99
(IEC 60079-18:1992) according to 6.6 of standard GOST R 51330.0-99
(IEC 60079-0-98) by its level of protection is referred to electrical equipment of higher explosion protection. The field of application of electrical equipment with this level of protection in coal mines is as follows:

- shaft signaling in mines of any category as to firedamp;

- portable periodically used electrical devices in underground workings of mines susceptible to firedamp or dust;

- haulage ways with fresh air of I and II category mines as to firedamp and coal dust explosion;

- charging rooms with separate ventilation in mines susceptible to firedamp or dust, including those susceptible to sudden outbursts;

- all mine workings in mines non-hazard as to firedamp, but hazard as to coal dust explosion.

Therefore, foreign explosion-proof electrical equipment supplied to Russian coal mines must incorporate the types of protection corresponding to the levels of explosion protection regulated by the above mentioned Safety Rules.

Section 3
National Differences

US **United States of America** National differences to IEC 60079-7 (2001)		
IEC Standard	**Clause**	**Difference**
IEC 60079-7 (2001)	General	The U.S. version of IEC 60079-7 does not include requirements for Group I electrical apparatus.
IEC 60079-7 (2001)	General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 60079-7 (2001)	2.	ANSI C78.1, Fluorescent Lamps – Rapid Start Types - Dimensional and Electrical Characteristics ANSI C81.61, Electric Lamp Bases ANSI C81.62, Lampholders for Electric Lamps ANSI/IEEE 515, IEEE Recommended Practice for the Testing, Design, Installation, and Maintenance of Electrical Resistance Heat Tracing for Industrial Applications ANSI/NFPA 70:1999, National Electrical Code □ ANSI/NFPA 497, Classification of Flammable Liquids, Gases or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas. ANSI/UL 94, Tests for Flammability of Plastics Materials ANSI/UL 486E, Equipment Wiring Terminals for use with Aluminum and/or Copper Conductors ANSI/UL 508, Industrial Control Equipment ANSI/UL 746A, Standard for Polymeric Materials - Short Term Property Evaluations ANSI/UL 746C, Standard for Polymeric Materials - Use in Electrical Equipment Evaluations

Section 3
National Differences

US United States of America National differences to IEC 60079-7 (2001)		
IEC Standard	**Clause**	**Difference**
		ANSI/UL 840, Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment ANSI/UL 1059, Terminal Blocks

Section 3
National Differences

US **United States of America** National differences to IEC 60079-7 (2001)

IEC Standard	Clause	Difference
IEC 60079-7 (2001)	4.2	**4.2 Terminals for external connections** Terminals for connections to external circuits shall be generously dimensioned to permit the effective connection of copper conductors of cross-section equal to at least that corresponding to the rated current of the electrical apparatus. The number and sizes of copper conductors that can be safely connected to terminals shall be specified in the descriptive documents according to 23.2 of IEC 60079-0 1998. NOTE — ANSI/UL 1059 and ANSI/UL 486E require terminals to be identified with the permitted wire sizes in AWG or kcmil. ~~NOTE 1 — Attention is drawn to the use of aluminum wire because of the difficulties associated with controlling critical creepage and clearance distances with the application of anti-oxidant materials. The connection of aluminum wire to external terminals may be accomplished by the use of suitable bi-metallic ferrule connections providing a copper connection to the terminal.~~ Terminals shall be subjected to the terminal insulating material tests of 6.9. Terminals with a rated voltage greater than 1500 V shall be subjected to the applicable examination and tests of ANSI/UL 1059 and ANSI/UL 486E except for the dielectric tests which shall be conducted in accordance with 6.10. **{Remaining text of this section is technically unchanged}**

Section 3
National Differences

US United States of America National differences to IEC 60079-7 (2001)		
IEC Standard	**Clause**	**Difference**
IEC 60079-7 (2001)	6.10	6.10 Terminal Dielectric Tests The test sample shall be mounted as in normal use. The voltage specified in Table 13 shall be applied for one minute, first between any adjacent terminal assemblies and then between all the terminal assemblies connected together and the mounting means. There shall be no dielectric breakdown during any of the tests.

Section 3
National Differences

IEC Standard: 60079-7:2nd Ed 1990 Amend 1 1991 and Amend 2 1993 Electrical apparatus for explosive gas atmospheres - Part 7: Increased safety "e"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**NO**	CAN/CSA-E60079-7-95
CH Switzerland	**YES Without Amds**		EN 50019 1994
CN China		**YES**	GB3836.3 2000
CZ Czech Republic	**YES Without Amds**		EN 50019 1994
DE Germany	**YES Without Amds**		EN 50019 1994
DK Denmark	**YES Without Amds**		EN 50019 1994
FI Finland	**YES Without Amds**		EN 50019 1994
FR France	**YES Without Amds**		EN 50019 1994
GB United Kingdom	**YES Without Amds**		EN 50019 1994
HR Croatia	**N/A**		
HU Hungary	**YES Without Amds**		EN 50019 1994
IN India		**N/R**	
IT Italy	**YES Without Amds**		EN 50019 1994
JP Japan		**NO**	
KR Republic of Korea		**YES Without Amds**	Labor Ministry Notice1998-65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES Without Amds**		EN 50019 1994
NO Norway	**YES Without Amds**		
NZ New Zealand		**NO**	
PL Poland	**N/A**		
RO Romania	**YES Without Amds**		EN 50019 1994
RU Russia		**N/R**	
SE Sweden	**YES Without Amds**		EN 50019 1994
SG Singapore		**N/R**	
SI Slovenia	**YES Without Amds**		EN 50019 1994
TR Turkey*		**TBA**	
US United States		N/R See IEC 60079-7 2001	
ZA South Africa		**NO**	SANS 60079-7:1990

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for details

Section 3
National Differences

Countries that have declared adoption of IEC 60079-7 2nd Ed 1990 Amend 1 1991 and Amend 2 1993 Without National Differences
AU Australia
CA Canada
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-7 2nd Ed 1990 Amend 1 1991 and Amend 2 1993 With National/Group Differences- Differences follow.

CN China

IEC Clause or Annex
Number **Details of difference IEC 60079-7:1990 (edition 2) Amendment 1: 1991 and Amendment 2:1993**

4.3 Add the following extra note to table 1:
Minimum creepage distances and clearances for electrical apparatus Group I with rated voltage 1140 V may be determined in accordance with interpolation.

5.1.3 Add the following extra note:
For electrical motor with sliding bearing, a hole should be designed for measuring of minimum radial air gap.

Annex Add Annex F (informative): Example for material group of general insulation material according to comparative tracking index.

Add Annex G (informative): Additional requirements for construction and testing of electrical motorwith high voltage

Section 3
National Differences

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

EN Clause or Annex Number details of differences
1 No differences but editorial

1 Refers to EN 50014, 50018, 50020, 50033, 50039, 60034-5 to HD series instead of a direct reference to relevant IEC

3 EN refers in many cases to IEV 426-08-01 not yet available for the relevant edition of IEC

3.1 EN note 1 adds complementary measures are those requested to comply with the present standards.
3.2 EN refers in b) to IEV 426-08-02 there is no note as in §3.2 of IEC

3.3 No differences with IEC

3.4 No differences with same IEC paragraph

3.5 EN refers to IEV-426-08-03 and does not refer to figure B as in §3.5 of IEC

3.6 No differences with IEC

3.7 No differences with IEC

3.8 No note as in same § of IEC

3.9 No differences with IEC

3.10 No differences with IEC, editorial

3.11 Similar to IEC + two notes

3.12 EN refers to IEV 486-8-01
3.12.2 EN refers to IEV 486-01-03

3.12.3 EN refers to IEV 486-02-20

3.12.4 No differences with IEC excepted for the note included in IEC

3.12.5 Refers to IEV 486-03-01 + note

3.12.6 Refers to 486-02-15

3.12.7 No differences with IEC

3.12.8 No differences with IEC

3.12.9 refers to IEV 486-02-15

3.13 Similar to IEC but does not include the IEC note

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National Differences

3.13.1 No differences with IEC, editorial

3.13.2 No differences with IEC, editorial

3.13.3 No differences with IEC

3.13.4 No differences with IEC

3.13.5 No differences with IEC

4 No differences with IEC

4.1 No major differences with IEC which refers to IEC 364-5-523 in its *notes*

4.2 EN has a general note concerning electrolytical corrosion where IEC has no note but refers directly to aluminium

4.3 No differences with IEC, but editorial

4.4 No differences with IEC excepted for reference to HD 214 S2 (IEC 112) + editorial

4.5 No differences

4.6 No differences but EN adds a § 4.6.4

4.7 EN has a different writing of §4.7.1

4.7.3 EN has only two notes in table 3 instead of 3 in IEC

4.8 No differences with IEC

4.9 Different: EN has a provision in §4.9.3 dealing with "i" circuits and system.

4.10 No differences with IEC

5.1 Differences are editorial. § 5.1.4.1 has two notes where IEC has only one.

5.2 EN has a note specifying that warning luminaires and other small lamps are not covered by this article. Otherwise differences are editorial.

5.3 Differences are editorial.

5.4 No Differences with IEC

5.5 No Differences with IEC

5.6 No Differences with IEC. § 5.6.2.7 has no note

§ 5.6.3.1 has an extended point c) and a very detailed note dealing with what has to be understood by "copper"

Section 3
National Differences

§5.6.3.4 is different from the relevant IEC article but is equivalent to article 3.6.3.5 of IEC, no §5.6.3.5.

5.7 Not equivalent to IEC, refers also to annex C (informative)

5.8.1 No Differences with IEC

5.8.2 Equivalent to IEC 5.8.3

5.8.3 Equivalent to IEC 5.8.5

5.8.4 Equivalent to IEC 5.8.6

5.8.5 Equivalent to IEC 5.8.7

5.8.6 Equivalent to IEC 5.8.8

5.8.7 Equivalent to IEC 5.8.9

5.8.8 Equivalent to IEC 5.8.10

5.8.9 Not totally Equivalent to IEC 5.8.11

5.8.10 Equivalent to IEC 5.8.12

5.9 Equivalent to IEC

6.1 Similar to IEC but has a specification regarding equipment with a galvanic insulation

6.2 No differences but editorial.

6.3 No differences

6.4 No differences

6.5 No differences

6.6 No differences EN adds a note regarding the choice of the method

6.7 Equivalent to IEC but what is a prescription in IEC is now note2 in EN

6.8 Differences are editorial but EN in §6.8.4 refers to table 4 of EN 50014 for impacts energy where IEC quote them.

6.8.5 is different from IEC

6.8.6 Does no exist in EN: Is included in 6.8.5 of EN

7 Equivalent to IEC but adds a §7.3 dealing with transformers

8 Differences are not only editorial but marking for terminal boxes or junction boxes is more developed, considerations for clean environment or for heating equipment are added.

Section 3
National Differences

Figure 1 and 2 Equivalent to IEC

Figure 4 Equivalent to IEC Figure B1

Figure F1 Equivalent to IEC F1

Annex A Equivalent to IEC annex D

Annex B Not totally equivalent to IEC Annex as more developed

Annex C No equivalent part in IEC

Annex D Equivalent to IEC Annex E

Annex E Equivalent to IEC Annex B but EN part has a very developed note in § E.2.3

KR Korea

IEC Clause or Annex
Number Details of difference IEC 60079-7:1990 (edition 2) Amendment 1 1991

The requirements of Group I do not apply.

4.7.3 Replace the requirement of Table 4 with the following table.

	Method of temp. Measurement	Thermal class of insulating material				
		A	E	B	F	H
Permissible temp. in rated service	R T	90 80	105 95	110 100	130 115	155 135
The limit of permissible temp. rise in rated service (For the amb. Temp. of 40ºC)	R T	50 40	65 55	70 60	90 75	115 95
Permissible temp. at end of time tE	R	160	175	185	210	235
Permissible temp. at end of time tE(for the ambient temp. tE)	T	120	135	145	170	195
Notes The letter “R” means resistance and the letter “T” means thermometer In this case, measurement by thermometer is permissible only when measurement by change of resistance is not permissible.						

4.10 The requirement of a) does not apply.

Replace the minimum radial gap with the following table.

Poles	Minimum radial air gap		
	D≤ 75	75<D≤ 750	D> 750
2	0.25	0.25+(D-75)/300	2.7

Section 3
National Differences

4	0.2	0.2+(D-75)/500	1.7
Not less than 6	0.2	0.25+(D-75)/800	1.2

5.2.6This requirement does not apply.

5.6 This requirement does not apply.

5.7 This requirement does not apply.

5.8 This requirement does not apply.

6.3 This requirement does not apply.

6.4 This requirement does not apply.

6.6 This requirement does not apply.

6.7 This requirement does not apply.

8. The requirement of i) does not apply.

Appendix C Requirements do not apply

Amendment 1 These requirements do not apply

Amendment 2 These requirements do not apply

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-11

IECEx Bulletin:2011(E)

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About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-11

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 60079-11 5th Edition 2006 Electrical apparatus for explosive gas atmospheres - Part 11: Intrinsic safety "i"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-11:11
CH Switzerland	**NO**		EN 60079-11:2007
CN China		**YES**	GB 3836.4 –2010 Explosive atmospheres Part4: Equipment protection by intrinsic safety "i" IEC60079-11:2006, MOD
CZ Czech Republic	**NO**		EN 60079-11:2007
DE Germany	**NO**		EN 60079-11:2007
DK Denmark	**NO**		EN 60079-11:2007
FI Finland	**NO**		EN 60079-11:2007
FR France	**NO**		EN 60079-11:2007
GB United Kingdom	**NO**		EN 60079-11:2007
HR Croatia	**NO**		EN 60079-11:2007
HU Hungary	**NO**		EN 60079-11:2007
IN India		**NO**	Adopted as IS /IEC 60079-11: 2006
IT Italy	**NO**		EN 60079-11:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-11:2007
NL The Netherlands	**NO**		EN 60079-11:2007
NO Norway	**NO**		EN 60079-11:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-11:2007
RO Romania	**NO**		EN 60079-11:2007
RU Russia		**NO**	GOST R 52350.11-2005
SE Sweden	**NO**		EN 60079-11:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-11:2007
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-11 2009 ANSI/UL 60079-11 2009
ZA South Africa		**NO**	SANS 60079-11:2006

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-11 5th Edition 2006 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-11 5th Edition 2006 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-11:11
Explosive atmospheres –
Part 11: Equipment protection by intrinsic safety "i"

This is the third edition of CAN/CSA-C22.2 No. 60079-11, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-11 (Fifth edition, 2006-07). It replaces the CAN/CSA-C22.2 No.60079-11:02 (adopted CEI/IEC 60079-11: Fourth edition, 1999-02: Entitled *Electrical apparatus for explosive gas atmospheres – Part 11: Intrinsic safety "i"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Section 3
National Differences

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

5.4 Level of protection “ic”

[Add the following paragraph]

This protection technique has not been acknowledged in the Canadian Electrical Code, Part I (C22.1); product evaluated to the “ic” level of protection may not have a market in Canada.

6.1 Enclosures

[Expand the principle in the 2nd paragraph in the Standard to the following effect]

The degree of protection required will vary according to the intended use, for example, a degree of protection of IP54 in accordance with IEC 60529 may be required for Group I apparatus. *In the same token, where the equipment is intended for use in Group III explosive dust atmospheres, the degree of protection by the enclosure shall be compatible with the EPL (Equipment Protection Level); for example, IP64 may be required for “Da” as specified in the application. Alternatively, consideration shall be given to encapsulation and conformal coating that dust would not be in contact with live-parts.*

7.4.1 General (Primary and secondary cells and batteries)

[Add the following paragraph]

In addition, where lithium types of cells and batteries are used, there shall be a resistor in series with cell (or battery) to limit the current to two-thirds or lower of the maximum pulse current of the cell (or battery), and a fuse in series of a rating not exceeding half of the maximum continuous discharge current of the cell (or battery).

NOTE: *The requirement that the resistor and the fuse be in series with the cell (or battery) is to ensure that a direct short-circuit of the cell (or battery) cannot be imposed. The resistor is for shark-ignition consideration, while the fuse is for temperature consideration. The two-third value is based on the 1.5 factor on current for spark-ignition assessment, and the ½ value is based on the indefinite arcing time of the fuse at twice its current rating.*

12.3 Warning markings

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

Section 3
National Differences

CN **China**

IEC Standard Number IEC 60079-11:2006

IEC Clause or Annex Number	Details of Differences
4	Add the following extra requirement to this Clause as note. *Group III equipment is not included in this standard.*
6.3.11	Add the following extra requirement to this Clause as note 2 *IS wiring and Non-IS wiring should be arranged separately.*
6.5	Add the following extra requirement to this Clause as note *In general, the loop of Group I equipment should not be grounded, only when ground protection is required.*
10.1.5.2 b)	This requirement does not apply.

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US United States of America

Clause	**US Differences to IEC 60079-11, 5th Ed, 2006-07**
General	The U.S. version of IEC 60079-11 does not include requirements for Group I electrical apparatus.
General	Where references are made to other IEC 60079-standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
General	Where references are made to hazardous areas, this is changed to the U.S. terms unclassified locations or hazardous (classified) locations as appropriate.
General	Where requirements call for the application of an "X" appended to the certificate number, this is replaced with a requirement to document this in the manufacturer's instructions.
1	<u>Class I, Zone 0, 1, or 2 hazardous (classified) locations as defined by the National Electrical Code, ANSI/NFPA 70</u> an explosive gas atmosphere
1	The requirements for intrinsically safe systems are provided in IEC 60079-25.
2	Additional references to align with US practice and the National Electrical Code: <u>UL746A, *Polymeric Materials B Short Term Property Evaluations, Fifth edition, (Edition Date November 1, 2000)*</u> <u>ANSI/UL 840, Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment</u>
3.12	Additional note added to provide clarification **internal wiring** wiring and electrical connections that are made within the apparatus by its manufacturer <u>NOTE Within a rack or panel, interconnections between separate pieces of apparatus made in accordance with detailed instructions from the manufacturer are considered to be internal wiring.</u>
3.23	Additional note added to provide clarification **creepage distance**

	shortest distance along the surface of an insulating medium in contact with air between two conductive parts NOTE The distance concerned here is, for example, applicable to printed circuits that have no coating in accordance with 6.4.8, where the insulation across which the creepage distance is measured is in direct contact with the air.
3.25	Additional definition added as IEC 60079-25 is not adopted in the US. **3.25** **intrinsically safe system** assembly of interconnected items of apparatus that may comprise intrinsically safe apparatus and associated apparatus, and interconnecting cables in which the circuits within those parts of the system that may be exposed to explosive gas atmospheres are intrinsically safe circuits.
3.26	Additional definition added as IEC 60079-25 is not adopted in the US. **3.26** **apparatus and systems of level of protection "ia"** electrical apparatus and systems containing intrinsically safe circuits that are incapable of causing ignition, with the appropriate safety factor: when up to two countable faults are applied and, in addition, those non-countable faults that give the most onerous condition.
3.27	Additional definition added as IEC 60079-25 is not adopted in the US. **3.27** **apparatus and systems of level of protection "ib"** electrical apparatus and systems containing intrinsically safe circuits that are incapable of causing ignition with the appropriate safety factor: when up to one countable fault is applied and, in addition, those noncountable faults that give the most onerous condition.
3.28	Additional definition added as IEC 60079-25 is not adopted in the US. **3.28** **apparatus and systems of level of protection "ic"** electrical apparatus and systems containing intrinsically safe circuits that are incapable of causing ignition with the appropriate safety factor: in normal operation.
5.1	Modification of text to reflect that level of protection "ic"is not currently recognized in the US.

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	Intrinsically safe apparatus and intrinsically safe parts of associated apparatus shall be placed in levels of protection "ia", "ib" or “ic”. NOTE: The National Electrical Code® (NEC®), through the 2008 edition only recognizes levels of protection “ia” for Class I, Zone 0 and “ib” for Class I, Zone 1 hazardous (classified) locations. No recognition is made for level of protection “ic”.
5.2	Additional text added as IEC 60079-25 is not adopted in the US and to provide clarification. In the case of systems, the countable faults specified will be applied to the system as a whole and not simultaneously to each item of electrical apparatus in the system. For example, a level of protection 'ia' system comprising two pieces of apparatus will have up to two countable faults applied, not four.
5.3	Additional text added as IEC 60079-25 is not adopted in the US and to provide clarification. In the case of systems, the countable fault specified shall be applied to the system as a whole and not simultaneously to each item of electrical apparatus in the system. For example, a level of protection 'ib' system comprising two pieces of apparatus will have only one countable fault applied, not two.

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<table>
<tr><td>Table 4</td><td>Table 4 modified to align with Draft 6th edition of IEC 60079-11

<table><tr><th>T6
A</th></tr><tr><td>0,5
0,7
0,8
1,0
1,2
<s>1,9</s> <u>1.7</u>
2,1
2,5
3,2
4,1
5,6
6,9
8,1
9,3
11,4
13,5
15,4</td></tr></table></td></tr>
<tr><td>6.3.1

6.3.1.2</td><td>Text in the IEC standard was incorrectly located as it applies to all cases not just those for Annex F.

6.3.1 Separation of conductive parts
Separation of conductive parts between
– intrinsically safe and non-intrinsically safe circuits, or
– different intrinsically safe circuits, or
– a circuit and earthed or isolated metal parts,

shall conform to the following if the type of protection depends on the separation.</td></tr>
</table>

	Separation distances shall be measured or assessed taking into account any possible movement of the conductors or conductive parts. Manufacturing tolerances shall not reduce the distances by more than 10 % or 1 mm, whichever is the smaller. Separation distances that comply with the values in 6.1.1 or 6.1.2 shall not be subject to a fault. Separation requirements shall not apply where earthed metal, for example tracks of a printed circuit board or a partition, separates an intrinsically safe circuit from other circuits, provided that breakdown to earth does not adversely affect the type of protection and that the earthed conductive part can carry the maximum current that would flow under fault conditions. NOTE 1 For example, the type of protection does depend on the separation to earthed or isolated metallic parts if a current-limiting resistor can be bypassed by short-circuits between the circuit and the earthed or isolated metallic part. An earthed metal partition shall have strength and rigidity so that it is unlikely to be damaged and shall be of sufficient thickness and of sufficient current-carrying capacity to prevent burnthrough or loss of earth under fault conditions. A partition either shall be at least 0,45 mm thick and attached to a rigid, earthed metal portion of the device, or shall conform to 10.6.3 if of lesser thickness. Where a non-metallic insulating partition having a thickness and appropriate CTI in accordance with Table 5 is placed between the conductive parts, the clearances, creepage distances and other separation distances either shall be measured around the partition provided that the partition has a thickness of at least 0.9 mm, or shall conform to 10.6.3 if of lesser thickness. NOTE 2 Methods of assessment are given in Annex C. 6.3.1.2 Distances according to Annex F For levels of protection “ia” and “ib”, if separation distances are less than the values specified in Annex F, they shall be considered as subject to non-countable short-circuit faults if this impairs intrinsic safety. For level of protection “ic”, if separation distances are less than the values specified in Annex F, they shall be considered as short-circuits if this impairs intrinsic safety. The fault mode of failure of segregation shall only be a short-circuit. ~~Separation requirements shall not apply where earthed metal, for example tracks of a printed circuit board or a partition, separates an intrinsically safe circuit from other circuits, provided that breakdown to earth does not adversely affect the type of protection and that the earthed conductive part can carry the maximum current that would flow under fault conditions.~~

	~~NOTE 1 For example, the type of protection does depend on the separation to earthed or isolated metallic parts if a current-limiting resistor can be bypassed by short-circuits between the circuit and the earthed or isolated metallic part.~~ ~~An earthed metal partition shall have strength and rigidity so that it is unlikely to be damaged and shall be of sufficient thickness and of sufficient current-carrying capacity to prevent burnthrough or loss of earth under fault conditions. A partition either shall be at least 0,45 mm thick and attached to a rigid, earthed metal portion of the device, or shall conform to 10.6.3 if of lesser thickness.~~ ~~Where a non-metallic insulating partition having an appropriate CTI is placed between the conductive parts, the clearances, creepage distances and other separation distances either shall be measured around the partition provided that the partition has a thickness of at least 0,9 mm, or shall conform to 10.6.3 if of lesser thickness.~~ ~~NOTE 2 Methods of assessment are given in Annex C.~~
Figure 2	Figure 2 is also applicable to Annex F 1 Chassis 2 Load 3 Non-intrinsically safe circuit defined by U_m 4 Part of intrinsically safe circuit not itself intrinsically safe 5 Intrinsically safe circuit 6 Dimensions to which Table 5 or Annex F ~~is~~ where applicable 7 Dimensions to which general industrial standards are applicable 8 Dimensions to 7.3 9 Dimensions to 6.2.1 for output terminals between separate Intrinsically safe circuits and between Intrinsically safe to non intrinsically safe circuits 10 If necessary
6.3.3	Additional text added to provide clarification In measuring or assessing clearances between conductive parts, insulating partitions of less than 0,9 mm thickness, or which do not conform to 10.6.3, shall be ignored. Other insulating parts shall conform to column 4 of Table 5 or column 2 of Table F.1 and column 2 of Table F.2 where applicable.
6.3.4	Additional text added to provide clarification Casting compound shall meet the requirements of 6.6. For those parts that require encapsulation, the minimum separation distance between encapsulated conductive parts and components, and the free surface of the casting compound shall be at least

	half the values shown in column 3 of Table 5, with a minimum of 1 mm, or column 3 of Table F.1 and column 3 of Table F.2 where applicable. When the casting compound is in direct contact with and adheres to an enclosure of insulating material conforming to column 4 of Table 5, no other separation is required (see Figure D.1).
6.3.7	Additional text added to provide clarification For the creepage distances specified in column 5 of Table 5 or Columns 2 of Table F.1 and column 5 of Table F.2 where applicable, the insulating material shall conform to column 7 of Table 5 or material Group IIIa or IIIb in accordance with IEC 60664-1, which specifies the minimum comparative tracking index (CTI) measured in accordance with IEC 60112 or ANSI/UL 746A. The method of measuring or assessing these distances shall be in accordance with Figure 3.
6.3.8	Additional text added to provide clarification Where bare conductors or conductive parts emerge from the coating the comparative tracking index (CTI) in column 7 of Table 5 or column 7 of Table F.2 or material Group Table F.1 shall apply to both insulation and coating.
6.3.12	Modification of text as follows to indicate Specific Conditions of Use with ANSI/ISA 60079-0 Where the circuit does not satisfy this requirement the ~~apparatus shall be marked with the symbol “X” the~~ documentation shall indicate the necessary information regarding the correct installation.
6.6	Additional text added to provide clarification b) have at its free surface a CTI value of at least that specified in Table 5 or Table F.2, or the material Group specified in Table F.1 if any bare conductive parts protrude from the casting compound;
7.3	External creepage and clearance distance used for fuses are considered no different to any other creepage and clearance distances. Fuses shall have a rated voltage of at least *U*m (or *U*i in intrinsically safe apparatus and circuits) ~~although~~ and the external creepage distances and clearances ~~they do not have to~~ shall conform to Table 5. General industrial standards for the construction of fuses and fuseholders shall be applied and their method of mounting including the connecting wiring shall not reduce the clearances, creepage distances and separations afforded by the fuse and its holder. Where required for intrinsic safety, the distances to other parts of the circuit shall comply with 6.1.1 or 6.1.2. NOTE 1 Microfuses conforming to IEC 60127 are acceptable provided they also conform to the external creepage

	distances and clearances of Table 5.
7.3	Creepage and clearance is based on the maximum voltage at one end of the fuse and not the voltage dropped across the fuse. Creepage and clearance distances across the current limiting resistor and its connecting tracks ~~shall be calculated using the voltage of 1,7 × I_n × maximum resistance of the current limiting resistor. The transient voltage shall not be considered. The separation distances between the resistor and other parts of the circuit~~ shall comply with 6.1.1 or 6.1.2.
7.4.7	Text modified to align with the deviation taken in the adoption of IEC 60079-0 If the cell or battery, requiring current-limiting devices to ensure the safety of the battery itself, is not intended to be replaced in the explosive gas atmosphere, it shall either be protected in accordance with 7.4.6 or alternatively it may be housed in a compartment capable of being opened or removed only with the aid of a tool ~~with special fasteners, for example those specified by IEC 60079-0.~~ It shall also conform to the following:
8.5	Additional text added to provide requirements for filter capacitors. **8.5 ~~Blocking~~ capacitors** **8.5.1 Blocking capacitors** Either of the two series capacitors in an infallible arrangement of blocking capacitors shall be considered as being capable of failing to short or open circuit. The capacitance of the assembly shall be taken as the most onerous value of either capacitor and a safety factor of 1,5 shall be used in all applications of the assembly. Blocking capacitors shall be of a high reliability solid dielectric type. Electrolytic or tantalum capacitors shall not be used. The external connections of each capacitor and of the assembly shall comply with 6.3 but these separation requirements shall not be applied to the interior of the blocking capacitors. The insulation of each capacitor shall conform to the dielectric strength requirements of 6.3.12 applied between its electrodes and also between each electrode and external conducting parts. Where blocking capacitors are used between intrinsically safe circuits and non-intrinsically safe circuits, the blocking capacitors are to be assessed as a capacitive coupling between these circuits. The energy transmitted is calculated using Um and the most onerous value of either capacitor and shall be in accordance with the permissible ignition energy of 10.7. All possible transients shall be taken into account, and the effect of the highest nominal operating frequency (as that supplied by the manufacturer) in that part of the circuit shall be considered.

	Where such an assembly also conforms to 8.8, it shall be considered as providing infallible galvanic separation for direct current. **8.5.2 Filter capacitors** Capacitors connected between the frame of the apparatus and an intrinsically safe circuit shall conform to 6.3.12. Where their failure by-passes a component on which the intrinsic safety of the circuit depends, they shall also conform to the requirements for blocking capacitors found in 8.5.1. A capacitor meeting the infallible separation requirements of 6.3 both externally and internally shall be considered to provide infallible separation and only one is required. NOTE The normal purpose of capacitors connected between the frame and circuit is the rejection of high frequencies.
10.1.4.1	Tolerance limits added for the supply in mains voltage. The circuit to be tested shall be based on the most incendive circuit that can arise, toleranced in accordance with Clause 7 and taking into account between 0 and 110% ~~a 10 % variation in~~ of the mains supply voltage.
10.1.4.2	Text deleted as a modiied requirement was added to 10.1.4.1. a) use the appropriate test mixtures in accordance with Table 7 and adjust ~~increase the mains (electrical supply system) voltage to 110 % of the nominal value to allow for mains variations, or set other voltages, for example batteries,~~ power supplies and voltage limiting devices at the maximum value in accordance with Clause 7, then: 1) for inductive and resistive circuits, increase the current to 1,5 times the fault current by decreasing the values of limiting resistance, if the 1,5 factor cannot be obtained, further increase the voltage; 2) for capacitive circuits, increase the voltage to obtain 1,5 times the fault voltage. Alternatively when an infallible current-limiting resistor is used with a capacitor, consider the capacitor as a battery and the circuit as resistive. When using the curves in Figures A.1 to A.6 or Tables A.1 and A.2 for assessment, this same method shall be used. b) use the appropriate ~~more easily ignited explosive~~ test mixtures in accordance with Table 8.
10.1.5.2	Text added to provided limits for IIB and IIC. – for connection of the combined inductance and capacitance where both are greater than 1 % of the allowed value (excluding the cable), allow up to 50 % each of the values of L and C, determined by the ignition curves and tables given in Annex A, when read with a safety factor of 1,5 on the current or voltage, as applicable. The maximum

	capacitance allowed shall not be more than Co = 1µF for IIB and Co = 600 nF for IIC
10.5.3	Text modified for clarification. If a battery comprises a number of discrete cells or smaller batteries combined in a well-defined construction conforming to the segregation and other requirements of this standard, then each discrete element shall be considered as an individual component for the purpose of testing. Except for specially constructed batteries ~~cells~~ where it can be shown that short circuits between cells cannot occur, the failure of each element shall be considered as a single fault. In less well-defined circumstances, the battery shall be considered to have a short-circuit failure between its external terminals.
10.5.3 a)	Text modified for clarification. a) When the internal resistance of a cell or battery is to be included in the assessment of intrinsic safety, its minimum resistance value shall be specified. If the cell/battery manufacturer is unable to confirm the minimum value of internal resistance, the most onerous value of short-circuit current from a test of 10 samples of the cell/battery together with the peak open-circuit voltage in accordance with 7.4.3 of the cell/battery shall be used to determine the internal resistance.
10.5.3 b)	Text modified for clarification. b) The maximum surface temperature shall be determined as follows. All current-limiting devices external to the cell or battery shall be short-circuited for the test. Any external sheath (of paper or metal, etc.) not forming part of the actual cell enclosure shall be removed for the test. The temperature shall be determined on the hottest surface that may be exposed to the explosive atmosphere ~~outer enclosure of each cell or battery~~ and the maximum figure taken. The test shall be carried out both with internal current-limiting devices in circuit and with the devices short-circuited using 10 cells in each case. The 10 samples having the internal current-limiting devices short-circuited shall be obtained from the cell/battery manufacturer together with any special instructions or precautions necessary for safe use and testing of the samples. For 'ic' the maximum surface temperature shall be determined by testing in normal operating conditions with all protection devices in place. NOTE 2 When determining the surface temperature of most batteries, the effect of built-in protective devices, for example fuses or PTC resistors, is not taken into account because this is an assessment of a possible internal fault, for example failure of a separator. NOTE 3 While determining the maximum surface temperature of a battery comprising more than one cell, provided

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	that the cells are adequately segregated from each other, only one cell should ~~must~~ be shorted at one time to determine this maximum surface temperature. (This is based on the extreme unlikelihood of more than one cell shorting at one time.)
10.10	The text was deleted as the US ordinary location requirements for transformers with self-resetting thermal trips are more severe that those quoted in this document. The test shall continue for at least 6 h or until the non-resetting thermal trip operates. ~~When a self-resetting thermal trip is used, the test period shall be extended to at least 12 h.~~
12.1	Additional marking requirements added for Live Maintenance. Where the manufacturer does not specify live maintenance procedures, the word "WARNING" and the following: "To prevent ignition of explosive atmospheres, disconnect power before servicing" Where the manufacturer specifies and provides live maintenance procedures, the word "WARNING" and the following: "To prevent ignition of explosive atmospheres, read understand and adhere to the manufacturer's live maintenance procedures." Alternatively the above warnings for live maintenance may be included in the documentation. (See Clause 13)

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IEC Standard IEC 60079-11 4th Edition 1999 Electrical apparatus for explosive gas atmospheres - Part 11: Intrinsic safety "i"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-11 2000
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-E60079-11-2001
CH Switzerland	**YES**		EN 50020 1994
CN China		**YES**	GB 3836.4 2nd Edition
CZ Czech Republic	**YES**		ČSN EN 50020 2002
DE Germany	**YES**		EN 50020 2002
DK Denmark	**YES**		DS/EN 50020 2002
FI Finland	**YES**		EN 50020 2002
FR Franc	**YES**		EN 50020 2002
GB United Kingdom	**YES**		BS EN 50020 2002
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 50020 2002
IN India		**NO**	Adopted as IS 5780:2002/IEC 60079-11:1999
IT Italy	**YES**		EN 50020 2002
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-11:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**YES**		EN 50020 2002
NO Norway	**YES**		EN 50020 2002
NZ New Zealand		**NO**	AS/NZS 60079-11 2000
PL Poland	**N/A**		
RO Romania	**YES**		EN 50020 2002
RU Russia		**YES**	GOST R 51330.10-99
SE Sweden	**YES**		
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 50020 2002
TR Turkey*		**TBA**	
US United States		**YES**	
ZA South Africa		**NO**	SANS 60079-11:1999

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

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Countries that have declared adoption of IEC 60079-11 4th Edition 1999 Without National Differences
AU Australia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-11 4th Edition 1999 With National/Group Differences- Differences follow.

CA Canada

Clause 7.4.1
[Add the following text]

In addition, where lithium types of cell and battery are used, there shall be a resistor in series with the cell (battery) to limit the current to two-third (2/3) or lower of the maximum pulse current of the cell (battery), and a fuse in series of rating not exceeding half (1/2) of the maximum continuous discharge current of the cell (battery).

Note: *The requirement of the resistor and the fuse in series with the cell (battery) is to ensure that a direct short-circuit of the cell (battery) cannot be imposed. The resistor is for spark-ignition consideration, while the fuse is for temperature consideration. The 2/3 value is based on the 1.5 factor on current for spark-ignition assessment, and the 1/2 value is based on the indefinate arcing time of the fuse at twice its current rating.*

CN China

IEC Clause or Annex Number **Details of Differences to IEC 60079-11:1999 (edition 4)**

6.4.11 Add: "Note 2: The conductors of intrinsically safe circuit and that of non-intrinsically safe circuit should be arranged separately as fully as possible.

6.6 Add: " Note: The earth wire should not be used usually as circuits in the intrinsically safe circuits for electrical apparatus group I, but except for earthing protection if necessary.

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Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Differences between EN 50020:2002 and IEC 60079-11:1999

Clause 5.1 " Level of protection" in EN

"In the determination of the permitted output parameters C_o, L_o, L_o/R_o for a source of power a safety factor of 1,5 shall be used in all circumstances.

Where the test apparatus specified in Annex B is used for high currents, then the permitted inductance is very low. In these circumstances the inductance shall be accurately specified in the certification documentation and controlled. The use of high currents in field wiring is not permitted.

The application of U_m includes any voltage up to that value and these values have to be taken into account during assessment and testing. However a slow increase of the voltage from the rated value to U_m shall not be assumed."

Clause 5.2:
In testing or assessing the circuits for spark ignition, the following safety factors shall be applied in accordance with 10.4.2:

for both a) and b)	1,5
for c)	1,0
for output parameters, C_o, L_o, L_o/R_o	1,5

Clause 6.1
First two paras different

Clause 6.2.4 Small components

Small components for example transistors or resistors, whose temperature exceeds that permitted for the temperature classification, shall be acceptable providing that when tested in accordance with 10.7, small components do not cause ignition. Alternatively, where no catalytic or other chemical reactions can result, one of the following is acceptable:

a) for Group II T4 and Group I temperature classification components shall conform to Table 3, including the relevant reduction of permitted maximum dissipation with increased ambient temperature listed in Table 3b;

b) for Group II T5 classification the surface temperature of a component with a surface area smaller than 10 cm^2 shall not exceed 150 °C.

In addition the permitted higher temperature shall not invalidate the type of protection for example by causing the component or adjacent parts of the apparatus to exceed any safety related rating, or to deteriorate or be distorted so as to invalidate creepage and clearance distances. In particular, with the higher levels of permitted power, extra care shall be taken in the selection of the materials to be used adjacent to these high temperature components, for example to prevent burning of the printed circuit boards.

Migration of components due to solder melting shall not be taken into account.

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Table 3a - Requirements at 40 °C ambient

Total surface area excluding lead wires	Group II T4		Group I Dust excluded	
	Maximum surface temperature °C	Maximum power dissipation W	Maximum surface temperature °C	Maximum power dissipation W
< 20 mm^2	275		950	
≥ 20 mm^2 ≤ 10 cm^2	200 or	1,3	450 or	3,3
> 10 cm^2	135 or	1,3	450 or	3,3

Table 3b - Variation in maximum power dissipation with ambient temperature

Maximum ambient temperature	**°C**	**Apparatus group**	**40**	**60**	**70**	**80**
Maximum power dissipation	W	Group II	1,3	1,2	1,1	1,0
		Group I	3,3	3,15	3,07	3,0

Where the outer surface of a component is not continuous, for example a component made up of tightly wound wire which is not consolidated or a tube with a central hole, then the surface to be considered for the purpose of Table 3a is the outer envelope of the component. The surface temperature of any part of such a component to which the gas has access shall be in accordance with the maximum surface temperature requirement of Table 3a and the maximum power criterion is not applicable in these circumstances.

Clause 6.4.8 Creepage distance under coating

A double solder mask is also acceptable as a conformal coating

Clause 7.3 Fuses

Creepage and clearance distances associated with the current limiting resistor and its immediate connecting wiring shall be calculated using the voltage of 1,7 x I_n x resistance of the current limiting resistor. The transient values shall not be considered.

Clause 7.4.8 External contacts for charging batteries

Cell or battery assemblies with external charging contacts shall be provided with means to prevent short-circuiting or to prevent the cells and batteries from delivering ignition-capable energy to the contacts when any pair of the contacts is accidentally short-circuited. This shall be accomplished by placing blocking diodes or an infallible series resistor in the charging circuits. For level of protection “ib” two diodes and for level of protection “ia” three diodes shall be used. To protect these diodes or resistors against excess currents during charging an appropriate battery charger shall be defined or an appropriately rated fuse shall be part of the circuit. The fuse shall either be encapsulated or shall not carry any current when situated in a potentially explosive atmosphere and the circuit is assessed as required by clause 5.

Clause 7.6 Failure of components and connections

d) diodes (including LEDs and Zener diodes) shall be considered at their rated voltages including tolerances and as failed to short and open circuit. Other semiconductors shall be considered to fail to short and open circuit and to any state to which they can be driven including the state in which maximum power is dissipated. Integrated circuits can fail so that any combination of short and open circuits can exist between their external connections (counting only one fault). Although any combination can be assumed, once that fault has been applied it cannot be changed, for example by application of a second fault. Under this fault situation any capacitance and inductance connected to the device shall be considered in their most onerous connection as a result of the applied fault;

Clause 8.3 Inductors

8.3.1 Damping windings

Damping windings used as short-circuited turns to minimise the effects of inductance shall be considered not to be subject to open-circuit faults if they are of reliable mechanical construction, for example seamless metal tubes or windings of bare wire continuously short-circuited by soldering.

8.3.2 Inductors made by insulating conductors

Inductors made by insulated conductors are not considered to fail to a lower resistance value than the nominal resistance (taking into account tolerances) if they comply with the following:

- the nominal conductor diameter of wires used for inductor windings shall be at least 0,05 mm;
- the conductor shall be covered with at least two layers of insulation, or be made of enamelled round wire in accordance with

a) grade 1 of EN 60317-3, EN 60317-7 or EN 60317-8 there shall be no failure with the minimum values of breakdown voltage listed for grade 2 and that when tested in accordance with clause 14 of EN 60317-3, EN 60317-7 or EN 60317-8 there shall be no more than six faults per 30 m of wire irrespective of diameter, or

b) grade 2 of EN 60317-3, EN 60317-7 or EN 60317-8;

- windings after having been fastened or wrapped shall be dried to remove moisture before impregnation with a suitable substance by dipping, trickling or vacuum impregnation. Coating by painting or spraying is not recognized as impregnation;
- the impregnation shall be carried out in compliance with the specific instructions of the manufacturer of the relevant type of impregnating substance and in such a way that the spaces between the conductors are filled as completely as possible and that good cohesion between the conductors is achieved;
- if impregnating substances containing solvents are used, the impregnation and drying process shall be carried out at least twice.

Clause 8.7 Wiring and connections

b 4) where the current carrying capacity of the single track or combination of tracks is tested for one hour with a current 1,5 times the maximum continuous current, which can flow in the track under normal and fault conditions. Where the maximum fault current cannot be established a test current of 10 A shall be used. The application of this test current should not cause the tested track to fail to open circuit or to be separated from its substrate at any point.

Where temperature classification of such a track is to be experimentally determined the maximum continuous current with a unity safety factor shall be used. Where the fault current is undetermined then a test current of 6,6 A shall be used.

Infallible wiring and connections may be used within an apparatus and not necessarily as part of an infallible assembly.

Clause 8.8 Galvanically separating components

Clause 8.8.2 Isolating components between intrinsically safe and non-intrinsically safe circuits
The requirements of Table 4 shall also apply to the isolating element except that inside sealed devices, e.g. opto-couplers, lines 5, 6 and 7 shall not apply. The non-intrinsically safe circuit connections shall be provided with protection to ensure that the ratings of the devices in accordance with 7.1 (the exceptions still being applicable) are not exceeded unless it can be shown that the circuits connected to these terminals cannot invalidate the infallible separation of the devices. Typically the inclusion of a single shunt Zener diode protected by a suitably rated fuse capable of interrupting the prospective peak current of the supply shall be considered as sufficient protection. For this purpose Table 4 shall not be applied to the fuse and Zener diode. The Zener diode power rating shall be at least $1,7I_n$ times the diode maximum Zener voltage. General industrial standards of construction shall be considered adequate for fuses and the method of mounting, e.g. in a fuse holder, shall not reduce the clearances and creepage distances afforded by the fuse itself. In some applications the intrinsically safe circuit connections may require the application of similar protective techniques to avoid exceeding the rating of the opto-couplers. The components shall withstand an electric strength test in accordance with 6.4.12 between the non-intrinsically safe circuit terminals and the intrinsically safe terminals. The manufacturers rated isolation voltage for the infallible separation of the component shall be not less than the test voltage required by 6.4.12.

Clause 8.8.3 Isolating components between separate intrinsically safe circuits
Isolating elements shall be considered to provide infallible separation of separate intrinsically safe circuits if the following conditions are satisfied:

a) the rating of the device shall be according to 7.1 (the exceptions still being applicable). Protective techniques (such as those indicated in 8.8.2) may be necessary to avoid exceeding the rating of the opto-coupler.

b) the device shall withstand an electric strength test as described in 6.4.12. The manufacturer's rated isolation voltage for the infallible separation of the component under test shall be not less than the test voltage required by 6.4.12.

Clause 10.1 Spark ignition test
Clause 10.1.1 General

Where voltages and currents are specified without specific tolerances, a tolerance of ± 2 % (including measurement uncertainty) is to be used.

Annex B1

B.1.7 Modification of test apparatus for use at higher currents

Test currents of 3 A to 10 A may be tested in the test apparatus when it is modified as follows:
The tungsten wires are replaced by wires with diameter increased to 0,37 mm to 0,43 mm and the free length reduced to 10,5 mm.

NOTE The reduction in free length reduces the wear on the cadmium disc.

The total resistance of the apparatus including the commutation contact resistance shall be reduced to less than 10 mΩ. Brushes of the type used in the automobile industry combined with brass sleeves on the apparatus shafts so as to increase the contact area have been found to be one practical solution.

The total inductance of the test apparatus and the inductance of the interconnection to the circuit under test must be minimised. A maximum value of 1 µH must be achieved.

The apparatus can be used for higher currents but special care in interpreting the results is necessary.

Annex C.1
To use line 2 of Table 4, multiply the measured values by the following factors:

Voltage difference	*U* < 10 V	10 V ≤ *U* < 30 V	*U* ≥ 30 V
A	1	1	1
B	3	3	3
C	3	4	6

To use line 3 of Table 4, multiply the measured values by the following factors:

Voltage difference	*U* < 10 V	10 V ≤ *U* < 30 V	*U* ≥ 30 V
A	0,33	0,33	0,33
B	1	1	1
C	1	1,4	2

To use line 4 of Table 4, multiply the measured values by the following factors:

Voltage difference	*U* < 10 V	10 V ≤ *U* < 30 V	*U* ≥ 30 V
A	0,33	0,25	0,17
B	1	0,75	0,50
C	1	1	1

Annex E
(normative)
Certification requirements for torches

This annex shall only be applied when torches, handlamps and caplights are being considered. A filament bulb shall not be considered as an infallible resistor in any other application. The cold resistance of the filament bulb may be taken into account in determining the maximum circuit current for the assessment of the intrinsic safety of the circuit provided that the mechanical protection of the bulb is provided by another method of protection, for example increased safety in accordance with EN 50019, the intrinsically safe circuit is limited to level of protection “ib” only,the supply voltage is limited to 10 V maximum, the construction of the bulb is such that it is not considered possible for its resistance to be reduced by bridging the filament with a lower resistance connection if the filament becomes disconnected. The internal construction of the bulb need not conform to Table 4, however, the external creepage and clearance distances on the bulb and associated holder shall meet Table 4, the cold resistance of the bulb is measured at the lowest ambient temperature for which the apparatus is rated,the replacement of the correct type of bulb must be ensured, for example by the lamp fitting or appropriate marking, assessment of the intrinsic safety of the circuit shall take into account the total inductance of the circuit.

RU Russia

**IEC Clause
or Annex
Number**

Details of Differences IEC 60079-11-99 (“i”)

5.2 To word the first paragraph as follows:

With Um and Ui applied, the intrinsically safe circuits of *ia* level shall not be capable of causing ignition of explosive mixture by thermal action under the test conditions stipulated in the present standard, and by sparking with probability over 10-3 in any of the following cases:…

a)during normal operation and introduction of all unaccountable faults creating most hazardous conditions;

a) during normal operation and introduction of one accountable and all unaccountable faults creating most hazardous conditions;

b) during normal operation and introduction of two accountable and all unaccountable faults creating most hazardous conditions.
Note: In each of the above cases unaccountable faults can be different.

5.2. New text is added:

In testing or assessing the circuits for intrinsic safety a safety factor equal to 1.5 shall be applied to voltage and current or their combination for sparking device of Type I and that equal to 2 for sparking devices of Type II and Type III in conformity with 10.4.2.

10.4.1. In each test mode at least 16000 accountable circuit closures and interruptions are made which are most hazard for the circuit being tested and can be realized by the present sparking device. The circuit is considered as intrinsically safe if after 16000 closures and interruptions and at the specified intrinsic safety factor the probability of ignition does not exceed 10^{-3} .On testing direct-current (rectified current) circuits the polarity of the supply source across the contacts of sparking device should change every 8000 closures and interruptions of the circuit.

When carrying out tests the measures should be taken so that the current in inductive circuits with closed contacts has a steady-state value, and the capacitor has enough time for recharging. The corresponding recommendations for standard sparing devices are given in Appendix B.

During the tests and after their completion the correct operation of sparking device should be checked by calibration. For a sparking device of Type I calibration is performed every 1000 rotations of the wire holder and after completion of tests. The correct operation of sparking devices of Types II and III should be checked in accordance with the instructions contained in Appendix B. If the calibration does not meet the requirements of p.10.3 the intrinsic safety tests of the circuit should be declared invalid.

Appendix B

(Paragraphs as per GOST R 51330.10-99)
B.1.3 Sparking device of Type II

B.1.3.1 Construction

B.1.3.1.1 Sparking device of Type II (Figure B.9) consists of a rotating metal disc with 10 zinc-plated steel wires of 0.4 mm diameter located on its periphery. 0.25-0.3 mm thick steel sawing with a tooth height of

0.4-0.5 mm serves as fixed electrode. Free length of a wire is ~ 25 mm. The sawing is fixed rigidly at two points on a special clamp. The distance between the fixing points is ~ 70 mm, the rounding-off radius of the sawing is ~ 100 mm.

B.1.3.1.2 On counting the number of sparks every contact with the sawing is considered as one spark. Rotational speed of moving contacts is about 40-60 rev/min.

B.1.3.2 Calibration of sparking device

B.1.3.2.1 Sensitivity of sparking device should be checked before, during and after every test run in accordance with p.10.3 and p. 10.4

B.1.3.2.2 The correct operation of sparking device of Type II is checked by connecting it to the pilot direct-current circuit every 4000 closures and interruptions of the test circuit. The sparking device is considered to be adjusted correctly if the probability of ignition of representative explosive mixture (preliminary activated) is at least 0.05.

B.1.3.2.3 The parameters of the pilot circuit for a sparking device of Type II are analogous to those of the pilot circuit for a sparking device of Type I specified in p. 10.3.

B.1.3.3 Field of application of sparking device

B.1.3.3.1 Sparking device of Type II is intended for intrinsical safety test of inductive and capacitive circuits and also other combined circuits, if it reproduces most hazardous discharge conditions for them. The sparking device of Type II is used to test electrical circuits with currents over 3 A, where the sparking device of Type I *cannot be used.*

The test circuits should have the following limiting values of parameters:
a) test current - not more than 10 A;
b) working voltage- not more than 1000 V;
c) circuit inductance – not more than 1 H;
d)current frequency – not more than 1.5 MHz.

Notes:
1 Sparking device of Type II does not provide low speed opening of contacts and cannot be used for intrinsical safety test of ohmic circuits.

2 During the tests the measures should be taken so that the current in inductive circuits has a steady-state value when the contacts are closed, and the capacitor completely discharges in the period when the contacts are open.

B.1.3.3.2 When a sparking device of Type II is used to test circuits having parameters that exceed the limits specified in B.1.3.3.1, its sensitivity should be checked and, if necessary, special measures should be taken so as to restore the sensitivity or take into account its change in the test results.

Notes
1 If the test current exceeds 5A, heating of contacts can become an additional cause of ignition rendering the test results not authentic.

2 Capacitive and inductive circuits with high time constants can be tested by reduction of rotational speed of the sparking device. If a sparking device of Type II is used to test capacitive circuits, constant contact of several wires with the disc of the sawing should be excluded. For example, a certain number of wires may be mounted so that the complete discharge of the capacitor could take place in the intervals between the series of sparks. On evaluating capacitive circuits every contact of a wire with the sawing disc is recorded as one

spark. At the same time, it is necessary to take account of the fact that the reduction of rotational speed of a sparking device can change its sensitivity.

B.1.4 Sparking device of Type III
B.1.4.1 Construction

B.1.4.1.1 Sparking device of Type III (Figure B.10) consists of two pairs of rollers. The rollers of each pair are pressed against each other. Rotational speed of the upper pair of rollers is four times as low as that of the lower pair of rollers.

B.1.4.1.2 Copper tinned wire of 0.26 mm diameter is fed from the drum and the upper pair of rollers to the lower pair of rollers through a small glass flask washed by explosive mixture. After leaving the small flask the wire is caught by the lower pair of rollers and, because of the difference in rotational speed of the upper and the lower rollers, it tears in the flask.

B.1.4.1.3 The test circuit is connected to the lower and the upper pair of rollers. The circuit closing occurs outside the flask at the moment of the contact with the lower pair of rollers. The circuit interruption occurs in explosive mixture at the moment of the wire breaking in the flask. The speed of contacts breaking varies from 0.2 to 3.0 m/s. The frequency of sparking is from 1 to 30 times a second. On contacts breaking at a rate of 0.6 m/s the time when the circuit is closed is about 10 ms, and the time when the circuit is open is about 114 ms.

B.1. 4.1.4 The resistance of two pairs of rollers with a wire gripped between them before it breaks is not more than 0.03 Ohm. The flow rate of explosive mixture is set at 2 - 5 cm^3/s.

B.1.4.2 Calibration of sparking device

B.1.4.2.1 1 Sensitivity of sparking device should be checked before, during and after every test run in accordance with p.10.3 and p. 10.4

B.1.4.2.2 The correct operation of sparking device of Type III is checked by connecting it to the pilot direct-current circuit every 4000 closures and interruptions of the test circuit. The sparking device is considered to be adjusted correctly if the probability of ignition of representative (appropriate activated test explosive mixture) explosive mixture is at least 0.05.

B.1.4.2.3 The parameters of the pilot circuit for a sparking device of Type III are analogous to those of the pilot circuit for a sparking device of Type I specified in p. 10.3.

B.1.4.3 Field of application of sparking device
B.1.4.3.1 Sparking device of Type III is intended for intrinsical safety test of inductive and capacitive circuits and also other combined circuits, if it reproduces most hazardous discharge conditions for them. Sparking device of Type III is used to test electrical circuits with currents over 10 A, where sparking devices of Type I and Type II cannot be used.

The test circuits should have the following limiting values of parameters:

a)switched current of the test circuit should not exceed the values at which the inductance of the sparking device and connecting wires starts exerting influence on the results of tests;

b) working voltage – not more than 1000 V;

c) inductance – not more than 1 H;

d) current frequency - not more than 1.5 Mhz.

Notes:

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1 Sparking device of Type III does not provide low speed opening of contacts and cannot be used for intrinsical safety test of ohmic circuits.

2 During the tests the measures should be taken so that the current in inductive circuits has a steady-state value when the contacts are closed.

B.1. 4.3.2 When a sparking device of Type III is used to test circuits having parameters that exceed the limits specified in B.1.8.3.1, its sensitivity should be checked and, if necessary, special measures should be taken so as to restore the sensitivity or take into account its change in the test results.

Note - At high currents in the test circuit heating of contacts can become an additional cause of ignition rendering the test results not valid

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	General	The U.S. version of IEC 60079-11 does not include requirements for Group I electrical apparatus.
IEC 60079-11 (1999)	General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 60079-11 (1999)	1.2	23.4.7.1 to 23.4.7.5 and 23.4.7.7 Tests on non-metallic enclosures Yes Yes
IEC 60079-11 (1999)	2	UL248-1:2000, Low Voltage Fuses – Part 1: General Requirements
IEC 60079-11 (1999)	3.5	3.5 normal operation] Operation of intrinsically safe apparatus or associated apparatus such that it conforms electrically and mechanically with the design specification produced by its manufacturer. ~~NOTE — Normal operation includes the application of U_m (see 3.15) from unspecified apparatus.~~
IEC 60079-11 (1999)	3.6	3.6 fault Any defect of any component, separation, insulation, or connection between components, not defined as infallible by this standard, upon which the intrinsic safety of a circuit depends. NOTE 1 — If a fault leads to consequential faults in other parts upon which the type of protection of the circuit depends, the primary and subsequent failures are considered to be a single fault. NOTE 2 — For the application of such faults, see 5.2 and 5.3.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	3.7	3.7 countable fault Fault which occurs in parts of <u>intrinsically safe apparatus or associated apparatus</u> ~~electrical apparatus~~ conforming to the constructional requirements of this standard.
IEC 60079-11 (1999)	3.8	3.8 non-countable fault Fault that occurs in parts of <u>intrinsically safe apparatus or associated apparatus</u> ~~electrical apparatus~~ not conforming to the constructional requirements of this standard.
IEC 60079-11 (1999)	3.9	3.9 infallible component or infallible assembly of components Component or assembly of components that is considered as not subject to certain fault modes as specified in this standard. The probability of such fault modes occurring in service or storage is considered to be so low that they are not to be taken into account. <u>NOTE — Such a component or assembly is considered as not subject to fault when assessments or tests for the type of protection are made.</u>
IEC 60079-11 (1999)	3.11	3.11 simple apparatus ~~An electrical component or combination of components of simple construction with well-defined electrical parameters which is compatible with the intrinsic safety of the circuit in which it is used.~~ <u>Passive components such as light-emitting diodes (LEDs), resistors, resistance temperature detectors (RTDs), and switches. Themocouples, and similar sources of generated energy that will not generate more than 1.5 volts, 0.1 ampere, and 25 milliwatts (NFPA 70-National Electrical Code 2002, 504-2)</u>

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	3.12	3.12 internal wiring Wiring and electrical connections that are made within the intrinsically safe apparatus or associated apparatus ~~apparatus~~ by its manufacturer. NOTE — Within a rack or panel, interconnections between separate pieces of apparatus made in accordance with detailed instructions from the manufacturer are considered to be internal wiring.
IEC 60079-11 (1999)	3.16	3.16 maximum input voltage (U_i or V_{max})
IEC 60079-11 (1999)	3.17	3.17 maximum output voltage (U_o or V_{oc}) Maximum output voltage (peak a.c. or d.c.) in an intrinsically safe circuit that can appear under open circuit conditions at the connection facilities of the apparatus at any applied voltage up to the maximum voltage, including U_m and U_i (or V_{max}).
IEC 60079-11 (1999)	3.18	3.18 maximum input current (I_i or I_{max})
IEC 60079-11 (1999)	3.19	3.19 maximum output current (I_o or I_{SC})
IEC 60079-11 (1999)	3.22	3.22 maximum external capacitance (C_o or C_a) Maximum capacitance in an intrinsically safe circuit that can be connected to the connection facilities of the associated apparatus without invalidating intrinsic safety.
IEC 60079-11 (1999)	3.23	3.23 maximum internal capacitance (C_i) Total equivalent internal capacitance of the intrinsically safe apparatus that is considered as appearing across the connection facilities of the apparatus.
IEC 60079-11 (1999)	3.24	32.4 maximum external inductance (L_o or L_a) Maximum value of inductance in an intrinsically safe circuit that can be connected to the connection facilities of the associated apparatus without invalidating intrinsic safety.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	3.25	3.25 maximum internal inductance (L_i) Total equivalent internal inductance of the intrinsically safe apparatus which is considered as appearing at the connection facilities of the apparatus.
IEC 60079-11 (1999)	3.26	3.26 maximum external inductance to resistance ratio (L_o/R_o or L_a/R_a) Ratio of inductance (L_o or L_a) to resistance (R_o or R_a) of any external circuit that can be connected to the connection facilities of the associated ~~electrical~~ apparatus without invalidating intrinsic safety.
IEC 60079-11 (1999)	3.27	3.27 maximum internal inductance to resistance ratio (L_i/R_i) Ratio of inductance (L_i) to resistance (R_i) which is considered as appearing at the external connection facilities of the intrinsically safe ~~electrical~~ apparatus.
IEC 60079-11 (1999)	3.31	3.31 creepage distance in air Shortest distance along the surface of an insulating medium in contact with air between two conductive parts. NOTE — The distance concerned here is, for example, applicable to printed circuits that have no coating in accordance with 6.4.8, where the insulation across which the creepage distance is measured is in direct contact with the air.
IEC 60079-11 (1999)	3.37 NEW	3.37 intrinsically safe system assembly of interconnected items of apparatus that may comprise intrinsically safe apparatus, associated apparatus and other apparatus, and interconnecting cables in which the circuits within those parts of the system that may be exposed to explosive gas atmospheres are intrinsically safe circuits.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	3.38 NEW	3.38 apparatus and systems of category 'ia' electrical apparatus and systems containing intrinsically safe circuits that are incapable of causing ignition, with the appropriate safety factor: when up to two countable faults are applied and, in addition, those non-countable faults that give the most onerous condition.
IEC 60079-11 (1999)	3.39 NEW	3.39 apparatus and systems of category 'ib' electrical apparatus and systems containing intrinsically safe circuits that are incapable of causing ignition with the appropriate safety factors: when up to one countable fault is applied and, in addition, those non-countable faults that give the most onerous condition.
IEC 60079-11 (1999)	3.40 NEW	3.40 infallible connections connections, including joints and interconnecting wiring, that are not considered as becoming open-circuited in service or storage.
IEC 60079-11 (1999)	3.41 NEW	3.41 control drawing a drawing or other document provided by the manufacturer of the intrinsically safe or associated apparatus that details the allowed interconnections between other circuits or apparatus. If the intrinsically safe or associated apparatus is investigated under the entity concept, the control drawing shall include the applicable electrical parameters to permit selection of apparatus for interconnection.
IEC 60079-11 (1999)	3.42 NEW	3.42 entity evaluation a method used to determine acceptable combinations of intrinsically safe apparatus and associated apparatus that have not been investigated in such combinations.
IEC 60079-11 (1999)	3.43 NEW	3.43 maximum output current - multiple channel apparatus (I_t) the maximum dc or peak ac current that can be extracted from any combination of terminals of a multiple channel associated apparatus.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	3.44 NEW	3.44 maximum output voltage - multiple channel apparatus (V_t or U_t) the maximum dc or peak ac open circuit voltage that can appear across any combination of terminals of a multiple channel associated apparatus.
IEC 60079-11 (1999)	3.45 NEW	3.45 source resistance (R_s) the minimum resistance of the power source that defines the output short-circuit current I_o.
IEC 60079-11 (1999)	3.46 NEW	3.46 source capacitance (C_s) the capacitance appearing at the output terminals of the power source.
IEC 60079-11 (1999)	3.47 NEW	3.47 source inductance (L_s) the inductance appearing at the output terminals of the power source.
IEC 60079-11 (1999)	3.48 NEW	3.48 live maintenance maintenance activities other than normal operation that are allowed by the manufacturer's documentation, such as calibration or programming, which are carried out while the intrinsically safe apparatus and circuits are energized.
IEC 60079-11 (1999)	3.49 NEW	3.49 maximum output power multiple channel apparatus (Pt) the maximum electrical power that can be taken from any combination of terminals of a multiple-channel associated apparatus.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	5.2	**Category 'ia'** With U_m and U_i applied, the intrinsically safe circuits in electrical apparatus of category 'ia' shall not be capable of causing ignition in each of the following circumstances: a) in normal operation and with the application of those non-countable faults that give the most onerous condition; b) in normal operation and with the application of one countable fault plus those non-countable faults that give the most onerous condition; in normal operation and with the application of two countable faults plus those non-countable faults that give the most onerous condition. The non-countable faults applied may differ in each of the above circumstances. In testing or assessing the circuits for spark ignition, the following safety factors shall be applied in accordance with 10.4.2: for both a) and b) 1.5 for c) 1.0 The safety factor applied to voltage or current for determination of surface temperature classification shall be 1.0 in all cases. If only one countable fault can occur, the requirements of b) are considered to give a category of 'ia' if the test requirements for 'ia' can then be satisfied. If no countable faults can occur, the requirements of a) are considered to give a category of 'ia' if the test requirements for 'ia' can then be satisfied. <u>In the case of systems, the countable faults specified will be applied to the system as a whole and not simultaneously to each item of electrical apparatus in the system. For example, a category 'ia' system comprising two pieces of apparatus will have up to two countable faults applied, not four.</u>

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	5.3	**Category 'ib'** With U_m and U_i applied, the intrinsically safe circuits in electrical apparatus of category 'ib' shall not be capable of causing ignition in each of the following circumstances: in normal operation and with the application of those non-countable faults which give the most onerous condition; in normal operation and with the application of one countable fault plus the application of those non-countable faults which give the most onerous condition. The non-countable faults applied may differ in each of the above circumstances. In testing or assessing the circuits for spark ignition, a safety factor of 1.5 shall be applied in accordance with 10.4.2. The safety factor applied to the voltage or current for the determination of surface temperature classification shall be 1.0 in all cases. If no countable fault can occur, the requirements of a) are considered to give a category of 'ib' if the test requirements for 'ib' can be satisfied. NOTE — Guidance on the assessment of intrinsically safe circuits for spark ignition is contained in annex A. Details of the spark test apparatus are given in annex B. In the case of systems, the countable fault specified shall be applied to the system as a whole and not simultaneously to each item of electrical apparatus in the system. For example, a category 'ib' system comprising two pieces of apparatus will have only one countable fault applied, not two.
IEC 60079-11 (1999)	5.4	**Simple apparatus** The following apparatus shall be considered to be simple apparatus: passive components, for example switches, junction boxes, resistors, RTDs and ~~simple semiconductor devices~~ LEDs; ~~sources of stored energy with well defined parameters, for example capacitors or inductors, whose values shall be considered when determining the overall safety of the system;~~ sources of generated energy, for example thermocouples and photocells, which do not generate

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
		more than 1.5 V, 100 mA and 25 mW. ~~Any inductance or capacitance present in these sources of energy shall be considered as in b).~~ Simple apparatus shall conform to all relevant requirements of this standard but need not be approved ~~certified~~ and need not comply with clause 12. In particular, the following aspects shall always be considered. simple apparatus shall not achieve safety by the inclusion of voltage and/or current limiting and/or suppression devices; simple apparatus shall not contain any means of increasing the available voltage or current, for example, circuits for the generation of ancillary power supplies; where it is necessary that the simple apparatus maintains the integrity of the isolation from 'earth' of the intrinsically-safe circuit, it shall be capable of withstanding the test voltage to earth in accordance with 6.4.12. Its terminals shall conform to 6.3.1; nonmetallic enclosures and enclosures containing light metals when located in the hazardous (classified) location ~~area~~ shall conform to 7.3 and 8.1 of IEC 60079-0 1998; when simple apparatus is located in the hazardous (classified) location ~~area~~, it shall be temperature classified. When used in an intrinsically safe circuit within their normal rating and at a maximum ambient temperature of 40°C, switches, plugs and sockets, and terminals are allocated a T6 temperature classification for Group II. ~~applications and considered as having a maximum surface temperature of 85°C for Group I applications. Other types of simple apparatus shall be temperature classified in accordance with clauses 4 and 6 of this standard~~. Where simple apparatus forms part of an apparatus containing other electrical circuits, the whole shall be ~~certified~~ approved. NOTE — Sensors that utilize catalytic reaction or other electro-chemical mechanisms are not normally simple apparatus. Specialist advice on their application should be sought.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	6.2.5 NEW	6.2.5 Fusion-sealed components If components (other than those complying with 6.2.4 and incandescent lamps) have envelopes sealed by fusion, and the envelope or surrounding enclosure can withstand the impact test in 23.4.3.1 of IEC 60079-0 1998 without the seal being broken, the relevant temperature is the maximum surface temperature measured on the envelope. NOTE — This requirement is not intended to apply to components, such as transistors and resistors, for which the temperature is always measured on the outer surface.
IEC 60079-11 (1999)	6.3.3	$\cancel{\frac{L_0}{R_0} = \frac{8eR_i + (64e^2R_i^2 - 72U_o^2\, e\, L_i)^{1/2}}{4.5\, U_o^2}}$ ~~H/Ω~~ $\frac{L_0}{R_0} = \frac{8e\,R_s + (64e^2R_s^2 - 72U_o^2\, e\, L_s)^{1/2}}{4.5\, U_o^2}$ H/Ω where *e* is the minimum spark-test apparatus ignition energy in joules, and is for: ~~Group I apparatus: 525 μJ~~ Group IIA apparatus: 320 μJ Group IIB apparatus: 160 μJ Group IIC apparatus: 40 μJ ~~R_i~~ R_s is the minimum output resistance of the power source, in ohms; U_o is the maximum open circuit voltage, in volts; ~~L_i~~ L_s is the maximum inductance present at the power source terminals, in henries. If ~~L_i~~ L_s = 0 then $\cancel{\frac{L}{R} = \frac{32eR_i}{9U_o^2}}$ ~~H/Ω~~ $\frac{L_o}{R_o} = \frac{32\,e\,R_S}{9U_o^2}$ H/Ω

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	6.4.8	The method used for coating the board shall be specified in the ~~certification~~ manufacturer's documentation. Where the coating is considered adequate to prevent conductive parts, for example soldered joints and component leads, protruding through the coating this shall be stated in the documentation and confirmed by examination.
IEC 60079-11 (1999)	6.4.12	Either the current flowing during the test shall not increase above that which is expected from the design of the circuit and shall not exceed 5 mA r.m.s. at any time, or where the circuit does not satisfy this requirement, the manufacturer's documentation shall make note of this.~~apparatus shall be marked with the symbol X~~.
IEC 60079-11 (1999)	6.6	**6.5 Earth (Ground) conductors, connections, and terminals** Where earthing, for example, of enclosures, conductors, metal screens, printed wiring board conductors, segregation contacts of plug-in connectors and diode safety barriers, is required to maintain the type of protection, the cross-sectional area of any conductors, connectors and terminals used for this purpose shall be such that they are rated to carry the maximum possible current to which they could be continuously subjected under the conditions specified in clause 5. Components shall also conform to clause 7. Where a connector carries ~~carrier's~~ earthed circuits and the type of protection depends on the earthed circuit, the connector shall comprise at least three independent connecting elements for 'ia' circuits and at least two for 'ib' circuits (see figure 2). These elements shall be connected in parallel. Where the connector can be removed at an angle, one connection shall be present at, or near to, each end of the connector. Terminals shall be fixed in their mountings without possibility of self-loosening, and shall be constructed so that the conductors cannot slip out from their intended location. Proper contact shall be assured without deterioration of the conductors, even if multi-stranded cores are used in terminals that are intended for direct clamping of the cores. The contact made by a terminal shall not be appreciably impaired by temperature changes in normal service. Terminals that are intended for clamping stranded cores shall include a resilient intermediate part. Terminals for conductors of cross-sections up to 4 mm² shall also be suitable for the effective connection of conductors having a smaller cross-section. Terminals that comply with the requirements of IEC 60079-7 are considered to conform to these requirements. The following shall not be used: a) terminals with sharp edges, which could damage the conductors;

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
		b) terminals that may turn, be twisted, or be permanently deformed by normal tightening; c) insulating materials that transmit contact pressure in terminals; d) Soldering lugs, push-in conductors, quick connects, or similar friction-fit connectors.
IEC 60079-11 (1999)	7.3	Fuses shall have a rated voltage of at least U_m (or U_i in intrinsically safe apparatus and circuits) ~~although~~ and the external creepage distances and clearances ~~they do not have to~~ shall conform to table 4. General industrial standards for the construction of fuses and fuseholders shall be applied, and their method of mounting shall not reduce the clearances, creepage distances, and separations afforded by the fuse and its holder. NOTE— Microfuses conforming to IEC 60127 are acceptable. A fuse shall be capable of interrupting the maximum prospective current of the circuit in which it is installed. For mains electricity supply systems not exceeding 250 V a.c r.m.s.., the prospective current shall normally be considered to be 1500 A a.c. The breaking capacity of the fuse is determined according to IEC 60127 or ~~an equivalent standard~~ UL-248-1.
IEC 60079-11 (1999)	7.4.7	If the cell or battery requiring current limiting devices to ensure the safety of the battery itself is not intended to be replaced in the hazardous (classified) location ~~area~~, it either shall be protected in accordance with 7.4.6 or alternatively it may be housed in a compartment capable of being opened or removed only with the aid of a tool ~~with special fasteners, for example those specified by IEC 60079-0~~. It shall also conform to the following: c) the apparatus shall have a warning label against changing the battery in the potentially explosive atmosphere, for example 'WARNING: DO NOT REMOVE BATTERY IN A POTENTIALLY EXPLOSIVE ATMOSPHERE'.
IEC 60079-11 (1999)	7.6	Apparatus shall be subject to special investigation when the manufacturer specifies live maintenance that requires access to the internal equipment. NOTE: Special investigation may include introduction of the spark test apparatus across distances complying with Table 4. Also, see markings , section 12.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	8.1.2	Fuses, fuseholders, circuit-breakers, and thermal devices shall conform to an appropriate recognized standard. ~~Conformance with the recognized standard shall not be verified by the certification body.~~
IEC 60079-11 (1999)	8.1.4	The test shall continue for at least 6 h or up to when the non-resetting thermal trip operates. ~~When a self-resetting thermal trip is used, the test period shall be extended to at least 12 h~~.
IEC 60079-11 (1999)	8.7	b) for printed board tracks: 1. where two tracks of minimum width 1 mm are in parallel; or 2. where a single track is at least 2 mm wide or has a width of 1% of its length, whichever is greater; and 3. where each track is formed from copper cladding having a nominal thickness of not less than 35 µm; NOTE — Infallible printed board tracks of material other than copper are permitted but will require special consideration.
IEC 60079-11 (1999)	10.11	Where it is necessary to protect the apparatus from external physical impact in order to prevent the impact energy exceeding these values, details of the requirements shall be specified by marking in accordance with 27.2 i) of IEC 60079-0 1998 or by adding the information to the Control Drawing. ~~as special conditions for safe use and the apparatus shall be marked with the sign X.~~

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	10.15 NEW	**10.15 Lamp Breakage Test** 10.15.1 The heated filament of a tungsten-filament lamp shall not ignite a surrounding explosive atmosphere when the glass envelope is broken. 10.15.2 The lamp is to be mounted in a test chamber connected to a power supply adjusted to deliver normal voltage at which the lamp will be operating in the product. Additional consideration is to be given to raising the voltage to the lamp to the maximum fault voltage that can be applied to the lamp. 10.15.3 The test chamber is to be filled with an explosive mixture as specified in Section10.2, or a test gas mixture having the lowest ignition temperature (see Section 10.7), whichever is more easily ignitable, in accordance with applicable fault procedures in Section 5. The lamp envelope is to be broken quickly and completely (not just cracked) while the lamp is immersed in the in the mixture to expose the glowing filament to the explosive mixture. The filament is not to be broken by the breakage of the envelope. Six samples are to be tested. Note – If the glass envelope cannot be broken inside the test chamber without also breaking the filament, such as very small lamp, the glass envelope may be broken outside the test chamber and the lamp with the broken or removed glass envelope placed in the test chamber for the tests.
IEC 60079-11 (1999)	11.2	The test voltage shall be applied for a period of at least 60 s. Alternatively the voltage tests specified in Table 9 may be conducted at 1.2 times the test voltage, but with a duration of at least 1 s.
IEC 60079-11 (1999)	12.1	Intrinsically safe apparatus and associated apparatus shall carry at least the minimum marking specified in IEC 60079-0 1998 and a reference to the Control Drawing. The marking of the serial number may be achieved by using a date or batch number code, which is sufficient to ensure traceability for quality-control purposes. NOTE — The serial number marking may be separate from the other marking. For associated apparatus the symbol AEx ia or AEx ib (or ia or ib if AEx is already marked) shall be enclosed in square brackets. All relevant parameters should be marked, for example U_m, L_i, C_i, L_o, C_o, wherever practicable.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
		NOTE — Standard symbols for marking and documentation are given in clause 3. Practical considerations may restrict or preclude the use of italic characters or of subscripts, and a simplified presentation may be used, for example, U0 rather than U_0. For voltage marking, the letter V may replace U. Where the manufacturer does not specify live maintenance procedures, the word "WARNING" and the following: "To prevent ignition of explosive atmospheres, disconnect power before servicing" Where the manufacturer specifies and provides live maintenance procedures, the word "WARNING" and the following: "To prevent ignition of explosive atmospheres, read understand and adhere to the manufacturer's live maintenance procedures." Alternatively the above warnings may be in the manufacturers instruction manual or on the Control Drawing. Where cautionary markings are required on the apparatus the text as described, following the word “CAUTION” or “WARNING” may be replaced by technically equivalent text. In addition to the marking requirements of 27.6 of IEC 60079-0 1998 for small electrical apparatus, the control drawing shall be marked on the apparatus or on the smallest unit packaging.

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US United States of America National differences to IEC 60079-11 (1999)		
IEC Standard	**Clause**	**Difference**
IEC 60079-11 (1999)	12.3	**12.1 Control drawings** A control drawing shall be provided for all intrinsically safe apparatus or associated apparatus that requires connection to other circuits or apparatus. An intrinsically safe system could consist of apparatus investigated as a system or apparatus investigated under the entity concept. If the intrinsically safe and associated apparatus are investigated as a system, the control drawing shall provide information for proper connection and installation. If the intrinsically safe or associated apparatus is investigated under the entity concept, the control drawing shall include applicable electrical parameters to permit selection of apparatus for interconnection. The control drawing shall contain notes to explain the following if applicable: a) The polarity requirements for associated apparatus. b) That associated apparatus must not be connected in parallel unless this is permitted by the associated apparatus approval. c) How to calculate the allowed capacitance and inductance values for the field wiring used in the intrinsically safe circuit. d) The hazardous (classified) locations in which the apparatus may be located. e) Permissible connections to simple apparatus. The control drawing for the apparatus investigated under the entity evaluation concept shall provide the following information: Either the ISA Marking, the IEC marking, or both, as shown in Table 12.3, may be used to designate circuit parameters on the apparatus and in the installation documents and control drawing. Table 12.3 is added material

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IEC Standard IEC 60079-11: 3rd Edition 1991 Electrical apparatus for explosive gas atmospheres - Part 11: Intrinsic safety "i"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**N/R Refer to IEC 60079-11:1999-02 (Fourth Edition)**	
CH Switzerland	**YES**		EN 50020 1994
CN China*		**TBA**	
CZ Czech Republic	**YES**		EN 50020 1994
DE Germany	**YES**		EN 50020 1994
DK Denmark	**YES**		EN 50020 1994
FI Finland	**YES**		EN 50020 1994
FR France	**YES**		EN 50020 1994
GB United Kingdom	**YES**		EN 50020 1994
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 50020 1994
IN India		**N/R**	
IT Italy	**YES**		EN 50020 1994
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998 65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 50020 1994
NO Norway	**YES**		EN 50020 1994
NZ New Zealand		**NO**	
PL Poland	**N/A**		
RO Romania	**YES**		EN 50020 1994
RU Russia		**N/R**	
SE Sweden	**YES**		EN 50020 1994
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 50020 1994
TR Turkey*		**TBA**	
US United States		N/R Refer to IEC 60079-11:1999-02 (4th Edition)	
ZA South Africa		**NO**	SANS 60079-11:1991

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

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Countries that have declared adoption of IEC 60079-11 3rd Edition 1991 Without National Differences
AU Australia
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-11 3rd Edition 1991 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number Details of Differences

	[ ] References in square bracket refer to clauses and figures in EN References without square brackets refer to IEC 79-11
6.1.3.1 Table 1	[Table 1] - Temperature classification of copper wiring

Diameter (see note 4)	Cross-sectional area (see note 4)	Maximum permissible current for temperature classification		
		T1-T4 & Group I	T5	T6
.. mm	mm²	A	A	A
0,035	0,000962	0,53	0,48	0,43
0,05	0,00196	1,04	0,93	0,84
0,1	0,00785	2,1	1,9	1,7
0,2	0,0314	3,7	3,3	3,0
0,35	0,0962	6,4	5,6	5,0
0,5	0,196	7,7	6,9	6,7

NOTE 1: The value given for maximum permissible current is the r.m.s. a.c. or d.c. value.

NOTE 2: For stranded conductors the cross-sectional area is taken as the total area of all strands of the conductor.

NOTE 3: The table also applies to flexible flat conductors, such as in ribbon cable, but not to printed circuit conductors for which see 6.1.3.1 [6.2.2.]

NOTE 4: Diameter and cross sectional area are the nominal dimensions specified by the wire manufacturer.

NOTE 5: Where P_i does not exceed 1,3 W the wiring can be awarded

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<table>
<tr><td></td><td>¦ a temperature classification of T4 and is acceptable for
¦ Group I.
The maximum current in insulated wiring shall not exceed the rating specified by the manufacturer of the wire.</td></tr>
<tr><td>6.1.3.2</td><td>Add the following requirements:
[6.2.3] <u>Printed circuit wiring</u>
On printed circuit boards of at least 0,5 mm thickness, having a conducting track of at least 35 μm thickness on one or both sides, a temperature classification T4 or Group I shall be given to the printed tracks if they have a minimum width of 0,3 mm and the continuous current in the tracks does not exceed 0,518 A. Similarly for minimum track widths of 0,5 mm, 1,0 mm and 2,0 mm, T4 shall be given for corresponding maximum currents of 0,814 A, 1,388 A and 2,222 A respectively. Track lengths of 10 mm or less shall be disregarded for temperature classification purposes.
For other applications, the temperature classification of copper wiring of printed boards shall be determined from table 2.

Manufacturing tolerances shall not reduce the values stated in this clause by more than 10% or 1mm whichever is the smaller.

Where P_i does not exceed 1,3 W, the printed wiring shall be given a temperature classification of T4 or Group I.</td></tr>
<tr><td>6.1.3.3</td><td>Replace by:
[6.2.4] <u>Small components</u>
Small components e.g. transistors or resistors whose temperature exceeds that permitted for the temperature classification shall be acceptable providing that they conform to one of the following:

a) When tested in accordance with [10.7], small components shall not cause ignition of the flammable mixture and any deformation or deterioration by the higher temperature shall not impair this type of protection.

b) For T4 and Group I classification small components shall conform to table 3.

c) For T5 classification, the surface temperature of components with a surface area smaller than 10 cm^3 (excluding lead wires) shall not exceed 150 ¦C.

Table 3 - Assessment for T4 classification according to component size and ambient temperature<table><tr><td>Total surface area excluding lead wires</td><td>Requirement for T4 and Group I classification</td></tr><tr><td>< 20mm²</td><td>Surface temperature ≤ 275°C</td></tr><tr><td>≥ 20mm²</td><td>Power dissipation ≤ 1,3W*</td></tr><tr><td>≥ 20mm² < 10cm²</td><td>Surface temperature ≤ 200°C</td></tr><tr><td colspan="2">*Reduced to 1,2W with 60°C ambient temperature or 1,0W with 80°C ambient temperature</td></tr></table></td></tr>
</table>

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	For potentiometers the surface to be considered shall be that of the resistance element and not the external surface of the component. The mounting arrangement and heatsinking and cooling effect of the overall potentiometer construction shall be taken into consideration during the test. Temperature shall be measured on the track with that current which flows under conditions of 'ib' or 'ia' as appropriate. If this results in a resistance value of less than 10 % of the track resistance value the measurement shall be carried out at 10 % of the track resistance value.
6.1.3.4	not applicable
6.3.1	Add the note: Terminals for connection of external circuits to intrinsically safe apparatus and associated apparatus should be so arranged that components will not be damaged when making the connections Add the requirement: The clearances between bare conducting parts of terminals of separate intrinsically safe circuits shall be equal to or exceed the values given in table 4. In addition, the clearances between terminals shall be such that the clearances between the bare conducting parts of connected external conductors are at least 6 mm when measured in accordance with [figure 1]. Any possible movement of metallic parts which are not rigidly fixed shall be taken into account. The minimum clearance between the bare conducting parts of external conductors connected to terminals and earthed metal or other conducting parts shall be 3 mm unless the possible interconnection has been taken into account in the safety analysis.
6.3.2	Add the following requirements: Plugs and sockets used for connection of external intrinsically safe circuits shall be separate from and non-interchangeable with those for non-intrinsically safe circuits. Where a plug or a socket is not prefabricated with its wires, the connecting facilities shall conform to [6.3.1.] If however the connections require the use of a special tool, e.g. by crimping, such that there is no possibility of a strand of wire becoming free then the connection facilities need only conform to table 4.
6.4.1 Table 4	Change Table 4 as follows: Minimum CTI = 100 Changed order of lines [1] Voltage [2] Clearance [3] Separation distances through casting compound [4] Separation distances through solid insulation [5] Creepage distance in air [6] Creepage distance under coating [7] CTI
6.4.1	Replace by the following requirement: [6.4.1] <u>Separation of conductive parts</u> Separation of conductive parts between: - intrinsically safe and non-intrinsically safe circuits; or - different intrinsically safe circuits; or - a circuit and earthed or isolated metallic parts,

shall conform to the following if this type of protection depends on the separation. Separation distances shall be measured or assessed taking account of any possible movement of the conductors or conductive parts. Manufacturing tolerances shall not reduce the distances by more than 10 % or 1 mm whichever is the smaller.

If the separation conforms to table 4, it shall be considered as not subject to failure to a lower insulation resistance.

Smaller separation distances, which exceed one-third of the values specified in table 4, shall be considered as subject to countable short-circuit faults.

If separation distances are less than one-third of the values specified in table 4, they shall be considered as subject to non-countable short-circuit faults if this impairs intrinsic safety.

Separation requirements shall not apply where earthed metal e.g. printed wiring or a partition, separates an intrinsically safe circuit from other circuits, provided that breakdown to earth does not adversely affect the type of protection and that the earthed conductive part can carry the maximum current that would flow under fault conditions.

NOTE 1: For example the type of protection does depend on the separation to earthed or isolated metallic parts if a current limiting resistor can be by-passed by short circuits between the circuit and the earthed or isolated metallic part.

An earthed metal partition shall have strength and rigidity so that it is unlikely to be damaged and shall be of sufficient thickness and of sufficient current- carrying capacity to prevent burn-through or loss of earth under fault conditions. A partition either shall be at least 0,45 mm thick and attached to a rigid, earthed metal portion of the device or shall conform to [10.10.2] if of lesser thickness.

Where a non-metallic insulating partition having an appropriate CTI is placed between the conductive parts the clearances, creepage distances and other separation distances either shall be measured around the partition provided that the partition has a thickness of at least 0,9 mm or shall conform to [10.10.2] if of lesser thickness.

NOTE 2: Methods of assessment are given annex C.

[6.4.2] Voltage between conductive parts

The voltage which is taken into account when using table 4 shall be the voltage between any two conductive parts for which the separation has an effect on this type of protection of the circuit under consideration, that is for example (see figure 3) the voltage between an intrinsically safe circuit and:

- part of the same circuit which is not intrinsically safe; or
- non-intrinsically safe circuits; or
- other intrinsically safe circuits.

The value of voltage to be considered shall be either of the following as applicable:

a) For separate circuits: the highest numerical sum of the peak voltages derived from:

the rated voltages of the circuits; or
the maximum voltages specified by the manufacturer which may safely be supplied to the circuits; or

any voltages generated within the same apparatus.

Where one of the voltages is less than 20 % of the other, it shall be ignored. Mains supply voltages shall be taken without the addition of standard mains tolerances. For such sinusoidal voltages, peak voltage shall be considered to be:

$\sqrt{2}$ x r.m.s. value of the nominal voltage.

b) Between parts of a circuit: the maximum peak value of the voltage that can occur in either part of that circuit. This may be the sum of the voltages of different sources connected to that circuit. One of the voltages may be ignored if it is less than 20% of the other.

In all cases, voltages which arise during the fault conditions of clause 5 shall, where applicable be used to derive the maximum.

Any external voltage shall be assumed to have the value U_m or U_i declared for the connection facilities through which it enters. Transient voltages such as might exist before a protective device, e.g. a fuse, opens the circuit shall not be considered when evaluating the creepage distance, but shall be considered when evaluating clearances.

[6.4.3] Clearance

In measuring or assessing clearances between conductive parts, insulating partitions of less than 0,9 mm thickness, or which do not conform to [10.10.2], shall be ignored. Other insulating parts shall conform to [line 4] of table 4.
For voltages higher than 1575 V peak, an interposing insulating partition or earthed metal partition shall be used. In either case the partition shall conform to [6.4].

[6.4.4] Separation distances through and requirements of casting compound

The casting compound shall conform to the following:

a) it shall have a temperature rating, specified by the manufacturer of the casting compound or apparatus, which is at least equal to the maximum temperature achieved by any component under encapsulated conditions.

Alternatively higher temperatures than the rated casting compound temperature shall be accepted provided that they do not cause any damage to the casting compound that would adversely affect this type of protection.

b) have at its free surface a CTI value of at least that specified in table 4 if any bare conductive parts protrude from the casting compound.

Only hard material, e.g. epoxy resin shall have its free surface exposed and unprotected, thus forming part of the enclosure (see [figure D1]). It shall conform to [10.10.1].

c) be adherent to all conductive parts, components and substrates except when they are totally enclosed by the casting compound.

d) be specified by its generic name and type designation given by the manufacturer of the casting compound.

For intrinsically safe apparatus all circuits connected to the encapsulated conductive parts and/or components and/or bare parts protruding from the casting compound shall be intrinsically safe. Fault conditions within the casting compound shall be assessed but the possibility of spark ignition shall not be considered.

NOTE: If circuits connected to the encapsulated conductive parts and/or components and/or bare parts protruding the casting compound are not intrinsically safe, they shall be protected by some other appropriate technique, listed in EN 50014.

The minimum separation distance between encapsulated conductive parts and components, and the free surface of the casting compound, shall be at least half the values shown in [line 3] „Separation Distances through Casting Compound" of table 4 with a minimum separation distance of 1mm. When the casting compound is in direct contact with an enclosure of insulating material conforming to [line 4] of table 4 no other separation is required (see figure [D1]).

The insulation of the encapsulated circuit shall conform to [6.4.11].

The failure of a component which is encapsulated or hermetically sealed e.g. a semiconductor which is used in accordance with [7.1] and in which internal clearances and distances through encapsulant are not defined is to be considered as a single countable fault.

Further requirements are given in [annex D].

[6.4.5] <u>Separation distances through solid insulation</u>

Solid insulation is insulation which is extruded or moulded but not poured. It shall have an electrical strength which conforms to [6.4.12] when the separation distance is in accordance with table 4.

NOTE 1: If the insulator is fabricated from two or more pieces of electrical insulating material which are solidly bonded together, then the composite may be considered as solid.

NOTE 2: For the purposes of this standard, solid insulation is considered to be prefabricated, e.g. sheet or sleeving or elastomeric insulation on wiring.

NOTE 3: Varnish and similar coatings are not considered to be solid insulation.

[6.4.6] <u>Composite separations</u>

Where separations are composite e.g. through a combination of air and insulation, the total separation shall be calculated on the basis of referring all separations to one line of table 4.

For example at 60 V:

Clearance [line 2] = 6 x separation through solid insulation [line 4]

Clearance [line 2] = 3 x separation through casting compound [line 3]

Equivalent clearance = actual clearance + (3 x any additional separation through encapsulant) + (6 x any additional separation through solid insulation).

To be infallible the above result shall be not less than the clearance value specified in table 4.

Any clearance or separation which is below one third of the relevant value specified in table 4 shall be ignored for the purpose of calculation.

[6.4.7] Creepage distance in air

For the creepage distance in air specified in [line 5] of table 4, the insulating material shall conform to [line 7] of table 4 which specifies the minimum comparative tracking index (CTI) measured in accordance with IEC 112. The method of measuring or assessing these distances shall be in accordance with figure 4.

Where a joint is cemented then the cement shall have insulation properties, equivalent to those of the adjacent material.

Where the creepage distance is made up from the addition of shorter distances, e.g. where a conductive part is interposed, distances of less than one third the relevant value in line 5 of table 4 shall not be taken into account. For voltages higher than 1575V peak, an interposing insulating partition or earthed metallic partition shall be used. In either case the partition shall conform to [6.4.1].

[6.4.8] Creepage distance under coating

A conformal coating shall seal the conductors in question against ingress of moisture and shall give an effective lasting unbroken seal. It shall adhere to the conductive parts and to the insulating material. If the coating is applied by spraying, two separate coats shall be applied. Other methods of application require only one coat, for example: dip coating, brushing, vacuum impregnating. A solder mask shall be considered as one of the two coatings, provided it is not damaged during soldering.

Where bare conductors emerge from the coating the comparative tracking index (CTI) in [line 7] of table 4 shall apply to both insulation and coating.

NOTE: The concept of creepage distances under coating was developed for flat surfaces e.g. non-flexible printed circuit boards. Radical departures from this format require special consideration.

[6.4.9] Requirements for assembled printed circuit boards

Where creepage and clearance distances affect the intrinsic safety of the apparatus the printed circuit shall conform to the following (see [figure 5]).

a) When a printed circuit is covered by a conformal coating according to [6.4.8] the requirements of [6.4.3] and [6.4.7] shall apply only to any conductive parts which lie outside the coating, including, for example:

	- tracks which emerge from the coating; - the free surface of a printed circuit which is coated on one side only; - bare parts of components able to protrude through the coating. b) The requirements of [6.4.8] shall apply to circuits or parts of circuits and their fixed components when the coating covers the connecting pins, solder joints and the conductive parts of any components. [6.4.10] Separation by earth screens Where separation between circuits or parts of circuits is provided by a metallic screen, the screen, as well as any connection to it, shall be capable of carrying the maximum possible current to which it could be continuously subjected in accordance with clause 5. Where the connection is made through a connector, the connector shall be constructed in accordance with [6.6]
6.4.2	Replace by: [7.2] Connectors for internal connections, plug-in cards and components These connectors shall be designed in such a manner that an incorrect connection or interchangeability with other connectors in the same electrical apparatus is not possible unless it does not result in an unsafe condition or the connectors are identified in such a manner that incorrect connection is obvious. Where this type of protection depends on a connection, the failure to a high resistance or open circuit of a connection shall be a countable fault in accordance with clause 5. If a connector carries earthed circuits and this type of protection depends on the earth connection, then the connector shall be constructed in accordance with [6.6].
6.4.3	not applicable
6.4.4	Replace by: [6.4.13] Relays Where the coil of a relay is connected to an intrinsically safe circuit the contacts in normal operation shall not exceed their manufacturers rating and shall not switch more than 5 A r.m.s. or 250 V r.m.s. or 100 VA. When the values switched by the contacts do not exceed 10A or 500VA, the values for creepage distance and clearance from table 4 for the relevant voltage shall be doubled. For higher values, intrinsically safe circuits and non-intrinsically safe circuits shall be connected to the same relay only if they are separated by an earthed metal barrier or an insulating barrier conforming to [6.4.1]. The dimensions of such an insulating barrier shall take into account the ionisation arising from operation of the relay which would generally require creepage distances and clearances greater than those given in table 4. Where a relay has contacts in intrinsically safe circuits and other contacts in non-intrinsically safe circuits the intrinsically safe and non-intrinsically safe contacts shall be separated by an insulating or earthed metal barrier conforming to [6.4.1] in addition to table 4. The relay shall be designed such that broken or damaged contact arrangements cannot become dislodged and impair the integrity of the separation

	between intrinsically safe and non-intrinsically safe circuits.
6.5.1	Replace by relevant clauses listed under 6.4.1 above.
6.5.2, 6.5.3	Replace by: [6.7] Encapsulation used for exclusion of potentially explosive atmosphere Where casting compound is used to exclude a potentially explosive atmosphere from components and intrinsically safe circuits, e.g. fuses, piezo-electric devices with their suppression components and energy storage devices with their suppression components, it shall conform to [6.4.4]. In addition, where casting compound is used to reduce the ignition capability of hot components e.g. diodes and resistors, the volume and thickness of the casting compound shall reduce the maximum surface temperature of the casting compound to the desired value.
6.5.4	
6.6	Replace by: [6.4.12] Electric strength tests The insulation between an intrinsically safe circuit and the frame of the electrical apparatus or parts which may be earthed shall normally be capable of withstanding an r.m.s., a.c. test voltage of twice the voltage of the intrinsically safe circuit or 500 V whichever is the greater. Either the current flowing during the test shall not increase above that which is expected from the design of the circuit and shall not exceed 5mA r.m.s. at any time, or where the circuit does not satisfy this requirement the apparatus shall be marked with the symbol X. The insulation between an intrinsically safe circuit and a non-intrinsically safe circuit shall be capable of withstanding an r.m.s. a.c. test voltage of 2U + 1000V with a minimum of 1500V r.m.s. where U is the sum of the r.m.s. values of the voltages of the intrinsically safe circuit and the non-intrinsically safe circuit. Where breakdown between separate intrinsically safe circuits could produce an unsafe condition, the insulation between these circuits shall be capable of withstanding an r.m.s. test voltage of 2U with a minimum of 500V, where U is the sum of the r.m.s. values of the voltages of the circuits under consideration. The test method used in the above tests shall be in accordance with [10.6].
6.7	Replace by: [6.4.11] Internal wiring Insulation, except for varnish and similar coatings, covering the conductors of internal wiring shall be considered as solid insulation (see [6.4.5]). The separation of conductors shall be determined by adding together the radial thicknesses of extruded insulation on wires that are lying side by side either as separate wires or in a cable form or in a cable. The distance between the conductors of any core of an intrinsically safe circuit and that of any core of a non-intrinsically safe circuit shall be in accordance with [line 4] of table 4 taking account of the requirements of [6.4.6] except when one of the following apply:

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	the cores of either the intrinsically safe or the non-intrinsically safe circuit are enclosed in an earthed screen; or, in category 'ib' electrical apparatus, the insulation of the intrinsically safe cores is capable of withstanding an r.m.s. a.c. test voltage of 2000V. NOTE: One method of achieving insulation capable of withstanding this test voltage is to add an insulating sleeve over the core.
6.7	Add the following requirement: [6.5] Protection against polarity reversal Protection shall be provided within intrinsically safe apparatus to prevent invalidation of this type of protection as a result of reversal of the polarity of supplies to that apparatus or at connections between cells of a battery where this could occur. For this purpose a single diode shall be acceptable.
6.8	Replace by: [6.6] Earth conductors, connections and terminals Where earthing e.g. of enclosures, conductors, metal screens, printed wiring board conductors, segregation contacts of plug-in connectors and diode safety barriers is required to maintain this type of protection, the cross-sectional area of any conductors, connectors and terminals used for this purpose shall be such that they are rated to carry the maximum possible current to which they could be continuously subjected under the conditions specified in clause 5. Components shall also conform to [clause 7]. Where a connector carries earthed circuits and this type of protection depends on the earthed circuit, the connector shall comprise at least three independent connecting elements for 'ia' circuits and at least two for 'ib' circuits (see [figure 2]). These elements shall be connected in parallel. Where the connector can be removed at an angle one connection shall be present at, or near to, each end of the connector. Terminals shall be fixed in their mountings without possibility of self-loosening; and shall be constructed so that the conductors cannot slip out from their intended location. Proper contact shall be assured without deterioration of the conductors, even if multi-stranded cores are used in terminals which are intended for direct clamping of the cores. The contact made by a terminal shall not be appreciably impaired by temperature changes in normal service. Terminals which are intended for clamping stranded cores shall include a resilient intermediate part. Terminals for conductors of cross sections up to 4 mm^2 shall also be suitable for the effective connection of conductors having a smaller cross section. Terminals which comply with the requirements of EN50019 are considered to conform to these requirements. The following shall not be used: a) terminals with sharp edges which could damage the conductors; b) terminals which may turn, be twisted or permanently deformed by normal tightening; c) insulating materials which transmit contact pressure in terminals.
6.9.1	Replace by: [6.3.3] Determination of maximum external inductance to resistance ratio (L_O/R_O) for resistance limited power source The maximum external inductance to resistance ratio (L_O/R_O) which may be connected to a resistance limited power sources shall be calculated using the

	following formula. This formula takes account of a 1,5 factor of safety on current and shall not be used where C_i for the output terminals of the power source exceeds 1 % of C_o. $L_o/R_o = (8eR_i + (64e^2R_i^2 - 72U_o^2 e L_i)^{1/2}) / 4,5 (U_o^2)$ (H/Ω) where e is the minimum spark-test apparatus ignition energy in joules and is for: Group I apparatus : 525 µJ Group IIA apparatus : 320 µJ Group IIB apparatus : 160 µJ Group IIC apparatus : 40 µJ R_i is the minimum output resistance of the power source in ohms; U_o is the maximum open circuit voltage in volts; L_i is the maximum inductance present at the power source terminals in henries. If L_i = O then $L_o/R_o = \frac{32}{9} \frac{e\ Ri}{U_o^2}$ (H/Ω) Where a safety factor of 1 is required, this value for L_o/R_o shall be multiplied by 2,25. NOTE 1: The normal application of the L_o/R_o ratio is for distributed parameters, e.g. cables. Its use for lumped values of inductance and resistance requires special consideration. NOTE 2: The calculation of L_o/R_o for power sources with non-linear output characteristics requires special consideration. [6.3.4] <u>Permanently connected cables</u> Apparatus constructed with a permanently connected cable shall be tested in accordance with [10.13].
6.9.2	Replace by EN 50039
6.10	Replace by: [10.11] <u>Tests for apparatus containing piezoelectric devices</u> Measure both the capacitance of the device and also the voltage appearing across it when any part of the apparatus which is accessible in service is impact tested in accordance with the 'high' column of table 4 in EN 50014:1992 carried out at 20 ± 10 °C using the test apparatus in annex D of EN 50014:1992. For the value of voltage, the higher figure of two tests on the same sample shall be used. When the apparatus containing the piezoelectric device includes a guard to prevent a direct physical impact, the impact-test shall be carried out on the guard with both it and the apparatus mounted as intended by the manufacturer. The maximum energy stored by the capacitance of the crystal at the maximum measured voltage shall not exceed the following;

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	for Group I apparatus:1500 μJ for Group IIA apparatus: 950 μJ for Group IIB apparatus: 250 μJ for Group IIC apparatus: 50 μJ Where the electrical output of the piezoelectric device is limited by protective components (including guards) these components shall not be damaged by the impact in such a way as to allow this type of protection to be invalidated. Where it is necessary to protect the apparatus from external physical impact in order to prevent the impact energy exceeding these values details of the requirements shall be specified as special conditions for safe use and the apparatus shall be marked with the sign X.
7.1	Replace by: [7.1] Ratings In both normal operation and after application of the fault conditions given in clause 5 any remaining components on which this type of protection depends, except such devices as transformers, fuses, thermal trips, relays and switches shall not operate at more than two thirds of their maximum current, voltage and power related to the rating of the device, the mounting conditions and the temperature range specified. These maximum rated values shall be the normal commercial ratings specified by the manufacturer of the component. Detailed testing or analysis of components and assemblies of components to determine the parameters e.g. voltage and current to which the safety factors are applied shall not be performed, since the factors of safety of 5.1 and 5.2 obviate the need for detailed testing or analysis. For example a zener diode stated by its manufacturer to be 10 V + 10% at 40 ¦C shall be taken to be 11V maximum without the need to take into account effects such as voltage elevation due to rise in temperature. Account shall also be taken of the effects of the mounting conditions and ambient temperature range specified by the manufacturer of the apparatus and by [5.2 of EN 50014:1992]. For example in the case of a semiconductor the power dissipation shall not exceed two thirds of that which will cause the maximum junction temperature to be reached under the particular mounting conditions.
7.2	[7.3] Fuses Where fuses are used to protect other components, 1,7In shall be assumed to flow continuously. The fuse time-current characteristics shall ensure that the transient ratings of protected components are not exceeded. Where the fuse time-current characteristic is not available from the manufacturer's data, a type test shall be carried out in accordance with [10.12] on at least ten samples. This test shows the capability of the sample to withstand 1,5 times any transient which can occur when U_m is applied through a fuse. Fuses located in the hazardous area shall be protected in accordance with [6.7]. Where fuses are encapsulated the casting compound shall not enter the fuse interior. This requirement shall be satisfied by testing samples or by a declaration from the fuse manufacturer confirming acceptability of the fuse for encapsulation. Alternatively the fuse shall be sealed prior to encapsulation. Fuses used to protect components shall be replaceable only by opening the apparatus enclosure. The type designation and In, or the characteristics important to intrinsic safety shall be marked adjacent to the fuses.

	Fuses shall have a rated voltage of at least U_m (or U_i in intrinsically safe apparatus and circuits) although they do not have to conform to table 4. General industrial standards for the construction of fuses and fuseholders shall be applied and their method of mounting shall not reduce the clearances, creepage distances and separations afforded by the fuse and its holder. NOTE 1: Microfuses conforming to IEC127 are acceptable. A fuse shall be capable of interrupting the maximum prospective current of the circuit in which it is installed. For mains electricity supply systems not exceeding 250 V a.c. the prospective current shall normally be considered to be 1500 A a.c. The breaking capacity of the fuse is determined according to IEC 127 or an equivalent standard. NOTE 2: Higher prospective currents may be present in some installations e.g. at higher voltages. If a current limiting device is necessary to limit the prospective current to a value not greater than the rated breaking capacity of the fuse, this device shall be infallible in accordance with [clause 7 and the rated values shall be at least: current rating: 1,5 x 1,7 x I_n voltage rating: U_m or U_i power rating: 1,5 x $(1,7 \times I_n)^2$ x resistance of limiting device
7.3	see 6.4.2 above
7.4	Replace by: [7.4] <u>Primary and secondary cells and batteries</u> [7.4.1.] <u>General</u> Some types of cells and batteries, e.g. some lithium types, may explode if short circuited or subjected to reverse charging. Where such an explosion could adversely affect intrinsic safety the use of such cells and batteries must be confirmed by their manufacturer as being safe for use in any particular intrinsically safe or associated apparatus when 5.1 or 5.2 as appropriate is applied. The documentation, and if practicable the marking, for the apparatus shall draw attention to the safety precautions to be observed. NOTE: Attention is drawn to the fact that the cell or battery manufacturer often specifies precautions for the safety of personnel. [7.4.2] <u>Electrolyte leakage</u> Either cells and batteries shall be of a type from which there can be no spillage of electrolyte or they shall be enclosed to prevent damage by the electrolyte to the components upon which safety depends. Cells and batteries declared as sealed (gas tight) or sealed (valve regulated) by their manufacturer (see [7.4.9]) shall be deemed to satisfy this requirement. Other cells and batteries shall be tested in accordance with [10.9.2] or written confirmation shall be obtained from the cell/battery manufacturer that the product conforms to [10.9.2]. If cells and batteries which leak electrolyte are encapsulated in accordance with [6.7], they shall be tested in accordance with [10.9.2] after encapsulation. Compartments containing cells or batteries which are charged within them shall be ventilated directly to the outside of the apparatus.

[7.4.3] Cell and battery voltages
For the purpose of evaluation and test, the cell/battery voltage shall be considered to be the maximum open circuit voltage attainable from either a new primary cell/battery or a secondary cell/battery just after a full charge as specified in [table 5]. When the cell or battery is not covered by [table 5] it shall be tested in accordance with [10.9] to determine the maximum open circuit voltage and the nominal voltage shall be that specified by the cell or battery manufacturer.

[Table 5] - Cell Voltages

K	Nickel-cadmium	1,5	1,3
	Lead-acid (dry)	2,35	2,2
	Lead-acid (wet)	2,67	2,2
L	Alkaline-manganese	1,65	1,5
M	Mercury-zinc	1,37	1,35
N	Mercury-manganese Dioxide-zinc	1,6	1,4
S	Silver-zinc	1,63	1,55
A	Zinc-air	1,55	1,4
C	Lithium-manganese dioxide	3,7	3,0
	Zinc-manganese dioxide (zinc-carbon Leclanche)	1,725	1,5
	Nickel-hydride	1,6	1,3

[7.4.4] Internal resistance of battery and cell
The internal resistance of a battery or cell shall be determined in accordance with [10.9.3].

[7.4.5] Current limiting devices for batteries in associated apparatus
The battery housing or means of attachment of associated apparatus shall be constructed so that the battery can be installed and replaced without adversely affecting the intrinsic safety of the apparatus.

NOTE: Where a current limiting device is necessary to ensure the safety of the battery output there is no requirement for the current limiting device to be an integral part of the battery.

[7.4.6] Current limiting devices for batteries to be used and replaced in hazardous areas
Where a battery requires current limiting devices to ensure the safety of the battery itself and is intended for use, and to be replaced in a hazardous atmosphere it shall form a completely replaceable unit with its current limiting devices. The unit shall be encapsulated or enclosed so that only the intrinsically safe output terminals and suitably protected intrinsically-safe terminals for charging purposes (if provided) are exposed.

[7.4.7] Current limiting devices for batteries to be used but not replaced in hazardous areas
If the cell or battery requiring current limiting devices to ensure the safety of the battery itself is not intended to be replaced in the hazardous area it either shall be protected in accordance with [7.4.6] or alternatively it may be housed in a compartment with special fasteners, e.g. those specified by EN 50014, It shall also conform to the following:

a) The cell or battery housing or means of attachment shall be arranged so that the cell or battery can be installed and replaced without reducing the intrinsic safety of the apparatus.

b) For portable electrical apparatus, such as radio receivers and transceivers the construction shall prevent the ejection or separation of the cells or batteries from the apparatus without invalidating the intrinsic safety of the apparatus when subjected to the drop test, in accordance with [23.4.3.2 of EN50014:1992] except that the prior impact test shall be omitted.

c) The apparatus shall have a warning label against changing the battery in the potentially explosive atmosphere, e.g. "DO NOT OPEN BATTERY ENCLOSURE IN A POTENTIALLY EXPLOSIVE ATMOSPHERE".

[7.4.8] External contacts for charging batteries

Cell or battery assemblies with external charging contacts shall be provided with means to prevent short circuiting or to prevent the cells and batteries from delivering ignition-capable energy to the contacts when any pair of the contacts is accidentally short circuited. This shall be accomplished in one of the following ways:

a) Blocking diodes or an infallible series resistor shall be placed in the charging circuits. For category 'ib' two diodes and for category 'ia' three diodes shall be used. Either the battery charger shall be associated apparatus or the diodes or resistor shall be protected by an appropriately rated fuse. The fuse shall either be encapsulated or shall not carry any current when situated in a potentially explosive atmosphere and the circuit is assessed as required by [clause 5].

b) For Group II electrical apparatus a degree of protection by enclosure of at least IP 20 for the charging circuit and a label warning against charging in the potentially explosive atmosphere shall be provided.

[7.4.9] Battery containment

Spark ignition capability and surface temperature of cells and batteries shall be tested or assessed in accordance with [10.9.3]. If the cell or battery satisfies one of the following requirements these properties shall be determined by the use of external terminals and surfaces:

a) They shall be sealed (gas-tight) cells or batteries
b) They shall be sealed (valve-regulated) cells or batteries

c) They shall be cells or batteries which are intended to be sealed in a similar manner to items a) and b) apart from a pressure relief device. Such cells or batteries shall not require addition of electrolyte during their life and shall have a sealed metallic or plastic enclosure conforming to one or more of the following:

1) Without seams or joints e.g. solid-drawn, spun or moulded, joined by fusion, eutectic methods, welding or adhesives sealed with elastomeric or plastics sealing devices retained by the structure of the enclosure and held permanently in compression e.g. washers and 'O' rings.

2) Swaged, crimped, shrunk on or folded construction of parts of the enclosure which do not conform with the above or parts using

	materials which are permeable to gas e.g. paper based materials shall not be considered to be sealed . 3) Seals around terminals shall be either constructed as above or be poured seals of thermosetting or thermoplastic compound. d) Encapsulated in a casting compound specified by the manufacturer of the casting compound as being suitable for use with the electrolyte concerned and conforming to [6.7]. A declaration of conformance to a) or b) shall be obtained from the manufacturer of the cell or battery and this shall not be checked by the testing station. Conformance to c) or d) shall be determined by physical examination of the cell or battery and where necessary its constructional drawings.
add to 7 after 7.5	[7.5] Semiconductors [7.5.1] Transient effects In associated apparatus semiconductor devices shall be capable of withstanding the peak of the a.c. voltage divided by any infallible series resistance. In an intrinsically safe apparatus any transient effects generated within that apparatus and from its power sources shall be ignored. [7.5.2] Shunt voltage limiters Semiconductors may be used as shunt voltage limiting devices provided that they conform to the following requirements and provided that relevant transient conditions are taken into account. They shall be capable of carrying, without open-circuiting, 1,5 times the current which would flow at their place of installation if they failed in the short circuit mode. In the following cases this shall be confirmed from their manufacturer's data by ; a) diodes, diode connected transistors, thyristors and equivalent semiconductor devices having a forward current rating of at least 1,5 times the maximum possible short-circuit current. b) zener diodes being rated: 1) in the zener direction at 1,5 times the power that would be dissipated in the zener mode, and; 2) in the forward direction at 1,5 times the maximum current that would flow if they were short-circuited. For category 'ia' the application of controllable semiconductor components as shunt voltage limiting devices e.g. transistors, thyristors, voltage/current regulators etc. is permitted if both the input and output circuits are intrinsically safe circuits or where it can be shown that they cannot be subjected to transients from the power supply network. In circuits complying with the above, two devices are considered to be an infallible assembly. For category 'ia' three thyristors may be used in associated apparatus provided the transient conditions of [7.5.1] are met. Circuits using shunt thyristors shall also be tested in accordance with [10.4.3.3].

[7.5.3] Series current limiters

The use of three series blocking diodes in circuits of category 'ia' is permitted, however other semiconductors and controllable semiconductor devices shall be used as series current-limiting devices only in category 'ib' apparatus.

NOTE: The use of semiconductors and controllable semiconductor devices as current- limiting devices is not permitted for category 'ia' apparatus because of their probable future use in areas in which a continuous or frequent presence of an explosive atmosphere may coincide with the possibility of a brief transient which could cause ignition.

[7.6] Failure of components and connections

The application of 5.1 and 5.2 shall include the following:

a) Where a component is not rated in accordance with [7.1] its failure shall be a non countable fault. Where a component is rated in accordance with [7.1] its failure shall be a countable fault.

b) Where a fault can lead to a subsequent fault or faults then the primary and subsequent faults shall be considered to be a single fault.

c) The failure of resistors to any value of resistance between open circuit and short circuit shall be considered (but see [8.4]).

d) Semiconductor devices shall be considered to fail to short circuit or to open circuit and to the condition to which they can be driven by failure of other components. For surface temperature classification failure of any semiconductor device to a condition where it dissipates maximum power shall be considered. Integrated circuits can fail so that any combination of short and open circuits can exist between their external connections. Although any combination can be assumed, once that fault has been applied it cannot be changed e.g. by application of a second fault. Under this fault situation any capacitance and inductance connected to the device may be considered in their most onerous connection as a result of the applied fault.

e) Connections shall be considered to fail to open circuit, and if free to move, may connect to any part of the circuit within the range of movement. The initial break is one countable fault and the reconnection is a second countable fault (but see [8.7]).

f) Clearances, creepage and separation distances shall be considered in accordance with [6.4].

g) Failure of capacitors to open circuit, short circuit and any value less than the maximum specified value shall be considered (but see [8.5]).

h) Failure of inductors to open circuit and any value between nominal resistance and short circuit but only to inductance to resistance ratios lower than that derived from the inductor specification.

i) Open circuit failure of any wire or printed circuit track, including its connections, shall be considered as a single countable fault.

Insertion of the spark test apparatus to effect an interruption, short circuit or earth fault

	shall not be considered as a countable fault but as a test in normal operation. Infallible connections and separations in accordance with [clause 8] shall not be considered as producing a fault and the spark test apparatus shall not be inserted in series with such connections or across such separations. However, where infallible connections and separations are not encapsulated or covered by a coating in accordance with clause 6 or do not maintain an enclosure integrity of at least IP20 when exposing connection facilities the spark test apparatus shall be inserted in series with such connections or across such separations.
8.1	Replace by: [8.1] Mains transformers [8.1.1] Winding faults Infallible mains transformers shall be considered as not being capable of failing to a short circuit between any winding supplying an intrinsically safe circuit and any other winding. Short circuits within windings and open circuits of windings shall be considered to occur. [8.1.2] Protective measures The input circuit of mains transformers intended for supplying intrinsically safe circuits shall be protected either by a fuse conforming to [7.3], or by a suitably rated circuit breaker. If the input and output windings are separated by an earthed metal screen (see type 2b construction in [8.1.3]), each non-earthed input line shall be protected by a fuse or circuit breaker. Where an embedded thermal fuse or other thermal device is used for additional protection against overheating of the transformer, a single device shall be sufficient. Fuses, fuseholders, circuit breakers and thermal devices shall conform to an appropriate recognised standard. Conformance with the recognised standard shall not be verified by the testing station. [8.1.3] Transformer construction All windings for supplying intrinsically safe circuits shall be separated from all other windings by one of the following types of construction: For type 1 construction the windings shall be placed either a) on one leg of the core, side by side; or b) on different legs of the core. The windings shall be separated in accordance with table 4. For type 2 construction the windings shall be wound one over another with either: a) solid insulation in accordance with table 4 between the windings, or b) an earthed screen (made of copper foil) between the windings or an equivalent wire winding (wire screen). The thickness of the copper foil or the wire screen, shall be in accordance with [table 6]. NOTE: This ensures that in the event of a short circuit between any winding

	and the screen, the screen will withstand, without breakdown, the current which flows until the fuse or circuit breaker functions. [Table 6] Minimum foil thickness or minimum wire diameter of the screen in relation to the rated current of the fuse.

Rating of fuse	A	0,1	0,5	1	2	3	5
Thickness of foil screen	mm	0,05	0,05	0,075	0,15	0,25	0,3
Diameter of the wire of the screen	mm	0,2	0,45	0,63	0,9	1,12	1,4

Manufacturer's tolerances shall not reduce the values given in [table 6] by more than 10% or 0,1 mm whichever is the smaller.

The foil screen shall be provided with two mechanically separate leads to the earth connection, each of which is rated to carry the maximum continuous current which could flow before the fuse or circuit breaker operates, e.g. 1,7In for a fuse.

A wire screen shall consist of at least two electrically independent layers of wire, each of which is provided with an earth connection rated to carry the maximum continuous current which could flow before the fuse or circuit breaker operates. The only requirement of the insulation between the layers is that it shall be capable of withstanding a 500 V test in accordance with [10.6].

The cores of all mains supply transformers shall be provided with an earth connection, except where earthing is not required for this type of protection e.g. when transformers with insulated cores are used.

The transformer windings shall be consolidated e.g. by impregnation or encapsulation.

[8.1.4] Transformer type tests

The transformer together with its associated devices e.g. fuses, circuit breakers, thermal devices and resistors connected to the winding terminations shall maintain a safe electrical isolation between the power supply and the intrinsically safe circuit even if any one of the output windings is short circuited and all other output windings are subjected to their maximum rated electrical load.

Where a series resistor is either incorporated within the transformer, or encapsulated with the transformer so that there is no bare live part between the transformer and the resistor, or mounted so as to provide creepage distances and clearances conforming to table 4 and the resistor remains in circuit after the application of clause 5, then the output winding shall not be considered as subject to short circuit except through the resistor.

The requirement for safe electrical isolation is satisfied if the transformer passes the type test described below and subsequently withstands a test voltage (see [10.6]) of $2U_n$ + 1000 V or 1500 V, whichever is the greater, between any winding(s) used to supply intrinsically safe circuits and all other windings where U_n is the highest rated

<table>
<tr><td></td><td>voltage of any winding under test.

The input current shall be adjusted to 1.7I_n or to the maximum continuous current which the circuit breaker will carry without operating. During the test this current shall be maintained within ± 10% of this value. The current shall be adjusted by varying the input voltage up to the rated input voltage of the transformer. Where this limit is reached the test shall proceed using the rated input voltage.

The test shall continue for at least 6 hours or up to when the non resetting thermal trip operators. When a self-resetting thermal trip is used the test period shall be extended to at least 12 hours.

For type 1 and type 2a transformers the transformer winding temperature shall not exceed the permissible value for the class of insulation given in IEC85. The winding temperature shall be measured in accordance with [10.5].

For type 2b transformers where insulation from earth of the windings used in the intrinsically safe circuit is required then the requirement shall be as above. However if insulation from earth is not required then the transformer shall be accepted providing that it does not burst into flames.

[8.1.5] <u>Routine test of mains transformers</u>

Each mains transformer shall be tested in accordance with [11.2].</td></tr>
<tr><td>8.2</td><td>Replace by:
[8.2] <u>Transformers other than mains transformers</u>

The infallibility and failure modes of these transformers shall conform to [8.1].

NOTE: These transformers can be coupling transformers such as those used in signal circuits or transformers for other purposes, e.g. those used for inverter supply units.

The construction and testing of these transformers shall conform to [8.1] except that they shall be tested at their maximum load. Where it is not practicable to operate the transformer under alternating current conditions each winding shall be subjected to a direct current of 1.7 I_n in the type test [8.1.4]. However, the routine test in accordance with [11.2] shall use a reduced voltage between the input and output windings of 2Un + 1000V r.m.s. or 1500V whichever is the greater.

When such transformers are connected to non-intrinsically safe circuits derived from mains voltages then either protective measures in accordance with [8.1.1] or a fuse and zener diode shall be included at the supply connection in accordance with [8.8] so that unspecified power shall not impair the infallibility of the transformer creepage distances and clearances. The rated input voltage of [8.1.4] shall be that of the zener diode.</td></tr>
<tr><td>8.4</td><td>Replace by:
[8.4] <u>Current limiting resistors</u>

Current limiting resistors shall be one of the following types:

a) film type

b) wire wound type with protection to prevent unwinding of the wire in the event</td></tr>
</table>

	of breakage c) printed resistors as used in hybrid and similar circuits covered by a coating conforming to [6.4.8] or encapsulated in accordance with [6.4.4]. An infallible current limiting resistor shall be considered as failing only to an open circuit condition which shall be considered as one countable fault. A current limiting resistor shall be rated, in accordance with the requirements of [7.1], to withstand at least 1,5 times the maximum voltage and to dissipate at least 1,5 times the maximum power that can arise in normal operation and under the fault conditions defined in [clause 5]. Faults between turns of correctly rated wire wound resistors with coated windings shall not be considered. The coating of the winding shall be assumed to comply with the required CTI value in accordance with table 4 at its manufacturer's voltage rating.
8.5	Not applicable
8.6	Replace by: [8.5] <u>Blocking capacitors</u> Either of the two series capacitors in an infallible arrangement of blocking capacitors shall be considered as being capable of failing to short or open circuit. The capacitance of the assembly shall be taken as the most onerous value of either capacitor and a safety factor of 1,5 shall be used in all applications of the assembly. Blocking capacitors shall be of a high reliability solid dielectric type. Electrolytic or tantalum capacitors shall not be used. The external connections of the assembly shall comply with [6.4] but these separation requirements shall not be applied to the interior of the blocking capacitors. The insulation of each capacitor shall conform to the electrical strength test of [6.4.12]. Where blocking capacitors are used between intrinsically safe circuits and non-intrinsically safe circuits all possible transients shall be considered. Where such an assembly also conforms to [8.8] it shall be considered as providing infallible galvanic separation for direct current. Capacitors connected between the frame of the apparatus and an intrinsically safe circuit shall conform to [6.4.12]. Where their failure by-passes a component on which the intrinsic safety of the circuit depends they shall also conform to the requirements for blocking capacitors. NOTE: The normal purpose of these capacitors is the rejection of high frequencies.
8.7	Replace by: [8.6] <u>Shunt safety assemblies</u> [8.6.1] <u>General</u> An assembly of components shall be considered as a shunt safety assembly when it ensures the intrinsic safety of a circuit by the utilisation of shunt components. Where diodes or zener diodes are used as the shunt components in an infallible shunt safety assembly they shall form at least two parallel paths of diodes. Diodes shall be rated to carry the current which would flow at their place of installation if they failed in the short-circuit mode. The connections of the shunt components shall either be in accordance with [8.7] or

	be arranged so that if either of the shunt paths becomes disconnected the circuit or component being protected becomes disconnected at the same time. NOTE 1: To prevent spark ignition when a connection breaks encapsulation in accordance with [6.4.4] may be required. NOTE 2: The shunt components used in these assemblies may conduct in normal operation. Where shunt safety assemblies are subjected to power faults specified only by a value of U_m, the components of which they are formed shall be rated in accordance with [7.1]. Where the components are protected by a fuse, the fuse shall be in accordance with [7.3] and the components shall be assumed to carry a continuous current of 1.7In of the fuse. The ability of the shunt components to withstand transients shall either be tested in accordance with [10.12] or be determined by comparison of the fuse-current time characteristic of the fuse and the performance characteristics of the device. Where a shunt safety assembly is manufactured as an individual apparatus rather than as part of a larger apparatus then the construction of the assembly shall be in accordance with [9.2]. When considering the utilisation of a shunt safety assembly and an infallible assembly the following shall be considered: a) Either of the two shunt paths in the assembly shall be considered as failing to an open circuit condition. b) The voltage of the assembly shall be that of the highest voltage shunt path. c) The failure of either shunt path to short circuit shall be considered as one fault. d) A safety factor of 1,5 shall be used in the application of all the fault counts of 5.1 and 5.2. e) Circuits using shunt thyristors shall be tested in accordance with [10.4.3.3]. [8.6.2] <u>Safety shunts</u> A shunt safety assembly shall be considered as a safety shunt when it ensures that the electrical parameters of a specified component or part of an intrinsically safe apparatus is controlled to values which do not invalidate intrinsic safety. Safety shunts shall be subjected to the analysis of transients required when connected to power supplies defined only by U_m in accordance with [8.6.1] except when used as follows: a) for the limitation of the discharge from energy storing devices e.g. inductors or piezo-electric devices; b) for the limitation of voltage to energy storing devices e.g. capacitors. An assembly of suitably rated bridge-connected diodes shall be considered as an infallible safety shunt.

	[8.6.3] Shunt voltage limiters A shunt safety assembly shall be considered as a shunt voltage limiter when it ensures that a defined voltage level is applied to an intrinsically safe circuit. Shunt voltage limiters shall be subjected to the analysis of transients required when connected to power supplies defined only by U_m in accordance with [8.6.1] except when the assembly is fed from one of the following: a) an infallible transformer in accordance with [8.1]; b) a diode safety barrier in accordance with clause [9]; c) a battery in accordance with [7.4]; d) an infallible shunt safety assembly in accordance with [8.6].
8.7	Add the following requirements: [9] Diode safety barriers [9.1] General NOTE: Diode safety barriers are assemblies incorporating shunt diodes or diode chains (including zener diodes) protected by fuses or resistors or a combination of these, manufactured as an individual apparatus rather than as part of a larger apparatus. The diodes limit the voltage applied to an intrinsically safe circuit and a following infallible current limiting resistor limits the current which can flow into the circuit. These assemblies are intended for use as interfaces between intrinsically safe circuits and non-intrinsically safe circuits, and shall be subject to the routine test of [11.1]. The ability of the safety barrier to withstand transient faults shall be tested in accordance with [10.12]. Safety barriers containing only two diodes or diode chains and used for category 'ia' shall be acceptable as infallible assemblies in accordance with [8.6] provided the diodes have been subjected to the routine tests specified in [11.1.2]. In category 'ia' two diode barriers, only the failure of one diode shall be taken into account in the application of clause 5. [9.2] Construction [9.2.1] Mounting The construction shall be such that when groups of barriers are mounted together any incorrect mounting is obvious e.g. by being asymmetrical in shape or colour in relation to the mounting. [9.2.2] Facilities for connection to earth In addition to any circuit connection facility which may be at earth potential the barrier shall have at least one more connection facility or shall be fitted with an insulated wire having a cross sectional area of at least 4 mm^2 for the additional earth connection.

	[9.2.2] Protection of components The assembly shall be protected against access to prevent repair or replacement of any components on which safety depends either by encapsulation in accordance with [6.4.4] or by an enclosure which forms a non-recoverable unit. The entire assembly shall form a single entity.
8.8	Replace by: [8.8] Galvanically separating components An infallible isolating element conforming to the following shall be considered as not being capable of failing to a short circuit across the infallible separation. Isolating elements other than transformers and relays, e.g. optocouplers, shall be considered to provide infallible separation of separate intrinsically safe circuits if the following conditions are satisfied. a) The rating of the device shall be according to [7.1]. b) Before application of U_m and U_i the device shall withstand an electric strength test as described in [6.4.12] where voltage U is the manufacturer's highest rated isolation voltage for the infallible separation of the component under test. Where separation is between intrinsically safe and non-intrinsically safe circuits the requirements of table 4 shall also apply to the isolating element except that inside sealed devices, e.g. optocouplers, [lines 5, 6 and 7] shall not apply. The non-intrinsically safe circuit terminals shall be provided with protection to ensure that the ratings of the devices in accordance with [7.1] are not exceeded unless it can be shown that the circuits connected to these terminals cannot invalidate intrinsic safety of the devices. Typically the inclusion of a single shunt zener diode protected by a suitably rated fuse capable of interrupting the prospective peak current of the supply shall be considered as sufficient protection. For this purpose table 4 shall not be applied to the fuse and zener diode. The zener diode power rating shall be at least $1.7I_n$ times the diode maximum zener voltage. General industrial standards of construction shall be considered adequate for fuses and the method of mounting, e.g. in a fuse holder, shall not reduce the clearances and creepage distances afforded by the fuse itself. Galvanically separating relays shall conform to [6.4.13] and any winding shall be capable of dissipating the maximum power to which it is connected. NOTE: Derating of the relay winding in accordance with [7.1] is not required.
8.9	Replace by: [8.7] Wiring and connections Wiring including its connections which forms part of the apparatus shall be considered as infallible against open circuit failure in the following cases: a) for wires 1) where two wires are in parallel; or 2) where a single wire has a diameter of at least 0,5 mm and has an unsupported length of less than 50 mm or is mechanically secured adjacent to its point of connection; or 3) where a single wire is of stranded or flexible ribbon type construction, has a cross-sectional area of at least 0,125 mm² (0,4 mm diameter), is not flexed

	in service and is either less than 50 mm long or is secured adjacent to its point of connection. b) for printed board tracks 1) where two tracks of minimum width 1 mm are in parallel; or 2) where a single track is at least 2 mm wide or has a width of 1 % of its length whichever is greater; and 3) each track shall be formed from copper cladding having a nominal thick ness of 35µm or greater. c) for connections (excluding plugs, sockets and terminals) 1) where there are two connections in parallel; or 2) where there is a single soldered joint in which the wire passes through the board (including through-plated holes) and is either bent over before soldering or, if not bent over, machine soldered or has a crimped connection or is brazed or welded; or 3) where there is a single connection which is screwed or bolted and conforms to [6.6].
9.1.1	Replace by: [10.1.2] <u>Spark test apparatus</u> The spark test apparatus shall be that described in annex B except where annex B indicates that it is not suitable. In these circumstances an alternative test apparatus of equivalent sensitivity shall be used and justification for its use shall be included in the definitive documentation. The use of the spark-test apparatus to produce short circuits, interruptions and earth faults shall be a test of normal operation and is a non-countable fault. at connection facilities; at internal connections or across internal creepage distances, clearances, distances through casting compound and distances through solid insulation, not conforming to table 4. The spark-test apparatus shall not be used: - across infallible separations, or in series with infallible connections - across creepage distances, clearances, distances through casting compound and distances through solid insulation conforming to table 4, - within associated apparatus other than at its intrinsically safe circuit terminals, - and between terminals of separate circuits conforming to [6.3.1], apart from the exceptions described in [7.6 i]. [10.4.3] <u>Testing considerations</u> [10.4.3.1] <u>General</u>

	Spark ignition tests shall be carried out with the circuit arranged to give the most incendive conditions. For simple circuits of the types for which the curves in figures 1 to 7 apply a short circuit test is the most onerous. For more complex circuits the conditions vary and a short circuit test may not be the most onerous, e.g. for constant voltage current limited power supplies the most onerous condition usually occurs when a resistor is placed in series with the output of the power supply and limits the current to the maximum which can flow without any reduction in voltage. [10.4.3.2] <u>Circuits with both inductance and capacitance</u> Where a circuit contains energy stored in both capacitance and inductance it may be difficult to assess such a circuit from the curves in figures 1 to 7 e.g. where the capacitive stored energy may reinforce the power source feeding an inductor. The circuit shall be tested with the combination of capacitance and inductance. [10.4.3.3] <u>Circuits using shunt short circuit (crowbar) protection</u> After the output voltage has stabilised, the circuit shall be incapable of causing ignition for the appropriate category of apparatus in the conditions of [clause 5]. Additionally, where the type of protection relies on operation of the crowbar caused by other circuit faults, the let-through energy of the crowbar during operation shall not exceed the following value for the appropriate Group: for Group IIC apparatus 20 µJ Group IIB apparatus 80 µJ Group IIA apparatus 160 µJ Group I apparatus 260 µJ As ignition tests with the spark test apparatus are not appropriate for testing the crowbar let-through energy, this let-through energy shall be assessed e.g. from oscilloscope measurements.
9.1.4	Replace c) by: c) for capacitive circuits - 400 revolutions (5 min), 200 revolutions at each polarity. Care shall be taken to ensure that the capacitor has sufficient time to recharge (at least three time constants). The normal time for recharge is about 20ms and where this is inadequate it shall be increased by removing one or more of the wires or by slowing the speed of rotation of the spark test apparatus. When wires are removed the number of revolutions shall be increased to maintain the same number of sparks.
9.1.5	replace by: [10.4.2] <u>Safety factors</u> NOTE:The purpose of the application of a safety factor is to ensure either that a type test or assessment is carried out with a circuit which is demonstrably more likely to cause ignition than the original, or that the original circuit is tested in a more readily ignited gas mixture. In general, it is not possible to obtain exact equivalence between different methods of achieving a specified factor of safety, but the following methods provide acceptable alternatives. The safety factor of 1,5 shall be obtained by one of the following methods: a) Increase the mains (electrical supply system) voltage to 110% of the nominal value to allow for mains variations, or set other voltages e.g. batteries, power supplies and voltage limiting devices at the maximum value in

	accordance with clause [7] then: 1) for inductive and resistive circuits, increase the current to 1,5 times the fault current by decreasing the values of limiting resistance, if the 1,5 factor cannot be obtained, further increase the voltage; 2) for capacitive circuits, increase the voltage to obtain 1,5 times the fault voltage. Alternatively when an infallible current limiting resistor is used with a capacitor, consider the capacitor as a battery and the circuit as resistive. NOTE: When using the curves 1 - 7 for assessment, this same method shall be used. b) Use the more easily ignited explosive test mixtures in accordance with table 8.
9.5	Replace by: [10.7] <u>Small component ignition test</u> A small component tested to demonstrate that it shall not cause temperature ignition of a flammable mixture in accordance with [6.2.4. a)] shall be tested as described below. The appearance of a cool flame shall be considered as an ignition. Detection of ignition shall either be visual or by measurement of temperature e.g. by a thermocouple. The component shall be tested under normal operation or the fault conditions in accordance with clause 5 which produces the highest value of surface temperature. The test shall be continued either until thermal equilibrium of the component and the surrounding parts is attained or until the component temperature drops. Where component failure causes the temperature to fall, the test shall be repeated five times using five additional samples of the component. Where in normal operation or the fault conditions in accordance with clause 5, the temperature of more than one component exceeds the temperature class of the apparatus the test shall be carried out with all such components at their maximum temperature. The component under test may either be mounted in the apparatus as intended and precautions taken to ensure that the test mixture is in contact with the component. Alternatively the test shall be carried out on a model which ensures representative results. Such a simulation shall consider the effect of other parts of the apparatus in the vicinity of the component being tested which affect the temperature of the mixture and the flow of the mixture around the component as a result of ventilation and thermal effects. The safety factor required by [5.3 of EN50014:1992] shall be achieved either by raising the ambient temperature at which the test is carried out or where this is possible by raising the temperature of the component under test and other relevant adjacent surfaces by the required margin. For T4 temperature classification the mixture shall be either: a) a homogeneous mixture of 23 % V/V diethyl ether and air; or b) a mixture of diethyl ether and air obtained by allowing a small quantity of diethyl ether to evaporate within a test chamber while the ignition test is being carried out. For other temperature classifications, the choice of suitable test mixtures shall be at

	the discretion of the testing station. If no ignition occurs during a test, the presence of the flammable mixture shall be verified by igniting the mixture by some other means.
9.6.2	Replace by: [10.10.1] <u>Casting compound</u> A force of 30 N shall be applied perpendicular to the surface of the casting compound with a 6 mm diameter flat ended metal rod for 10 s. No damage to or permanent deformation of the encapsulation or movement greater than 1 mm shall occur. Where a free surface of casting compound occurs, in order to ensure that the compound is rigid but not brittle, one of the following impact tests shall be carried out on the surface of the casting compound at (20 $\pm$ 10) °C using the test apparatus described in annex D of EN50014:1992. a) for Group I applications where casting compound forms part of the external enclosure and is used to exclude a potentially explosive atmosphere an impact energy of 20 J shall be used, b) for all other applications an impact energy of 2 J shall be used. The casting compound shall remain intact and no permanent deformation shall occur. Minor surface cracks shall be ignored.
9	Add the following tests: [10.5] <u>Temperature tests</u> All temperature data shall be referred to a reference ambient temperature of 40 °C or the maximum ambient temperature marked on the apparatus. Tests to be based on a reference ambient temperature shall be conducted at any ambient temperature between 20 °C and the reference ambient temperature. The difference between the ambient temperature at which the test was conducted and the reference ambient temperature shall then be added to the temperature measured unless the thermal characteristics of the component are non-linear e.g. batteries. If the temperature rise is measured at the reference ambient temperature, that value shall be used in determining the temperature classification. Temperatures shall be measured by any convenient means. The measuring element shall not substantially lower the measured temperature. An acceptable method of determining the rise in temperature of a winding is as follows: measure the winding resistance with the winding at a recorded ambient temperature apply the test current or currents and measure the maximum resistance of the winding and record the ambient temperature at the time of measurement calculate the rise in temperature from the following equation: $t = \frac{R}{r}(k + t_1) - (k + t_2)$ where

t is the temperature rise in degree Kelvin;

r is the resistance of the winding at the ambient temperature t, in ohms;

R is the maximum resistance of the winding under the test current conditions in ohm;

t_1 is the ambient temperature in degrees Celsius when r is measured;

t_2 is the ambient temperature in degrees Celsius when R is measured;

k is the inverse of the temperature coefficient of resistance of the winding at 0 °C and has the value of 234,5 for copper.

[10.8] Determination of parameters of loosely specified components

Ten unused samples of the component shall be obtained from any source or sources of supply and their relevant parameters shall be measured. Tests shall normally be carried out at or referred to the specified maximum ambient temperature e.g. 40 °C but where necessary temperature sensitive components e.g. nickel cadmium cells/batteries shall be tested at lower temperatures to obtain their most onerous conditions.

The most onerous values for the parameters, not necessarily taken from the same sample, obtained from the tests on the ten samples shall be taken as representative of the component.

[10.9] Test for cells and batteries

[10.9.1.] General

Rechargeable cells or batteries shall be fully charged and then discharged at least twice before any tests are carried out. On the second discharge, or subsequent one as necessary, the capacity of the cell or battery shall be confirmed as being within its manufacturer's specification to ensure that tests can be carried out on a fully charged cell or battery which is within its manufacturer's specification.

When a short circuit is required for test purposes the resistance of the short circuit link, excluding connections to it, either shall not exceed 3 mΩ or have a voltage drop across it not exceeding 200 mV or 15 % of the cell e.m.f. The short circuit shall be applied as close to the cell or battery terminals as practicable.

[10.9.2] Electrolyte leakage test for cells and batteries

The test samples shall be placed with any case discontinuities e.g. seals facing downward or in the orientation specified by the manufacturer of the device, over a piece of blotting paper. Ten test samples shall be subjected to the most onerous of the following:

a) short circuit until discharged;

b) application of input or charging currents within the manufacturer's recommendations;

	c) charging a battery within the manufacturer's recommendations with one cell fully discharged or suffering from polarity reversal. The conditions above shall include any reverse charging due to conditions arising from the application of 5.1 and 5.2. They shall not include the use of an external charging circuit which exceeds the charging rates recommended by the manufacturer of the cell or battery. There shall be no visible sign of electrolyte on the blotting paper or on the external surfaces of the test samples when they have cooled. Where casting compound has been applied to achieve conformance to [7.4.9] examination of the cell at the end of the test shall show no damage which would invalidate conformance with [7.4.9]. [10.9.3] Spark ignition and surface temperature of cells and batteries If a battery comprises a number of discrete cells or smaller batteries combined in a well defined construction conforming to the segregation and other requirements of this standard, then each discrete element shall be considered as an individual component for the purpose of testing. Except for specially constructed cells where it can be shown that short circuits between cells cannot occur the failure of each element will be considered as a single fault. In less well defined circumstances the battery will be considered to have a short circuit failure between its external terminals. Cells and batteries which conform to [7.4.9] shall be tested or assessed as follows. a) Spark ignition assessment or testing shall be carried out at the cell or battery external terminals except that where a current limiting device is included and the junction of this device and the cell or battery conforms to clause [6.7], the test or assessment shall include the current limiting device. When the internal resistance of a cell or battery is to be included in the assessment of intrinsic safety its minimum resistance value shall be obtained in writing from the manufacture of the cell/battery and that correspondence shall be made available to the testing station. If the cell/battery manufacturer is unable to confirm the minimum value of internal resistance the testing station shall use the most onerous value of short circuit current from a test of 10 samples of the cell battery together with the peak open circuit voltage in accordance with [7.4.3].of the cell/battery to determine the internal resistance. b) The maximum surface temperature shall be determined as follows. All current limiting devices external to the cell or battery shall be short circuited for the test. Any external sheath (of paper or metal etc) not forming part of the actual cell enclosure shall be removed for the test. The temperature shall be determined on the outer enclosure of each cell or battery and the maximum figure taken. The test shall be carried out both with internal current limiting devices in circuit and with them short circuited using 10 cells in each case. The 10 samples having the internal current limiting devices short circuited shall be obtained from the cell/battery manufacturer together with any special instructions or precautions necessary for safe use and testing of the samples. NOTE: When determining the surface temperature of most batteries, the effect of built in protective devices, e.g. fuses or PTC resistors, is not taken into account because this is an assessment of a possible internal fault e.g. failure of a separator. 10.11 Tests for apparatus containing piezoelectric devices

	Measure both the capacitance of the device and also the voltage appearing across it when any part of the apparatus which is accessible in service is impact tested in accordance with the 'high' column of [table 4 in EN 50014:1992] carried out at 20 ± 10 °C using the test apparatus in [annex D of EN 50014:1992]. For the value of voltage, the higher figure of two tests on the same sample shall be used. When the apparatus containing the piezoelectric device includes a guard to prevent a direct physical impact, the impact-test shall be carried out on the guard with both it and the apparatus mounted as intended by the manufacturer. The maximum energy stored by the capacitance of the crystal at the maximum measured voltage shall not exceed the following; for Group I apparatus: 1500 µJ for Group IIA apparatus: 950 µJ for Group IIB apparatus: 250 µJ for Group IIC apparatus: 50 µJ Where the electrical output of the piezoelectric device is limited by protective components (including guards) these components shall not be damaged by the impact in such a way as to allow this type of protection to be invalidated. Where it is necessary to protect the apparatus from external physical impact in order to prevent the impact energy exceeding these values details of the requirements shall be specified as special conditions for safe use and the apparatus shall be marked with the sign X. [10.12] Type tests for diode safety barriers & safety shunts The following tests are used to demonstrate that the safety barrier or safety shunt can withstand the effects of transients. Infallibly rated resistors shall be considered to be capable of withstanding any transient to be expected from the specified supply. The diodes shall be shown to be capable of withstanding the peak U_m divided by the value (at 20 °C) of the fuse resistance and any infallible resistance in series with the fuse, either by the diode manufacturer's specification or by the following test: Subject each type of diode in the direction of utilisation (for zener diodes the zener direction) to five rectangular current pulses each of 50 µs duration repeated at 20ms intervals. With a pulse amplitude of the peak of the U_m divided by the 'cold' resistance value of the fuse at 20 °C (plus any infallible series resistance which is in circuit). Where the manufacturer's data shows a pre-arcing time greater than 50µs at this current the pulse width will be changed to represent the actual pre-arcing time. Where the pre-arcing time cannot be obtained from the available manufacturers data, ten fuses shall be subjected to the calculated current and their pre-arcing time measured. This value, if greater than 50µs shall be used. The diode voltage shall be measured at the component manufacturer's test current before and after this test. The measured voltages shall not differ by more than 5% (the 5 % includes the uncertainties of the test apparatus). The highest voltage elevation observed during the test shall be used as the peak value of a series of pulses to be applied in a similar manner as above to any semiconductor current limiting devices. After testing these devices shall again be checked for conformance to the component

manufacturer's specification.

From a generic range manufactured by a particular manufacturer it is necessary to test only a representative sample of a particular voltage to demonstrate the acceptability of the generic range.

[10.13] Cable pull test

Apparatus which is constructed with an integral cable for external connections shall be subjected to a pull test on the cable if breakage of the terminations inside the apparatus could result in intrinsic safety being invalidated; for example where there is more than one intrinsically safe circuit in the cable and breakage could lead to an unsafe interconnection. The test shall be carried out as follows.

Apply a tensile force of 30 N on the cable in the direction of the cable entrance into the apparatus for the duration of 1 h.

Although the cable sheath may be displaced no visible displacement of the cable terminations shall be observed.

This test shall not be applied to individual conductors which are permanently connected and do not form part of a cable.

[11 Routine verifications and tests

[11.1] Routine tests for diode safety barriers

[11.1.1] Completed barriers

A routine test shall be carried out on each completed barrier to check correct operation of each barrier component and the resistance of any fuse. The use of removable links to allow this test shall be acceptable provided that intrinsic safety is maintained with the links removed.

[11.1.2] Diodes for 2-diode 'ia' barriers

The voltage across the diodes shall be measured as specified by their manufacturer at ambient temperature before and after the following tests.

a) Subject each diode to a temperature of 150 °C for 2 h.

b) Subject each diode to the pulse current test in accordance with [10.12].

[11.2] Routine test for mains transformers

In routine tests, the voltages applied to mains transformers shall conform to the values given in [table 9] where U_n is the highest rated voltage of any winding under test.

[Table 9] - Routine test voltages for mains transformers

Where applied	r.m.s. test voltage
Between input and output windings	$4U_n$ or 2500V, whichever is greater

	Between all the windings and the core or screen ¦ $2U_n$ or 1000V, whichever is greater --+--- Between each winding which supplies an intrinsically safe circuit and any other output winding ¦ $2U_n$ + 1000v OR 1500v, whichever is greater --- During these tests there shall be no breakdown of the insulation between windings or between any winding and the core or screen.
10	replace by: 12. Marking 12.1 <u>General</u> Intrinsically safe apparatus and associated apparatus shall carry at least the minimum marking specified in EN 50014. The marking of the serial number may be achieved by using a date or batch number code which is sufficient to ensure traceability for quality control purposes. NOTE: The serial number marking may be separate from the other marking. For associated apparatus the symbol EEx ia or EEx ib (or ia or ib if EEx is already marked) shall be enclosed in square brackets. All relevant parameters should be marked e.g. U_m, L_i, C_i, L_o, C_o, wherever practicable. NOTE: Standard symbols for marking and documentation are given in clause 3. Practical considerations may restrict or preclude the use of italic characters or of subscripts and a simplified presentation may be used, for example Uo rather than U_o. [12.2] <u>Marking of connection facilities</u> Connection facilities, terminal boxes, plugs and sockets of intrinsically safe apparatus and associated apparatus shall be clearly marked and shall be clearly identifiable. Where a colour is used for this purpose, it shall be light blue. Where parts of an apparatus or different pieces of apparatus are interconnected using plugs and sockets, these plugs and sockets shall be identified as containing only intrinsically safe circuits. Where a colour is used for this purpose it shall be light blue. In addition, sufficient and adequate marking shall be provided to ensure correct connection for the continued intrinsic safety of the whole. NOTE: It may be necessary to include additional labels, e.g. on or adjacent to plugs and sockets, to achieve this. If clarity of intention is maintained the apparatus label may suffice. The following are examples of marking.

Section 3

National Differences

a) Self-contained intrinsically safe apparatus

C. TOME LTD
PAGING RECEIVER TYPE 3
EEx ia IIC T4
-25 °C ≤ Ta ≤ +50 °C
ACB Ex95****
Serial No: XXXX

c) Associated apparatus

J SCHMIDT AG
STROMVERSORGUNG TYP 4
[EEx ib] I
ACB 95*.****
Um: 250 V   Po: 0,9 W
Io: 150 mA Uo: 24 V
Lo: 20 mH   Co: 5,5 µF

b) Intrinsically safe apparatus designed to be connected to other apparatus

M HULOT
TRANSDUCTEUR TYPE 12
EEx ib IIB T4
ACB No: Ex95****
Li: 10 µH   Ci: 1200 pF
Ui: 28 V   Ii: 250 mA
Pi: 1,3 W

d) Associated apparatus protected by an EEx d enclosure

PIZZA ELECT. SpA
EEx d [ia] IIB T6
ACB No: Ex95****
Um: 250 V   Po: 0,9 W
Uo: 36 V   Io: 100 mA
Co: 0,31 µF Lo: 15 mH
Serial No: XXXX

where:
ACB represents the initials of the certifying body.

13 Documentation

The descriptive documentation required by [EN 50014, clause 23.2] shall include the following information:

a) electrical parameters for the apparatus

1) <u>power</u> <u>sources</u> - output data such as - U_o, I_o, P_o and, if applicable. C_o, L_o and/or the permissible L_o/R_o ratio

2) <u>power</u> <u>receivers</u> - input data such as - U_i, I_i, P_i, C_i, L_i and the L_i/R_i ratio

b) any special requirements for installation and use.

c) the maximum value of U_m which may be applied to terminals of non-intrinsically safe circuits of associated apparatus.

d) any special conditions which are assumed in determining the type of

	protection, for example that the voltage is to be supplied from a protective transformer or through a diode safety barrier. e) conformance or non-conformance with [6.4.12]. f) the designation of the surfaces of any enclosure only in circumstances where this is relevant to intrinsic safety.
Annex B	Replaced by [6.4.4] and [6.7] - see above.
Figure 4	To be replaced by [Figure A2]
Include Figures:	[Figure 1] Clearances and creepage distance requirements for terminals carrying separate intrinsically circuits [Figure 2] Examples of independent and non-independent connecting elements [Figure 3] Separation of conductive parts [Figure 4] Determination of creepage distances (in air) [Figure D1] Examples of encapsulated assemblies con forming to [6.4.4] and [6.7] [Figure D2] Application of encapsulation without enclosure

KR Korea

IEC Clause or Annex Number	Details of Differences to IEC 60079-11 3rd Edition

The requirements of Group I do not apply.

6.1.1 This requirement does not apply.

6.1.3.2 This requirement does not apply.

6.1.3.3 This requirement does not apply.

6.1.3.4 This requirement does not apply.

8.5 This requirement does not apply.

8.7.1 This requirement does not apply.

8.8 The requirement of optocuplers does not apply.

8.9.2 This requirement does not apply.

8.9.3 This requirement does not apply.

9.1.5 The requirement of the safety factor achieved by using explosive test mixture does not apply.

9.5 This requirement does not apply.

Annex A This requirements does not apply.

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-15

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IECEx Bulletin – Section 3: National Differences for 60079-15

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3

National Differences

IEC Standard 60079-15 4th Edition 2010 Electrical apparatus for explosive gas atmospheres Part 15: Electrical apparatus with type of protection "n"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-15:2010
CN China		**YES**	GB3836.9-2006 modified IEC 60079-15: 2001
CZ Czech Republic	**NO**		EN 60079-15:2010
DE Germany	**NO**		EN 60079-15:2010
DK Denmark	**NO**		EN 60079-15:2010
FI Finland	**NO**		EN 60079-15:2010
FR France	**NO**		EN 60079-15:2010
GB United Kingdom	**NO**		EN 60079-15:2010
HR Croatia	**NO**		EN 60079-15:2010
HU Hungary	**NO**		EN 60079-15:2010
IN India		**N/R**	
IT Italy	**NO**		EN 60079-15:2010
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-15:XXXX (AWAITING TO BE PUBLISHED AS A NEW STANDARD FOR PART 15)
NL The Netherlands	**NO**		EN 60079-15:2010
NO Norway	**NO**		EN 60079-15:2010
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-15:2010
RO Romania	**NO**		EN 60079-15:2010
RU Russia*		**TBA**	Included in the national standardization program for 2009
SE Sweden	**NO**		EN 60079-15:2010
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-15:2010
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-15:2009

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Countries that have declared adoption of IEC 60079-15 4th Edition 2009 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-15 4th Edition 2009 With National/Group Differences- Differences follow.

CN China

IEC Standard Number IEC 60079-15:2009 Ed4.0

IEC Clause or Annex Number	Details of Differences

The details of national differences are differences between GB3836.8-2006 and IEC60079-15 Ed3.0 plus the technical changes from IEC60079-15 Ed3.0 to IEC60079-15 Ed4.0.

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IEC Technical Report IEC 60079-15 3rd Edition 2005 Electrical apparatus for explosive gas atmospheres Part 15: Electrical apparatus with type of protection "n"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-15 2006
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-15:11
CH Switzerland	**NO**		EN 60079-15:2005
CN China		**YES**	GB3836.9-2006 modified IEC 60079-15: 2001
CZ Czech Republic	**NO**		EN 60079-15:2005
DE Germany	**NO**		EN 60079-15:2005
DK Denmark	**NO**		EN 60079-15:2005
FI Finland	**NO**		EN 60079-15:2005
FR France	**NO**		EN 60079-15:2005
GB United Kingdom	**NO**		EN 60079-15:2005
HR Croatia	**N/A**		
HU Hungary	**NO**		EN 60079-15:2005
IN India		**NO**	Adopted as IS/IEC 60079-15: 2005
IT Italy	**NO**		EN 60079-15:2005
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-15:2005
NL The Netherlands	**NO**		EN 60079-15:2005
NO Norway	**NO**		EN 60079-15:2005
NZ New Zealand		**NO**	AS/NZS 60079-15 2006
PL Poland	**N/A**		
RO Romania	**NO**		EN 60079-15:2005
RU Russia		**NO**	GOST R 52350.15-2005
SE Sweden	**NO**		EN 60079-15:2005
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 60079-15:2005
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-15 2009 ANSI/UL 60079-15 2009
ZA South Africa		**NO**	SANS 60079-15:2005

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Countries that have declared adoption of IEC 60079-15 3rd Edition 2005 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
ZA South Africa

Countries that have declared adoption of IEC 60079-15 3rd Edition 2005 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-15:11
Electrical apparatus for explosive atmospheres –
Part 15: Construction, test and marking of type of protection "n" electrical apparatus

This is the third edition of CAN/CSA-C22.2 No. 60079-15, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-15 (Third edition, 2005-03). It replaces the CAN/CSA-C22.2 No.60079-15:02 (adopted CEI/IEC 60079-15: Second edition, 2001-02: Entitled *Electrical apparatus for explosive gas atmospheres – Part 15: Type of protection "n"*).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

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[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11.

6.7.3 Conformal coating

[Add a paragraph/note to the following effect]

NOTE: A solder mask can be considered as one the two coatings required as far as the printed traces are of concern. The soldering pads/points (where the solder flows in the soldering process) has only one coating only; therefore, unless the spacing between soldering pads complies with the "in-air" spacing requirements, the solder mask cannot be considered as one of the two overall (covering all live parts) coatings .

14.3 Internal connection facilities

[Add a NOTE to the following effect]

NOTE: Soldering is not permitted as a means of connection for the earth (ground) conductor.

17.1 General – Supplementary requirements for non-sparking electrical machines

[Add a paragraph to the following effect]

Currently, the Canadian Electric Code, Part I (C22.1) permit the use of general-purpose rotating machines that do not incorporate arcing, sparking or excessive heat-generating parts. This type of "general-purpose" rotating machines may not be compatible or meeting the requirements of an "n"-protected rotating machines. Care is to be taken when evaluating a rotating machine to the "n" requirements and so marked.

35.3 Examples of marking

[Add a NOTE to Example 2]

NOTE: The marking of "II" only, (without the A, B and/or C designations), has been removed in the IEC 60079-0 (Edition 5.0, 2007-10) standard. For the gas group markings, it is to be IIA, IIB and/or IIC.

35.3.1 Warning markings

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

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CN China

IEC Standard Number IEC 60079-15:2005 Ed3.0…..

IEC Clause or Annex Number	Details of Differences

1 Scope
Linking the standard to IEC 60079-0definition of associated energy limiting apparatus [nL] and [Ex nL] do not apply

12 Materials used for cementing
Requirements of thermal endurance test for cement do not apply

14.3 Internal connection facilities
Provision for internal connection method does not apply

17.7.3 Assessment for possible air gap sparking
Air gap spark test requirement for motors over 100 kW does not apply

17.8.2 Operation with a frequency convertor or a non-sinusoidal supply
Type test methodsdo not apply

17.9 Additional requirements for machines with rated voltage greater than 1 kV ..
This clause does not apply

20.2 Maintaining degree of protection
The requirement relaxation for plugs and sockets for rated currents not exceeding 10 A and rated voltage not exceeding either 250 V a.c. or 60 V d.c. does not apply

21.2.5.3 Electronic starters and ignitors
the endurance test for ignitors does not apply

21.2.5.5 Ballasts
Test requirements for ballasts are different.

23 Supplementary requirements for non-sparking low power apparatus
Table 10 – Minimum creepage distances, clearances and separations for low power apparatus does not apply

33.8 Test for screw lamp holders
Minimum removal torque for E40 in Table 12 does not apply.

33.10.4.1 Ignitor thermal endurance test
Test condition for Ignitor thermal endurance test does not apply

33.14 Additional ignition tests for large or high-voltage machines
Ignition tests for large or high-voltage machines do not apply

Add supplementary requirements for “n-pressure”
Add high voltage test for ballast used with ignitors
Add third party test

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US United States of America

Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
General	Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
General	All references to "X" conditions have been replaced with references to ANSI/ISA 60079-0 where the requirements for the "X" marking after the certificate reference have been replaced with specific instructions or by reference to specific instructions
General	To not conflict with the National Electrical Code® definition of "device", references made to "device" or "devices", are replaced with "apparatus", "component", or "apparatus or component", as applicable.
General	Where references are made to the designation, "Ex nL", for associated energy-limited apparatus, the reference is replaced by "AEx nL", noting that the marking (per 35.1) is specified as "AEx nC" to align with the National Electrical Code®.
1	Modified the text of the first paragraph as follows to align with the National Electrical Code®: This standard ~~part of IEC 60079~~ specifies requirements for the construction, testing and marking for Group II electrical apparatus with type of protection, "n" intended for use in ~~explosive gas atmospheres~~ Class I, Zone 2 hazardous (classified) locations as defined by the National Electrical Code® (NEC®), ANSI/NFPA 70.
1 Table 1	Where reference is made to "part", that reference is replaced with "standard". Delete note b) and add Note 4 at the end of the Table to account for the fact that the National Electrical Code® does not include type of protection "nL" and marks the energy limitation concept as "nC": ~~Type of protection nC includes encapsulated devices, enclosed break devices, non-incendive components, sealed devices and hermetically sealed devices.~~ NOTE 4 The National Electrical Code® (NFPA 70) through the 2008 edition only recognizes types of protection "nA", "nC" and "nR"; and not "nL". See note in 35.1.
2	References to the following standards are added / removed to align with changes throughout the document: *ANSI/NFPA 70, National Electrical Code®*

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
	ANSI/UL 248-1, Low Voltage Fuses, Part 1 General Requirements *ANSI/UL 486A, Wire Connectors* *ANSI/UL 486B, Wire Connectors* *ANSI/UL 746A, Standard for Polymeric Materials - Short Term Property Evaluations* ANSI/IEC 60529:2004:~~1989~~ ~~IEC 60034 (all parts), *Rotating electrical machines*~~ ~~IEC 60034-7, *Rotating electrical machines – Part 7: Classification of type of construction, mounting arrangements and terminal box position (IM Code)*~~ ~~IEC 60034-25, *Rotating electrical machines – Part 25: Guide for the design and performance of cage induction motors specifically designed for converter supply*~~ ~~EN 50262, *Metric cable glands for electrical installations*~~
3.8	The term and definition for “sealing device” were removed to eliminate confusion with “sealed device” (3.9.2.5). **3.8** ~~**sealing device**~~ ~~device to prevent the flow of a gas or a liquid between apparatus and a conduit by providing sealing facilities~~

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
3.10	The following term and definition were added based on the definition in IEC 60079-0, Edition 5. **normal operation** operation of apparatus conforming electrically and mechanically with its design specification and used within the limits specified by the manufacturer [based on ANSI/ISA-60079-0] NOTE 1 The limits specified by the manufacturer may include persistent operational conditions, e.g. operation of a motor on a duty cycle. NOTE 2 Variation of the supply voltage within stated limits and any other operational tolerance is part of normal operation.
6.7.1	The following was added to the list of exceptions to clarify the requirement: portions of circuit boards meeting the voltage and power requirements of Clause 23 and separated from other circuits by the values given in Table 2.
6.7.4	The text in the second paragraph was modified as follows to recognize the US standard for CTI: Table 3 gives the grouping of electrical insulating materials according to the CTI determined in accordance with IEC 60112 or ANSI/UL 746A.
14.3	The following bullet was modified as follows to recognize the US standards for twist-on connecting devices: twist-on connecting devices meeting the requirements of IEC 60998-2-4 or that comply with ANSI/UL 486A-486B
17	The following note was added before Section 17.1 to highlight the permission granted by the National Electrical Code®: NOTE Article 505 of the National Electrical Code® (NFPA 70) permits some general purpose, squirrel-cage induction motor constructions to be used in Zone 2 locations, see 505.20 (C) Exception 4.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
17.1	Notes 1, 4, 5, 6 and 7 were added as follows to add clarification and are based on the notes proposed to appear in IEC 60079-15, 4th Edition: NOTE 1 The requirements of this standard assume that the occurrence of a flammable gas atmosphere and a motor start sequence do not occur simultaneously, and may not be suitable in those cases where these two conditions do occur simultaneously. NOTE ~~1~~2 If certification (third party) is sought, it is not a requirement of this standard that the certification body confirm conformance to IEC 60034 (series). The manufacturer should state the basis of compliance in the documentation, see Clause 36. NOTE ~~2~~3 The requirements of IEC 60034-5 replace those of 6.6. NOTE 4 Type 'n' high-voltage motors should not be used where the probability of a gas release cannot be totally disassociated with the start sequence as an independent event. The seal systems of centrifugal compressors are known to produce such releases during starting and should be subject to assessment. Seal or lubricating oil systems shared between a motor and its driven compressor are not recommended. NOTE 5 Normal operating conditions for electric machines are assumed to be rated full-load steady conditions. NOTE 6 Starting (acceleration) of electric machines is excluded as part of normal operation under duty S1 or S2. NOTE 7 Due to the potential for more frequent starts of motors with duty S3 to S10, the requirements for rotor sparking address the risk of rotor sparking during starting as a "normal" condition.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
17.4	The text of the first paragraph was modified as follows to add clarification and is based on the text proposed to appear in IEC 60079-15, 4th Edition: ~~The minimum~~ A radial air gap between stator and rotor ~~(in mm), when the rotating electrical machine is at rest, shall not be less than the value calculated using the following equation~~ shall be specified to avoid contact and demonstrated by one of the following means: a) Measurement of the radial air gap of the test sample b) Calculation of the minimum radial air gap NOTE 1 It is acknowledged that, with assemblies, all parts will not exist at the worst case dimensions simultaneously. A statistical treatment of the tolerances, such as “RMS”, may be required gap verification. NOTE 2 It is not a requirement of this standard that the manufacturer’s gap calculations be verified. Also, it is not a requirement of this standard that gap be verified by measurement. c) Construction in accordance with the following equation:
17.5	The entire section was removed to align with the existing permission in the National Electrical Code®.
17.6	The entire section was removed to align with the existing permission in the National Electrical Code®.
17.7.1	The text was modified as follows to add clarification and is based on the text proposed to appear in IEC 60079-15, 4th Edition: ~~Precautions shall be taken to guard against incendive arcs or sparks during normal operation of the rotating electrical machine. In particular, t~~The joints between bars and short-circuiting rings shall be brazed or welded and compatible materials shall be used to enable high quality joints to be made.
17.7.2	The text was modified as follows to add clarification and is based on the text proposed to appear in IEC 60079-15, 4th Edition: Cast rotor cages shall be made by pressure die-casting or centrifugal casting or equivalent techniques ~~designed to ensure the complete filling of the slot~~.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
17.7.3	The text was modified as follows to add clarification and is generally based on the text proposed to appear in IEC 60079-15, 4th Edition: Rotating electrical machines <u>of other than a duty type S1 or S2,</u> with a rated output exceeding 100 kW shall be assessed for possible air gap sparking <u>as follows</u>. ~~Motors with a duty type S1 and S2 which are running continuously with an average starting frequency in normal operation not exceeding one start per week are excluded from these requirements.~~ If the total sum of the factors determined by Table 6 is greater than ~~5,~~6<u>, one of the following shall be applied:</u> a) the machine or a representative sample shall be tested in accordance with ~~33.14.1~~<u>33.14.2</u>; or b) the machine shall ~~be constructed to~~ allow special measures to be ~~employed~~<u> applied</u> to ensure that its enclosure does not contain an explosive gas atmosphere at the time of starting. ~~T~~<u>In this case</u>, the machine marking shall ~~include the symbol "X",~~ <u>be</u> in accordance with ~~IEC~~ <u>29.2 (i) of ANSI/ISA</u>~~IEC~~-60079-0~~,~~ and the ~~special~~<u>Specific</u> ~~c~~<u>Conditions of Use</u> to be employed shall be specified in the documentation as required by Clause 36~~.~~<u>; or</u> c) the starting current of the machine is required to be limited to 300% of rated current, I*N*. In this case, the machine marking shall be in accordance with 29.2 (i) of <u>ANSI/ISA</u>~~IEC~~-60079-0 and the Specific Conditions of Use shall include that the motor is suitable only for reduced voltage starting which limits the starting current to 300% of the rated current. <u>NOTE 1 The use of a converter to provide the current limitation is generally an accepted solution. For other reduced voltage starting methods, the motor and the reduced-voltage starter need to be carefully coordinated.</u> NOTE <u>2</u> Special measures that can be applied include pre-start ventilation <u>to remove any ignitable accumulation of flammable gases</u> or the application of fixed gas detection <u>(see 60079-29-2)</u> inside the machine enclosure <u>to confirm that the machine is free of ignitable concentrations of flammable gases</u>. Other methods may be applied with the agreement of the manufacturer and the user.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
17.7.3 Table 6	The text of the first three entries in the "Value" column was modified as follows to add clarification and is generally based on the text proposed to appear in IEC 60079-15, 4th Edition: Uninsulated bar f ~~F~~abricated rotor cage Open slot c ~~C~~ast aluminium rotor cage ≥ 200 kW per pole Open slot c ~~C~~ast aluminium rotor cage < 200 kW per pole
17.8.2.3	Note 2 was removed based on IEC 60079-15, 4th Edition: NOTE ~~1~~ The determination of the temperature class by calculation should be agreed between the manufacturer and the user as appropriate. ~~NOTE 2 The temperature differential between stator and rotor of a machine operating with a non-sinusoidal supply, or generating into a thyristor load, may vary greatly from the temperature differential that would occur on the same machine operating with a sinusoidal supply, or generating into a linear load. Therefore special attention needs to be paid to the rotor temperature which may be a limiting feature of the machine, particularly in the case of rotor cage windings.~~
17.9	The entire section, including Table 7, was removed as is proposed for IEC 60079-15, 4th Edition.
17.10	The text of the section was modified as follows to add clarification and is generally based on the text proposed to appear in IEC 60079-15, 4th Edition: **17.10 Stator winding insulation system** If the rated voltage exceeds 1 kV for random-wound stators or 6,6 kV for form-wound stators, type tests in accordance with ANSI/ISA-60079-0 shall be conducted, and the machine shall be fitted with anti-condensation heaters. NOTE It is recommended that partial discharges be minimized for all high-voltage windings. For windings with a rated voltage of 6,6 kV, or greater, the use of partial discharge suppressant materials is recommended.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
19.1	The text of the second paragraph was modified as follows to recognize the US standard for fuses: Fuses shall be deemed non-sparking apparatus devices if they are non-rewirable, non-indicating cartridge types or indicating cartridge types, according to IEC 60269-3 or ANSI/UL 248-1, operating within their rating.
20.1	The text of paragraph b) was modified as follows to clarify the text: b) ~~if they are allocated and connected to only one apparatus,~~ they shall be secured mechanically to prevent unintentional separation and the apparatus shall be marked with the warning given in item b) of Table 13.
21.2.8.3	The text of paragraph c) was modified as follows to recognize the US standards for twist-on connecting devices: c) screwless terminals of the following types: 1) those that comply with the relevant clause of IEC 60598-1, except for spring-leaf terminals of the type shown in Figure 2b) of this standard; 2) the "acceptable" type of spring-leaf terminal with the conductor clamped between metal surfaces as shown in Figure 2a) for circuits meeting the relevant requirements for nonpermanent connections using spring type terminals complying with 15.5 of IEC 60598-1, together with an additional type test consisting of pulling on the conductor with a force of 15 N for 1 min during which period the conductor shall not move from the terminals, damage to the conductor being disregarded; 3) twist-on connecting devices meeting the requirements of IEC 60998-2-4 or that comply with ANSI/UL 486A-486B; 4) insulated crimped connectors.
23	The text of the first paragraph was modified as follows as the IP54 enclosure was considered suitable for pollution degree 3: Electronic and allied low power apparatus, assemblies and sub-assemblies used, for example, for measurement, control or communication purposes, ~~used in an area of not more than pollution degree 2, as defined in IEC 60664-1, and~~ which do not comply with 6.7 and 6.8.2 shall comply with the following:

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
33.5.1	The text of the first paragraph was modified as follows to add clarification and is generally based on the text proposed to appear in IEC 60079-15, 4th Edition: The ~~device~~ apparatus shall be conditioned energized at rated voltage in an air oven for 7 days at a temperature at least 10 K higher than *T*amb max, or ~~at a temperature which achieves~~ the maximum service temperature ~~*Tc*~~ + 10 K or un-energized at (80 ±2) °C, whichever is the ~~greater~~ greatest, followed by 1 day at 10 K lower than the minimum rated service temperature.
33.14.2.4	The entire section was removed as test was considered too severe for Zone 2.
34.2.3	The entire section was removed as a routine test was not considered necessary.
35.1	The text of the second paragraph was modified as follows as a non-incednive component also requires marking of the gas group: Gas groups shall be marked for apparatus protected by types of protection energy-limited, associated energy-limited, non-incendive components and enclosed-break apparatus devices. The paragraph and note were added to account for the fact that the National Electrical Code® does not include type of protection "nL" and marks it as "nC": All components and apparatus complying with the requirements for "nL", "[nL]|, or "nAnL" shall be marked "nC", "[nC]", and "nC" respectively NOTE Type of protection nL is not recognized in NEC® editions through 2008 and therefore, an accepted marking such as "nC" is used.

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
35.3	Example 1 was modified as follows to align with the changes made to IEC 60079-0 marking: ABC Industries Ltd Type HXR Class I Zone 2 AEx nA d IIB T3 –20 °C ≤ *T*a ≤ +60 °C Certificate number: 045673X Example 2 was modified as follows to align with the changes made to IEC 60079-0 marking: XYZ Ltd Type 1456 Class I Zone 2 AEx nR II Certificate number: 986U Example 3 was modified as follows to align with the changes made to Clause 35 and to IEC 60079-0 marking: G. Schwarz A.G. Model FUb69 Class I Zone 2 AEx nA ~~nL~~ nC IIC T4 IECEx-04.3412 Example 4 was modified as follows to align with the changes made to Clause 35 and to IEC 60079-0 marking: G. Smith Inc 1 000 CV transmitter Class I Zone 2 AEx ~~nL~~ nC IIB T5 *U*i= 30 V ; *I*i= 20 mA ; *P*i= 1 W ; *C*i= 30 nF ; *L*i= 1 mH Certificate No. Ex 04.16542 Example 5 was modified as follows to align with the changes made to Clause 35 and to IEC 60079-0 marking: K Chambers LLC PSU Type 54 Class I Zone 2 AEx nA ~~[nL]~~ [nC] IIC T5 *U*o= 30 V ; *I*o= 20 mA ; *P*o= 1W ; *C*o= 100 nF ; *L*o= 10 mH Certificate No. IECEx 05.9876

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Clause	US Differences to IEC 60079-15, 3rd Ed, 2005
36	The following requirement for additional documentation was modified to align with changes to 17.7.3: ~~information where special measures are to be employed to ensure that the enclosure of a large rotating machine rated over 100 kW does not contain an explosive gas atmosphere at the time of starting (see 17.7.3);~~ For motors with duty of S3 to S10, information where special measures are to be employed to ensure that the enclosure of a large rotating machine rated over 100 kW does not contain an explosive gas atmosphere at the time of starting (see 17.1)
37	The text of the second paragraph was modified as follows as a only the equipment that includes a test port can be checked and it was not correct for the “product” standard to set the “user” requirement for a checking interval when the actual operating conditions are not known: The instructions for restricted breathing apparatus <u>equipped with a test port</u> shall include requirements for routine checking ~~at intervals of six months or as experience dictates~~ using the test requirements in 33.7.1.

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IEC Technical Report IEC 60079-15 2nd Edition 2001 Electrical apparatus for explosive gas atmospheres Part 15: Electrical apparatus with type of protection "n"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**N/R**	
BR Brazil		**N/R**	
CA Canada		**NO**	CAN/CSA-E6007915-2001
CH Switzerland	**YES**		EN 60079-15:2003
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 60079-15:2004
DE Germany	**YES**		EN 60079-15:2003
DK Denmark	**YES**		EN 60079-15:2003
FI Finland	**YES**		EN 60079-15:2003
FR France	**YES**		EN 60079-15:2003
GB United Kingdom	**YES**		BS EN 60079-15:2003
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 60079-15:2003
IN India		**N/R**	
IT Italy	**YES**		EN 60079-15:2003
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-15:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**YES**		
NO Norway	**YES**		
NZ New Zealand		**N/R**	
PL Poland	**N/A**		
RO Romania	**YES**		EN 60079-15:2003
RU Russia		**NO**	GOST R 51330.14:99
SE Sweden	**YES**		SS-EN 60079-15:2003
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 60079-15:2003
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-15:2001

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-15 2nd Edition 2001 Without National Differences
CA Canada
JP Japan
MY Malaysia
RU Russia
ZA South Africa

Countries that have declared adoption of IEC 60079-15 2nd Edition 2001 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Endorsement notice

The text of the International Standard IEC 60079-15:2001 was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

1 Scope

Replace the text of Clause 1 by:

This European Standard specifies requirements for the construction, testing and marking for Group II electrical apparatus with type of protection 'n', intended for use only in areas where explosive atmospheres of gas, vapour and mist are unlikely to occur or if they do occur, are likely to do so infrequently or for a short period only.

This standard is applicable to non-sparking electrical apparatus and also to apparatus with parts or circuits producing arcs or sparks or having hot surfaces which, if not protected in one of the ways specified in this standard, could be capable of igniting a surrounding explosive atmosphere.

A non-incendive component is limited in use to the particular circuit for which it has been shown to be non-ignition capable and, therefore, cannot be separately assessed as complying with this standard.

This standard is applicable to electrical equipment and components of Group II, Category 3G. Such equipment is designed to be capable of functioning in conformity with the operating parameters established by the manufacturer and ensuring a normal level of protection.

Equipment in this category ensures the requisite level of protection during normal operation.

NOTE 1 In this standard the word 'apparatus' has the same meaning as the word 'equipment' used in the Directive.

Compliance with this European Standard does not imply any removal of, or lowering of, the requirements of any other European Standard with which the electrical apparatus complies.
This standard supplements, and may enhance, the requirements for apparatus for normal industrial applications.

NOTE 2 The application of this standard may require the exercise of engineering judgement because of the wide range of apparatus and technologies covered. If apparatus is to be certified the relevant requirements may need to be agreed between the manufacturer and the testing station.

NOTE 3 This standard makes several specific references to IEC 60079-0. It is not intended that apparatus with type of protection "n" should comply with IEC 60079-0 in its entirety, or that the level of protection achieved by compliance with this standard should be equal to the level of protection achieved by compliance with IEC 60079-0 and any of the types of protection listed therein.

3 Definitions

3.10 Ex component

Add the following note to the definition:

NOTE Directive 94/9/EC defines a component as:
Any item essential to the safe functioning of equipment and protective systems but with no autonomous function.
In this standard, the words 'Ex component' have the same meaning as the word 'component used in the Directive'.

19 Supplementary requirements for hermetically sealed devices producing arcs, sparks or hot surfaces

Add the following note after the first paragraph.

NOTE A leakage rate equivalent to a He-leakage rate less than 10^{-2} Pa*l/sec (10^{-4} mbar*l/sec) at a pressure difference of 10^5 Pa (1 bar) is sufficient.

21 Supplementary requirements for energy-limited apparatus and circuits producing arcs, sparks or hot surfaces

21.8.1 Ratings of components

Replace the text of 21.8.1 by:
Any component on which the type of protection depends, except such devices as transformers, fuses, thermal trips, relays and switches, shall either

- have a failure mode such that protection is maintained or,

- not operate in normal conditions at more than two thirds of their maximum current, voltage and power related to the rating of the device, the mounting conditions and the temperature range specified. These maximum rated values shall be those specified by the manufacturer of the component.

NOTE Normal operation may include open-circuit, short-circuit and earth fault conditions at the field terminals.

22 Supplementary requirements for restricted-breathing enclosures protecting apparatus producing arcs, sparks or hot surfaces

Replace the text of Clause 22 by:
Protection by restricted breathing enclosures may be applied under the following two circumstances with differing test and provision for maintenance according to the particular type.

a) Enclosures containing sparking contacts but with a limitation in dissipated power such that the averaged air temperature within the enclosure does not exceed the external ambient temperature by more than 10 K. However the internal air temperature may exceed the external ambient temperature by up to 20 K if the rate of temperature decay, when the apparatus is de-energized, is limited to not more than 10 K/h.

b) Enclosures not containing sparking contacts with limitation only in external surface temperature.

NOTE 1 The use of a restricted-breathing enclosure to protect against ignition from sparking contacts is not allowed where, because of high internal air temperatures, there is an increased risk of drawing the hazardous atmosphere into the enclosure when the apparatus is de-energized.

NOTE 2 The effects of the sun's direct heating on the exterior of the enclosure should be taken into account. This can cause a larger internal temperature change than the 10 K allowed.

NOTE 3 Restricted breathing is not suitable for equipment operated on a short time duty cycle because of the increased probability that the apparatus might be de-energized when flammable gas or vapour surrounds the enclosure.

22.2 Equipment of type 22.1 a) shall be provided with a test point to enable routine testing of the restricted breathing properties to be carried out after installation and during routine maintenance. It shall be subject to the type test of 26.8.1.
22.3 Equipment of type 22.1 b) shall either be considered as being of type 22.1 a) or shall omit the test point and be subject to the type test of 26.8.2
22.4 Resilient gasket seals shall be positioned so that they are not subject to mechanical damage under normal operating conditions and they shall retain their sealing properties over the expected life of the device. Alternatively the manufacturer shall recommend a nominated replacement frequency.
22.5 Poured seals and encapsulating compounds shall have a continuous operating temperature (COT) at least 10 K higher than that occurring when operating in the most onerous rated service conditions.
22.6 Restricted-breathing enclosures without the provision for carrying out checks after installation or maintenance shall be type tested, including the cable entry devices.
NOTE The installation instructions provided with the apparatus should contain information on the selection of entry devices and cables.
22.7 If internal fans are fitted, the suction shall not induce a depression at a potential source of leakage.

26 Type tests
26.3.1 Order of tests
Replace the text of 26.3.1 by:
26.3.1.1 Non-metallic enclosures and non-metallic parts of enclosures (other than glass)
Tests shall be made on two samples which shall be submitted first to the tests of thermal endurance to heat (see 26.3.2.1), then to the tests of thermal endurance to cold (see 26.3.2.2), then to the mechanical tests (see 26.3.3), then to the tests for degrees of protection (IP) (see 26.3.4) or the tests for restricted breathing (see 26.8) and finally, when necessary, to any other test specified in this standard or any product standard.
26.3.1.2 Metallic enclosures, metallic parts of enclosures and glass parts of enclosures
Tests shall be made on the number of samples specified for each test, first the mechanical tests (26.3.3), then to the tests for degrees of protection (IP) (see 26.3.4) or the tests for restricted breathing (see 26.8) and finally, when necessary, any other test specified in this standard or any product standard.

26.5.1 Preparation of enclosed-break device samples
Delete the second paragraph.
Add the following note:
NOTE Any remaining non-metallic parts of the enclosure will have been subjected to the conditioning test described in 26.3.2.

26.6.3.1 d) **Replace** "10 °C" by "–10 °C".

28 Marking
28.2 c) **Replace** the existing text of c) by:
c) the symbol EEx n;
28.4 c) **Replace** the existing text of c) by:
c) the symbol EEx n;
28.5 b) **Replace** the existing text of b) by:
b) the symbol EEx n;
28.6.1 **Replace** line 3 of the marking example by:
EEx nA II T3 X

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28.6.2 Replace line 3 of the marking example by:
EEx nR II U

28.6.3 Replace line 2 of the marking example by:
EEx nL IIC T4

29 Documentation

Delete the existing Clause 29 and replace it by:

29 Instructions

29.1 All electrical apparatus shall be accompanied by instructions, including at least the following particulars:

a recapitulation of the information with which the electrical apparatus is marked, except for the serial number (see Clause 28), together with any appropriate additional information to facilitate maintenance (e.g. address of the importer, repairer, etc.);

instructions for safe:

putting into service;

use;

assembling and dismantling;

maintenance (servicing and emergency repair);

installation;

adjustment;

where necessary, training instructions;

details which allow a decision to be taken as to whether the apparatus can be used safely in the intended area under the expected operating conditions;

electrical and pressure parameters, maximum surface temperatures and other limit values;

where necessary, special conditions of use, including particulars of possible misuse which experience has shown might occur. For luminaires, this shall include a warning that type "n" luminaires shall not be operated in an ambient temperature in excess of T_a even for a short period;

additional special conditions requiring the use of the symbol "X" (see 28.2 k and 28.5 g);

NOTE It is important to ensure that the requirements of special conditions for safe use are passed to the purchaser together with any other relevant information.

where necessary, the essential characteristics of tools which may be fitted to the apparatus;

a list of the standards, including the issue date, with which the apparatus is declared to comply. For certified apparatus, the inclusion of the certificate can be used to satisfy this requirement.

29.2 The instructions shall contain the drawings and diagrams necessary for the putting into service, maintenance, inspection, checking of correct operation and, where appropriate, repair of the apparatus, together with all useful instructions, in particular with regard to safety.

29.3 Where it is necessary for the user to replace cells or batteries contained within an enclosure, the relevant parameters to allow correct replacement shall be included in the instructions, including either the manufacturer's name and part number, or the electrochemical system, nominal voltage and rated capacity.

Bibliography

Add the following notes for the standards indicated:

IEC 60068-2-6 NOTE Harmonized as EN 60068-2-6:1995 (not modified).

IEC 60068-2-75 NOTE Harmonized as EN 60068-2-75:1997 (not modified).

IEC 60345 NOTE Harmonized as HD 438 S1:1984 (not modified).

IEC 60947-1 NOTE Harmonized as EN 60947-1:1999 (not modified).

ISO 4892 NOTE Harmonized in the series EN ISO 4892 (not modified).

Section 3
National Differences

IEC Technical Report IEC 60079-15 1st Edition 1987 Electrical apparatus for explosive gas atmospheres Part 15: Electrical apparatus with type of protection "n"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil	**N/R**		
CA Canada		**N/R**	
CH Switzerland	**N/R**		
CN China*		**TBA**	
CZ Czech Republic	**N/R**		
DE Germany	**N/R**		
DK Denmark	**N/R**		
FI Finland	**N/R**		
FR France	**N/R**		
GB United Kingdom	**N/R**		
HR Croatia	**N/A**		
HU Hungary	**N/R**		
IN India		**N/R**	
IT Italy	**N/R**		
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998 65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**N/R**		
NO Norway	**N/R**		
NZ New Zealand		**NO**	
PL Poland	**N/A**		
RO Romania	**N/R**		
RU Russia		**N/R**	
SE Sweden	**N/R**		
SG Singapore		**N/R**	
SI Slovenia	**N/R**		
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/UL 60079-15 2002
ZA South Africa		**NO**	SANS 60079-15:1987

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-15 1st Edition 1987 Without National Differences
AU Australia
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-15 1st Edition 1987 With National/Group Differences- Differences follow.

KR Korea

Number	IEC Clause or Annex Details of Differences to IEC Standard IEC 60079-15 Edition: 1 (1987)

The requirements for Group I do not apply.
4.3.5 This requirement does not apply.

Section 3
National Differences
US United States of America

US National differences to IEC 60079-15 (1987)		
IEC Standard	**Clause**	**Difference**
IEC 60079-15 (1987)	7	General – Connection facilities for earthing or equipotential bonding conductors, and for cable and conduit entries, shall comply with the applicable requirements of clauses 15 and 16 of IEC 60079-0, and the US National Differences for IEC 60079-0 (1998) + A1 (2000), clauses 15 and 16.
IEC 60079-15 (1987)	10.1	Add a new sub-clause as follows: (k) Class I, Zone 2. Delete item i).
IEC 60079-15 (1987)	10.1(c) and 10.2(a)	Modify as follows, "the symbol AEx...."
IEC 60079-15 (1987)	10.1	~~Items c), d) and e) shall be marked in the same sequence as given above, e.g:~~ ~~Ex nA II T4 or Ex nA II T4 X (where marking according to item i) is to be included)~~ Marking should be in the following sequence, item k), c), d) and e) e.g: Class I, Zone 2, AEx nA II T4
IEC 60079-15 (1987)	10.2	Add new sub-clauses as follows: (e) Cl. 1, Zn. 2, or equivalent; (f) II, IIA, IIB, or IIC as appropriate; and (g) Temperature class or maximum surface temperature as appropriate. Delete item c).
IEC 60079-15 (1987)	10.3 (new)	When there are special conditions necessary for safe use, the conditions shall be marked in a cautionary statement on the product, or the conditions shall be included in the installation documents and a reference to the instructions shall be marked on the product.
IEC 60079-15 (1987)	10.4 (new)	Metric threaded entries shall be identified to indicate the thread type.
IEC 60079-15 (1987)	10.5 (new)	10.5 Class I, Division 2 Equivalency Marking

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National Differences

US National differences to IEC 60079-15 (1987)		
IEC Standard	**Clause**	**Difference**
		Refer to US National Differences for IEC 60079-0 (1998) + A1 (2000), clause 27.11.
IEC 60079-15 (1987)	10.6 (new)	10.6 Abbreviated Markings On very small electrical apparatus and on components where there is limited space, the following abbreviations are permitted to be used: *Class – Cl* *Division – Div* *Group – Gp or Grp*
IEC 60079-15 (1987)	10.7 (new)	Where cautionary markings are required on the apparatus, the text as described, following the word “CAUTION” or “WARNING,” may be replaced by technically equivalent text. NOTE — The preferred format of cautionary markings includes three elements: a key word such as WARNING or CAUTION, the hazard risk, and the action required to avoid the hazard.
IEC 60079-15 (1987)	14.1	Fuses shall be considered non-sparking devices if they comply with any of the following: a) Non-rewirable sand-filed non-indicating cartridge type fuses capable of interrupting a prospective current of at least ~~4000 A~~ 1500 A. b) Single cartridge fuses rated at no more than 6 A and 250 V to earth and installed in fuse holders which provide degree of protection IP64 c) Non-rewirable fuses rated at not more than 6 A and 250 V capable of interrupting a prospective current of at least ~~4000 A~~ 1500 A at mains voltage and installed in apparatus such as meters, instruments and relays.
IEC 60079-15 (1987)	14.3	Change warning label requirement to read: “WARNING – DO NOT OPEN WHEN ENERGIZED”

Section 3

National Differences

<table>
<tr><th colspan="3">US National differences to IEC 60079-15 (1987)</th></tr>
<tr><th>IEC Standard</th><th>Clause</th><th>Difference</th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>IEC 60079-15 (1987)</td><td>15.1</td><td>Plugs and sockets and similar connectors for internal connections shall be considered to be not normally sparking if they are arranged so that they cannot become loose nor become separated due to vibration, or so that they require a separating force of at least 15N. Where a socket is provided for the mounting of a light-weight component (for example a fuse or connection jumper) the separating force (in Newtons) need not be greater than 10 times the mass of the component (in kg).</td></tr>
<tr><td>IEC 60079-15 (1987)</td><td>15.4</td><td>Change warning label requirement to read:

“WARNING – DO NOT CONNECT OR DISCONNECT WHEN ENERGIZED”</td></tr>
<tr><td>IEC 60079-15 (1987)</td><td>16.4.2</td><td>Replace existing tables with the following:
<table><tr><th>Lamp cap</th><th>Torque Nm</th></tr><tr><td>E14/E13</td><td>1,0 ± 0,1</td></tr><tr><td>E27/E26</td><td>1,5 ± 0,1</td></tr><tr><td>E40/E39</td><td>3,0 ± 0,1</td></tr></table><table><tr><th>Lamp cap</th><th>Torque Nm</th></tr><tr><td>E14/E13</td><td>0,3</td></tr><tr><td>E27/E26</td><td>0,5</td></tr><tr><td>E40/E39</td><td>1,0</td></tr></table></td></tr>
<tr><td>IEC 60079-15 (1987)</td><td>23.1.4</td><td>Delete and replace with the following:

"The following statement (or equivalent wording) shall be marked adjacent to the conduit opening: ’CAUTION – Restricted breathing device. To maintain restricted breathing properties, a seal must be provided on all conduit or cable entries.’ "</td></tr>
</table>

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IECEx Bulletin

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IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-18

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-18

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3

National Differences

IEC Standard: IEC 60079-18: 3rd Edition 2009 Electrical apparatus for explosive gas atmospheres - Part 18: Construction, test and marking of type of protection encapsulation 'm' electrical apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-18:11
CH Switzerland	**NO**		EN 50079-18:2009
CN China		**YES**	GB 3836.9-2006 identical IEC 60079-18:2004
CZ Czech Republic	**NO**		EN 50079-18:2009
DE Germany	**NO**		EN 50079-18:2009
DK Denmark	**NO**		EN 50079-18:2009
FI Finland	**NO**		EN 50079-18:2009
FR France	**NO**		EN 50079-18:2009
GB United Kingdom	**NO**		EN 50079-18:2009
HR Croatia	**NO**		EN 50079-18:2009
HU Hungary	**NO**		EN 50079-18:2009
IN India		**N/R**	To be adopted (approved for wide circulation)
IT Italy	**NO**		EN 50079-18:2009
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-18:XXXX (AWAITING TO BE PUBLISHED AS A NEW STANDARD FOR PART 18)
NL The Netherlands	**NO**		EN 50079-18:2009
NO Norway	**NO**		EN 50079-18:2009
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 50079-18:2009
RO Romania	**NO**		EN 50079-18:2009
RU Russia		**NO**	GOST R IEC 60079.18-2008
SE Sweden	**NO**		EN 50079-18:2009
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 50079-18:2009
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-18:2009

(N/R Not recognised)

***** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-18 3rd Edition 2009 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-18 3rd Edition 2009 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-18:11
Explosive atmospheres –
Part 18: Equipment protection by encapsulation "m"

This is the second edition of CAN/CSA-C22.2 No. 60079-18, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-18 (Edition 3.0, 2009-05) together with Corrigendum 1 (June 2009). It replaces the CAN/CSA-C22.2 No. 60079-18:95 (adopted CEI/IEC 60079-18: First edition, 1992-10: Entitled *Electrical apparatus for explosive gas atmospheres – Part 18: Encapsulation "m"*).

It shall be noted that this IEC 60079-18 (Edition 3.0) supersedes and replaces both earlier standards IEC 60079-18:2004 (Explosive "Gas" atmospheres) and IEC 61241-18:2004 (Explosive "Dust" atmospheres).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this*

Section 3
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Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle."

July 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)
C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

7.3.1 Group III "m" equipment
Table 2 – Minimum thickness of compound adjacent to free space for Group III "m" equipment

[Remove all requirements where reference is made to "mc" level of protection in the Table.]

7.4.2 Windings for electrical machines

[Add a paragraph to the following effect]

In addition, rotating machines shall also comply with the applicable requirements in CAN/CSA-C22.2 No. 60079-7:09 (Protection by increased safety – "e".

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7.6 External connections
7.6.1 General

[Add a paragraph to the following effect]

Where encapsulated electrical apparatus, devices and/or components that are not designed to be mounted within an additional enclosure, integral threaded entries or hubs for field wiring shall be provided. (For example: A solenoid valve actuator.)

7.8.5 Current limitation

[Add a paragraph to the following effect]

Where lithium types of cells and batteries are used, there shall be a resistor in series with the cell (or battery) to limit the current to two-thirds or lower of the maximum pulse current of the cell (or battery), and a fuse in series of a rating not exceeding half of the maximum continuous discharge current of the cell (or battery).

NOTE: See CAN/CSA-C22.2 No. 60079-11:11 for the above Canadian deviation.

10 Marking

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

CN China

IEC Standard Number IEC 60079-18:2009 Ed3.0…..

IEC Clause or Annex Number	Details of Differences

The details of national differences are the technical changes from IEC60079-18 Ed2.0 to IEC60079-2 Ed3.0.

Section 3
National Differences

IEC Standard: IEC 60079-18 2nd Edition 2004 Electrical apparatus for explosive gas atmospheres - Part 18: Construction, test and marking of type of protection encapsulation 'm' electrical apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-18: 2005
BR Brazil		**NO**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-18:2004
CN China		**NO**	GB 3836.9-2006 identical IEC 60079-18:2004
CZ Czech Republic	**NO**		EN 60079-18:2004
DE Germany	**NO**		EN 60079-18:2004
DK Denmark	**NO**		DS/EN 60079-18:2004
FI Finland	**NO**		EN 60079-18:2004
FR France	**NO**		EN 60079-18:2004
GB United Kingdom	**NO**		EN 60079-18:2004
HR Croatia	**N/A**		
HU Hungary	**NO**		EN 60079-18:2004
IN India		**NO**	Adopted as IS/IEC 60079-18: 2004
IT Italy	**NO**		EN 60079-18:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-18:2005
NL The Netherlands	**NO**		EN 60079-18:2004
NO Norway	**NO**		EN 60079-18:2004
NZ New Zealand		**NO**	AS/NZS 60079-18: 2005
PL Poland	**N/A**		
RO Romania	**NO**		EN 60079-18:2004
RU Russia*		**TBA**	
SE Sweden	**NO**		SS-EN 60079-18
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-18:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 60079-18 2009 ANSI/UL 60079-18 2009
ZA South Africa		**NO**	SANS 60079-18:2004

(N/R Not recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-18 2nd Edition 2004 Without National Differences
AU Australia
BR Brazil
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-18 2nd Edition 2004 With National/Group Differences- Differences follow.

US Difference on following page.

Section 3
National Differences

US United States of America

Clause	US Differences to IEC 60079-18, 2nd Ed, 2004
General	The U.S. version of IEC 60079-18 (ANSI/ISA 60079-18:2009 and ANSI/UL 60079-18:2009) does not include requirements for Group I electrical apparatus. Where references are made to other IEC 60079 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
1	Added clarification that this standard not only supplements Part 0, but that it also, in some cases, “modifies” Part 0 in some cases. Modification of text as follows: This standard supplements and modifies the general requirements in ANSI/ISA-60079-0 ~~IEC 60079-0~~. Where a requirement of this standard conflicts with a requirement of ANSI/ISA-60079-0, the requirement of this standard will take precedence.
3.11	Addition of 3.11 to clarify the intended meaning of the term used in the standard: **3.11 solid insulation** solid insulation is insulation material which is extruded or molded but not poured. Insulators fabricated from two or more pieces of electrical insulating material which are solidly bonded together may be considered as solid. Varnish and similar coatings are not considered to be solid insulation.
4.2	Modification of text as follows to reflect US fuse standard: If a non-resetting protective device according to ~~IEC 60127~~UL 248-1 or IEC 60691 is used, then only one device is necessary for both levels of protection.

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Clause	US Differences to IEC 60079-18, 2nd Ed, 2004
5.2	Modification of text as follows to align with US implementation of "X" marking: **Specification** ~~The manufacturer shall attest on his own responsibility that the material complies with the compound specification.~~ The material specification shall include: a) the name and address of the manufacturer of the compound, b) the exact and complete reference of the material, and, if relevant percentage of fillers and any other additives, the mixture ratios and the type designation, c) if applicable, any treatment of the surface of the compound(s), for example varnishing, d) if applicable, to obtain correct adhesion of the compound to a component any requirement for pre-treating of the component, for example cleaning, etching, e) if appropriate, the result of the water absorption test in accordance with 8.1. Where this test has not been performed, to indicate this special condition of use the apparatus shall be marked ~~"X"~~ in accordance with item i) of 29.2 of ANSI/ISA 60079-0 or ANSI/UL 60079-0 ~~IEC 60079-0~~. f) the dielectric strength in accordance with IEC 60243-1 at the maximum temperature of the apparatus determined according to 8.2.2, g) temperature range of the compound(s) (upper and lower continuous operating temperature), h) in the case of "m" apparatus where the compound is part of the external enclosure the temperature index TI value as defined by 7.1.3 d) of ANSI/ISA 60079-0 or ANSI/UL 60079-0 ~~IEC 60079-0~~. As an alternative to the TI, the relative thermal index (RTI-mechanical impact) may be determined in accordance with ANSI/UL 746B[1], i) the colour of the compound used for the test samples where the compound specification will be influenced by changing the colour.
6.1	Modification of text as follows to clarify intent:

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Clause	**US Differences to IEC 60079-18, 2nd Ed, 2004**
	The maximum surface temperature and the maximum value of the continuous operating temperature of the compound shall not be exceeded during normal operation. The "m" apparatus shall be protected in such a way that the encapsulation "m" is not adversely affected under specified fault conditions.
6.2	Modification of text in second paragraph as follows to reflect US fuse standard: If the non-resetting protective device complies with ~~IEC 60127~~UL 248-1 or IEC 60691, only one device is necessary for both levels of protection.
7.4.5	Modification of text in second paragraph as follows to clarify the specific dielectric test to be conducted For both levels of protection "ma" and "mb", the end of the slot and the end-winding shall be protected by the minimum thickness of compound in accordance with 7.4.1. A dielectric strength test, in accordance with 8.2.4, shall be passed with $U = (2U + 1\ 000\ \text{V})$ a.c. with a minimum of 1 500 V a.c.
7.4.6.1	Modification of text as follows to clarify that this refers toi "working" voltgage, and not "rated" voltage: Multi-layer printed wiring boards complying with the requirements of IEC 62326-4-1, performance level C, with the minimum distances in 7.4.6.2 and operated at working voltages less than or equal to 500 V, shall be considered to be encapsulated providing they meet 7.4.6.2.

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National Differences

Clause	**US Differences to IEC 60079-18, 2nd Ed, 2004**
7.4.6.2	Modification of text in second paragraph as follows to align with values in Table 6 The minimum distance between the printed circuit conductors and the edge of the multi-layer printed wiring board or any hole in it shall be at least 3 mm. If the edges or holes are protected with metal or insulating material extending at least 1 mm for "mb" or 3 mm for "ma", along the surface of the board from the edges or holes, the distance of the printed wiring conductors may be reduced to 1 mm. Insulating material shall comply with the requirements for conformal coating in accordance with ANSI/ISA-60079-11 ~~IEC 60079-11~~. Metal coating shall have a minimum thickness of 35 µm, see also Figure 4 and Table 6.
7.7	Modification of text as follows to align with US adoption of 60079-26: Bare live parts that pass through the surface of the compound shall be protected by another type of protection as follows: ~~a) for level of protection "ma", one of the types of protection listed in IEC 60079-26;~~ b) for levels of protection "ma" or "mb", one of the types of protection listed in ANSI/ISA 60079-0 and ANSI/UL 60079-0 ~~IEC 60079-0~~, excluding type of protection "n".
7.8.6	Modification of text in second paragraph as follows to reflect US fuse standard: A resistor, a current limiting device or a fuse according to ~~IEC 60127~~UL 248-1 or an equivalent standard, may be used to ensure the safe current specified by the manufacturer of the cells or battery is not exceeded. If replaceable fuses are used ~~they~~ the apparatus shall be marked to show ~~their~~ the fuse rating and function.
7.9.1	Modification of text in second paragraph as follows to clarify which rating is applicable: The protective device shall be capable of interrupting the maximum fault current of the circuit in which it is installed. The rated voltage of

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Clause	US Differences to IEC 60079-18, 2nd Ed, 2004
	the protective device shall at least correspond to the working voltage of the circuit in which it is installed.
7.9.2.1	Modification of text as follows to clarify which rating is applicable: Fuses shall have a voltage rating not less than that of the circuit in which they are installed and shall have a breaking capacity not less than the fault current of the circuit. Unless otherwise specified, a fuse shall be assumed to be capable of passing 1,7 x rated current continuously. The time-current characteristic of the fuse shall ensure that the COT of the encapsulant or the temperature class are not exceeded. The time-current characteristics of the fuses, in accordance with ~~IEC 60127 or~~ ANSI/UL 248-1, shall be stated by the manufacturer of the fuses.
7.9.2.2	Modification of text in first paragraph as follows: Where the encapsulation is unable to withstand a single fault, the “m” apparatus may be connected to separate protective devices. To indicate this special condition of use, the apparatus shall be marked ~~“X”~~ in accordance with item i) of 29.2 of ANSI/ISA 60079-0 or ANSI/UL 60079-0 ~~IEC 60079-0~~.
8.2.4.1	Modification of text in third paragraph as follows to clarify requirement: The test voltage shall be 500 V r.m.s. at 48 Hz to 62 Hz for apparatus where the sum of the supply voltages does not exceed 90 V peak. If the sum of the supply voltages exceeds 90 V peak, the test voltage shall be 2*U* + 1 000 V, with a minimum of 1 500 V a.c. at 48 Hz to 62 Hz. Where an alternating test voltage would damage electronic components in the encapsulation, the test voltage shall be 2*U* + 1 400 V d.c., with a minimum of 2 100 V d.c.

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Clause	US Differences to IEC 60079-18, 2nd Ed, 2004
8.2.5.2	Modification of text in third paragraph as follows to clarify the requirement: The tensile force (in Newton) applied shall either be 20 times the value in millimetres of the diameter of the cable or 49 times the mass (in kilograms) ~~5 times the weight~~ of the "m" apparatus, whichever is the lower value. This value can be reduced to 25 % of the required value in the case of permanent installations. The minimum tensile force shall be 1 N and the minimum duration shall be 1 h. The force shall be applied in the least favourable direction.
10	Modification of text as follows to clarify that the intended type of protection marking be "ma" or "mb" as appropriate: ………………..the marking shall include – the rated voltage, – the rated current or rated power (for apparatus at other than unity power factor both values shall be marked), – the prospective short circuit current of the external electric supply source if different from 1 500 A, – other information necessary for safe operation of the particular apparatus; and – in place of the marking 'm' for the type of protection specified in ANSI/ISA-60079-0, the levels of protection "ma" or "mb" shall be marked as appropriate.

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National Differences

IEC Standard: IEC 60079-18 1st Edition 1992 Electrical apparatus for explosive gas atmospheres - Part 18: Encapsulation "m"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**N/R**	
CA Canada		**YES**	CAN/CSA-E60079-18-95
CH Switzerland	**YES**		EN 50028 1987
CN China		**TBA**	
CZ Czech Republic	**YES**		EN 50028 1987
DE Germany	**YES**		EN 50028 1987
DK Denmark	**YES**		EN 50028 1987
FI Finland	**YES**		EN 50028 1987
FR France	**YES**		EN 50028 1987
GB United Kingdom	**YES**		EN 50028 1987
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 50028 1987
IN India		**NO**	Adopted as IS 15451:2004/IEC 60079-18: 1992
IT Italy	**YES**		EN 50028 1987
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998-65
MY Malaysia		**NO**	MS IEC 60079-18:2003 (ALREADY WITHDRAWN)
NL The Netherlands	**YES**		EN 50028 1987
NO Norway	**YES**		EN 50028 1987
NZ New Zealand		**NO**	
PL Poland	**N/A**		
RO Romania	**YES**		EN 50028 1987
RU Russia		**YES**	GOST R 51330.17-99
SE Sweden	**YES**		EN 50028 1987
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 50028 1987
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-18:1992

(N/R Not recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

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National Differences

Countries that have declared adoption of IEC 60079-18 1st Edition 1992 Without National Differences
AU Australia
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-18 1st Edition 1992 With National/Group Differences- Differences follow.

CA Canada

Sub-Clause 5.5.3
[Add the following clause]
Where encapsulated electrical apparatus, devices, and/or components are not designed to be
mounted within an additional enclosure, integral threaded entries or hubs shall be provided.

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National Differences

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number	Details of Differences
3.2	Add the following extra requirement to this Clause: *"epoxy resin (cold curing)"* – is included in the definition of compounds
3.8	The definition of *"Ex Component"* does not appear in EN 50028. This definition appears in EN 50014. IEC 79-18, Clause 3.8 states that the definition of Ex component will be transferred in due course to IEC 79-0
5.1.2	The following requirement does not apply: "Loose filling materials shall not be used for the purpose of reducing the free volume inside a void to the specified upper limit"
5.1.5	The requirement contained in this clause is varied by EN 50028: *EN 50028 considers components such as resistors, coils and capacitors not subject to fault* *IEC 79-18 considers the components not subject to a short circuit or failure to a lower resistance* or to a higher Capacitance, (μF) *IEC 79-18 contains specific requirements for Relays which are not contained in EN 50028*
5.2.3	The following requirement does not apply: *"Where a protective housing or a part of a protective housing of non-metallic material serves directly to support live bare parts the resistance to tracking and creepage distances on the surface of the walls of the protective housing shall comply with the requirements of 4.4 of IEC 79-7"*
5.5	Add the following extra paragraph to this clause: *Non- portable encapsulated electrical of Group I," may have a permanently connected cable." (Note that it is mentioned in EN 50028 it is contrary to clause 14.1 of the General Requirements EN 50014)*
6	The contents of this clause are varied by EN 50028: *EN 50028 clause 5 is the equivalent clause to 6 of IEC 79-18, however the details of test sequencing and tests not performed that are contained in clause 5 of EN 50028 are located in Appendix B of IEC 79-18.*

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6.1	The contents of this clause are varied by EN 50028: *IEC 79-18 specifies using only cells which are ”not expected to release gas......”.* *EN 50028 clause 4.8 specifies using only cells which ”do not release gas ....”.*
7	This clause does not apply in EN 50028. *There is no equivalent single clause in EN 50028 the requirements are contained in various clauses of EN 50028 and EN 50014.*
8.1.2	Where the water absorption test is not undertaken IEC 79-18 requires that the apparatus be X marked and the associated documentation shall indicate the restriction of use. EN 50028 does not have this requirement.
8.2.3	The following requirement does not apply: *The test voltage shall be 500 V r.m.s. for apparatus with supply voltages not exceeding 90 V peak.*
9.1	Add the following extra requirement to this clause: *No visible damage of the compound shall be evident such as discolouration*
9.3	Replace the clause with the following: *The electrical data shall be checked by measurement of voltage, current and active power.*
9.4	Replace the clause with the following: In the case of voltage transformers having voltages above 1 kV, the loss angle shall be measured for the measurement of partial discharges defined in IEC publication 44-4.
10.2.5	The marking U for an *”Ex Component”* does not appear in EN 50028 it appears in EN 50014. IEC 79-18, Clause 10.2.5 states that the marking U for an *”Ex Component”* will be transferred in due course to IEC 79-0
Annex B	Replace annex B with Annex B of EN 50028 but the test clause numbers that are referenced should be replaced by the equivalent test clause numbers in this ”Difference Document”.
Annex C.1.1	The test in this annex is contained in EN 50014 and has the following technical difference. IEC 79-18 Annex C.1.1 (Thermal endurance), where the service temperature exceeds 75°C specifies a two week test temperature range of 90°C to 97°C. EN 50014 (Thermal endurance) specifies a range of 93°C to 97°C
Annex	The test in this annex is contained in EN 50014 and has the following

Section 3
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C.2.1	technical differences. IEC 79-18 Annex C. 2.1 (Resistance to Chemical agents for group I apparatus) refers to the "samples" being submitted to the test. EN 50014 (Resistance to Chemical agents for group I apparatus) refers to plastic enclosures and plastic parts of enclosures being submitted to the test. IEC 79-18 Annex C.2.1 (Resistance to Chemical agents for group I apparatus), requires one sample for each of the two tests, totalling two samples. EN 50014 (Resistance to Chemical agents for group I apparatus) requires two samples for each of the two tests totalling four samples. IEC 79-18 Annex C. 2.1 specifies the duration of two tests to be 24 h without any tolerance; EN 50014 specifies a tolerance of ± 2 h.
Annex C.3.1	Replace this annex with clause 23.4.7.5 (Resistance to Light) of EN 50014:1992. However note the following differences: IEC 79-18 Annex C.3.1 doesn't mention that the test is appropriate to plastics, whereas in EN 50014 it specifically mentions the test is undertaken only on plastics materials that are not protected from light. The black panel temperature of 55°C to 58°C in IEC 79-18 should be replaced by, 52°C to 58°C in EN 50014.

KR Korea

IEC Clause or Annex Number	Details of Differences to IEC 60079-18:1992 (edition 1)
	The requirements for Group I do not apply.
9.4	This requirement does not apply.
10.2.5	This requirement does not apply.
Appendix C C.2	This requirement does not apply
C.3.1	This requirement does not apply

Section 3
National Differences

RU Russia

Group I electrical equipment with types of protection "e" and "m" (standards GOST R 51330.8-99 (IEC 60079-7:2001) and GOST R 51330.8-99 (IEC 60079-18:1992) according to 6.6 of standard GOST R 51330.0-99

(IEC 60079-0-98) by its level of protection is referred to electrical equipment of higher explosion protection.
The field of application of electrical equipment with this level of protection in coal mines is as follows:

- shaft signalling in mines of any category as to firedamp;
- portable periodically used electrical devices in underground workings of mines susceptible to firedamp or dust;
- haulageways with fresh air of I and II category mines as to firedamp and coal dust explosion;
- charging rooms with separate ventilation in mines susceptible to firedamp or dust, including those susceptible to sudden outbursts;
- all mine workings in mines non-hazard as to firedamp, but hazard as to coal dust explosion..

Therefore, foreign explosion-proof electrical equipment supplied to Russian coal mines must incorporate the types of protection corresponding to the levels of explosion protection regulated by the above mentioned Safety Rules.

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-19

IECEx Bulletin:2011(E)

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Edition 4.0 2011-02

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-19

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COMMISSION

Section 3
National Differences

IEC Standard 60079-19 2nd Edition 2006 Explosive atmospheres - Part 19: Equipment repair, overhaul and reclamation			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		
CN China*		**TBA**	
CZ Czech Republic	**NO**		
DE Germany	**NO**		
DK Denmark	**NO**		
FI Finland	**NO**		
FR France	**NO**		
GB United Kingdom	**NO**		
HR Croatia	**NO**		
HU Hungary	**NO**		
IN India		**NO**	Adopted as IS/IEC 60079-19: 2006
IT Italy	**NO**		
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia*		**TBA**	
NL The Netherlands	**NO**		
NO Norway	**NO**		
NZ New Zealand		**NO**	
PL Poland	**NO**		
RO Romania	**NO**		
RU Russia*		**TBA**	
SE Sweden	**NO**		
SG Singapore		**NO**	
SI Slovenia	**NO**		
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-19:2006

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-19 2nd Edition 2006 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-19 2nd Edition 2006 With National/Group Differences- Differences follow.

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IECEx Bulletin

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-25

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National Differences

IEC Standard 60079-25: 2nd Edition 2009 Electrical apparatus for explosive gas atmospheres – Part 25: Intrinsically safe systems			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA C22.2 No. 60079-25:11 *(For Details contact Member Body)*
CH Switzerland	**NO**		EN 60079-25:2010
CN China		**NO**	GB 3836.18 –2010
CZ Czech Republic	**NO**		EN 60079-25:2010
DE Germany	**NO**		EN 60079-25:2010
DK Denmark	**NO**		EN 60079-25:2010
FI Finland	**NO**		EN 60079-25:2010
FR France	**NO**		EN 60079-25:2010
GB United Kingdom	**NO**		EN 60079-25:2010
HR Croatia	**NO**		EN 60079-25:2010
HU Hungary	**NO**		EN 60079-25:2010
IN India		**N/R**	
IT Italy	**NO**		EN 60079-25:2010
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	NATIONAL TC IS GOING TO ADOPT AND THE DRAFT STANDARD IS READY TO BE ISSUED FOR PUBLIC COMMENT.
NL The Netherlands	**NO**		EN 60079-25:2010
NO Norway	**NO**		EN 60079-25:2010
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-25:2010
RO Romania	**NO**		EN 60079-25:2010
RU Russia		**NO**	GOST R IEC 60079-25-2008
SE Sweden	**NO**		EN 60079-25:2010
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-25:2010
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-25:2009

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-25 2nd Edition 2009 Without National Differences
AU Australia
BR Brazil
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-25 2nd Edition 2009 With National/Group Differences- Differences follow.

Section 3
National Differences

IEC Standard 60079-25 1st Edition 2003 Electrical apparatus for explosive gas atmospheres – Part 25: Intrinsically safe systems			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-25: 2004
BR Brazil		**NO**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-25:2004
CN China*		**TBA**	
CZ Czech Republic	**NO**		ČSN EN 60079-25:2004
DE Germany	**NO**		EN 60079-25:2004
DK Denmark	**NO**		DS/EN 60079-25:2004
FI Finland	**NO**		EN 60079-25:2004
FR France	**NO**		EN 60079-25:2004
GB United Kingdom	**NO**		BS EN 60079-25:2004
HR Croatia	**N/A**		
HU Hungary	**NO**		EN 60079-25:2004
IN India		**NO**	Adopted as IS/IEC 60079-25: 2003
IT Italy	**NO**		EN 60079-25:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-25:2003
NL The Netherlands	**NO**		EN 60079-25:2004
NO Norway	**NO**		EN 60079-25:2004
NZ New Zealand		**NO**	AS/NZS 60079-25: 2004
PL Poland	**N/A**		
RO Romania	**NO**		EN 60079-25:2004
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 60079-25:2004
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 60079-25:2004
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-25:2003

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-25 1st Edition 2003 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
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Countries that have declared adoption of IEC 60079-25 1st Edition 2003 With National/Group Differences- Differences follow.

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IECEx Bulletin

Edition 4.0 2011-02

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-26

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Section 3
National Differences

IEC Standard 60079-26: 2nd Edition 2006 Electrical apparatus for explosive gas atmospheres – Part 26: Construction, test and marking of Group II Zone 0 electrical apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-26:11
CH Switzerland	**NO**		EN 60079-26:2007
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-26:2007
DE Germany	**NO**		EN 60079-26:2007
DK Denmark	**NO**		EN 60079-26:2007
FI Finland	**NO**		EN 60079-26:2007
FR France	**NO**		EN 60079-26:2007
GB United Kingdom	**NO**		EN 60079-26:2007
HR Croatia	**NO**		EN 60079-26:2007
HU Hungary	**NO**		EN 60079-26:2007
IN India		**N/R**	
IT Italy	**NO**		EN 60079-26:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-26:2007
NL The Netherlands	**NO**		EN 60079-26:2007
NO Norway	**NO**		EN 60079-26:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-26:2007
RO Romania	**NO**		EN 60079-26:2007
RU Russia		**NO**	GOST R 52350.26-2007
SE Sweden	**NO**		EN 60079-26:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-26:2007
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-26:2006

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-26 2nd Edition 2006 Without National Differences
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BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-26 2nd Edition 2006 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-26:11
Explosive atmospheres –
Part 26: Equipment with equipment protection level (EPL) Ga

This is the first edition of CAN/CSA-C22.2 No. 660079-26, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-26 (Second edition, 2006-08).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

Section 3
National Differences

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to gas explosion, dust explosion and layer ignition, hazards only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety Standard for Electrical Installations

4.2 Protection measures against ignition hazards of the electrical circuits
4.2.3 Encapsulation as a sole means of protection
4.2.4 Application of two independent types of protection providing EPL Gb

[Add a NOTE to the following effect]

NOTE 1: The penetration of electrical circuits between a Zone 0 area and a lesser hazardous area, and the sealing requirements (to prevent migration of gas or transition of an explosion through the wire-way) at the boundary of transit, shall be in accordance with this Standard or in compliance with the requirements of the Canadian Electrical Code, Part I.

NOTE 2: The wiring method within the Zone 0 area shall be in accordance with this Standard or in compliance with the requirements of the Canadian Electrical Code, Part I.

7 Information for use

[Add a paragraph to the following effect]

It is a requirement in Canada that any warning and cautionary statements, as well as safety instruction for the proper installation and operation of the equipment, must be in both English and French. In addition, all installation and operation manual must accompany each package and shipment of the equipment.

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National Differences

IEC Standard 60079-26 1st Edition 2004 Electrical apparatus for explosive gas atmospheres – Part 26: Construction, test and marking of Group II Zone 0 electrical apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-26: 2005 (Int)
BR Brazil		**N/R**	
CA Canada*		**TBA**	
CH Switzerland	**YES**		EN 60079-26:2004
CN China*		**TBA**	
CZ Czech Republic	**YES**		EN 60079-26:2004
DE Germany	**YES**		EN 60079-26:2004
DK Denmark	**YES**		DS/prEN 60079-26:2005
FI Finland	**YES**		EN 60079-26:2004
FR France	**YES**		EN 60079-26:2004
GB United Kingdom	**YES**		EN 60079-26:2004
HR Croatia	**N/A**		
HU Hungary	**YES**		EN 60079-26:2004
IN India		**N/R**	
IT Italy	**YES**		EN 60079-26:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-26:2005
NL The Netherlands	**YES**		EN 60079-26:2004
NO Norway	**YES**		EN 60079-26:2004
NZ New Zealand		**NO**	AS/NZS 60079-26: 2005
PL Poland	**N/A**		
RO Romania	**YES**		EN 60079-26:2004
RU Russia		**N/R**	
SE Sweden	**YES**		EN 60079-26:2004
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 60079-26:2004
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**NO**	SANS 60079-26:2004

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-26 1st Edition 2004 Without National Differences
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Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

COMMON MODIFICATIONS

Replace the title by:
Electrical apparatus for explosive gas atmospheres – Part 26: Construction, test and marking of Group II Category 1 G electrical apparatus.

Replace Clause 6 by:

6 Marking

6.1 General

The apparatus shall be marked according to the requirements of the Directive 94/9/EC for equipment
Group II, Category 1 G and according to the type of protection as defined in the applicable standard.
Apparatus intended for installation in the boundary wall between areas requiring different categories
shall have both categories marked on the label separated by a "/" and the corresponding symbols for
each type of protection separated by a "/". In the case where the apparatus group or temperature class
differ for the two types of protection, the complete designation of each rating shall be used and
separated by a "/".
- 3 - EN 60079-26:2004
Where more than one type of protection is used in accordance with 4.2.4, the symbols for the types of
protection shall be joined with a "+".

6.2 Examples of marking

a) Apparatus which is intended to be completely installed inside the Zone 0 area for example:
II 1 G Ex ia IIC T6

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or
II 1 G Ex d+e IIB T4
b) Associated apparatus, which is installed outside the hazardous area and providing external
electrical circuits protected by intrinsic safety “ia” according to EN 50020, which can be connected
to Category 1 apparatus, for example:
II (1) G [Ex ia] IIC

NOTE 1 No designation of the temperature class is required, as this apparatus is located outside the hazardous area.
c) Equipment which is installed in the boundary wall between areas requiring different categories,
both categories are marked on the label separated by a slash for example:
II1/2 G Ex d IIC T6
or
II 1/2 G Ex ia/d IIC T6

NOTE 2 Intrinsic safety “ia” apparatus in Zone 0 with a flameproof “d” compartment in Zone 1.
or
II 1/2 G Ex d+e/d IIB T4

NOTE 3 Two independent types of protection flameproof “d” and increased safety “e” in Zone 0 with a flameproof “d” compartment in Zone 1.
The documentation shall state which parts of the apparatus are suitable for installation in each zone.

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National Differences

IEC Standard 60079-27 2nd Edition 2008 Electrical apparatus for explosive gas atmospheres – Part 27: Fieldbus intrinsically safe concept (FISCO)			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-27: 2006
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-27:11
CH Switzerland	**NO**		EN 60079-27:2008
CN China		**NO**	GB 3836.19 –2010
CZ Czech Republic	**NO**		EN 60079-27:2008
DE Germany	**NO**		EN 60079-27:2008
DK Denmark	**NO**		EN 60079-27:2008
FI Finland	**NO**		EN 60079-27:2008
FR France	**NO**		EN 60079-27:2008
GB United Kingdom	**NO**		EN 60079-27:2008
HR Croatia	**NO**		EN 60079-27:2008
HU Hungary	**NO**		EN 60079-27:2008
IN India		**N/R**	
IT Italy	**NO**		EN 60079-27:2008
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-27:2009
NL The Netherlands	**NO**		EN 60079-27:2008
NO Norway	**NO**		EN 60079-27:2008
NZ New Zealand		**NO**	AS/NZS 60079-27: 2006
PL Poland	**NO**		EN 60079-27:2008
RO Romania	**NO**		EN 60079-27:2008
RU Russia		**YES**	GOST R IEC 60079-27-2008
SE Sweden	**NO**		EN 60079-27:2008
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 60079-27:2008
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 60079-27:2008

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Section 3
National Differences

Countries that have declared adoption of IEC 60079-27 2nd Edition 2008 Without National Differences
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JP Japan
MY Malaysia
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-27 2nd Edition 2008 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-27:11
Explosive atmospheres –
Part 27: Fieldbus intrinsically safe concept (FISCO)

This is the first edition of CAN/CSA-C22.2 No. 60079-27, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-27 (Edition 2.0, 20086-01).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

Section 3
National Differences

1 Scope

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to gas explosion, dust explosion and layer ignition, hazards only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE 1: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11 (General requirements) and CAN/CSA-.60079-11.11 (Intrinsic safety)

NOTE 2: The technique of intrinsic safety “ic” protection is not currently recognized in CE Code, Part I. The use of “ic” rated equipment, (under this Standard, *Fieldbus intrinsically safe concept (FISCO)* system), is considered as equivalent to the “nL” equipment previously acknowledged for use in the *Filedbus non-incendive concept (FNICO)* system.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety Standard for Electrical Installations

5 System requirements
5.1 General

[Add a NOTE to the following effect]

While the parameters (the resistance, inductance, capacitance and maximum lengths) of the cable or wire shall be in compliance with the specified values in this Standard; the selection and installation of the types of cable or wire shall be in accordance with the Canadian Electrical Code, Part I.

Section 3
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RU Russia

60079-27 2008 2nd ed.	GOST R IEC 60079.27-2008	MOD	The following text is added: - in 4.3.3 page 4 – the values of minimum input parameters of field devices shall be I_i 380 mA and P_i 5,32 W; - in 5.3 – minimum input parameters of the device shall be – Ii=380 mA and Pi=5.32 W; - in 4.7 page. 5 – Note – Each unit of electrical equipment shall be marked with a symbol of explosion protection level placed in front of the symbol Ex in accordance with GOST 52350.0. Added text is italicised. Replace the references to IEC by the references to GOST R.

Section 3
National Differences

IEC Standard 60079-27 1st Edition 2004 Electrical apparatus for explosive gas atmospheres – Part 27: Fieldbus intrinsically safe concept (FISCO)			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 60079-27 2003 (Int)
BR Brazil		**N/R**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-27:2006
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-27:2006
DE Germany	**NO**		EN 60079-27:2006
DK Denmark	**NO**		EN 60079-27:2006
FI Finland	**NO**		EN 60079-27:2006
FR France	**NO**		EN 60079-27:2006
GB United Kingdom	**NO**		EN 60079-27:2006
HR Croatia	**NO**		EN 60079-27:2006
HU Hungary	**NO**		EN 60079-27:2006
IN India		**N/R**	
IT Italy	**NO**		EN 60079-27:2006
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 60079-27:2006
NO Norway	**NO**		EN 60079-27:2006
NZ New Zealand		**NO**	AS/NZS 60079-27 2003 (Int)
PL Poland	**NO**		EN 60079-27:2006
RO Romania	**NO**		EN 60079-27:2006
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 60079-27:2006
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 60079-27:2006
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**NO**	SANS 60079-27:2004

(N/R Not Recognised)

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Section 3
National Differences

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NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 60079-27 1st Edition 2004 With National/Group Differences- Differences follow.

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National Differences

IEC Standard 60079-28 1st Edition 2006 Explosive atmospheres - Part 28: Protection of equipment and transmission systems using optical radiation			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 60079-28:2007
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-28:2007
DE Germany	**NO**		EN 60079-28:2007
DK Denmark	**NO**		EN 60079-28:2007
FI Finland	**NO**		EN 60079-28:2007
FR France	**NO**		EN 60079-28:2007
GB United Kingdom	**NO**		EN 60079-28:2007
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HU Hungary	**NO**		EN 60079-28:2007
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JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
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RU Russia*		**TBA**	
SE Sweden	**NO**		EN 60079-28:2007
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SI Slovenia	**NO**		EN 60079-28:2007
TR Turkey*		**TBA**	
US United States*		**TBA**	
ZA South Africa		**NO**	SANS 60079-28:2006

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JP Japan
MY Malaysia
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Countries that have declared adoption of IEC 60079-28 1st Edition 2006 With National/Group Differences- Differences follow.

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-29-1

INTERNATIONAL
ELECTROTECHNICAL
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Section 3
National Differences

IEC Standard 60079-29-1 1st Edition 2007 Explosive atmospheres - Part 29-1: Gas detectors - Performance requirements of detectors for flammable gases			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-29-1:11
CH Switzerland	**YES**		EN 60079-29-1:2007
CN China*		**TBA**	
CZ Czech Republic	**YES**		EN 60079-29-1:2007
DE Germany	**YES**		EN 60079-29-1:2007
DK Denmark	**YES**		EN 60079-29-1:2007
FI Finland	**YES**		EN 60079-29-1:2007
FR France	**YES**		EN 60079-29-1:2007
GB United Kingdom	**YES**		EN 60079-29-1:2007
HR Croatia	**YES**		EN 60079-29-1:2007
HU Hungary	**YES**		EN 60079-29-1:2007
IN India		**NO**	
IT Italy	**YES**		EN 60079-29-1:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-29-1:2009
NL The Netherlands	**YES**		EN 60079-29-1:2007
NO Norway	**YES**		EN 60079-29-1:2007
NZ New Zealand		**NO**	
PL Poland	**YES**		EN 60079-29-1:2007
RO Romania	**YES**		EN 60079-29-1:2007
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 60079-29-1:2007
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 60079-29-1:2007
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Please Note that IEC 60079-29-1 First Edition (2007) has replaced IEC 61779-1 to IEC 61779-5:1998 Series.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-29-1 1st Edition 2007 Without National Differences
AU Australia
BR Brazil
IN India
JP Japan
MY Malaysia
NZ New Zealand
SG Singapore

Countries that have declared adoption of IEC 60079-29-1 1st Edition 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-29-1:11
Explosive atmospheres –
Part 29-1: Gas detectors – Performance requirements of detectors for flammable gases

This is the first edition of CAN/CSA-C22.2 No. 60079-29-1, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-29-1 (Edition 1.0, 2007-08).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

Section 3
National Differences

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Electrical equipment of **Group I**, (intended for use in mines susceptible to firedamp), is under the jurisdiction of the Canadian Federal Government. Such types of equipment shall be referred to the respective AHJ (authority having jurisdiction).

[Add a NOTE to the following effect]

NOTE: Equipment shall be marked "IEC 60079-29-1" to signify conformance with this performance Standard [4.3(a)]. This does not represent equivalent to equipment conforming to CSA standard C22.2 No. 152 which requirements are specifically geared to the Canadian climate, environment and industrial applications.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

4.3 Labelling and marking

[Add the following NOTE]

NOTE: It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

4.4 Instruction manual

[Add a paragraph to the following effect]

Instructions, relevant to the safe installation, usage and operation of the equipment, shall be in both English and French.

5 Test methods

[See following comments]

Section 3
National Differences

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Group Differences declared for the European Countries EN 60079-29-1:2007 / IEC 60079-29-1:2007

1 Scope

Delete NOTE 1 - Renumber NOTES 2, 3 and 4 as NOTES 1, 2 and 3

2 Normative References

Delete:
IEC 61000-4-1: *Electromagnetic compatibility (EMC) – Part 4-1: Testing and measurement techniques – Overview of IEC 61000-4 series.* Basic EMC publication
IEC 61000-4-3: *Electromagnetic compatibility (EMC) – Part 4-3: Testing and measurement techniques – Radiated, radio-frequency, electromagnetic field immunity test.* Basic EMC publication
IEC 61000-4-4: *Electromagnetic compatibility (EMC) – Part 4-4: Testing and measurement techniques – Electrical fast transient/burst immunity test* – Basic EMC publication

Insert:
EN 50270, Electromagnetic compatibility – Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen
EN 50271, Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen – Requirements and tests for apparatus using software and/or digital technologies

3 Terms and Definitions

Delete IEC 60079-0 and insert EN 60079-0 (This substitution is made throughout the document but not separately listed in this comparison document.)

4 General Requirements

4.1.2
Delete: … as specified in the other relevant parts of IEC 60079 series.
Insert: …. as specified in the appropriate regulations for explosion protection.

Delete: …. the required temperature limits of these other parts of IEC 60079 ….
Insert: …. the required temperature limits of the appropriate regulations for explosion protection ….

4.1.3 Delete entire sub-clause

4.2.9
Delete sub-clauses 4.2.9.1 through to 4.2.9.6
Insert: The apparatus shall fulfil the requirements of EN 50271

4.3 Delete the NOTE

4.4 c) NOTE Delete IEC 60079-29-2 and Insert EN 60079-29-2

5 Test Methods

Section 3

National Differences

5.4.21 Power supply interruptions, voltage transients and step changes of voltage
Delete entire sub-clause

Insert new clause
5.4.21 Electromagnetic compatibility
The apparatus shall be set up under normal conditions, in accordance with with 5.3 and then shall be subjected to the tests specified in EN 50270.

5.4.25 Electromagnetic immunity
Delete entire sub-clause

Annex A1

5.4.21 Delete row contents and insert:
Electromagnetic compatibility / According to EN 50270 / According to EN 50270 / According to EN 50270 / According to EN 50270

5.4.25 Delete row

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-29-4

IECEx Bulletin:2011(E)

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IECEx Bulletin –
Section 3: National Differences for 60079-29-4

INTERNATIONAL
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Section 3
National Differences

IEC Standard 60079-29-4 1st Edition 2009 Explosive atmospheres - Part 29-4: Gas detectors - Performance requirements of open path detectors for flammable gases			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil*		**TBA**	
CA Canada*		**TBA**	
CH Switzerland	**YES**		EN 60079-29-4:2010
CN China*		**TBA**	
CZ Czech Republic	**YES**		EN 60079-29-4:2010
DE Germany	**YES**		EN 60079-29-4:2010
DK Denmark	**YES**		EN 60079-29-4:2010
FI Finland	**YES**		EN 60079-29-4:2010
FR France	**YES**		EN 60079-29-4:2010
GB United Kingdom	**YES**		EN 60079-29-4:2010
HR Croatia	**YES**		EN 60079-29-4:2010
HU Hungary	**YES**		EN 60079-29-4:2010
IN India*		**TBA**	
IT Italy	**YES**		EN 60079-29-4:2010
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 60079-29-4:2010
NO Norway	**YES**		EN 60079-29-4:2010
NZ New Zealand		**NO**	
PL Poland	**YES**		EN 60079-29-4:2010
RO Romania	**YES**		EN 60079-29-4:2010
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 60079-29-4:2010
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 60079-29-4:2010
TR Turkey*		**TBA**	
US United States*		**TBA**	
ZA South Africa*		**TBA**	

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed. Please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 60079-29-4 1st Edition 2009 Without National Differences
AU Australia
JP Japan
NZ New Zealand
SG Singapore

Countries that have declared adoption of IEC 60079-29-4 1st Edition 2009 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Group Differences declared for the European Countries EN 60079-29-4:2010 / IEC 60079-29-4:2009

Scope
Replace "IEC 60079-29" with "EN 60079-29" in paragraph 1.
Replace "IEC 60079-29-1" with "EN 60079-29-1" in Note 1.
Delete note 3.

2 Normative references
Replace "IEC 60079 (all parts)" with "EN 60079 (all parts)" in paragraph 2.
Replace "IEC 60079-0" with "EN 60079-0" in paragraph 3.
Replace "IEC 60079-29-1" with "EN 60079-29-1" in paragraph 4.
Delete references to IEC 61000-4-1 and IEC 61000-4-3 in paragraphs 6 and 7.
Add the following undated references:
EN 50270, *Electromagnetic compatibility - Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen*
EN 50271, *Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen - Requirements and tests for apparatus using software and/or digital technologies*

3 Terms and definitions
In the 1st paragraph:
Replace "IEC 60079-0" with "EN 60079-0".
Replace "IEC 60079-29-1" with "EN 60079-29-1".

4 General requirements
4.1.1
In the 2nd paragraph, line 2, **replace** "… of these other parts of IEC 60079" with "… of the appropriate regulations for explosion protection".
4.3
Add "The apparatus shall fulfil the requirements of EN 50271."
Delete subclauses 4.3.1 to 4.3.5.

5 Test methods
5.2.4.2
In the second paragraph, **replace** reference to test 5.4.16 by 5.4.20.
5.4.12
In the 3rd paragraph, **replace** "A 90% of the change......." with "90% of the change.......".
5.4.15 Power supply interruptions and transients

Section 3

National Differences

Replace 5.4.15.1, 5.4.15.2 and 5.4.15.3 with “Text deleted”.

5.4.17

Replace this subclause by:

5.4.17 Electromagnetic compatibility

The apparatus shall be set up under normal conditions, in accordance with 5.3, and then shall be subjected to the tests specified in EN 50270.

NOTE For this test the operating distance may be relaxed to suit the requirements of the EMC test facility.”

7 Information for use

7.1 Labelling and marking

Delete the text and replace with:

“The equipment shall comply with the marking requirements of EN 60079-0.

In addition, the equipment shall also be marked:

a) “EN 60079-29-4” (to represent conformance with this performance standard);

b) year of construction (may be encoded within the serial number).”

7.2 Instruction manual

Add “EN” before “60079-0” in paragraph 1, line 2.

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IECEx PUBLICATION

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IECEx Bulletin – Section 3: National Differences for 60079-30-1

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IECEx Bulletin – Section 3: National Differences for 60079-30-1

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Section 3
National Differences

IEC Standard 60079-30-1 1st Edition 2007 Explosive atmospheres - Part 30-1: Electrical resistance trace heating - General and testing requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-30-1:11
CH Switzerland	**NO**		EN 60079-30-1:2007
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-30-1:2007
DE Germany	**NO**		EN 60079-30-1:2007
DK Denmark	**NO**		EN 60079-30-1:2007
FI Finland	**NO**		EN 60079-30-1:2007
FR France	**NO**		EN 60079-30-1:2007
GB United Kingdom	**NO**		EN 60079-30-1:2007
HR Croatia	**NO**		EN 60079-30-1:2007
HU Hungary	**NO**		EN 60079-30-1:2007
IN India		**N/R**	
IT Italy	**NO**		EN 60079-30-1:2007
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**NO**	MS IEC 60079-30-1:2009
NL The Netherlands	**NO**		EN 60079-30-1:2007
NO Norway	**NO**		EN 60079-30-1:2007
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-30-1:2007
RO Romania	**NO**		EN 60079-30-1:2007
RU Russia		**NO**	GOST R 60079-30-1 - 2009
SE Sweden	**NO**		EN 60079-30-1:2007
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-30-1:2007
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/IEEE 515-2004
ZA South Africa		**NO**	SANS 60079-30-1:2007

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Please Note IEC 60079-30-1 First Edition (2007) has replaced IEC 62086-1 First Edition (2001).

Section 3
National Differences

Countries that have declared adoption of IEC 60079-30-1 1st Edition 2007 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
MY Malaysia
NZ New Zealand
RU Russia
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-30-1 1st Edition 2007 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-30-1:11
Explosive atmospheres –
Part 30-1: Electrical resistance trace heating – General and testing requirements

This is the first edition of CAN/CSA-C22.2 No. 60079-30-1, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-30-1 (First edition, 2007-01).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

Section 3
National Differences

[Add the following paragraphs]

The requirements of IEC 60079 series Standards cover the protection techniques with respect to explosion hazard only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

[Add a NOTE to the following effect]

NOTE: See also CAN/CSA-C22.2 No. 60079-9:11 for further Canadian deviations.

2 Normative references

[Add the following]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety standard for electrical installations

4 General requirements
4.1 General

[Add a NOTE to the following effect]

NOTE: Any cautionary or warning statement, (e.g.: "This mechanical covering shall not be removed and the trace heaters shall not be operated without the mechanical covering being in place"), that relates to the safety of use of the equipment shall be in both English and French.

4.3 Circuit protection requirements for branch circuits

[Add a NOTE to the following effect]

NOTE: In consideration that the trace heaters under this Standard are based on the principle of the protection technique of Increased Safety "e"; where the circuit protection requirements in this Standard exceed the general requirements in CE Code, Part I, the requirements in this Standard shall take precedence.

4.4.3 Controlled design

[Add the following sentence to the existing NOTE]

NOTE: If the control devices ………. should be provided. In this case, the instructions shall be in both English and French.

6 Marking

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements, relating to the safe installation, operation and maintenance of the equipment, must be in both English and French.

Section 3
National Differences

US United States of America

US National differences to IEC 60079-30-1 (2007) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 60079-30-1 (2007)	1	~~IEC 60079-10~~ ANSI/NFPA 70 Art. 505
IEC 60079-30-1 (2007)	2	ASTM D 5025-1999, A Laboratory Burner Used for Small-Scale Burning Tests on Plastic Materials ASTM D 5207-2003, Calibration of 20 and 125 mm Test Flames for Small-Scale Burning Tests on Plastic Materials. ANSI/NFPA 70, National Electrical Code® (NEC®) NFPA 325-1994, Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids.
IEC 60079-30-1 (2007)	4.1	~~70 %~~ 80 % of the surface. For surface heating devices (panels) an integral metallic screen grid or equivalent electrically conductive covering, covering at least 80% of the exposed surface opposite the surface to be heated shall be incorporated into the construction. ~~Trace heaters which are identified for use only in areas with a low risk of mechanical damage are subjected to a reduced load in the impact test in 5.1.5 and a reduced force in the deformation test in 5.1.6, and shall be clearly marked as specified in 6.3.~~ ~~Trace heaters may be supplied with additional mechanical protection to meet the requirements of this standard if they are supplied as an integral assembly (prefabricated), and that contain the following statement in the instructions: This mechanical covering shall not be removed and the trace heaters shall not be operated without the mechanical covering in place.~~
IEC 60079-30-1 (2007)	4.2	Terminations and connections may be identified as an integral part of a trace heater, or may be identified separately, in which case they are regarded as Ex components in accordance with clause 13 of IEC 60079-0. In addition, these Ex components shall be evaluated in accordance with national and international standards relevant to their construction and use.
IEC 60079-30-1 (2007)	4.4.3	~~Controlled design applications, which require the use of a temperature control device to limit the maximum surface temperature, shall comply with a) for zone 1 and either a) or b) for zone 2:~~ ~~a) for zone 1 applications: a protective device such as a temperature limiter shall be provided~~

Section 3
National Differences

		~~that will de-energize the system to prevent exceeding the maximum permissible surface temperature. In case of an error by, or damage to the sensor, the heating system shall be de-energized before replacing the defective equipment. The protective device shall operate independently from the temperature controller. The protective device shall have the following characteristics:~~
IEC 60079-30-1 (2007)	4.4.3 Continued	~~1. Resetting by hand only~~ ~~2. Resetting possible only after the normal operating conditions have returned, or if the switching state is monitored continuously~~ ~~3. Tool or keyed lock required for resetting~~ ~~4. Temperature setting secured and locked to prevent manipulation~~ ~~5. A control that will de-energize the circuit if the sensor fails~~ ~~b) for zone 2 applications: a single temperature controller with failure annunciation may be used. If so, provision of adequate monitoring of such annunciation, such as 24 h surveillance, shall be made.~~ Controlled design – This approach requires the use of a temperature control device to limit the maximum pipe temperature for Zone 2 applications. When using a temperature controller without failure annunication, a separate high-temperature limit controller to de-energize the heating device shall be included in the design with either a manual reset or annunciatioon. Alternately, a single temperature controller with failure annunciation can be used. Controlled design may not be used for Zone 1 applications. Trace heating devices may not be used in Zone 0 applications.
IEC 60079-30-1 (2007)	5.1.3	by means of a d.c. source voltage (nominal) of ~~500 V~~ 1000 Vdc for mineral insulated heaters and 2500 Vdc for polymer insulated heaters.
IEC 60079-30-1 (2007)	5.1.4	…the distance from the cotton to the point of the flame application is 250 mm. A laboratory burner described in ASTM D 5025-1999 shall be used for the test. The gas flame produced by the burner is to be calibrated as described in ASTM D 5207-2003. The fuel shall be methane, propane, or natural gas, and shall be of a grade suitable for calibration to the ASTM D 5207-2003 procedure. The height of the ~~natural gas~~ flame of the burner shall be adjusted…

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IEC 60079-30-1 (2007)	5.1.5	~~A sample approximately 200 mm in length is placed on a rigid flat steel plate and positioned underneath an intermediate piece of hardened steel in the shape of a horizontal cylinder with a diameter of 25 mm. This cylinder is required to have a length of 25 mm with smoothly rounded edges to a radius of approximately 5 mm when used to test heating pads and heating panels (see Figure 2). For the test, the cylinder is laid horizontally on the sample and, in the case of a trace heating cable, its axis is placed across the sample. A trace heating cable having a non-circular cross-section shall be so positioned that the impact is applied along the minor axis (that is to say the trace heating cable is positioned flat on the steel plate).~~ ~~Other than in tests on electrical trace heating intended for use in applications with low risk of mechanical damage, a hammer with a mass of 1 kg shall be allowed to fall once onto the horizontal cylinder from a height of 700 mm (that is to say with an impact load of 7 J).~~ ~~For trace heating intended for use in applications with a low risk of mechanical damage in accordance with 4.1, the height may be reduced to 400 mm (that is to say an impact load of 4 J). Electrical resistance trace heating submitted to such a test shall be clearly marked as specified in 6.3 to caution the user as to its reduced mechanical capability.~~ ~~Immediately after impact conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3 while the steel cylinder and hammer are still in place on the sample. For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to $\sqrt{2}$ U + 500 V d.c. for cables rated over 60 V d.c~~. [Note: Figure 2 is also deleted] The impact test shall be conducted on a sample that is positioned on a hardened rigid steel plate and the assembly conditioned for a minimum of 4 h at the manufacturer's minimum recommended installation temperature. After conditioning and while still at the minimum recommended installation temperature, a sample shall be subjected to the impact forces of a 50.8 mm diameter cylindrical steel plunger with smoothly rounded edges, having a mass of 1.8 kg and allowed to free fall from a height of 762 mm, resulting in an impact force of 13.6 J. Then the impacted portion of the sample shall be immersed in tap water at 10C to 25C for 5 minutes, and the dielectric test outlined in 5.1.2 shall be conducted. For surface heating devices, both the heating region and cold leads shall be impacted. After testing, the sheath or dielectric insulation shall have no visible cracks when examined with normal vision.

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IEC 60079-30-1 (2007)	5.1.6	A sample ~~is~~ of the heating device as well as a sample of each type of integral components and non-heating lead (if applicable), shall be placed on a rigid flat steel plate. A crushing force of 1500 N is then applied for 30 s, without shock, by means of a 6 mm diameter steel rod with hemispherical ends and a total length of 25 mm. For the test, the rod is laid flat on the sample and in the case of a trace heating cable it is placed across a specimen at right angles. For cable that is oval or rectangular in shape, the widest surface shall be the surface on which the load is applied. In the case of a pad, it is necessary to ensure that the cylinder rests across the active element. ~~For electrical trace heating intended for use in applications with low risk of mechanical damage, the crushing force may be reduced to 800 N. Electrical resistance trace heating submitted to such a test shall be clearly marked as specified in 6.3 to caution the user concerning its reduced mechanical capability~~ After the deformation load is applied for 30 s, conformity is verified by testing the electrical insulation in accordance with 5.1.2 ~~and 5.1.3~~ while the horizontal steel rod is still in place on the sample and the load applied. ~~For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to~~ $\cancel{\sqrt{2}U}$ ~~+ 500 V d.c. for cables rated over 60 V d.c.~~ NOTE Trace heating cable samples need only be approximately 200 mm in length.
IEC 60079-30-1 (2007)	5.1.7	~~This test applies only to trace heaters that have a stated minimum bending radius of less than 300 mm.~~ At the end of this period, and with the sample maintained at the minimum recommended installation temperature, the sample… For heating panels the heating region shall be bent around a mandrel equivalent to the manufacturer's minimum bending radius. ~~Conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3. For MI cables, the required test voltage in 5.1.2 is reduced to 2*U* + 500 V a.c. for cables rated over 30 V a.c., or to~~ $\cancel{\sqrt{2}U}$ ~~+ 500 V d.c. for cables rated over 60 V d. c.~~ When this has been completed, the sample shall be immersed in tap water at 10C to 25C for 5 min, and then the dielectric test outlined in 5.1.2 shall be conducted.

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IEC 60079-30-1 (2007)	5.1.8	~~Conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3. For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to $\sqrt{2}\,U$ + 500 V d.c. for cables rated over 60 V d.c.~~ Within 1 h after this conditioning and while still immersed in the water, the sample shall be subjected to the dielectric test outlined in 5.1.2.
IEC 60079-30-1 (2007)	5.1.9	~~For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to $\sqrt{2}\,U$ + 500 V d.c. for cables rated over 60 V d.c.~~
IEC 60079-30-1 (2007)	5.1.10	The rated output of the heating cable or heating panel or pad shall be verified by one of the following ~~two~~ three methods, as selected by the manufacturer: a) conductance: the measured ac conductance or conductance per unit length, at a specified temperature, shall be within the manufacturer's declared tolerance; ~~a)~~ b) resistance: the measured d.c. resistance per unit length at a specified temperature shall be within the manufacturer's declared tolerance; ~~b)~~ c) thermal: …
IEC 60079-30-1 (2007)	5.1.11	~~The thermal stability of the electrical insulating materials of trace heaters shall be verified on a sample or prototype after it has been stored at a temperature of the manufacturer's declared maximum withstand temperature +20 K, but not less than 80 °C, for at least four weeks. Compliance of the sample or prototype shall be verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3. For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to $\sqrt{2}\,U$ + 500 V d.c. for cables rated over 60 V d.c.~~ ~~Terminations that insure the vapour tightness of trace heaters made of hygroscopic materials (for example the cold end seals of MI cable sets) are subjected to a temperature of (80 ± 2) °C for 4 weeks at not less than 90 % R.H. Compliance of the sample or prototype shall be verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3. For MI cables, the required test voltage in 5.1.2 is reduced to 2U + 500 V a.c. for cables rated over 30 V a.c., or to $\sqrt{2}\,U$ + 500 V d.c. for cables rated over 60 V d.c.~~ A sample of heating device shall be placed in a forced-circulation air oven. The oven shall be heated to, and maintained at, a temperature of 25 °C ± 5 °C above the highest exposure temperature declared by the manufacturer for a period of 14 days.

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IEC 60079-30-1 (2007)	5.1.11 Continued	The sample shall be removed from the air oven and cooled to room temperature. Heating cable samples shall be wound six close turns around a mandrel having a radius equal to twelve times the radius of the primary bending plane or thickness of the heating cable. Surface heating devices, if flexible, shall be wrapped on a mandrel with a radius equivalent to the manufacturer's minimum recommended bending radius. While still on the mandrel the sample, except at terminations or ends where the conductor is exposed, shall be submerged in tap water at 10 °C to 25 °C for 5 min. While still in the tap water, the dielectric test outlined in 5.1.2 shall be performed. Rigid heating devices shall also be submerged in tap water and tested. Upon completion of the test, the sample shall have no visible cracks when examined with normal vision.
IEC 60079-30-1 (2007)	5.1.12	~~In explosive gas atmospheres, it is critical to ensure that the maximum surface temperature of a trace heater is lower than the ignition temperature of the explosive gas atmosphere. The manufacturer's quality program shall demonstrate the thermal safety of trace heating products with respect to time.~~ **Thermal performance benchmark** One of the following tests shall be used to verify power output stability for all parallel heating devices. **Primary Test** Three randomly selected samples representing the maximum output of all cables or surface heating devices under evaluation shall be tested. If the type of cable or surface heating device has different levels of rated voltages and wattages, then three samples each shall be selected that represent 1) the lowest rated voltage level and the maximum rated output, and 2) the maximum rated voltage and the minimum rated output. Samples shall be terminated according to the manufacturer's specifications, such that a heating length of at least 600 mm or representative surface heating device dimensions is provided. The aging temperature of the test shall be the maximum declared maintain temperature of the heating device. The samples shall be conditioned, while energized, at the aging temperature for 120 h ± 24 h. The initial output of the samples is then to be determined by one of the three methods given in 5.1.9, with the exceptions of sample length and number of test temperature points for procedure 5.1.9 b). For this case, the samples shall be evaluated at rated voltage and at the manufacturer's stated reference temperature for the rated output. The output shall be within the manufacturer's stated output range.

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IEC 60079-30-1 (2007)	5.1.12 Continued	The samples shall be attached to a fixture or suitable heat sink as described in 5.1.9 b), and insulated accordingly. The pipe or heat sink temperature shall be set to the specified aging temperature and maintained with ± 3 °C plus 1% of the temperature reading in degrees Celsius. Circulating fluid or external heating may be used to raise the fixture to the aging temperature. The samples shall be operated at rated voltage. The power supply shall be attached to a 15 min cycle timer such that the samples are energized for 12 min and de- energized for 3 min. The samples shall be exposed to this conditioning for 32 weeks (5376 h). For heating devices with an intermittent temperature exposure rating, the samples are exposed to the same conditioning for 32 weeks, except for an 8 h excursion once each week. At the beginning of the 8 h, the samples shall be disengaged from the cycle timer. The pipe or heat sink temperature shall be increased to a temperature equal to the manufacturer's stated intermittent exposure temperature. The time allowed to increase the temperature should be no greater than 1 h. After 7 h from the beginning of the excursion, the pipe or heat sink temperature shall be decreased back to the aging temperature, again allowing no more than 1 h for the operation. Where the intermittent temperature rating is based on the heating device being energized, then the heating device shall be continuously energized during this temperature excursion, except during cool down back to the maximum maintain temperature. Where the intermittent temperature rating is based on the heating device being de-energized, the exposure cycle is conducted in a de-energized condition. At the end of the 8 h excursion, the samples shall be re-engaged to the cycle timer. The excursions should occur on the same day each week. At the end of the 32 weeks of operation at the continuous rated temperature, the output of the samples shall be determined by the same procedure as utilized for the initial readings. The heating device samples shall maintain a power level within plus 20% or minus 25% of the initial measured output.

IEC 60079-30-1 (2007)	5.1.12 Continued	**Alternative Test** The test apparatus shall consist of a metal platen(s) with the ability to change temperature within specified levels. The platen(s) shall be sized to expose all parts of the heating device samples, which would be exposed under normal installation conditions, to the temperature levels required by this procedure. The test apparatus shall insure that the heating device samples are in intimate contact with the platen. The test apparatus may be supplied with a sample-mounting fixture. Offsets may be built into the fixture or platen(s) to accommodate end termination/power transition fittings/boots, if provided, where their size profile exceeds the heating device profile. The apparatus shall allow energizing of the heating device samples as required during the test procedure. The samples shall be thermally insulated on the side not facing the platen so as to assure effective heat transfer from the platen to the heating device samples. The temperature of the platen(s) shall be uniformly controlled to a maximum tolerance of ± 5 °C for platen temperatures less than100 °C or 5% of the maximum continuous operating temperature if above 100 °C. The platen described above may be a flat metal plate, a metal pipe, or a metal surface typical of the majority of applications for the heating device being tested. The heating device samples shall be randomly selected and shall be a minimum of 0.3 m in length. Where the heating device is irregular in shape, such as a surface heating device, the heating device sample shall consist of at least one heating unit. If the heating devices are part of a heating device product range, with common materials (with materials having the same performance ratings) and construction, which have different levels of rated voltages and power outputs, then three samples each shall be selected that represent: (1) the lowest rated voltage level and the maximum rated power output; (2) the highest rated voltage and the minimum rated power output. Heating device samples may be conditioned, at the maximum rated voltage for up to 150 hours at the manufacturer's declared maximum continuous operating temperature prior to starting the test.

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IEC 60079-30-1 (2007)	5.1.12 Continued	The heating device samples shall be installed on the sample-mounting fixture or directly applied to the platen. The samples shall be powered at the maximum rated voltage. The temperature of the platen shall be 23±5°C. The initial power output of the samples shall be determined by measuring voltage and current after the device has reached equilibrium. The heating device samples of continuous parallel construction, while still installed on the sample mounting fixture or platen and energized at the maximum rated voltage, shall be temperature cycled by alternately exposing the samples to platen(s) temperatures corresponding to 23 ±5°C and the maximum continuous operating temperature. The samples may be de-energized during the cool down period. The heating device samples of zone type parallel construction shall be temperature cycled in the same manner with the exception that the samples shall be de-energized when not being held at the maximum continuous operating temperature. If the cycle temperature range exceeds 350 C, the lower temperature may be set at 350 C below the maximum continuous operating temperature. The energized samples shall be exposed to each of these temperature extremes for a minimum of 15 minutes and a transition time between extremes shall not exceed 15 minutes with a cycle being one complete exposure at both temperature extremes. A minimum of 1500 cycles shall be performed. Following the temperature cycling, the temperature of the platen(s) shall be raised to the maximum exposure temperature (the higher of the continuous or intermittent value) declared by the manufacturer and held for a period of no less than 250 hours. Where the maximum exposure temperature is declared as "power on", the samples shall be energized at the maximum rated voltage. After completion of the maximum exposure testing, the heating device samples power output shall be measured, using the same method and platen temperature as used during the initial measurements. The heating device samples shall maintain a power level within plus 20% or minus 25% of the initial measured output.
EC 60079-30-1 (2007)	5.1.13.2	These procedures are used to validate a manufacturer's design methodology and calculations. The procedures may be repeated with varied parameters, such as insulation type and thickness, to the satisfaction of the certifying agency.

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IEC 60079-30-1 (2007)	5.1.14	This time-current characteristic shall not be more than the value declared by the manufacturer. The samples shall be in the upper 1/3 of the manufacturer's declared power output tolerances.
IEC 60079-30-1 (2007)	None	**[Procedure from IEEE-515, Sections 4.3.1.3 and 4.3.5.3 – no IEC 60079-30-1 equivalent]** **Chemical exposure tests** The heating device and any integral components shall be exposed (completely immersed except for the connections) to the following chemicals: acetone, ethyl acetate, isooctane, hexane, methanol, methyl ethyl ketone, methylene chloride, and toluene. The exposure duration shall be 24 h. The chemical shall be maintained at a temperature not less than 5 °C below the boiling point. Boiling points shall be obtained from NFPA 325-1994. A separate heater sample including any polymeric sheath (overjacket) shall be stressed by wrapping it six close turns around a mandrel having a radius equal to 12 times the diameter of the primary bending plane of the device, and shall be exposed (completely immersed except for the ends) to each of the above chemicals. The test sample shall be removed from the fluid, allowed to stabilize for at least 1 h, and then subjected to the dielectric test outlined in 5.1.2 with the sample (except for the exposed ends) immersed in tap water at 10C to 25C for 5 minutes. The test voltage shall be applied between the water and the metallic covering for heating device constructions with a polymeric jacket, or between the water and the heating device conductors for heater constructions without a polymeric jacket. No dielectric breakdown shall occur. Additionally, stressed samples shall have no visible cracks or other damage when examined. This test confirms the integrity of the heater jacket or sheath after chemical exposure. An additional set of unstressed samples shall be exposed to the same chemicals in a like manner. After the exposure, the test samples shall be removed from the fluid, allowed to stabilize for at least 1 h, and then subjected to the following tests: *Deformation*—The deformation test described in 5.1.6 shall be conducted on the heating device. Proof loads shall be 204 kg (2000 N). *Impact*—The impact test described in 5.1.5 shall be conducted, retaining the same impact areas. The impact force shall be 27.1 J. *Cold bend*—The cold bend test described in 5.1.7 shall be conducted. *Ground path integrity*—Ground path conductance shall be measured based on 5.1.15. After the chemical exposure, the continuous metallic sheath, metallic braid, or other equivalent electrically conductive material shall have a conductivity equal to or greater than the manufacturer's

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		declared value.

IEC 60079-30-1 (2007)	5.2.1	~~Each supplied length or item, whether in bulk or individually fabricated items, shall be subjected to the dielectric test specified in 5.1.2, except that the test voltage for MI shall be 2*U* + 500 V a.c. for cables rated over 30 V a.c., or $\sqrt{2}\,U$ + 500 V d.c. for cables rated over 60 V d.c.~~ ~~The polymeric sheath (overjacket) used for corrosion resistance over the metallic braiding or continuous metallic covering shall be subjected to the dielectric test of 1000 Va.c. whilst immersed in water. As an alternative to immersing in water, the electrical insulation of trace heaters may be subjected to a dry spark test, with a minimum test voltage of 3 000 V a.c. r.m.s. that has a substantially sinusoidal waveform at 2 500 Hz to 3 500 Hz. For a 3 000 Hz supply, the rate of travel of the trace heater through the dry test device, in metres per second, shall not be more than 3,3 times the length of the electrode measured in centimetres.~~ The primary electrical insulation jacket of the heating device shall withstand a dry-spark test at a minimum of 6000 Vac. The dry-spark test shall have a substantially sinusoidal waveform at 2500 Hz to 3500 Hz. For a 3000 Hz supply, the speed of the wire in meters per second shall not be more than 33 times the length of the electrode in centimeters; this requirement is proportional to frequency. As an alternative to the dry-spark test, the dielectric tests in 5.1.2 may be conducted. After the installation of a metallic covering, metallic braid, or other equivalent electrically conductive material, ground plane, or continuous metal sheath the heating device shall have the dielectric test described in 5.1.2 conducted. Nonmetallic overjackets shall withstand an additional dry-spark test with a minimum test voltage of 3000 Vac. As an alternative to the dry-spark test, the dielectric tests in 5.1.2 may be conducted.
IEC 60079-30-1 (2007)	6.1	…in accordance with ~~IEC 60079-0~~ ISA/UL 60079-0 and ANSI/NFPA 70 Art. 505, and shall additionally include… Tubing bundles shall be marked on the bundle jacket. a) the symbol provided for the type of protection used for trace heating ~~shall be “e” for increased safety, which~~ does not preclude the use of additional types of protection appropriate for the components…

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IEC 60079-30-1 (2007)	6.2	… shall be marked in accordance with ~~IEC 60079-0~~ ISA/UL 60079-0 and ANSI/NFPA 70 Art. 505, and shall…
IEC 60079-30-1 (2007)	6.3	~~Any special conditions for safe use, including the provision in item f) below, shall be outlined in the installation instructions and in the certificate of conformity, and the product and the certificate shall be marked with the symbol "X".~~ ~~c) the statement "Ground fault equipment protection is required for each circuit".~~ ~~d) the statement "De-energise all power circuits before installation or servicing".~~ ~~e) the statement "Keep ends of trace heaters and kit components dry before and during installation".~~ ~~f) for trace heaters investigated for reduced levels of impact and/or deformation, the statement "Caution: Only use in areas with low risk of mechanical damage".~~ ~~h) the statement "The presence of the trace heaters shall be made evident by the posting of caution signs or markings at appropriate locations and/or at frequent intervals along the circuit".~~

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IECEx Bulletin

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 60079-31

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Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 60079-31

INTERNATIONAL
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Section 3
National Differences

IEC Standard 60079-31 1st Edition 2008 Explosive atmospheres - Part 31: Equipment dust ignition protection by enclosure "t"			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 60079-31:11
CH Switzerland	**NO**		EN 60079-31:2010
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 60079-31:2010
DE Germany	**NO**		EN 60079-31:2010
DK Denmark	**NO**		EN 60079-31:2010
FI Finland	**NO**		EN 60079-31:2010
FR France	**NO**		EN 60079-31:2010
GB United Kingdom	**NO**		EN 60079-31:2010
HR Croatia	**NO**		EN 60079-31:2010
HU Hungary	**NO**		EN 60079-31:2010
IN India		**N/R**	
IT Italy	**NO**		EN 60079-31:2010
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 60079-31:2010
NO Norway	**NO**		EN 60079-31:2010
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 60079-31:2010
RO Romania	**NO**		EN 60079-31:2010
RU Russia*		**TBA**	Included in the national standardization program for 2009
SE Sweden	**NO**		EN 60079-31:2010
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 60079-31:2010
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**NO**	SANS 60079-31:2008

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Please Note that IEC 60079-31 has been developed from the first Edition of IEC 61241-1 (2004) which it now supersedes.

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Countries that have declared adoption of IEC 60079-31 1st Edition 2008 Without National Differences
AU Australia
BR Brazil
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 60079-31 1st Edition 2008 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 60079-31:11
Explosive atmospheres –
Part 31: Equipment dust ignition protection by enclosure "t"

This is the first edition of CAN/CSA-C22.2 No. 60079-31, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 60079-31 (Edition 1.0, 2008-11). Publication of this Standard supersedes CAN/CSA-C22.2 No. E61241-1-1.

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

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The requirements of IEC 60079 series Standards cover the protection techniques with respect to gas explosion, dust explosion and layer ignition, hazards only. The CAN/CSA-C22-2 No. 60079 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE 1: The IEC 61241 series of Standards, cover dust-explosion and dust-layer combustion, are being converted (merged) into the IEC 60079 series of Standards. Due to the publication schedules (dates), some IEC 61241 and IEC 60079 Standards may not be fully aligned.

NOTE 2: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11 and CAN/CSA-C22.2 No. 61241-0:11.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)
Where reference is made to CSA publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following publication and the years shown indicated the latest editions available at the time of printing:

C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

6.1.1 Type tests for dust exclusion by enclosures

[Add a NOTE to the following effect]

NOTE: The ingress protection test is now based on the “IP” rating tests. The alternative testing by the heat/cold cycling (as permitted in former IEC 61241-1, which is now superseded by this Standard) is no longer an “alternative”.

7 Marking

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

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Section 3: National Differences for 61241-0

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Section 3
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IEC Standard: IEC 61241-0 1st Edition 2004 Electrical apparatus for use in the presence of combustible dust – Part 0: General requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-0: 2005
BR Brazil		**NO**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 61241-0:11
CH Switzerland	**NO**		EN 61240 - 0:2004
CN China		**NO**	GB 12476.1-(draft) identical IEC 61241-0:2004
CZ Czech Republic	**NO**		EN 61240 - 0:2004
DE Germany	**NO**		EN 61240 - 0:2004
DK Denmark	**NO**		DS EN 61240 - 0:2004
FI Finland	**NO**		EN 61240 - 0:2004
FR France	**NO**		EN 61240 - 0:2004
GB United Kingdom	**NO**		EN 61240 - 0:2004
HR Croatia	**NO**		
HU Hungary	**NO**		EN 61240 - 0:2004
IN India		**NO**	Adopted as IS/IEC 61241-0: 2004
IT Italy	**NO**		EN 61240 - 0:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 61240 - 0:2004
NO Norway	**NO**		EN 61240 - 0:2004
NZ New Zealand		**NO**	AS/NZS 61241-0: 2005
PL Poland	**NO**		
RO Romania	**NO**		EN 61240 - 0:2004
RU Russia		**NO**	GOST R IEC 61241-0-2007
SE Sweden	**NO**		EN 61240 - 0:2004
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 61240 - 0:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 61241-0 2006
ZA South Africa		**NO**	SANS 61241-0:2004

(N/R Not Recognised)

***** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details***

Section 3
National Differences

Countries that have declared adoption of IEC 61241-0 1st Edition 2004 Without National Differences
AU Australia
BR Brazil
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
RU Russia
ZA South Africa

Countries that have declared adoption of IEC 61241-0 1st Edition 2004 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 61241-0:11
Electrical apparatus for use in the presence of combustible dust – Part 0: General requirements

This is the first edition of CAN/CSA-C22.2 No. 61241-0, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 61241-0 (First edition, 2004-07).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption b the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee* *interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 61241series Standards cover the protection techniques with respect to dust explosion and layer ignition hazard only. The CAN/CSA-C22-2 No. 61241 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE 1: Currently, the IEC 61241 series of Standards are being converted (merged) into the IEC 60079 series of Standards. Some Standards, relevant to combustible dust environments, may be under the IEC 61241 series or under the IEC 60079 series depending on the scheme of progress.

NOTE 2: See also Canadian deviations under CAN/CSA-C22.2 No. 60079-0:11.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)
Where reference is made to CSA publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following publication and the years shown indicated the latest editions available at the time of printing:

C22.1-09
Canadian Electrical Code, Part I

CAN/CSA-M421-00 (R2005)
Use of electricity in mines

19.1 Plugs and sockets construction – b)

[Add a paragraph to the following effect]

Fasteners conforming to the national standard of the respective country are also permitted; provided that their effectiveness is equal to, or exceed, the effectiveness and criteria (e.g. strength, tolerance and configuration) stipulated in the Standard. In this case, such fasteners shall be specified in the marking or instruction manual; and the equipment shall marked with an "X" in accordance with item l) of 29.2

Section 3
National Differences

20 Caplights, caplamps and handlamps

[Add the following NOTE]

NOTE: These types of equipment, (cap-lights, etc.) are generally for use in mines susceptible to firedamp. Such equipment is under the jurisdiction of the Canadian Federal Government and is outside the scope of CSA.

22.5 Limits - (Apparatus incorporating cells and batteries)

[Add the following paragraph]

Where lithium types of cells (or batteries) are used, there shall be an infallible resistor or fuse in series with the cell (or battery) to limit the short-circuit current to the maximum permissible current drain as defined by the manufacturer of the cell (or battery).

NOTE: This requirement is to obviate the likelihood of an explosion of lithium cells (or batteries) from exploding in a direct short-circuit.

27.2 Cable entries with clamping by filling compound

[Add a paragraph to the following effect]

Due to the extreme weather in the Canadian geography; and, the preparation of the compound, the pouring and curing procedures are so weather and labour dependent, *clamping by filling compound* is **not** an acceptable method in Canada.

29.1 General – (Marking)

[Add the following paragraph]

It is a requirement in Canada that any warning and cautionary statements must be in both English and French.

29.2 Marking of all electrical apparatus

[The NOTE under Clause 29.2 (e) is no longer valid. Add a paragraph to the following effect]

The method of testing by “Practice A” or “Practice B” under IEC 61241-1 have been consolidated into one “Practice” only in the IEC 60079-31 (Edition 1.0: 2008-11) standard. It is no longer a requirement to have the marking “A” or “B” to denote the testing method. This also applies to some of the marking examples in Clause 30.

NOTE: The dust ingress protection test is based on IP ratings only.

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National Differences

US United States of America

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
General	Where references are made to other IEC 60079 or IEC 61241 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences. The term “cable entry” has been replaced with the term “cable gland” for consistency with IEC 60079-0, 4th Ed.
1	<u>Add new ultimate paragraph and note as follows:</u> <u>Electrical apparatus and Ex components for use in explosive dust atmospheres shall also comply with the applicable requirements for similar apparatus for use in unclassified locations.</u> <u>NOTE Requirements for safety of electrical equipment in ordinary (unclassified) locations can be found in ANSI Standards, NEMA Standards, Federal Regulations, etc. Apparatus listed by a Nationally Recognized Testing Laboratory is considered to meet the applicable requirements found in these standards. A list of commonly applied standards is shown in informative Annex A.</u>
2	<u>Add additional US reference standards as follows:</u> <u>ANSI/UL 486E, *Equipment Wiring Terminals for Use with Aluminum and/or Copper Conductors*</u> <u>ANSI/UL 746B, *Polymeric Materials – Long-Term Property Evaluations*</u> <u>ANSI/UL 746C, *Polymeric Materials – Use in Electrical Equipment Evaluations*</u> <u>ANSI/NFPA 70:2005, *National Electrical Code ®*</u>
3.3	Revise term as follows to align with US practice **3.3** **conductive dust** ~~dust, fibres or flyings with electrical resistivity equal to or less than 103 Ωm~~ <u>metal dust</u>

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
3.4	Revised term to align with that shown in the 5th Ed of IEC 60079-0 under preparation: **3.4** **explosive dust atmosphere** mixture with air, under atmospheric conditions, of flammable substances in the form of dust, fibres or flyings in which, after ignition, ~~combustion spreads throughout the unconsumed mixture~~ permits self sustaining propagation
3.15	Term has been deleted as follows as the term is not used in 61241-0: **3.15** ~~**zones**~~ ~~areas classified for explosive dust atmospheres are divided into zones; based upon the frequency and duration of the occurrence of explosive dust/air atmospheres~~
3.16	Term has been revised as follows to align with the National Electrical Code, ANSI/NFPA 70: **3.16** **Zone 20** ~~place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently~~ an area where combustible dust or ignitable fibers and flyings are present continuously or for long periods of time in quantities sufficient to be hazardous
3.17	Term has been revised as follows to align with the National Electrical Code, ANSI/NFPA 70: **3.17** **Zone 21** ~~place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur, occasionally, in normal operation~~ an area where combustible dust or ignitable fibers and flyings are likely to exist occasionally under normal operation in quantities sufficient to be hazardous

Section 3
National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
3.18	Term has been revised as follows to align with the National Electrical Code, ANSI/NFPA 70: **3.18** **Zone 22** ~~place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only~~ an area where combustible dust or ignitable fibers and flyings are not likely to occur under normal operation in quantities sufficient to be hazardous
3.41	The term has been revised to align with the definition used in IEC 60079-0, 4th Ed. **3.41** **Ex component** part of electrical apparatus or a module (other than an Ex cable gland), which is not intended to be used alone and requires additional consideration when incorporated into electrical apparatus or systems for use in explosive atmospheres ~~part of electrical apparatus for potentially explosive atmospheres, which is not intended to be used alone in such atmospheres and requires additional certification when incorporated into electrical apparatus or systems for use in potentially explosive atmospheres~~
3.42	Term has been deleted as the “X” suffix is not currently recognized in the US marking system: **3.42** **~~"X" symbol~~** ~~symbol used as a suffix to a certificate reference to denote special conditions for safe use~~
3.43	Term has been deleted as the “U” suffix is not currently recognized in the US marking system: **3.43** **~~"U" symbol~~** ~~"U" is the symbol used as a suffix to a certificate reference to denote an Ex component.~~

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
3.44	The term has been revised to align with the definition used in IEC 60079-0, 4th Ed. **3.44** **certificate** document that assures the conformity of a product, process, system, person, or organisation with specified requirements NOTE 1 The certificate may be either the supplier's declaration of conformity, or the purchaser's recognition of conformity, or certification (as a result of action by a third party) as defined in ISO/IEC 17000. ~~document confirming that the apparatus is in conformity with the requirements, the type tests and, where appropriate, the routine tests in the standard referred to therein~~ ~~NOTE 1 A certificate can relate to Ex apparatus or an Ex component.~~ NOTE 2 A certificate may be produced by the manufacturer, the user, or a third party, for example, an IEC Ex Accepted Certification Body, a national certification body, or an authorized person.
3.45	A definition has been added as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **3.45** **continuous operating temperature (COT)** maximum temperature which ensures the stability and integrity of the material for the expected life of the apparatus, or part, in its intended application
3.46	A definition has been added as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **3.46** **service temperature** temperature reached when the apparatus is operating at rated conditions NOTE Each apparatus may reach different service temperatures in different parts.

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
4.2	The text has been deleted as follows as there was no enforceable text and the appropriate requirements for Zone 20 are addressed in 61241-11 and 61241-18: ~~**4.2 Principles for design and testing of apparatus for use in Zone 20**~~ ~~Apparatus for Zone 20 needs special consideration.~~ ~~The apparatus shall be designed to be capable of functioning in conformity with the operational parameters established by the manufacturer and ensuring a very high level of protection.~~ ~~Apparatus in Zone 20 is intended for use in areas in which explosive atmospheres caused by air/dust mixtures are present continuously, for long periods or frequently. Apparatus in this zone must ensure the requisite level of protection, even in the event of rare incidents relating to equipment, and is characterized by means of protection such that either~~ ~~- in the event of failure of one means of protection, at least an independent second means provided the requisite level of protection; or~~ ~~- the requisite level of protection is assured in the event of two faults occurring independently of each other.~~ ~~The special requirements for Zone 20 shall be investigated under simulated working conditions as stated by the manufacturer.~~ ~~NOTE 1 Apparatus for measurement and control techniques (e.g. instrumentation, sensors, controls) are typical applications under dust of excessive layers.~~ ~~NOTE 2 Power engineering apparatus (such as motors, luminaires, plugs and sockets) should wherever practicable be placed outside of such areas.~~

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
4.3	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **"WARNING - AFTER DE-ENERGIZING, DELAY X MINUTES BEFORE OPENING"** "X" being the value in minutes of the delay required. Alternatively the apparatus may be marked with the following or equivalent warning: **"WARNING - DO NOT OPEN WHEN AN EXPLOSIVE DUST ATMOSPHERE IS PRESENT".**
5.1	The text has been modified as follows to reflect the required temperature testing of the various types of protection: **5.1 Maximum surface temperature** For electrical apparatus, the maximum surface temperature shall be specified in relevant documentation according to 23.2. This maximum surface temperature shall be ~~arranged~~ determined by the methods specified for the specific types of protection employed, and marked according to 29.2 7) ~~g)~~ and shall be either: - defined by the actual maximum surface temperature, or, if appropriate, - restricted to the specific combustible dust for which it is intended.
5.2	The text has been deleted as shown to align with US practice for dust layers: ~~**5.2 Maximum surface temperature with respect to dust layers above 50 mm**~~ ~~In addition to the maximum surface temperature required in 5.1, the maximum surface temperature may be stated for a given depth of layer, *T*L, of dust surrounding all sides of the apparatus, unless otherwise specified in the documentation, and marked according to 29.2 h).~~

Section 3
National Differences

<table>
<tr><th>Clause</th><th>US Differences to IEC 61241-0, 1st Ed, 2004</th></tr>
<tr><td>5.3</td><td>The text has been modified as shown to add clarity and to acknowledge the lack of US marking for “X”:

5.3 Ambient temperature
Electrical apparatus shall normally be designed for use in the ambient temperature range between –20 ºC and +40 ºC; in this case, no additional ambient temperature marking is necessary.

When the electrical apparatus is designed for use in a different range of ambient temperatures, it is considered to be special; the ambient temperature range shall then be stated by the manufacturer and specified in the certificate; the marking shall then include either the symbol "Ta" or "Tamb" together with the special range of ambient temperatures <s>or, if this is impracticable, the symbol "X" shall be placed after the certificate reference, according to 29.2 l) (see Table 1).</s>

Table 1 – Ambient temperatures in service and additional marking
<table><tr><th>Electrical apparatus</th><th>Ambient temperature in service</th><th>Additional marking</th></tr><tr><td>Normal</td><td>Maximum: +40 ºC,
Minimum: –20 ºC</td><td>None</td></tr><tr><td>Special</td><td>Stated by the manufacturer and specified in the certificate</td><td>Ta or Tamb with the special range, for example "–30 ºC ≤ Ta ≤ +40 ºC"<s> or the symbol "X".</s></td></tr></table></td></tr>
<tr><td>6.1.1</td><td>Modify text as follows to note that documents are not submitted to the manufacturer:

6.1.1 Material specification
Documents <s>submitted to the manufacturer </s>shall specify both the material and the manufacturing process of the enclosure or part of the enclosure.</td></tr>
</table>

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
6.1.2	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **6.1.2 Plastic materials** The specification for plastic materials shall include a) the name of the manufacturer; b) the exact and complete reference of the material including its colour, percentage of fillers and any other additives if used; c) the possible surface treatments, such as varnishes, etc.; d) the temperature index "TI" corresponding to the 20 000 h point on the thermal endurance graph without loss of flexural strength exceeding 50 %, determined in accordance with IEC 60216-1 and IEC 60216-2 and based on the flexing property in accordance with ISO 178. If the material does not break in this test before exposure to the heat, the index shall be based on the tensile strength in accordance with ISO 527 with test bars of Type 1A or 1B. As an alternative to the Temperature Index TI, the Relative Thermal Index (RTI – mechanical impact) may be determined in accordance with ANSI/UL 746B. The data by which these characteristics are defined shall be supplied by the ~~apparatus~~ manufacturer.
6.1.3	Modify text as shown to reflect that 3rd party certification requirements are not appropriate in an equipment standard: ~~**6.1.3 Verification of compliance**~~ ~~The testing station is not required to verify compliance of the material with its specification.~~

Section 3
National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
6.1.4.1	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **6.1.4.1 Temperature index** For Zone 20 and 21 apparatus, the plastic materials shall have a temperature index “TI” corresponding to the 20 000 h point or RTI–mechanical of at least 20 K greater than the temperature of the hottest point of the enclosure or the part of the enclosure (see 23.4.7.1), having regard to the maximum ambient temperature in service. For Zone 22 apparatus, the plastic materials shall have a temperature index TI corresponding to the 20 000 hour-point or RTI–mechanical (see IEC 60216-1 and IEC 60216-2) or a continuous operating temperature (COT) of at least 10 K greater than the temperature of the hottest point of the enclosure having regard to the maximum ambient temperature in service according to data supplied by the apparatus manufacturer.
6.2.1	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **6.2.1 Composition** Materials used in the construction of enclosures of electrical apparatus to be used in explosive dust atmospheres shall not contain, by ~~weight~~ mass, more that 7,5 % in total of magnesium and titanium.
10.3	Modify text as shown to reflect that 3rd party certification requirements are not appropriate in an equipment standard: ~~**10.3 Verification**~~ ~~The testing station is not required to verify the characteristics listed in the documents mentioned in 10.1.~~

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
12.1	The text has been modified as shown to acknowledge the lack of US marking for "X": **12.1 Attached cables** Electrical apparatus intended for connection to external circuits shall include connection facilities, except where the electrical apparatus is manufactured with a cable permanently connected to it. All apparatus constructed with permanently connected unterminated cables shall be marked in accordance with 29.2 ~~with the symbol "X"~~ to indicate the need for appropriate connection of the free end of the cable.
13.2	The text has been deleted as shown as the US installation and construction requirements do not require an external earthing terminal. ~~**13.2 External connection**~~ ~~Electrical apparatus with a metallic enclosure shall have an additional external connection facility for an earthing or equipotential bonding conductor. This external connection facility shall be electrically in contact with the facility required in 13.1. The external connection facility is not required for electrical apparatus which is designed to be moved when energized and is supplied by a cable incorporating an earthing or equipotential bonding conductor.~~
13.3	The text has been modified as shown to clarify that some double-insulated equipment may require bonding to avoid electrostatic discharge: **13.3 ~~Facility not required~~ Apparatus not requiring earthing** ~~Neither an internal nor external earthing or~~ A bonding connection facility is not required for electrical apparatus for which earthing (or bonding) is not required, such as electrical apparatus having double or reinforced insulation, or for which supplementary earthing is not necessary. NOTE Double insulated apparatus, while not presenting a risk of electrical shock, may need to be earthed or bonded to reduce the risk of ignition.

Section 3
National Differences

<table>
<tr><th>Clause</th><th>US Differences to IEC 61241-0, 1st Ed, 2004</th></tr>
<tr><td>13.4</td><td>

The text has been modified as shown to align with the conductor sizing requirements of the National Electrical Code and the pullout and secureness requirements of ANSI/UL 486E:

13.4 ~~Effective connection~~ Size of conductor connection
Earthing or equipotential bonding connection facilities shall allow for the effective connection of at least one conductor with a cross-sectional area as ~~shown in Table 2~~ given in ANSI/NFPA 70 National Electrical Code Section 250.122.

~~Table 2 - Minimum cross-sectional areas of protective conductors~~

~~Cross-sectional area of phase conductors of the installation S~~ ~~mm^2~~	**~~Minimum cross-sectional area of the corresponding protective conductor Sp~~** ~~mm^2~~
~~$S \leq 16$~~	~~S~~
~~$16 < S \leq 35$~~	~~16~~
~~$S > 35$~~	~~0,5 S~~

~~In addition to meeting this requirement,~~ When provided, the supplemental earthing or bonding connection facilities on the outside of electrical apparatus shall provide for effective connection of a conductor of at least 4 mm^2(10 AWG). Effective connection shall be verified by compliance with the pullout and secureness tests in ANSI/UL 486E. Terminals shall be marked in accordance with 29.6.

NOTE The requirements for equipment grounding facilities and bonding are specified in the appropriate standard for electrical equipment in unclassified locations and may be more stringent than the requirements shown in 29.6. Refer to Annex B for further information.

</td></tr>
</table>

Section 3
National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
14 14.1	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **14 ~~Cable and conduit entries~~ Entries into enclosures** **14.1 ~~Intended use~~ Identification of entries** ~~The manufacturer shall specify in the documents submitted according to 23.2 the entries intended for use with cable or conduit, their position on the apparatus and the maximum number permitted.~~ The manufacturer shall specify in the documents prepared according to 23.2, the entries, their position on the apparatus, and the maximum number permitted. The thread form (for example, metric or NPT) of threaded entries shall be marked on the apparatus.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
14.2	The text has been revised as follows to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **14.2 ~~Construction~~ Cable glands** ~~Cable and conduit entries shall be constructed and fixed so that they do not alter the specific characteristics of the type of protection of the electrical apparatus on which they are mounted. This shall apply to the whole range of cable dimensions specified by the manufacturer of the cable entries as suitable for use with those entries.~~ Cable glands shall not invalidate the specific characteristics of the type of protection of the electrical apparatus on which they are mounted. This shall apply to the whole range of cable dimensions specified by the manufacturer of the cable glands as suitable for use with those glands. Cable glands may form an integral part of the apparatus, i.e. one major element or part forms an inseparable part of the enclosure of the apparatus. In such cases, the glands shall be tested with the apparatus. NOTE Cable glands, which are separate from, but installed with, the apparatus are usually tested separately from the apparatus but may be tested together with the apparatus if the apparatus manufacturer so requests. Cable glands, whether integral or separate, shall meet the relevant requirements of the specific type of protection. NOTE 1 The ordinary location requirements for cable glands can be found in ANSI/UL 514B, "Conduit, Tubing, and Cable Fittings". NOTE 2 The requirements for cable glands, type of protection "iD", can be found in ANSI/UL 514B, "Conduit, Tubing, and Cable Fittings". NOTE 3 The requirements for cable glands, type of protection "tD" can be found in ANSI/UL 2225, "Metal-Clad Cables and Cable-Sealing Fittings for use in Hazardous (Classified) Locations".

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
14.3	Text has been deleted as shown as the concept is now addressed in the modified 14.2: ~~**14.3 Integral part of the apparatus**~~ ~~Cable and conduit entries may form an integral part of the apparatus, i.e. one major element or part forms an inseparable part of the enclosure of the apparatus. In such cases the entries shall be tested and certified with the apparatus.~~ ~~NOTE Cable and conduit entries, which are separate from, but installed with the apparatus, are usually tested and certified separately from the apparatus but may be tested and certified together with the apparatus if the apparatus manufacturer so requests.~~
14.5	The text has been revised as follows to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **14.5 ~~Method of attaching~~ <u>Enclosure entry</u>** Entry by conduit or cable <u>gland</u> ~~entries~~ shall be either by screwing into threaded holes or by locking in plain holes - in the wall of the enclosure, or - in an adaptor plate designed to be fitted in or on the walls of the enclosure.~~, or~~ ~~- into a suitable stopping box, integral with or attached to the wall of the enclosure.~~
14.7	The text has been revised as follows to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **14.7 Branching point temperatures** ~~When the temperature under rated conditions, including any manufacturer's installation requirements, is higher than 70°C at the cable or conduit entry point, or 80°C at the branching point of the conductors, the outside of the electrical apparatus shall be marked as a guide for the selection by the user of the cable or of the wiring in the conduit, in order to ensure that the rated temperature of the cable is not exceeded (see Figure 1).~~<u>When the temperature under rated conditions is higher than 60 °C at the entry point or 60 °C at the branching point of the conductors, information shall be provided in the instructions to provide guidance to the user on the proper selection of cable and cable gland or conductors in conduit.</u>

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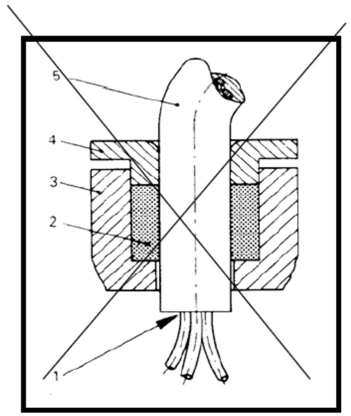
National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
Figure 1	Figure 1 has been replaced bu Figure 1A and Figure 1B as follows to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.:

~~Key~~
~~1) Branching point of the conductors~~
~~2) Sealing ring~~
~~3) Cable entry body~~
~~4) Clamping ring with curved rim~~
~~5) Cable~~



~~**Figure 1 - Illustration of entry points and branching points**~~

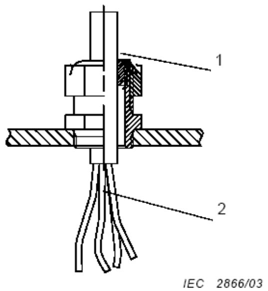


Figure 1a – Cable gland

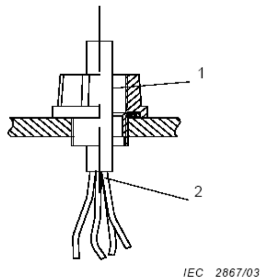


Figure 1b – Conduit entry

Key
1 Entry point
2 Branching point

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National Differences

Clause	US Differences to IEC 61241-0, 1st Ed, 2004
15	The text has been revised as follows to align the text with the current proposed text for IEC 60079-0, Ed. 5.: **15 Radiating equipment** ~~The energy levels of radiation generating equipment shall not exceed the values given below (see IEC 61241-14).~~ The threshold power of radio frequency (10 kHz to 300 GHz) for continuous transmissions and for pulsed transmissions whose pulse durations exceed a thermal initiation time of 200 µs shall not exceed 6W. Programmable or software control intended for setting by the user shall not be permitted. For pulsed radar and other transmissions where the pulses are short compared with the thermal initiation time, the threshold energy value shall not exceed 1500 µJ. The output parameters of lasers or other continuous wave sources shall not exceed 5 mW/mm^2 or 35 mW for continuous wave lasers and other continuous wave sources; and 0,1 mJ/mm^2 for pulse lasers or pulse light sources with pulse intervals of at least 5 s. The output parameters from ultrasonic sources shall not exceed 0,1W/cm^2 and 10MHz for continuous sources; or an average power density 0,1W/cm^2 and 2mJ/cm^2 for pulse sources. **~~15.1 Lasers and other continuous wave source~~** **~~15.1.1 Zone 20 and Zone 21~~** ~~Radiation-generating electrical equipment, if tested and permitted in accordance with this specification for Zone 20 or 21, may be used. Independently of this fact it shall be ensured that irradiation power or irradiation that may penetrate into or occur in Zone 20 or 21, even in the case of rare disturbances in the entire part of the radiation process proceeding in Zone 20 or 21 and at any point in the radiation cross-section shall not exceed the following values:~~

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	~~• 5 mW/mm^2 or 35mW for continuous wave lasers and other continuous wave sources, and~~ ~~• 0,1 mJ/mm^2 for pulse lasers or pulse light sources with pulse intervals of at least 5 s.~~ ~~Radiation sources with pulse intervals of less than 5 s are regarded as continuous light sources in this respect.~~ ~~**15.1.2 Zone 22**~~ ~~Equipment generating radiation may be used. The irradiation intensity or irradiation shall not exceed 10 mW/mm2 or 35 mW continuous and 0,5 mJ/mm^2 for pulse in normal operation.~~ ~~**15.2 Ultrasonic sources**~~ ~~For ultrasonic sources the power level shall not exceed a power density in the sound field of 0,1 W/cm^2 and a frequency of 10 MHz for continuous sources and 2 mJ/cm^2 for pulse sources. The average power density shall not exceed 0,1 W/cm^2.~~ ~~**15.2.1 Zone 20 and Zone 21**~~ ~~In Zone 20 or 21 ultrasonics sources shall not exceed a power density in the sound field of 0,1 W/cm^2 and a frequency of 10 MHz for continuous sources and 2 mJ/cm^2 for pulse sources. The average power density shall not exceed 0,1 W/cm^2 .~~ ~~**15.2.2 Zone 22**~~ ~~In Zone 22 no special safety measures against ignition hazards due to the use of ultrasonics themselves are necessary, insofar as the power density in the sound field generated does not exceed 0,1 W/cm^2 and an installed frequency of 10 MHz.~~
16.1	The text has been modified as shown so that it is applicable to all Zones: **16.1 Ventilation openings for external fans**

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	The degree of protection (IP) of ventilation openings for external fans of rotating electrical machines shall be at least - IP20 on the air inlet side, - IP10 on the air outlet side, according to IEC 60034-5. For vertical rotating machines ~~for use in Zone 20 or 21~~, foreign objects shall be prevented from falling into the ventilation openings.
16.3	The text has been modified as shown so that it is applicable to all Zones: **16.3 Clearances for the ventilating system ~~for use in Zone 20 or 21~~** In normal operation the clearances, taking into account design tolerances, between the external fan and its hood, ventilation screens and their fasteners shall be at least one hundredth of the maximum diameter of the fan, except that the clearances need not exceed 5 mm and may be reduced to 1 mm if the opposing parts are manufactured so as to have dimensional accuracy and stability. In no case shall the clearance be less than 1 mm.
16.4.2	The text has been modified as shown so that it is applicable to all Zones: **16.4.2 Thermal stability of plastic materials ~~for use in Zones 20 and 21~~** The thermal stability of plastic materials shall be considered adequate if the manufacturer’s specified operating temperature of the material exceeds the maximum temperature to which the material will be subjected in service (within the rating) by at least 20 K.
16.4.3	The text has been modified as shown so that it is applicable to all Zones and to align with the requirements of 6.2.1: **16.4.3 Materials containing light metals ~~for use in Zones 20 and 21~~**

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	The external fans, fanhoods and ventilation screens, of rotating electrical machines, manufactured from materials containing light metals shall not contain by weight mass more than 7,5% of magnesium and titanium.
17.2	The text has been modified as shown to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **17.2 Interlocking** Disconnectors (which are not designed to be operated under the intended load) shall - be electrically or mechanically interlocked with a suitable load breaking device, or - be marked at a place near the actuator of the disconnector, with the warning **“WARNING-DO NOT OPERATE UNDER LOAD”.**
17.4	The text has been modified as shown to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **17.4 Openings** Doors and covers giving access to the interior of enclosures containing remotely operated circuits with switching contacts which can be made or broken by non-manual influences (such as electrical, mechanical, magnetic, electro-magnetic, electro-optical, pneumatic, hydraulic, acoustic or thermal) shall either - be interlocked with a disconnector which prevents access to the interior unless it has been operated to disconnect unprotected internal circuits; or - the apparatus shall be marked with the warning: **"WARNING-DO NOT OPEN WHEN ENERGIZED".**
18	The text has been modified as shown to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **18 Fuses** Enclosures containing fuses shall either - be interlocked so that insertion or removal of replaceable elements can be carried out only with the supply disconnected

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	and so that the fuses cannot be energized until the enclosure is correctly closed; or alternatively - the apparatus shall be marked with the warning **"WARNING-DO NOT OPEN WHEN ENERGIZED"**
19 19.1	The text has been modified as shown to reflect the US practice that only interlocking is a suitable means of protection for plugs and sockets: **19 Plugs and sockets** ~~The requirements for plugs and sockets are not applicable for type of protection Ex iD.~~ **19.1 Plugs and sockets construction** Plugs and sockets shall comply with either a) or b) below: a) be interlocked mechanically, or electrically, or otherwise designed so that they cannot be separated when the contacts are energized and the contacts cannot be energized when the plug and socket are separated. ;or ~~b) be fixed together by means of special fasteners that shall conform to the following:~~ ~~- the thread shall be coarse pitch in accordance with ISO 262, with a tolerance fit of 6g/6H in accordance with ISO 965;~~ ~~- the head of the screw or nut shall be in accordance with ISO 4014, ISO 4017, ISO 4032 or ISO 4762, and in the case of hexagon socket set screws ISO 4026, ISO 4027, ISO 4028 or ISO 4029;~~ ~~- the holes of the electrical apparatus shall be threaded for a distance to accept a thread engagement, h, at least equal to the major diameter of the thread of the fastener (see figures 2 and 3).~~ ~~- The thread shall have a tolerance fit of 6H in accordance with ISO 965, and either a) the hole under the head of the associated fastener shall allow a clearance not greater than a medium tolerance fit of H13 in accordance with ISO 286-2 (see Figure 2 and ISO 273); or~~ ~~b) the hole under the head (or nut) of an associated reduced shank fastener shall be threaded to enable the fastener to be retained. The dimensions of the threaded hole shall be such that the surrounding surface in contact with the head of such a fastener shall be at least equal to that of a fastener without a reduced shank in a clearance hole (see Figure 3).~~ ~~In the case of hexagon socket set screws the screw head shall have a tolerance fit of 6H in accordance with ISO 965 but~~

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	~~shall not protrude from the threaded hole after tightening, and the apparatus marked with the warning:~~ ~~**"DO NOT SEPARATE WHEN ENERGIZED".**~~
19.2	The text has been modified as follows to add a signal word, allow technically equivalent text, and specify what the "hazard" is: **19.2 Bolted plugs and sockets** Where bolted types cannot be de-energized before separation because they are connected to a battery, the marking shall then state the following or equivalent **"WARNING-SEPARATE ONLY WHEN A COMBUSTIBLE DUST ~~HAZARD DOES NOT EXIST~~ ATMOSPHERE IS NOT PRESENT".**
20.1	The text has been modified as shown to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **20.1 Light transmitting covers** The source of light of luminaires shall be protected by a light-transmitting cover, which may be provided with an additional guard with no individual opening greater than 2 500 mm^2 ~~comprising a mesh of not greater than 50 mm squares~~. If any opening size exceeds this ~~mesh sizes exceed 50 mm squares~~ then the luminaire cover shall be considered as unguarded.
20.4	The text has been modified as shown to be applicable to all luminaires, not just intrinsically safe luminaires, as compliance with 60079-11 may not provide an adequate level of safety in an explosive dust atmosphere: **20.4 Covers** ~~Except in the case of intrinsically safe luminaries to IEC 60079-11,~~ Covers giving access to the lampholder and other internal parts of luminaires shall either a) be interlocked with a device which automatically disconnects all poles of the ~~lampholder~~ luminaire as soon as the cover opening procedure begins; or

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	b) be marked with the warning **"WARNING-DO NOT OPEN WHEN ENERGIZED".**
20.5	The text has been deleted as shown as compliance with the requirements as stated would not provide an adequate level of safety in an explosive dust atmosphere: ~~**20.5 Parts remaining energized**~~ ~~In the case of a) above, where it is intended that some parts other than the lampholder will remain energized after operation of the disconnecting device, then in order to minimize the risk of explosion, those energized parts shall be protected by:~~ ~~- clearances and creepage distances between phases (poles) and to earth in accordance with the requirements of IEC 60079-7 ; and~~ ~~- an internal supplementary enclosure (which can be the reflector for the light source) which contains the energized parts and provides a degree of protection of at least IP30, according to IEC 60529; and~~ ~~- marking on the internal supplementary enclosure with the warning :~~ ~~**DO NOT OPEN WHEN ENERGIZED".**~~
21	The title was revised as shown to align with the text of the US adoption of IEC 60079-0, Ed. 4.: **21 Caplights~~, caplamps~~ and ~~handlamps~~ handlights**

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
21.1	The text was modified as shown to identify that the leakage of electrolyte is from the battery. The note was considered extraneous and was deleted as the leakage is required to be prevented: **21.1 Leakage** Leakage of the electrolyte from the battery shall be prevented in all positions of the apparatus. ~~NOTE The materials used for handlamps and caplights, which may be exposed to the electrolyte, should be chemically resistant to the electrolyte.~~
21.2	The text was modified as shown to address the US practice of connection using a cord for this type of equipment: **21.2 Separate enclosures** Where the source of light and the source of supply are housed in separate enclosures, which are not mechanically connected other than by an electric cable, the cable ~~entries~~ glands and the connecting cable shall be tested as appropriate according to the specific type of protection ~~Clause 27 or Clause 28~~. The cable shall be a cord which is listed for extra-hard usage.
23.4.1	The text was modified as shown to align with the text of the US adoption of IEC 60079-0, Ed. 4.: **23.4.1 General** ~~The prototype or sample shall be tested in accordance with the requirements for type tests of this standard. However, the responsible party~~ ~~- may omit certain tests judged to be unnecessary. A record of all tests carried out and the justification for those omitted shall be kept;~~ ~~- shall not conduct the tests which have already been carried out by an accepted source on an Ex component.~~

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
	The prototype or sample shall be tested in accordance with the requirements for type tests of this standard and of the specific standards for the types of protection concerned. However, certain tests judged to be unnecessary may be omitted from the testing program. A record shall be made of all tests carried out and of the justification for those omitted. It is not necessary to repeat the tests that have already been carried out by an accepted source on an Ex component. ~~For tests required to be carried out by a testing station, the tests shall be made either in the laboratory of the testing station or, subject to agreement between the testing station and the manufacturer, elsewhere under the supervision of the testing station, for example at the manufacturer's works.~~ Each test shall be made in that configuration of the apparatus, which is considered to be the most unfavourable.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
23.4.2.1	The text has been modified as shown to correct the acceleration of gravity, remove the requirements related to certification, clarify the impact test requirements for light transmitting parts, and add a tolerance to the temperature test values: **23.4.2.1 Test for resistance to impact** In this test the electrical apparatus is submitted to the effect of test mass of 1 kg falling vertically from a height (h). The height (h) is dependent on the impact energy (E), which is specified in Table 5 according to the application of the electrical apparatus (h = E/~~10~~ 9,8; h in metres and E in joules). The test mass shall be fitted with an impact head in hardened steel in the form of a hemisphere 25 mm in diameter.Before each test, it is necessary to check that the surface of the impact head is in goodcondition. Normally the resistance to impact test is made on apparatus which is completely assembled and ready for use; however, if this is not possible (e.g. for light-transmitting parts) the test is made with the relevant parts removed but fixed in their mounting or an equivalent frame. Tests on an empty enclosure are permitted ~~only if there has been prior agreement between the manufacturer and testing station.~~ with appropriate justification in the documentation (see 23.2). For light-transmitting parts made of glass, the test shall be made on three samples but only once on each. In all other cases the test shall be made on two samples, at two separate places on each sample. The points of impact shall be the places considered ~~by the testing station, or as agreed by the manufacturer and purchaser,~~ to be the weakest. The electrical apparatus shall be mounted on a steel base so that the direction of the impact is normal to the surface being tested if it is flat, or normal to the tangent to the surface at the point of impact if it is not flat. The base shall have a mass of at least 20 kg or be rigidly fixed or inserted in the floor (secured in concrete, for example). **Table 5 - Tests of resistance to impact**

	Impact energy joules	
Risk of mechanical danger	**High**	**Low**
1. Guards, protective covers, fanhoods, cable entries	7	4
2. Plastic enclosures	7	4
3. Light metal or cast metal enclosures	7	4
4. Enclosures of materials not included in 3 with wall thickness less than 1 mm	7	4
5. Light-transmitting parts without guard	4	2
6. Light transmitting parts with guard having individual openings from 625 mm^2 to 2 500 mm^2 (tested without guard)	2	1
NOTE A guard for light transmitting parts having individual openings from 625 mm^2 to 2 500 mm^2 reduces the risk of impact, but does not prevent impact.		

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
	When an electrical apparatus is submitted to tests corresponding to the low risk of mechanical danger, it shall be marked with the symbol “X” according to 29.2. Normally the test is carried out at an ambient temperature of (20 ± 5) °C, except where the material data shows it to have a reduction in resistance to impact at lower temperatures within the specified ambient range, in which case the test shall be performed at <u>5 K to 10 K below</u> the lowest temperature within the specified range. When the electrical apparatus has an enclosure or a part of an enclosure in plastic material, including plastic fanhoods and ventilation screens in rotating electrical machines, the test shall be carried out at the upper and lower temperatures according to 23.4.7.1
23.4.2.2	The text has been modified as shown to remove the requirements related to certification and add a tolerance to the temperature test values: **23.4.2.2 Drop test** In addition to being submitted to the resistance to impact test according to 23.4.2.1, handheld electrical apparatus or electrical apparatus carried on the person, ready for use, shall be dropped four times from a height of 1 m onto a horizontal concrete surface. The position of the sample for the drop test shall be ~~selected by the testing station, or as agreed by the manufacturer and purchaser~~ <u>that which is considered to be the most unfavorable.</u> For apparatus with an enclosure in other than plastic material the test shall be carried out at a temperature of (20 ± 5) °C, except where the material data shows it to have a reduction in resistance to impact at lower temperatures within the specified ambient range, in which case the test shall be performed at <u>5 K to 10 K below</u> the lowest temperature within the specified range. For electrical apparatus, which has enclosures or parts of enclosures made of plastic material, the tests shall be carried out at <u>5 K to 10 K below</u> the lower ambient temperature according to 23.4.7.1.
23.4.3	The text was modified as follows to add acceptance criteria for IEC 60034-5:

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
	23.4.3 Test for dust exclusion (degree of protection) Depending on the environmental conditions likely to be encountered (such as area classification and conductivity of dust) two levels of dust exclusion efficiency have been adopted: “dust-tight” and “dust-protected” enclosures. Applicable levels of dust exclusion are dependant on the type of protection employed and specified in the applicable part of IEC 61241 series for that type of protection. For the purposes of acceptance criteria in accordance with IEC 60034-5, all dusts shall be considered to be conductive. Where a standard for electrical apparatus for explosive dust atmospheres specifies acceptance criteria for IPXX, these shall be applied instead of those in IEC 60529 or IEC 60034-5.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
23.4.5.1	Text was modified as follows to add an alternative method for temperature rise testing of motors, clarify the testing at the maximum rated ambient temperature, and delete the unenforceable paragraph on faults and malfunctions: **23.4.5.1 Measurement for maximum surface temperature** The thermal tests shall be made at the rating of the electrical apparatus ~~at an ambient temperature between 10 °C and 40 °C~~ and with the most unfavourable voltage between 90 % and 110 % of the rated voltage of the electrical apparatus, unless other IEC publications prescribe other tolerances for equivalent industrial electrical apparatus. For electrical machines, determination of the maximum surface temperature may alternatively be conducted at the worst case test voltage within "Zone A" per IEC 60034-1. In this case, the rating shall be marked on the equipment and included in the manufacturer's instructions. The test shall be made under the most adverse conditions, including overloads and recognized abnormal conditions that may be specified in an IEC standard giving specific requirements for the electrical apparatus concerned. Adverse conditions may also arise from the use of electrical apparatus on inverter supplies, frequent starting, etc. ~~For Zone 20 the adverse conditions must take into account 2 simultaneous faults or rare malfunctions, for Zone 21 foreseeable malfunctions and for Zone 22 normal operation applies.~~ The measurement of the surface temperatures shall be made with the electrical apparatus mounted in its normal service position. For electrical apparatus, which can be normally used in different positions, the temperature in each position shall be determined and the highest temperature considered. When the temperature is determined for certain positions only this shall be specified in the test report and the electrical apparatus shall be marked accordingly. The measuring devices (thermometers, thermocouple, etc.) and the connecting cables shall be selected and so arranged that they do not significantly affect the thermal behaviour of the electrical apparatus. The final temperature is considered to have been reached when the rate of rise of temperature does not exceed 2 K/h. Maximum surface temperature shall be corrected for the maximum ambient temperature specified in the rating.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
23.4.5.2	The text was deleted as shown as the specification of dust layer thickness is not US practice: ~~**23.4.5.2 Measurement for surface temperature under excess layer**~~ ~~If the requirements of 5.2 apply, then the electrical apparatus to be tested shall be mounted and surrounded by a layer depth *L* as stated by the manufacturer's specification. The measurement for the maximum surface temperature shall be made according to 23.4.4.1 using a dust having a thermal conductivity of no more than 0,003 kcal/m ºCh.~~
23.4.5.3	The text was deleted as requirements were considered unenforceable and the referenced unenforceable text of 23.4.5.1 has also been deleted: ~~**23.4.5.3 Temperature control**~~ ~~Some apparatus may require the provision of integral temperature sensitive devices. e.g. some electric motors, fluorescent luminaires, etc. The effect of such devices shall be tested under simulated working conditions. This protection shall be subject to fault consideration as 23.4.5.1 according to the intended zone of use.~~
23.4.7	The title was modified as shown as the non-metallic requirements apply to apply equipment: **23.4.7 Tests of non-metallic enclosures or of non-metallic parts of apparatus ~~for use in Zone 20 or 21~~**

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
23.4.7.5.1	The text was modified as shown to address current US practice for UV resistance: **23.4.7.5.1 General** A test of resistance of the material to light shall be made only if the enclosure or parts of enclosure made of plastics materials are not protected from light. The test is to be conducted as follows, unless the material is otherwise evaluated in accordance with the resistance to ultraviolet light test of ANSI/UL 746C. The test shall be made on six test bars of standard size 50 mm x 6 mm x 4 mm according to ISO 179. The test bars are to be made under the same conditions as those used for the manufacture of the enclosure concerned; these conditions are to be stated in the test report of the electrical apparatus.
23.4.7.5.2	The text was modified as shown to acknowledge the lack of US marking for "X": **23.4.7.5.2 Tests not applied** If the apparatus is protected from light (for example daylight or light from luminaires) when installed, and the test of 23.4.7.5.1 in consequence is not carried out, the apparatus shall be ~~'X'~~ marked according to 29.2.
23.4.7.7	The text of the 2nd paragraph was modified as follows to transfer the actual requirements from 60079-0 top this document: The test piece shall have an intact surface and shall be cleaned with distilled water, then with isopropyl alcohol (or any other solvent that can be mixed with water and will not affect the material of the test piece), then once more with distilled water before being dried. Untouched by bare hands, it shall then be conditioned for 24 h at (23°±°2)° C and (50°±°2)% relative humidity ~~the temperature and humidity according to 7.3 of IEC 60079-0~~. The test shall be carried out under ambient conditions.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
23.4.7.8	The text has been deleted as shown as the tests for cable glands in Clause 27 have been deleted. The US requirements for cable glands are specified in the notes to 14.2: ~~**23.4.7.8 Ageing test for material used for elastomeric sealing rings**~~ ~~The material used for the manufacture of the sealing rings is prepared in the form of test pieces in accordance with the standards ISO 48 and ISO 1818; the hardness is determined in accordance with the same standards at ambient temperature.~~ ~~The test piece is then placed in an oven in which the temperature is maintained at 100 ± 5 °C for at least 168 h without interruption; they are then kept for at least 24 h at ambient temperature, then placed in a refrigerator in which the temperature is maintained at - 20 ± 2 °C for at least 48 h without interruption; they are finally kept for at least 24 h at ambient temperature. The hardness is then determined again.~~ ~~At the end of the test procedure the variation in hardness, expressed in IRHD units as specified in the ISO standards given above, shall not exceed 20 % of the hardness before ageing.~~ ~~Where a cable entry is intended to be used at a temperature above that foreseen in 14.7 the ageing test shall be carried out at a temperature 20 ± 5 K above the declared maximum operating temperature of the cable. Where a cable entry is intended to be used in an ambient temperature below – 20 °C, the test in the refrigerator shall be carried out at the declared minimum ambient temperature with a tolerance of ± 2 K.~~

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
25	The text has been modified as shown to remove the requirements for 3rd party certification: **25 Manufacturer's responsibility** By marking the electrical apparatus in accordance with Clause 29, the manufacturer attests on his own responsibility that the electrical apparatus has been constructed in accordance with the applicable requirements of the relevant standards in safety matters the routine verifications and tests in Clause 24 have been successfully completed and that the product complies with the ~~specification submitted to the testing station~~ documentation.
26	The text has been deleted as shown as the US standard does not address repaired or modified equipment: ~~**26 Verifications and tests on modified or repaired electrical apparatus**~~ ~~Modifications made on the electrical apparatus affecting the integrity of the type of protection or the temperature of the apparatus shall be permitted only if the modified apparatus is resubmitted to a testing station.~~ ~~NOTE In the case of repairs to electrical apparatus affecting the type of protection, the parts that have been repaired should be subjected to new routine verifications and tests, which need not necessarily be made by the manufacturer.~~
27	The text has been deleted as shown as the US standard does not include requirements for cable glands. The US requirements for cable glands are specified in the notes to 14.2:
28	The text has been deleted as shown as the US standard does not include requirements for cable glands. The US requirements for cable glands are specified in the notes to 14.2:

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
29.2	The text has been modified as follows to add the US prefix “A” to the “Ex”, acknowledge types of protection maD and mbD, delete the IP marking, clarify the temperature marking, delete the TI dust layer marking, revise the certificate marking to align with US practice, and acknowledge the lack of US marking for “X”: **29.2 Marking of all electrical apparatus** The marking shall include the following: 1) the name of the manufacturer or the registered trade mark; 2) manufacturer’s type identification; 3) the symbol AEx, which indicates that the electrical apparatus corresponds to one or more of the types of protection which are the subject of the specific standards listed in Clause 1; 4) the symbol for each type of protection used: - "pD": protection by pressurization; - "tD": protection by enclosure; - "iaD" or “ibD”: protection by intrinsic safety; - "maD" or “mbD”: protection by encapsulation; 5) the zone in which the apparatus can be used; ~~NOTE In addition to marking the zone for enclosures complying with IEC 61241-1 the zone is prefixed with A for practice A and B For practice B~~ ~~6) the degree of protection (IP Code);~~ 7) the maximum surface temperature in degrees C, preceded by the letter T, (e.g. T90ºC), ~~marked as a temperature value~~; ~~8) where appropriate according to 5.2 the maximum surface temperature TL shall be shown on the certificate as a temperature value, with the layer depth indicated in mm, or with the symbol X;~~ 9) where appropriate according to 5.3 the marking shall include either the symbol "Ta" or "Tamb" together with the special range of ambient temperature ~~or the symbol X~~; 10) a serial or batch number, but not required for - connection accessories (cable glands ~~and conduit entries~~, blanking plates, adaptor plates, plugs and sockets and bushings); - very small electrical apparatus on which there is limited space; 11) ~~where a certificate has been issued,~~ the name or mark of the issuer and the certificate reference ~~in the following form: the last two figures of the year of certification followed by the serial number of the certificate in that year;~~ NOTE Listing by a Nationally Recognized Testing Laboratory (NRTL) is considered to meet the requirements

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	12) specific installation instructions or reference to a specific installation document when ~~if~~ it is necessary to indicate special conditions for safe use~~, the symbol "X" shall be placed after the certificate reference. The issuer may accept the use of a warning marking as an alternative to the requirement for the "X" marking~~; NOTE The manufacturer should ensure that the requirements of the special conditions for safe use are passed to the purchaser together with any relevant information. 13) any additional marking prescribed in the specific standards for the types of protection concerned, as in Clause 1; 14) any marking normally required by the standards of construction of the electrical apparatus. ~~This marking need not be verified by the certificate issuer~~ of a certificate.
29.3	The text has been modified as follows to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **29.3 Multiple protection techniques** Where different types of protection are used on different parts of an electrical apparatus, each respective part shall bear the symbols for the type of protection concerned. Where more than one type of protection is used in an electrical apparatus, the symbols for the ~~main~~ types of protection shall appear ~~first and be followed by the symbols the other types of protection used~~ in alphabetical order.

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Clause	US Differences to IEC 61241-0, 1st Ed, 2004
29.5	The text has been modified as follows to o add the US prefix “A” to the “Ex”, remove the requirements for 3rd party certification, and delete the “X” and “U” suffixes: **29.5 Reduced marking** On very small electrical apparatus and on Ex components where there is limited space, ~~the certificate issuer may allow~~ a reduction in the marking is permitted but will require at least: 1) the name or registered trade mark of the manufacturer; 2) the symbol AEx and the symbol of the type of protection; 3) the name or mark of the ~~testing station~~ issuer of the certificate; 4) the certificate reference if applicable; ~~5) for electrical apparatus, the symbol "X" if appropriate; or for Ex components, the symbol "U".~~
29.6	New text added as shown to address the marking of the supplementary external bonding terminal (when provided) to align the text with the text of the US adoption of IEC 60079-0, Ed. 4.: **29.6 External grounding or bonding terminal** If a supplementary external grounding or bonding terminal is identified by being either colored green or by being marked “G,” ″GR,″ ″Ground,″ ″Grounding”, “Protective Earth”, “PE”, or ″Ex ″; the instructions provided with the equipment shall indicate that the internal grounding terminal shall be used for the equipment grounding connection and that the external terminal is for a supplementary bonding connection where local codes or authorities permit or require such connection.
30	The marking examples have been modified to reflect the changes made to the requirements of 29.2.

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IECEx Bulletin –
Section 3: National Differences for 61241-1-1

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard: IEC 61241-1-1 2nd Edition 1999 Electrical apparatus for use in the presence of combustible dust - Part 1-1: Electrical apparatus protected by enclosures and surface temperature limitation- Specification for apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-1-1:1999
BR Brazil*		**TBA**	
CA Canada		**YES**	CAN/CSA-E61241-1-1-2001
CH Switzerland	**YES**		EN 50281-1-1 1998
CN China		**NO**	GB 12476.1-2000
CZ Czech Republic	**YES**		EN 50281-1-1 1998
DE Germany	**YES**		EN 50281-1-1 1998
DK Denmark	**YES**		EN 50281-1-1 1998
FI Finland	**YES**		EN 50281-1-1 1998
FR France	**YES**		EN 50281-1-1 1998
GB United Kingdom	**YES**		EN 50281-1-1 1998
HR Croatia	**YES**		
HU Hungary	**YES**		EN 50281-1-1 1998
IN India		**N/R**	
IT Italy	**YES**		EN 50281-1-1 1998
JP Japan		**NO**	
KR Republic of Korea		**YES**	Labor Ministry Notice1998-65
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 50281-1-1 1998
NO Norway	**YES**		EN 50281-1-1 1998
NZ New Zealand		**NO**	AS/NZS 61241-1-1;1999
PL Poland	**YES**		
RO Romania	**YES**		EN 61241-1-1 1998
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 50281-1-1 1998
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 50281-1-1 1998
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 61241-1-1

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Please Note IEC 61241-1-1 Second Edition (1999) has been superseded by IEC 61241-1 First Edition (2004)

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Countries that have declared adoption of IEC 61241-1-1 2nd Edition 1999 Without National Differences
AU Australia
CN China
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 61241-1-1 2nd Edition 1999 With National/Group Differences- Differences follow.

CA Canada

Clause 1 - Scope
[Add the following text]

As the requirements of IEC 61241-1-1 cover the protection method by enclosures and surface temperature limitation only, the CAN/CSA-E61241-1-1 must be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

Clause 2 - Normative references
[Add the following CSA Standard list]

CSA International Publications
Where reference is made to CSA Publications, such reference shall be considered to refer to the latest edition and all amendments published to that edition. This Standard refers to the following Standards, and the years shown indicated the latest editions available at the time of printing

CSA Standards
C22.1-98,
Canadian Electrical Code, Part I;

CAN/CSA-C22.2 No. 0-M91 (C1997),
General Requirements — Canadian Electrical code, Part II;

Sub-Clause 14.3
[Revise to read as follow]

For vertical rotating machines for use in zone 20, 21 or 22, foreign objects shall be prevented from falling into the ventilation openings.

Sub-Clause 20.1.1 - Table 6
[Delete the following]

- “MF or” in the last column of the Table.
- The “note” in the last row of the Table.

Clause 22 - Manufacturer’s responsibility
[Add the following text (which is not to be superseded by any other Clause in this Standard)]

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This standard applies to the safety of such equipment designed to be installed and used in accordance with the rules of the *Canadian Electrical Code (CEC), Part I*. It is the responsibility of the manufacturer to ensure that the equipment meets the applicable standards for general electrical safety and usage in accordance with the regulatory requirements.

Clause 23 - Verifications and tests on modified or repaired electrical apparatus
[Change the "NOTE" to read as follow]

Modifications, repairs and re-builds of electrical equipment shall be governed by Section 2 of the Canadian Electrical Code (CEC) - Part I.

Sub-Clause 26.2.2.1 - Marking for use in zone 21
[Change the sub-title to read as follow]

Marking for use in zone 20 or 21

[Add the following text]

The hazardous location designation:
Class II, Division 1, Groups E, F and/or G

Sub-Clause 26.2.2.2 - Marking for use in zone 22
[Add the following text]

The hazardous location designation:
Class II, Division 2, Groups E, F and/or G

Sub-Clause 26.2.3.1 - Marking for use in zone 21
[Change the sub-title to read as follow]

Marking for use in zone 20 or 21

[Add the following text]

The hazardous location designation:
Class II, Division 1, Groups E, F and/or G

Sub-Clause 26.2.3.2 - Marking for use in zone 22
[Add the following text]

The hazardous location designation:
Class II, Division 2, Groups E, F and/or G

CN China

GB12476.1-2000 is identical to IEC 61241-1-1:1999 without national differences

KR Korea

IEC Clause or Annex
Number Details of Differences to IEC 61241-1-1 1999 (edition 2)

5.1.4.1This requirement does not apply.

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Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

IEC Clause or Annex Number	Details of Differences
	[ ] References in square bracket refer to clauses and figures in EN 50281- References without square brackets refer to IEC 61241-1-1
5.1.5.1	[4.2.3] The requirements apply to outer plastic surfaces…
5.1.5.4	not applicable
10.3	[4.7.3] Terminal compartments shall be so designed that after proper connectio of the conductors the creepage distances and the clearances comply with the requirements, if any, of the specific European Standard.
12.4	Add the requirements: [4.9.4] Cable entries, whether integral or separate, shall meet the relevant requirements of Annex B of EN 50014 – Ex cable entries (= Annex B of IEC 60079-0). [4.9.5] Cable entries shall meet the requirements of the specific degree of dust ingress protection.
13	not applicable
-	Add: [4.10] Radiating equipment The energy levels of radiation generating equipment shall not exceed the values given below (see clauses 8 and 9 of EN 50281-1-2). [4.10.1] For lasers and other continuous wave sources: 5 mW/mm^2 for continuous wave lasers and other continuous wave sources, and 0,1 mJ/mm^2 for pulse lasers or pulse light sources with pulse intervals of at least 5s. Radiation sources with pulse intervals of less than 5s are regarded as continuous light sources in this respect. [4.10.2] For ultrasonic sources the power level shall not exceed a power density in the sound field of 0,1 W/cm^2 and a frequency of 10 MHz for continuous sources and 2mJ/cm^2 for pulse sources. The average power density shall not exceed 0,1 W/cm^2.
14.4	Add: [5.1.2]:… and shall have an electrical insulation resistance, measured according to 23.4.7.8 of EN 50014 not exceeding 10^9 Ohm.
15	Add: [5.2.4] Doors and covers giving access to the interior of enclosures containing remotely operated circuits with switching contacts which can be made broken by non-manual influences (such as electrical, mechanical, magnetic, electro-magnetic, electro-optical, pneumatic, hydraulic, acoustic or thermal) shall

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	either be interlocked with a disconnector which prevents access to the interior unless it has been operated to disconnect unprotected internal circuits; or the apparatus shall be marked with the warning: " DO NOT OPEN WHEN ENERGIZED" In the case of i) above, where it is intended that some internal parts will remain energized after operation of the disconnector, then in order to minimize the risk to maintenance personnel, those energized parts shall be protected as follows: clearances and creepage distances between phases (poles) and to earth in accordance with the requirements of the relevant standard; an internal supplementary enclosure which contains the energized parts and provides a degree of protection of at least IP20, according to EN 60529 8 IEC 60529), so arranged that a tool cannot contact the energized parts through any openings; marking on the internal supplementary enclosure with the warning: "DO NOT OPEN WHEN ENERGIZED"
17.4	Add: [5.4.4] Socket outlets shall be designed to prevent dust falling into the socket.
20 Table 6- Subclause 20.4.4	[8] Table clause 23.4.5 (according to EN 50014) Torque test for bushing : For apparatus for use in zone 22 the test has to be carried out by MF or TS
20 Table 6- Subclause 22	[8] Table clause 25 (according to EN 50014) Responsibility For apparatus for use in zone 20 or 21 the requirement or test has to be carried out by TS
20.4.3.4	not applicable
20.4.3.5	not applicable
20.4.5.2	Add: [10.2].. The effect of such devices shall be tested under simulated working conditions.
20.4.5.4	Add: [10.5] Category 1 (zone 20) apparatus In addition to the tests in 20.4.5 the thermal test for this apparatus shall be made under the conditions of rare incidents related to the apparatus and under dust layers of unknown or excessive thickness formed on the apparatus as agreed between specifier and manufacturer.
20.4.5.5	not applicable
26.2.2	[11.3.1] Marking for Category 1 and 2 (zone 20 and zone 21) the symbol εx in a hexagon, which indicates that the electrical apparatus has been constructed and tested for use in a combustible dust atmosphere or is specifically associated with such an apparatus;

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	where a certificate of compliance has been obtained, the name or mark of the national or other appropriate authority and the certification reference, preferably in the following form: the year of certification followed by the serial number of the certificate in that year; the symbols - 1 or 2 indicating the category of the apparatus, followed by - D for "Dust Ignition Protection", followed by the maximum surface temperature T to be marked as a temperature value. [11.3.2] Marking for Category 3 (zone 22) the symbol εx in a hexagon, which indicates that the electrical apparatus has been constructed and tested for use in a combustible dust atmosphere or is specifically associated with such an apparatus; the symbols - 3 indicating the category of the apparatus, followed by - D for "Dust Ignition Protection", followed by the maximum surface temperature T to be marked as a temperature value

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IEC Standard: IEC 61241-1-1 1st Edition 1993 Electrical apparatus for use in the presence of combustible dust - Part 1-1: Electrical apparatus protected by enclosures and surface temperature limitation- Specification for apparatus			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-1-1
BR Brazil*		**TBA**	
CA Canada*		**TBA**	
CH Switzerland	**N/R**		
CN China		**NO**	GB 12476.1-2000 identical IEC 61241-1-1:1999
CZ Czech Republic	**N/R**		
DE Germany	**N/R**		
DK Denmark	**N/R**		
FI Finland	**N/R**		
FR France	**N/R**		
GB United Kingdom	**N/R**		
HR Croatia	**N/A**		
HU Hungary	**N/R**		
IN India*		**TBA**	
IT Italy	**N/R**		
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	
NL The Netherlands	**N/R**		
NO Norway	**N/R**		
NZ New Zealand		**NO**	AS/NZS 61241-1-1
PL Poland	**N/A**		
RO Romania	**N/R**		
RU Russia*		**TBA**	
SE Sweden	**N/R**		
SG Singapore		**NO**	
SI Slovenia	**N/R**		
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**NO**	SANS 61241-1-1

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

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Countries that have declared adoption of IEC 61241-1-1 1st Edition 1993 Without National Differences
AU Australia
CN China
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 61241-1-1 1st Edition 1993 With National/Group Differences- Differences follow.

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-1

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-1

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

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IEC Standard: IEC 61241-1 1st Edition 2004 Electrical apparatus for use in the presence of combustible dust – Part 1: Protection by enclosures 'tD'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-1: 2005
BR Brazil*		**TBA**	
CA Canada*		**TBA**	
CH Switzerland	**NO**		EN 61241-1:2004
CN China		**NO**	GB 12476.5-(draft) identical IEC 61241-1:2004
CZ Czech Republic	**NO**		ČSN EN 61241-1:2005
DE Germany	**NO**		EN 61241-1:2004
DK Denmark	**NO**		DS/EN 61241-1:2004
FI Finland	**NO**		EN 61241-1:2004
FR France	**NO**		EN 61241-1:2004
GB United Kingdom	**NO**		BS EN 61241-1:2004
HR Croatia	**NO**		
HU Hungary	**NO**		EN 61241-1:2004
IN India		**NO**	Adopted as IS/IEC 61241-1: 2004
IT Italy	**NO**		EN 61241-1:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 61241-1:2004
NO Norway	**NO**		EN 61241-1:2004
NZ New Zealand		**NO**	AS/NZS 61241-1: 2005
PL Poland	**NO**		
RO Romania	**NO**		EN 61241-1:2004
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 61241-1:2004
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 61241-1:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 61241-1 2006
ZA South Africa		**NO**	SANS 61241-1:2004

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details***

Please Note that IEC 61241-1 first Edition (2004) has been superseded by IEC 60079-31 First Edition (2008)

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National Differences

Countries that have declared adoption of IEC 61241-1 1st Edition 2004 Without National Differences
AU Australia
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 61241-1 1st Edition 2004 With National/Group Differences- Differences follow.

US Differences on following page.

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National Differences

US United States of America

Clause	US Differences to IEC 61241-1, 1st Ed, 2004
General	Where references are made to other IEC 61241 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
1	The 1st and 3rd paragraphs have been modified as follows to align with the National Electrical Code, ANSI/NFPA 70: This ~~part of IEC 61241~~ standard is applicable to electrical apparatus protected by enclosures and surface temperature limitation for use in ~~areas where combustible dust may be present in quantities which could lead to a fire or explosion hazard~~ explosive dust atmospheres classified as zone 21 or zone 22 hazardous locations in accordance with Article 506 of the NEC®. It specifies requirements for design, construction and testing of electrical apparatus. This standard supplements the general requirements in ~~IEC 61241-0~~ ANSI/ISA-61241-0. NOTE IEC 61241-14 gives guidance on the selection and installation of the apparatus. Apparatus within the scope of this standard may also be subjected to additional requirements in other standards – for example, IEC 60079-0. The ignition protection is based on the limitation of the maximum surface temperature of the enclosure and on other surfaces which could be in contact with dust and on the restriction of dust ingress into the enclosure ~~by the use of "dust-tight" or "dust-protected" enclosures~~.
3.1	The definition has been revised as follows to reflect that proposed for IEC 60079-31: **dust ignition protection type "tD"** ~~all relevant measures specified in this standard (e.g. dust ingress protection and surface temperature limitation) applied to electrical apparatus protected by an enclosure to avoid ignition of a dust layer or cloud~~ type of protection for explosive dust atmospheres where electrical apparatus is provided with an enclosure providing dust ingress protection and a means to limit surface temperatures

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
4	In accordance with the concepts set down in the National Electrical Code, the requirements for Zone 20 have been deleted as follows as "tD" is not permitted in Zone 20 in the US: **4 Construction** ~~In addition to the requirements of Clause 4 of IEC 61241-0, the following applies.~~ ~~Apparatus for use in zone 20:~~ ~~If the enclosure of the equipment is in accordance with the requirements of this standard, and all gaskets preventing the ingress of dust are not stressed by moving parts (e.g. shaft, operating rod), the enclosure is considered as infallible and the fault analysis is only applicable to the electrical circuit.~~

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
6.1	The text of 6.1 has been modified as follows to remove the requirements for Zone 20, and to more clearly define the requirements for Zones 21 and 22: **6.1 General** For apparatus marked for use in zone 21, the tests of 8.2.1.2 for IP6X shall be conducted. For apparatus marked for use in zone 22, the tests of 8.2.1.3 for IP5X shall be conducted. ~~Dust-tight enclosures, IP6X, shall be applied for use in:~~ ~~– zone 20;~~ ~~– zone 21;~~ ~~– zone 22 with conductive dust.~~ ~~Dust-protected enclosures, IP5X, shall be applied for use in:~~ ~~– zone 22 with non-conductive dust.~~ ~~At present, there are no additional requirements for practice A to those given in IEC 61241-0 and in this standard.~~
6.2	Add construction requirements as follows for all joints: **6.2 Joints** **6.2.1 General** All joints in the structure of the enclosure between surfaces, whether permanently closed or designed to be opened from time to time, shall fit closely together within the tolerances specified in the documentation. They shall be effectively sealed against the ingress of dust and shall comply with the following particular requirements:

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
	The length of thread for all screw joints employing parallel threads shall be not less than five threads. Hinges shall be clear of joints and shall not be used as a means of obtaining a seal unless they are provided with facilities for positive location and correct operation is assured; and it is not possible for damage to occur to either the hinge or enclosure during normal operation and use; and they are not manufactured from materials that are subject to wear that would affect the correct operation; and correct compression of the gasket is achieved without causing undue movement, stress or distortion to the gasket. Where necessary a suitable register shall be provided for covers and the like to ensure correct replacement. **6.2.2 Gaskets and seals** Gaskets under compression between joint surfaces may be used to ensure the effectiveness of the enclosure sealing. All gaskets shall be of one-piece continuous construction, i.e. with an uninterrupted periphery. Gaskets, whether of the loose or secured variety, shall be of a form that will provide and maintain in service a dust-excluding seal between the joint surfaces of the enclosure. NOTE Unless all gaskets are either cemented or moulded into one face of the jointed surface, the design of the enclosure should be such that the inspection of the assembly to determine whether gaskets have been fitted correctly can be made without disturbing the enclosure. A gasket may be cemented to one face only of the joint surface. Gasketed joints shall not be supplemented by the application of a sealant material except on one face only of the joint surfaces.

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
	Gaskets shall be of material that is not subject to appreciable ageing or to plastic flow between the envisaged maintenance periods as determined by the tests in 8.1 through 8.2.3. A flexible seal, e.g. a bellows, shall be such that it is not over-stressed at any point and shall be protected from external mechanical damage and secured at each end by positive mechanical means. **6.2.3 Cemented joints** Cemented joints shall comply with the following requirements: • They shall not be used on lids or covers which have to be removed to gain access to field wiring connections or in-service adjusting facilities. • The width of a cemented joint shall be equal to or greater than 5 mm. **6.2.4 Operating rods and spindles** Openings in enclosures for rods or spindles shall have means, other than grease or compound, that will prevent the ingress of dust, both when the spindles or rods are in motion and when they are at rest. **6.2.5 Power shafts** Openings in enclosures for power shafts shall have means, other than grease or compound, that will prevent the ingress of dust, both when the shafts are in motion and when they are at rest. **6.2.6 Windows** **6.2.6.1 Cemented windows** A cemented window shall be cemented either directly into the wall of the enclosure so as to form with the latter an inseparable assembly, or into a metallic frame such that the assembly can be replaced as a unit.

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
	6.2.6.2 Gasketed windows A gasketed window may be mounted directly in the wall or cover of the enclosure, no separate detachable frame being necessary. It shall incorporate a suitable seal or gasket.
7.1 (New)	Add new sub-clause 7.1 as follows to provide calling text for the tests: **7.1 General** For apparatus marked for use in zone 21, the tests of 8.2.1.4 shall be conducted. For apparatus marked for use in zone 22, the tests of 8.2.1.5 shall be conducted.
7.1 (Existing)	Renumber 7.1 to 7.2 along with all subsequent sub-clauses.
7.3.1	Modify text as follows to include applicability for Zone 22, and remove applicability for Zone 20 as this is outside of the US scope: **7.3.1 General** For apparatus intended for use in zone ~~20 and~~ 21 and 22, dust exclusion shall not depend on running contact seals. The equipment shall meet the design details of Tables 3 and 4 and the tests of 8.2.1 without the running contact seal installed.

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
8.1	Delete existing text as follows as this document already supplements 61241-0 and does not require restating: **8.1 General** ~~In addition to the verification and testing requirements of IEC 61241-0, the following applies.~~
8.2.1.1	Delete 3rd paragraph as follows as this was unenforceable text: ~~The precautions that shall be taken where a combustible dust is used for the tests is left to the discretion of the testing authority or others concerned.~~
8.2.1.2	Modify text as follows to clarify the order of tests to be conducted and add a pressure test based on that proposed for IEC 60079-31: <u>The apparatus shall be subjected to the thermal endurance to heat, thermal endurance to cold, and mechanical tests of ANSI/ISA-61241-0.</u> <u>The apparatus shall then be subjected to the pressure test of 8.2.3.</u> ~~Enclosures~~ <u>Finally, the apparatus</u>, including rotating machines, shall satisfy the requirements for IP6X, as specified in IEC 60529, <u>or IEC 60034-5 as applicable. The apparatus shall be considered as belonging to “Category 1 enclosure” as specified in IEC 60529.</u>
8.2.1.3	Modify text as follows to clarify the order of tests to be conducted and add a pressure test based on that proposed for IEC 60079-31: <u>The apparatus shall be subjected to the thermal endurance to heat, thermal endurance to cold, and mechanical tests of</u>

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
	ANSI/ISA-61241-0. The apparatus shall then be subjected to the pressure test of 8.2.3. ~~All enclosures~~ Finally, the apparatus, including rotating machines, shall satisfy the test and acceptance requirements of IP5X, as specified in IEC 60529 or IEC 60034-5 as applicable. For the purposes of acceptance criteria in accordance with IEC 60034-5, all dusts shall be considered to be conductive and therefore IP5X shall be assessed against the acceptance criteria of IP6X. The apparatus shall be considered as belonging to “Category 1 enclosure” as specified in IEC 60529.
8.2.2	Modify text as follows to reflect the US practice for dust blanketing: **8.2.2 Thermal tests** ~~**8.2.2.1 Apparatus for practice A**~~ ~~This test shall be carried out as described in 23.4.5 of IEC 61241-0.~~ ~~**8.2.2.2 Apparatus for practice B**~~ This test shall be carried out in accordance with ~~23.4.5.1 to 23.4.5.3 of IEC 61241-0~~ ANSI/ISA-61241-0 with the additional requirement that the apparatus shall be covered with the maximum amount of dust that it can retain. As an alternative, a 12,5 mm thick layer of dust paste may be put on top of the apparatus to simulate the build-up conditions. NOTE The paste should consist of 45 % dust (e.g. wheat flour) and 55 % water by weight. The temperature value should be measured after the paste has dried.
8.2.3	Add pressure test as follows based on that proposed for IEC 60079-31:

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Clause	US Differences to IEC 61241-1, 1st Ed, 2004
	8.2.3 Pressure test A positive pressure of 27 kPa shall be applied to the apparatus for at least 1 minute. During this test, any breathing or draining device shall be removed and the entry plugged. There shall be no evidence of damage to the enclosure due to the pressure applied.
9	Modify text as follows to align with 61241-0 and account for no need to identify Practice A or Practice B: **9 Marking** ~~Clause 29 of IEC 61241-0~~ The marking requirements of ANSI/ISA-61241-0 are supplemented by the following ~~applies except as follows~~: ~~29.2, item 4:~~ The symbol for the type of protection used shall be “tD” (protection by enclosure). ~~29.2, item 5:~~ ~~The zone in which the apparatus can be used is prefixed by “A” for practice A and “B” for practice B.~~ ~~29.2, item 6:~~ ~~Practice B does not need to include the IP rating.~~

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-4

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-4

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 61241-4 1st Edition 2001 Electrical apparatus for use in the presence of combustible dust – Part 4: Type of protection 'pD'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-4:2002
BR Brazil*		**TBA**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 61241-4:11
CH Switzerland	**NO**		EN 60079-14:2006
CN China		**NO**	GB 12476.7-2010
CZ Czech Republic	**NO**		EN 60079-14:2006
DE Germany	**NO**		EN 60079-14:2006
DK Denmark	**NO**		EN 60079-14:2006
FI Finland	**NO**		EN 60079-14:2006
FR France	**NO**		EN 60079-14:2006
GB United Kingdom	**NO**		EN 60079-14:2006
HR Croatia	**NO**		
HU Hungary	**NO**		EN 60079-14:2006
IN India		**NO**	Adopted as IS/IEC 61241-4: 2001 (under printing)
IT Italy	**NO**		EN 60079-14:2006
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 60079-14:2006
NO Norway	**NO**		EN 60079-14:2006
NZ New Zealand		**NO**	AS/NZS 61241-4:2002
PL Poland	**NO**		
RO Romania	**NO**		EN 60079-14:2006
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 60079-14:2006
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 60079-14:2006
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 61241-2 2006 Note that ISA has adopted the proposed IEC number for the document
ZA South Africa		**NO**	SANS 61241-4

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details***

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Countries that have declared adoption of IEC 61241-4 1st Edition 2001 Without National Differences
AU Australia
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 61241-4 1st Edition 2001 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 61241-4:11
Electrical apparatus for use in the presence of combustible dust – Part 4: Type of protection "pD"

This is the first edition of CAN/CSA-C22.2 No. 61241-4, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 61241-4 (First edition, 2001-03).

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

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[… and the rest of that is necessary…]

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 61241series Standards cover the protection techniques with respect to dust explosion and layer ignition hazard only. The CAN/CSA-C22-2 No. 61241 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE 1: Currently, the IEC 61241 series of Standards are being converted (merged) into the IEC 60079 series of Standards. Some Standards, relevant to combustible dust environments, may be under the IEC 61241 series or under the IEC 60079 series depending on the scheme of progress.

NOTE 2: See also Canadian deviations under CAN/CSA-C22.2 No. 61241-0:11.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety standard for electrical installations

4.3 Cleaning

[Add a NOTE to the following effect]

NOTE: The degree of cleanliness is difficult to define. However, the minimum is as follow:
a) Remove all dust layer accumulated around the interface between the cover and the body of the enclosure before opening the equipment;
b) Ensure that the amount of residual dust (non-visible or non-detectable to the best of the human deficiency) remaining inside the enclosure would not result in a dust-cloud (becoming an explosive dust atmosphere) due to the turbulence of the purge medium (pressurization air/gas);
c) Before re-closing the cover, there shall be absolutely no dust cloud in the surrounding atmosphere.
d) Before energizing, (that is, switching-on the electrical power to the equipment), the equipment shall be purged with the pressurizing medium for at least 5 volume change (that is, equal to five times the internal volume of the enclosure). [The intent of the purge is to create turbulence and to remove whatever residual dust that was un-detected in the cleaning process.]

4.4 Discharge of protective gas

[Add a NOTE to the following effect]

NOTE: Protective gas being discharged into the hazardous area is not permitted. The reasons are:
a) The discharge of protective gas must not become the source of creating turbulence in the combustible dust atmosphere.
b) The opening of the discharge or the leaky joints of the enclosure must not become the channel of dust entries when relatively hot components or devices within the pressurized enclosure starts to cool and drawing dust through those opening or leaky joints.

5.5 Doors and covers

[Add the following paragraph]

It is a requirement in Canada that any warning or cautionary statements must be in both English and French.

7 Safety provisions and safety devices (except for static pressurization)

7.1 General

[Add a paragraph to the following effect]

All safety related devices, as determined by the certifier, shall be provided by the manufactured and shall be part-and-parcel of the overall delivery of the equipment.

7.5.1.1 Switching off electrical supply

[Add the following paragraph]

It is a requirement in Canada that any warning or cautionary statements must be in both English and French.

7.5.1.2 Alarm only

[Add the following paragraph]

It is a requirement in Canada that any warning or cautionary statements must be in both English and French.

8 Safety provisions and safety devices for static pressurization

[Add a NOTE to the following effect]

NOTE: Contrary to NOTE 1 and NOTE 2, it is the responsibility of the manufacturer and the certifier (not the user) to determine the suitability and provision of any safety related devices and their use.

10.6 Impact test

[Replace the contents in this Sub-clause with the following]

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An impact test in accordance with IEC 61241-0 shall be performed on all pressurized enclosures. If the “low risk” criteria are applied, the equipment shall be marked with the symbol “X” and shall be so defined in the test report.

11 Marking

[Add the following paragraph]

It is a requirement in Canada that any warning or cautionary statements must be in both English and French.

Further, any instruction relevant to the safe installation, usage and operation of the equipment, shall be in both English and French.

B.5 Switching off electrical supply

[Add a NOTE to the following effect]

NOTE: If there is any evidence that dust has entered the enclosure, a cleaning process shall be conducted, (see 4.3 for cleaning guidance.).

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US United States of America

Clause	US Differences to IEC 61241-4, 1st Ed, 2001
General	Where references are made to other IEC 61241 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
3.8	Revised definition as follows to change “outlet apertures” to “outlets as it is not intended that inlets be closed after purging completed: **3.8** **pressurization with leakage compensation** maintenance of an overpressure within a pressurized enclosure so that, when the ~~outlet apertures~~ outlets – if any – are closed, the supply of protective gas is sufficient to compensate for any leakage from the pressurized enclosure and its ducts
3.9	Term has been deleted as follows as the text in 5.3.3 using the term was revised to add clarity: **3.9** ~~**pressurization with continuous flow of the protective gas**~~ ~~maintenance of an overpressure within a pressurized enclosure with continuous flow of the protective gas through the enclosure~~

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National Differences

Clause	US Differences to IEC 61241-4, 1st Ed, 2001
3.10	Term has been deleted as follows as it appears in 61241-0 and is not necessary to repeat in 61241-4: **3.10** **~~electrical apparatus~~** ~~items applied as a whole or in part for the utilization of electrical energy. These include, among others, items for the generation, transmission, distribution, storage, measurement, regulation, conversion, and consumption of electrical energy and items for telecommunications~~
3.12	Term has been deleted as follows as the term is not used in 61241-4: **3.12** **~~self-revealing fault~~** ~~fault which would cause a malfunction of the apparatus necessitating correction before proceeding with further operation of the apparatus and which may be indicated, for example, by an audible or visible signal~~
3.14	Term has been revised as follows to align with the current proposed text for IEC 60079-2, Ed. 5.: **3.14** **~~protective~~ safety device** device provided to protect a system against conditions which could result in a fire or explosion
3.18	Term has been revised as follows to align with the current proposed text for IEC 60079-2, Ed. 5.: **3.18** **pressurization system** grouping of safety devices and other components used to pressurize and monitor or control a pressurized enclosure ~~grouping of components used to pressurize and monitor a pressurized enclosure~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
3.20	Term has been deleted as follows as it appears in 61241-0 and is not necessary to repeat in 61241-4: **3.20** ~~**zones**~~ ~~classified areas are divided into zones based upon the frequency and duration of the occurrence of explosive dust/air mixtures. Dust layers should also be taken into consideration [IEC 61241-3, definition 2.10]~~
3.21	Term has been deleted as follows as it appears in 61241-0 and is not necessary to repeat in 61241-4: **3.21** ~~**zone 20**~~ ~~area in which combustible dust, as a cloud, is present continuously or frequently, during normal operation, in sufficient quantity to be capable of producing an explosive concentration of combustible dust mixed with air, and/or where layers of dust of uncontrollable and excessive thickness can be formed.~~ ~~This can be the case inside dust containment areas where dust can form explosive mixtures frequently or for long periods of time. This occurs typically inside equipment [IEC 61241-3, definition 2.11]~~
3.22	Term has been deleted as follows as it appears in 61241-0 and is not necessary to repeat in 61241-4: **3.22** ~~**zone 21**~~ ~~area not classified as zone 20 in which combustible dust, as a cloud, is likely to occur during normal operation, in sufficient quantities to be capable of producing an explosive concentration of combustible dust mixed with air.~~ ~~This zone can include, among others, areas in the immediate vicinity of powder filling or emptying points and areas where dust layers occur and are likely, in normal operation, to give rise to an explosive concentration of combustible dust mixed with air [IEC 61241-3, definition 2.12]~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
3.23	Term has been deleted as follows as it appears in 61241-0 and is not necessary to repeat in 61241-4: **3.23** ~~**zone 22**~~ ~~areas not classified as zone 21 in which combustible dust clouds may occur infrequently, and persist for only a short period, or in which accumulations or layers of combustible dust may be present under abnormal conditions and give rise to combustible mixtures of dust in air. Where, following an abnormal condition, the removal of dust accumulations or layers cannot be assured, then the area is to be classified zone 21.~~ ~~This zone can include, among others, areas in the vicinity of equipment containing dust, where dust can escape from leaks and form deposits (such as milling rooms in which dust can escape from the mills and then settle)~~ ~~IEC 61241-3, definition 2.13]~~
4.2	The title has been modified as follows to add “or alarm” as the requirements include both disconnecting and alarming: **4.2 Automatic disconnection <u>or alarm</u>**

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
4.4	The text has been modified as follows to align the text with the current proposed text for IEC 60079-2, Ed. 5.: **4.4 Discharge of protective gas** ~~Preferably, the protective gas may be discharged into a non-hazardous area. In the case of pressurization with continuous flow, and where the protective gas is discharged into the hazardous area, means are to be provided to prevent hot particles or other ignition sources from equipment under normal or fault conditions entering the hazardous area.~~ The pressurized enclosure and the ducting, if any, for the protective gas shall be provided with a spark and particle barrier to guard against the ejection of incandescent particles into the hazardous area. Incandescent particles shall be assumed to be normally produced unless make/break contacts operate at less than 10 A and the working voltage does not exceed either 275 V a.c. or 60 V d.c., and the contacts have a cover. EXCEPTION 1: The spark and particle barrier is not required for a normally closed relief vent exhausting into a zone 21 area if incandescent particles are not normally produced. EXCEPTION 2: The spark and particle barrier is not required when exhausting into a zone 22 area if incandescent particles are not normally produced. If the manufacturer does not provide the spark and particle barriers, the apparatus shall be marked in accordance with 29.2 of ANSI/ISA-61241-0.

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
5.2	The text of the 2nd paragraph has been modified as follows to allow the safety device to be provided by the user: If a pressure can occur in service that can cause a deformation of the enclosure ducts, if any, or connecting parts, a safety device shall be fitted by the manufacturer <u>or by the user</u> to limit the maximum internal overpressure to a level below that which could adversely affect the type of protection. <u>If the safety device is to be fitted by the user, the apparatus shall be marked in accordance with 29.2 of ANSI/ISA-61241-0, and the description documents shall contain all the necessary information required by the user to ensure protection of the enclosure and ducts.</u>
5.3.3	The text has been modified as follows to eliminate the need to define the term "pressurization with continuous flow": **5.3.3** ~~In the case of pressurization with~~ <u>When a</u> continuous flow of protective gas <u>is required, e.g. for cooling,</u> the enclosure shall have one or more inlet apertures and one or more outlet apertures for the connection of the inlet and outlet ducts for the protective gas.
5.4	Text has been deleted as follows as the concept is more completely addressed in 61241-0 and is not necessary in 61241-4: **5.4 Electrical connections to enclosures** ~~The entry of electrical conductors shall be by means of a cable gland or conduit directly into the enclosure such that the method of protection is maintained, or by means of a separate terminal box which is protected by one of the types of protection for electrical equipment in combustible dust atmospheres in accordance with IEC 61241-1-2.~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
5.5	The text has been modified as follows to eliminate confusing text, provide text for the warning, provide the use of an alarm for Zone 22, eliminate markings that lready appear in the marking section, and simplify the text on cleaning: **5.5 Doors and covers** When it is necessary to delay the opening of an enclosure due ~~to a risk of explosion because of the existence of an external explosive dust atmosphere, and due,~~ for example, to the surface temperature of internal parts of that equipment or to residual charge on components, the doors and covers shall carry a warning: ~~giving the delay to be observed after switching off the electrical supply.~~ **WARNING – DO NOT OPEN ANY DOOR OR COVER FOR xxx MINUTES AFTER REMOVING POWER.** NOTE xxx is the time in minutes required for internal parts to cool below the permitted maximum value, see Clause 6. Doors and covers which can be opened without the use of tools or keys shall be interlocked with the electrical supply to disconnect all power to the electrical apparatus or provide an alarm. See Table 1 for requirements based upon area classification. In the case of static pressurization, doors and covers shall only be opened by the use of a tool, and shall ~~carry the following warning~~ be marked: **WARNING - REMOVE TO A NON-HAZARDOUS AREA BEFORE OPENING** ~~Where doors and covers are provided to permit inspection in service, they shall carry the following or equivalent warning:~~ ~~**DO NOT OPEN WHILE ENERGIZED**~~ ~~except where provision is made for adjustment during operation, in which case the warning shall indicate~~ ~~**SEE INSTRUCTIONS BEFORE OPENING**~~ A suitable number of doors or covers shall be provided for effective cleaning ~~The number of doors and covers are to be chosen with regard to the design and disposition of the apparatus, particular consideration being given to the needs including~~ of subcompartments. ~~into which the apparatus might be divided.~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
7.1	Modify text of 3rd paragraph as shown to align the "X" marking with US requirements: The safety devices shall be provided by the manufacturer of the apparatus or by the user. In the latter case, the apparatus shall be marked ~~"X"~~ in accordance with 29.2 of ANSI/ISA-61241-0 and the description documents shall contain all the necessary information required by the user to ensure conformity with the requirements of the standard.
7.2	Modify test as shown to eliminate the need to take action on exceeding overpressure: Pressure-activated devices provided for the operation of alarm and electrical trip devices shall operate when the pressure within the enclosure falls below the permitted minimum value~~ or exceeds the permitted maximum pressure~~. Flow-monitoring devices provided for the operation of alarm and electrical trip devices shall operate when the pressure within the enclosure falls below the permitted minimum value, ~~or exceeds the permitted maximum pressure,~~ and shall be located at the outlet.
7.3	Delete text as no specific marking is required and complete warning text is provided in Clause 11: **7.3 Electrical supply** ~~Where protection is not provided to prevent the ingress of dust following the shutdown or failure of the pressurization system, a warning label shall be fixed stating that dust shall be removed from the interior prior to switching on the electrical supply.~~
7.5.1.1.c	Delete text as no specific marking is required and complete warning text is provided in Clause 11: ~~c) A warning shall be prominently mounted on the apparatus stating the following or equivalent:~~ ~~**WARNING: REMOVE ALL DUST FROM THIS ENCLOSURE**~~ ~~**BEFORE CONNECTING OR RESTORING THE ELECTRICAL SUPPLY**~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
7.5.1.2.b	Modify text as shown to reflect that apparatus does not carry a warning, but is marked with a warning: b) All doors and covers which can be opened without the use of a tool shall be marked ~~carry the following warning~~: **WARNING: DO NOT OPEN WHILE ENERGIZED** **UNLESS IT IS OBVIOUS THAT NO COMBUSTIBLE DUST IS PRESENT**
7.5.1.2.c	Delete text as no specific marking is required and complete waring text is provided in Clause 11: ~~c) A warning shall be prominently mounted on the apparatus stating the following or equivalent:~~ ~~**WARNING: REMOVE ALL DUST FROM THE INSIDE OF THE ENCLOSURE**~~ ~~**BEFORE CONNECTING OR RESTORING THE ELECTRICAL SUPPLY**~~
7.9	Modify text as shown to align with new definition of “safety device”: **7.9 Separate enclosures** Where the apparatus is intended to form part of a multi-enclosure installation with a common protective gas supply, the resulting design shall take into account the most unfavourable conditions of the whole installation. If the ~~protective~~ safety devices are intended to be common, the opening of a door or cover need not necessarily switch off the whole electrical supply or initiate the alarm provided that a) the opening of a door or cover is preceded by switching off the supplies to the apparatus located in that particular enclosure that is not protected by a suitable concept of dust ignition protection, or the door is interlocked with the supplies to achieve the same end; and b) the common ~~protective~~ safety device continues to monitor the rest of the installation.

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
9.1	Modify text as shown to specify that the protective gas shall be non-flammable: **9.1 Type of gas** A protective gas shall be used for maintaining pressurization in the enclosure. The gas shall be non-flammable and shall not, by reason of its chemical characteristics or the impurities that it may contain, reduce the level of protection below that sought, or affect the satisfactory operation and integrity of the apparatus. Means for removing any oil or moisture or other undesirable impurities from the protective gas shall be provided as necessary.
10.1	Modify text as follows to clarify the requirements for impact testing: **10.1 General** In addition to the type verification and routine tests detailed in ~~IEC 61241-1-1~~ ANSI/ISA-61241-0, pressurized enclosures shall be subjected to tests detailed in 10.2 to 10.5 below and, where applicable, ~~10.6 and~~ 10.7. ~~If applicable, the~~ The impact test of ~~10.6~~ ANSI/ISA-61241-0 shall be conducted prior to all other tests.
10.2.a	Modify text as shown to eliminate the need to define “pressurization with continuous flow”: a) The enclosure design and protection measures are such that pressurization ~~with leakage compensation and pressurization by continuous flow of protective gas are~~ is achieved in accordance with this standard.
10.4.1	Modify text as shown to add a tolerance to the test period: **10.4.1 Test conditions** The enclosure and associated fittings concerned with pressurization shall be assembled for use and run under normal conditions of operation for at least 5 min with the minimum protective gas supply as stated by the manufacturer. During and on completion of this period, the minimum overpressure of the enclosure shall be not less than that specified in 7.6.

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
11.1	Modify text as shown to eliminate markings already addressed by 61241-0: **11.1 Pressurized enclosure** The enclosures, except pressurized enclosures protected by static pressurization, in addition to the marking required in ANSI/ISA-61241-0, shall be marked as follows. • The pressurized enclosure shall be marked "WARNING – PRESSURIZED ENCLOSURE - DO NOT OPEN WHEN AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT - REMOVE ALL DUST FROM THIS ENCLOSURE AND RESTORE MINIMUM OVERPRESSURE BEFORE CONNECTING OR RESTORING THE ELECTRICAL SUPPLY" ~~• The name or registered trade mark of the manufacturer.~~ ~~• Manufacturer's type identification.~~ ~~• Identification of the type of protection "pD".~~ ~~• The maximum surface temperature according to clause 6.~~ ~~• For use in zone 21, the number "21".~~ ~~• For use in zone 22, the number "22".~~ ~~• A serial number, if required.~~ • Any marking normally required by the standards of construction of the electrical apparatus. • Warning labels, as required by this standard, see 11.3 and 11.5. ~~• Where a certificate of compliance has been obtained, the name or mark of the national or other appropriate authority and the certificate reference.~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
11.2	Modify text as shown to eliminate reference to testing station, use the accepted term luminaire, and identify the need to mark the protective gas temperature: **11.2 Additional marking** In addition, the following, ~~(as agreed upon between the certificate applicant and testing station~~ if necessary~~)~~. a) The minimum and, if applicable, maximum pressure during operation, or the minimum rate of flow of protective gas. b) The type of protective gas (when this is not air). c) Any other limitations affecting the safe use of the apparatus. d) The position at which the pressure and flow are monitored shall be indicated either on the equipment enclosure or in the technical manual. e) For luminaires ~~light fittings~~, the maximum wattage of the lamp which may be used without exceeding the maximum surface temperature of the enclosure. f) The temperature of the protective gas when required by 9.3.
11.3	Modify text as shown to use consistent text for warning markings: **11.3 Pressurized enclosures protected by static pressurization** ~~A warning label (or labels) shall be fitted with the following wording.~~ The enclosure shall be marked: a) **WARNING - THIS ENCLOSURE IS PROTECTED BY STATIC PRESSURIZATION** b) **WARNING - THIS ENCLOSURE SHALL BE FILLED ONLY IN A NON-HAZARDOUS AREA ACCORDING TO THE MANUFACTURER'S INSTRUCTIONS**
11.4	Delete text as shown to eliminate reference to testing station: **11.4 Any other marking required** ~~As agreed upon between the certificate applicant and testing station if necessary.~~

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Clause	US Differences to IEC 61241-4, 1st Ed, 2001
11.5	Add text as shown to identify markings called out in the text and permit technically equivalent text: **11.5 Warnings required in other clauses** Where warning markings are required by this standard, the text following the word "WARNING" may be replaced by technically equivalent text. Multiple warnings may be combined into one equivalent warning. Clause Recommended warning 5.5 **WARNING – DO NOT OPEN ANY DOOR OR COVER FOR xxx MINUTES AFTER REMOVING POWER** 5.5 **WARNING - REMOVE TO A NON-HAZARDOUS AREA BEFORE OPENING** 7.5.1.2 b) **WARNING: DO NOT OPEN WHILE ENERGIZED UNLESS IT IS OBVIOUS THAT NO COMBUSTIBLE DUST IS PRESENT**
B.4	Modify text as shown to align the text with the current proposed text for IEC 60079-2, Ed. 5.: **B.4 Common supply of protective gas** When a source of protective gas is common to separate enclosures, the protective measures may be common to several, provided that the resulting protection takes account of the most unfavourable conditions in the whole assembly. If the ~~protective devices~~ safety devices are common, the opening of a door or cover need not switch off the electrical supply to the whole assembly or initiate the alarm provided that a) the said opening is preceded by switching off the electrical supply to that particular equipment, except to such parts as are protected by a suitable type of protection; b) the common ~~protective device~~ safety device continues to monitor the pressure in all the other enclosures of the group; and c) the subsequent switching-on of the electrical supply to that particular equipment is preceded by the applicable cleaning procedure.

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IECEx Bulletin

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-11

IECEx Bulletin:2011(E)

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 61241-11

INTERNATIONAL
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COMMISSION

Section 3
National Differences

IEC Standard 61241-11 1st Edition 2005 Electrical apparatus for use in the presence of combustible dust - Part 11: Protection by intrinsic safety 'iD'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil*		**TBA**	
CA Canada		**YES**	CAN/CSA-C22.2 No. 61241-11:11
CH Switzerland	**NO**		EN 61241-11:2006
CN China		**NO**	GB 12476.4-2010
CZ Czech Republic	**NO**		EN 61241-11:2006
DE Germany	**NO**		EN 61241-11:2006
DK Denmark	**NO**		EN 61241-11:2006
FI Finland	**NO**		EN 61241-11:2006
FR France	**NO**		EN 61241-11:2006
GB United Kingdom	**NO**		EN 61241-11:2006
HR Croatia	**NO**		EN 61241-11:2006
HU Hungary	**NO**		EN 61241-11:2006
IN India		**NO**	Adopted as IS/IEC 61241-11: 2005
IT Italy	**NO**		EN 61241-11:2006
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 61241-11:2006
NO Norway	**NO**		EN 61241-11:2006
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 61241-11:2006
RO Romania	**NO**		EN 61241-11:2006
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 61241-11:2006
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 61241-11:2006
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 61241-11 2006
ZA South Africa		**NO**	SANS 61241-11

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 61241-11 1st Edition 2005 Without National Differences
AU Australia
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
SG Singapore
ZA South Africa

Countries that have declared adoption of IEC 61241-11 1st Edition 2005 With National/Group Differences- Differences follow.

CA Canada

CAN/CSA-C22.2 No. 61241-11:11
Electrical apparatus for use in the presence of combustible dust – Part 4: Protection by intrinsic safety "iD"

This is the first edition of CAN/CSA-C22.2 No. 61241-11, which is an adoption, with Canadian deviations, of the identically titled CEI/IEC (*Commission Électrotechnique Internationale / International Electrotechnical Commission*) Standard 61241-11 (First edition, 2005-10) together with Corrigendum 1.

This Standard is considered suitable for use for conformity assessment within the stated scope of the Standard.

This Standard was reviewed for Canadian adoption by the CSA Technical Committee on International Standards, under the jurisdiction of the Strategic Steering Committee on Requirements for Electrical Safety, and has been formally approved by the Technical Committee. This Standard has been approved as a National Standard of Canada by the Standards Council of Canada.

Interpretations: The Strategic Steering Committee on Requirements for Electrical Safety has provided the following direction for the interpretation of standards under its jurisdiction: "*The literal text shall be used in judging compliance of products with the safety requirements of this Standard. When the literal text cannot be applied to the product, such as for new materials or construction, and when a relevant committee interpretation has not already been published, CSA's procedures for interpretation shall be followed to determine the intended safety principle.*"

February 2009

[… and the rest of that is necessary…]

Section 3
National Differences

Canadian deviations

1 Scope

[Add the following paragraphs]

The requirements of IEC 61241series Standards cover the protection techniques with respect to dust explosion and layer ignition hazard only. The CAN/CSA-C22-2 No. 61241 series of CSA Standards (based on the adoption of corresponding IEC Standards) are to be used in conjunction with other applicable standards containing the appropriate electrical safety requirements for general use equipment.

NOTE 1: Currently, the IEC 61241 series of Standards are being converted (merged) into the IEC 60079 series of Standards. Some Standards, relevant to combustible dust environments, may be under the IEC 61241 series or under the IEC 60079 series depending on the scheme of progress.

NOTE 2: See also Canadian deviations under CAN/CSA-C22.2 No. 61241-0:11, CAN/CSA-C22.2 No. 60079-0:11 and CAN/CSA-C22.2 No. 60079-11.

2 Normative references

[Add the following paragraph]

CSA (Canadian Standards Association)

C22.1-09
Canadian Electrical Code, Part I – Safety standard for electrical installations

Section 3
National Differences

US United States of America

Clause	*US Differences to IEC 61241-11, 1st Ed, 2005*
General	Where references are made to other IEC 61241 or IEC 60079-standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences. The term “category” has been replaced in all instances with the more y agreed term “level of protection”.
1	A title has been added to the untitled table of exclusions as follows: **Table A — Exclusion of specific clauses of ANSI/ISA-61241-0-2006**
1	The following rows in Table A have been deleted to align with ANSI/ISA 61241-0: 4.2, 5.2, 6.1.3, 23.4.5.2, 23.4.5.3, 27, and 28.
5	Text modified as follows to add specific requirements for levels of protection “iaD” and “ibD”: **5 ~~Categories~~ Levels of protection of electrical apparatus** Clause 5 of ANSI/ISA-60079-11 ~~IEC 60079-11~~ applies together with the following requirement: Intrinsically safe apparatus and intrinsically safe parts of associated apparatus shall be placed in level of protection “iaD” or “ibD”. Intrinsically safe apparatus complying with the requirements of 5.2 of ANSI/ISA-60079-11 and this standard shall be marked “iaD”. Intrinsically safe apparatus complying with the requirements of 5.3 of ANSI/ISA-60079-11 and this standard shall be marked “ibD”. Apparatus shall, at a minimum, meet the spark ignition energy level requirements for Group IIB apparatus.

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National Differences

Clause	*US Differences to IEC 61241-11, 1st Ed, 2005*
6.1	Text modified as follows to clarify the requirements for enclosures: **6.1 Enclosures** In principle, intrinsically safe apparatus does not need an enclosure as the method of protection is embodied within the circuits. However, where intrinsic safety can be impaired by access to conducting parts, for example if dust of any type can bridge infallible creepage distances, an enclosure of at least IP6x in accordance with IEC 60529 ~~and as required by IEC 61241-1~~ shall be provided. In all other circumstances, the degree of protection provided by the enclosure that is required will vary according to the intended purpose of the apparatus. Parts of apparatus not within an enclosure having a degree of protection of at least IP6x that are encapsulated to a depth of at least 1 mm and meet the requirements of 6.7 shall be considered to be adequately protected against dust contamination. Where parts of a circuit are not protected by an enclosure or encapsulation as indicated, for example uninsulated sensor circuits in direct contact with the explosive dust atmosphere, they shall be assessed or tested in accordance with this standard assuming that all spacings do not meet the creepage and clearance requirements of 6.4. The enclosure construction and/or the encapsulation shall be recorded in the definitive documentation (see Clause 13).
6.2 6.2.1 6.2.2	Text has been modified as follows to include requirements for apparatus not immersed in dust, delete the requirements for potentiometers as they are already addressed by 60079-11, and align with the temperature class requirements of the National Electrical Code: **6.2 Temperatures of apparatus ~~immersed in dust~~** **6.2.1 General** Apparatus shall meet the requirements of 6.2 of ANSI/ISA-60079-11, taking into consideration the requirements of 5.2 and 5.3 of ANSI/ISA-60079-11 applied as appropriate for the required level of protection, and 6.2.2.1 of this standard. **6.2.2 Supplementary requirements for apparatus immersed in dust** **~~6.2.2 Apparatus and component temperature~~** **6.2.2.1 General**

Section 3
National Differences

Clause	*US Differences to IEC 61241-11, 1st Ed, 2005*
	~~6.2.1 General~~ ~~Whenever possible intrinsically safe apparatus for dusts should be enclosed.~~ This clause applies to such apparatus as temperature sensors or parts of measuring devices which cannot be enclosed and are intended for immersion in a dust. All temperatures shall be measured or assessed under the most onerous fault conditions in this standard but without applying factors of safety to the current, voltage or power. ~~The following requirements are applicable to apparatus used with dusts having a minimum ignition temperature of a 5 mm layer of not less than 210 °C.~~ ~~For potentiometers, the surface to be considered shall be that of the resistance element and not the external surface of the component. The mounting arrangement and heat-sinking and cooling effect of the overall potentiometer construction shall be taken into consideration during the test. Temperature shall be measured on the track with that current which flows under conditions of "iaD" or "ibD" as appropriate. If this results in a resistance value of less than 10 % of the track resistance value, the measurement shall be carried out a 10% of the track resistance value.~~ **~~6.2.2~~ 6.2.2.2 Apparatus and component temperature** ~~Apparatus is considered suitable for total immersion, or being subjected to an uncontrolled layer of such dust, if it conforms to one of the following:~~ ~~a) the matched power dissipation in any component shall be in accordance with Table 1.~~ **~~Table 1 – Permitted dissipation within a component~~** ~~Maximum ambient temperature~~ | ~~°C~~ | ~~40~~ | ~~70~~ | ~~100~~ ~~Permitted dissipation~~ | ~~mW~~ | ~~750~~ | ~~650~~ | ~~550~~ ~~b) Where intrinsically safe electrical circuits are exposed to, or directly immersed in a flammable dust with a dust layer smouldering temperature greater than 200 °C, no testing of temperature rise is required provided that at the point of contact the available power is less than 750 mW and the short-circuit current is less than 250 mA.~~ ~~c) The apparatus shall satisfactorily comply with the standardized dust immersion test specified in 6.2.3. Apparatus which complies shall be considered as suitable for use at the maximum stated ambient temperature with any combustible dust having an ignition temperature no lower than 210 °C.~~ The maximum surface temperature of all components, enclosures and the wiring in contact with combustible dust atmospheres and determined during testing shall be marked on the product in accordance with 29.2 of ANSI/ISA-61241-0.

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National Differences

Clause	***US Differences to IEC 61241-11, 1st Ed, 2005***
6.2.3	The text of 6.2.3 is modified to add a suitable test: **6.2.3 Standardized dust immersion temperature test** ~~This test and the associated requirements are under further consideration.~~ Intrinsically safe apparatus having an enclosure meeting the requirements for at least IP6x in accordance with IEC 60529 shall be tested in accordance with 8.2.2 of ISA-61241-1 with the temperature measurement made on the outside of the enclosure there shall be no evidence of charring of the dust. Intrinsically safe apparatus having an enclosure which does not meet the requirements for IP6x in accordance with IEC 60529 shall be tested with a 12.5 mm layer of dust paste applied over the circuit and shall additionally comply with the requirements of 6.2.6. Temperature tests shall be conducted in accordance with 23.4.5.1 of ANSI/ISA-61241-0 with the temperature measurements made on the components and wiring, there shall be no evidence of charring of the dust.
6.2.4	Text deleted as it is already addressed by the text of 6.2.1. ~~**6.2.4 Wiring within apparatus**~~ ~~As for 6.2.2 of IEC 60079-11.~~
6.2.5	Text deleted as it is already addressed by the text of 6.2.1. ~~**6.2.5 Printed circuit wiring**~~ ~~As for 6.2.3 of IEC 60079-11~~.

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National Differences

Clause	*US Differences to IEC 61241-11, 1st Ed, 2005*
6.2.6	The text has been modified as follows to clarify the requirements: **6.2.6 Unenclosed circuits** Circuits of intrinsically safe apparatus intended to be in direct contact with the explosive dust atmosphere shall be capable of meeting the requirements for level of protection "iaD" and at least the spark ignition requirements for Group IIB apparatus. In this case, it shall be assumed that, unless encapsulated in accordance with 6.7, all spacings do not meet the creepage and clearance distance requirements specified in 6.4 and that all connections between intrinsically safe circuits or earthed parts and conductors are in the most unfavorable condition.
10.1	A second paragraph has been added as follows to align the requirements with those of 60079-11: A circuit may be exempted from a type test with the spark-test apparatus if its structure and its electrical parameters are sufficiently well defined for its safety to be deduced from the Group IIB reference curves figures A.1 to A.6 or tables A.1 and A.2 of ANSI/ISA-60079-11 by the methods described in Annex A of ANSI/ISA-60079-11.
10.4	The small component ignition test was deleted as follows as it not applicable to dust atmospheres: ~~**10.4 Small component ignition test**~~ ~~As for 10.7 of IEC 60079-11:1999.~~
10.7.1	The text of the second paragraph has been modified as shown as 60079-0 no longer refers to an impact energy in Joules: Where a free surface of casting compound occurs, in order to ensure that the compound is rigid but not brittle, the following impact test shall be carried out on the surface of the casting compound at (20 ± 10) °C using the test apparatus described in Annex C of ANSI/ISA-60079-0 ~~IEC 60079-0:1997~~. For all applications ~~a minimum impact energy of 2 J~~ a 1kg test mass falling from a height of 0.2m shall be used.

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Clause	***US Differences to IEC 61241-11, 1st Ed, 2005***
10.8	*The 3rd paragraph has been modified as shown to align the "X" marking with US requirements:* Where it is necessary to protect the apparatus from external physical impact in order to prevent the impact energy exceeding these values, details of the requirements shall be specified as ~~special~~ conditions <u>of</u> ~~for safe~~ use and the apparatus shall be marked ~~with an "X"~~ <u>in accordance with 29.2 i) of ANSI/ISA-61241-0</u>.
12.1	The 1st paragraph has been modified as shown to introduce the US requirement for marking of a Control Drawing number: Intrinsically safe apparatus and associated apparatus shall carry at least the minimum marking specified in <u>29.2 of ANSI/ISA-61241-0</u>, ~~IEC 61241-0 and~~ 12.2 of this standard <u>and a reference to the Control Drawing when required by 12.3</u>. The marking of the serial number may be achieved by using a date or batch number code which is sufficient to ensure traceability for quality control purposes.
12.1	The 2nd paragraph has been modified as shown to introduce the US requirement for the "A" prefix prior to the "Ex": For associated apparatus the symbol <u>A</u>Ex iaD or <u>A</u>Ex ibD (or iaD or ibD if <u>A</u>Ex is already marked) shall be enclosed in square brackets.
12.2	The marking examples are modified to remove the certificate number reference and add the prefix "A" to the "Ex"
12.3	New text is added as follows to address the US requirement for a Control Drawing: <u>**12.3 Control drawings**</u> <u>A control drawing shall be provided for all intrinsically safe apparatus or associated apparatus that requires connection to other circuits or apparatus. See 12.3 of ANSI/ISA-60079-11.</u>

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Clause	***US Differences to IEC 61241-11, 1st Ed, 2005***
13	The 2nd paragraph is modified as follows to align with the temperature testing requirements: In addition the documentation shall specify the maximum surface temperature of the apparatus and the conditions under which the temperature was determined. ~~For example "maximum surface temperature 135 °C derived from the corn dust immersion temperature test of 6.2.3".~~

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-18

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61241-18

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 61241-18 1st Edition 2004 Electrical apparatus for use in the presence of combustible dust – Part 18: Protection by encapsulation 'mD'			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61241-18: 2005
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**NO**		EN 61241-18:2004
CN China		**NO**	GB 12476.6-2010
CZ Czech Republic	**NO**		ČSN EN 61241-18:2005
DE Germany	**NO**		EN 61241-18:2004
DK Denmark	**NO**		DS/EN 61241-18:2004
FI Finland	**NO**		EN 61241-18:2004
FR France	**NO**		EN 61241-18:2004
GB United Kingdom	**NO**		BS EN 61241-18:2004
HR Croatia	**NO**		
HU Hungary	**NO**		EN 61241-18:2004
IN India		**NO**	Adopted as IS/IEC 61241-18: 2004
IT Italy	**NO**		EN 61241-18:2004
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 61241-18:2004
NO Norway	**NO**		EN 61241-18:2004
NZ New Zealand		**NO**	AS/NZS 61241-18: 2005
PL Poland	**NO**		
RO Romania	**NO**		EN 61241-18:2004
RU Russia*		**TBA**	
SE Sweden	**NO**		EN 61241-18:2004
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 61241-18:2004
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 61241-18 2006
ZA South Africa		**NO**	SANS 61241-18

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details***

Section 3
National Differences

Countries that have declared adoption of IEC 61241-18 1st Edition 2004 Without National Differences
AU Australia
CN China
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
IN India
JP Japan
NZ New Zealand
ZA South Africa

Countries that have declared adoption of IEC 61241-18 1st Edition 2004 With National/Group Differences- Differences follow.

US Differences on following page.

Section 3
National Differences

US United States of America

Clause	US Differences to IEC 61241-18, 1st Ed, 2004
General	Where references are made to other IEC 61241 or IEC 60079-standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
3.8	Deletion of term as term is not used in the document ~~**3.8**~~ ~~**void**~~ ~~unintentional space created as a consequence of the encapsulation process~~
3.9	Deletion of term as term is not used in the US adoption of the document ~~**3.9**~~ ~~**free space**~~ ~~intentionally created space surrounding components or space inside components~~
3.11	Addition of 3.11: **3.11 solid insulation** solid insulation is insulation material which is extruded or molded but not poured. Insulators fabricated from two or more pieces of electrical insulating material which are solidly bonded together may be considered as solid. Varnish and similar coatings are not considered to be solid insulation.
4.3	Modification of text as follows to identify which fault conditions apply and to use the defined term "compound": **4.3 Level of protection "maD"** Level of protection "maD" shall not be capable of causing ignition in each of the following circumstances:

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	a) in normal operation and installation conditions; b) ~~any specified abnormal conditions~~ under fault conditions as specified in 7.2; ~~c) in defined failure conditions.~~ For level of protection "maD" the working voltage at any point in the encapsulation circuit shall not exceed 1 kV. For level of protection "maD", components without additional protection shall only be used if they cannot damage the ~~encapsulation~~ compound mechanically or thermally in the case of any fault. Alternatively, where a fault of an internal component may lead to failure of the encapsulation system due to increasing temperature, the requirements of 6.2 shall apply. NOTE Certain components whose application is allowed according to this standard for level of protection "mbD" can possibly make the protective measure 'encapsulation' ineffective by mechanical or thermal damage as a consequence of internal reactions. This risk should be excluded for level of protection "maD" apparatus.
4.4	Modification of text as follows to identify which fault conditions apply: **4.4 Level of protection "mbD"** Level of protection "mbD" shall not be capable of causing ignition in each of the following circumstances: a) in normal operation and installation conditions; b) ~~in defined failure conditions~~ under fault conditions as specified in 7.2.
5.1	Modification of the 3rd paragraph of 5.1 to use the defined term "compound":

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	Due consideration shall be taken in the selection of ~~encapsulating materials~~ compounds to allow for the expansion of components during operation and in the event of allowable faults.
5.2	Modification of 5.2, first and second paragraphs: ~~The manufacturer shall attest on his own responsibility that the material complies with the compound specification.~~ The material specification shall include:
6.1	Modification as follows: **6.1 General** The maximum surface temperature and the maximum value of the continuous operating temperature of the compound shall not be exceeded during normal operation. The encapsulation "mD“ apparatus shall be protected in such a way that the encapsulation "mD“ is not adversely affected under specified fault conditions.
7.1	Modification of the 1st and 2nd paragraphs of 7.1 to use the defined term “compound” and align the “X” marking with US requirements: **7.1 General** Where the ~~encapsulation~~ compound forms part of the external enclosure it shall comply with the requirements for non-metallic enclosures and parts of non-metallic enclosures of ANSI/ISA-61241-0. If additional protective measures are required by the user in order to satisfy the requirements of this standard, for example, additional mechanical protection, to indicate this special condition of use, the apparatus shall be marked

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	~~"X"~~ in accordance with item i) of 29.2 of ~~IEC~~ ANSI/ISA-61241-0.
7.2.4	Modification of the 3rd paragraph of 7.2.4 to use the defined term "compound" In all cases the ~~encapsulant~~ compound is additionally subjected to the dielectric strength test of 8.2.4.
7.2.5	Modification of 7.2.5 to use the defined term "compound" **7.2.5 Encapsulation "mD" apparatus with free surface** The thickness of the compound between the free surface of the compound and the components or conductors in the ~~encapsulation~~ compound, as shown in Figure 1, shall comply with Table 2.

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<table>
<tr><th>Clause</th><th>US Differences to IEC 61241-18, 1st Ed, 2004</th></tr>
<tr><td>Table 2</td><td>
Modification of Table 2 to use the defined term “compound”:
<table>
<tr><th>Level of protection "maD"</th><th>Level of protection "mbD"</th></tr>
<tr><td rowspan="2">b ≥3 mm</td><td>Free surface < 2 cm2
b ≥distance according to Table 1, but not less than 1 mm</td></tr>
<tr><td>Free surface > 2 cm2
b ≥distance according to Table 1, but not less than 3 mm</td></tr>
<tr><td>c ≥distance according to Table 1</td><td>c ≥distance according to Table 1</td></tr>
<tr><td>d ≥3 mm</td><td>d ≥1 mm</td></tr>
<tr><td colspan="2">where
b is the distance between the component and the free su□face;
c is the distance between the component and non current-carrying parts inside the ~~encapsulation~~ compound;
d is the distance between a non current-carrying part and the free surface.</td></tr>
</table>
</td></tr>
<tr><td>7.2.6</td><td>
Modification of 7.2.6 to use the defined term “compound”:

7.2.6 Encapsulation "mD" apparatus with metal enclosure

The thickness of the compound between the wall or the free surface of the compound and the components or conductors in the ~~encapsulation~~ compound, as shown in Figure 2, shall comply with Table 3.
</td></tr>
<tr><td>7.2.7</td><td>Modification of 7.2.7 to use the defined term “compound”:</td></tr>
</table>

Section 3
National Differences

Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	7.2.7 Encapsulation "mD" apparatus with plastic enclosure The thickness of the compound between the wall or the free surface of the compound and the components or conductors in the ~~encapsulation~~ compound are shown in Figure 3 and shall comply with Table 4.
7.2.8	Modification of 2nd paragraph of 7.2.8 to identify the specific test to be conducted: For both levels of protection “maD” and "mbD" the end of the slot and the end-winding shall be protected by the minimum thickness of compound in accordance with 7.2.4. An electric strength test, in accordance with 8.2.4, shall be passed with $U = (2U + 1\ 000\ \text{V})$ a.c. with a minimum of 1 500 V a.c.
7.2.9.1	Modification of text as follows to clarify that requirement is based on working voltage: Multi-layer printed wiring boards complying with the requirements of IEC 62326-4-1, performance level C, with the minimum distances in 7.2.9.2 and operated at working voltages less than or equal to 500 V shall be considered to be encapsulated provided they meet the following clauses.
7.2.9.2	Modification of text of 2nd paragraph as follows to align with requirements shown in Table 5: The minimum distance between the printed circuit conductors and the edge of the multi-layer printed wiring board or any hole in it shall be at least 3 mm. If the edges or holes are protected with metal or insulating material extending at least 1 mm for “mbD” or 3 mm for “maD” along the surface of the board from the edge/hole, the distance of the printed wiring conductors may be reduced to 1 mm. Insulating material shall comply with the requirements for conformal coating in accordance with ~~IEC~~ ISA-61241-11. Metal coating shall have a minimum thickness of 35 µm, see also Figure 4 and Table 5.

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
7.4	Modification of text as follows to define what “suitable methods” are: **7.4 External connections** ~~With hard compounds, the sleeving of the connection cable shall be protected against damage by suitable methods. If the entry takes the form of a cable that is permanently connected to the encapsulation "mD“ apparatus, the pull test shall be carried out according to 8.2.5.~~ The entry of all electric conductors, including cables, into the compound shall be designed in such a way that the ingress of an explosive atmosphere into the encapsulation “mD” apparatus under normal operating or specified fault conditions is prevented. NOTE: This may generally be achieved by a conductor path in the compound that is at least 5 mm long. When compounds are used to secure the connection cable, the cable shall be suitably protected against damage from flexing. If the entry takes the form of a cable that is permanently connected to the encapsulation “mD” apparatus, the pull test shall be carried out according to 8.2.5.
7.6.4.b	Modification of 7.6.4.b to use the defined term “compound” b) they shall be provided with a safety device in accordance with 7.6.5 to 7.6.9 to prevent unacceptable overheating or gassing inside the ~~encapsulation~~ compound.
7.6.6	Modification of 2nd paragraph to add the US fuse standard, and to clarify that it is the apparatus that is marked, and not the replaceable fuse. A resistor, a current limiting device or a fuse according to ~~IEC 60127~~ ANSI/UL 248-1 or an equivalent standard, may be used to ensure the safe current specified by the manufacturer of the cells or battery is not exceeded. If replaceable fuses

Section 3
National Differences

Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	are used ~~they~~ the apparatus shall be marked to show ~~their~~ the fuse rating and function.
7.6.8	Revised text as follows to clarify that neither condition may be exceeded: **7.6.8 Charging of batteries** The charging circuits shall be fully specified as part of the apparatus. The charging system shall be such that the following is satisfied: a) with one fault condition of the charging system the charger voltage and current shall not exceed the limits specified by the manufacturer; or b) if, during charging, it is possible for the limit values specified by the manufacturer of the cells or battery for the cell voltage or the charging current to be exceeded, a separate safety device in accordance with 7.7 shall be provided to avoid a release of gas ~~and~~ or exceeding the manufacturer's maximum rated cell temperature.
7.7.1	Modification of 7.7.1 to clarify the protective device ratings and use the defined term "compound" ….. The purpose of the protective devices is the safe technical limitation of an inadmissible heat rise in encapsulation "mD" apparatus. The protective devices shall be capable of interrupting the maximum fault current of the circuit in which they are installed. The rated voltage of the protective device shall at least correspond to the working voltage of the circuit in which it is installed. Where the ~~encapsulation~~ compound contains a cell or battery and a safety device is provided to prevent excessive overheating (see 7.6.6), the safety device can also be considered as a protective device, providing it also protects all other components inside the same ~~encapsulation~~ compound from exceeding the COT or temperature class.
7.7.2.1	Modification of 7.7.2.1 to clarify the protective device ratings and add the US fuse standard:

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	7.7.2.1 General Fuses shall have a voltage rating not less than that of the circuit in which they are installed and shall have a breaking capacity not less than the fault current of the circuit. Unless otherwise specified, a fuse shall be assumed to be capable of passing 1,7 x rated current continuously. The time-current characteristic of the fuse shall ensure that the COT of the ~~encapsulant~~ compound or the maximum surface temperature are not exceeded. The time-current characteristics of the fuses, in accordance with ~~IEC 60127~~ or ANSI/UL 248-1 shall be stated by the manufacturer of the fuses.
7.7.2.2	Modification of 7.7.2.2 to clarify that it is the apparatus that is subjected to fault, not the compound, and align the “X” marking with US requirements: **7.7.2.2 Protective devices that are connected to the encapsulation "mD" apparatus** Where the encapsulation “mD” apparatus is unable to withstand a single fault, then the encapsulation "mD“ apparatus may be connected to separate protective devices. To indicate this special condition of use the apparatus shall be marked ~~“X”~~ in accordance with item i) of 29.2 of ~~IEC~~ ISA 61241-0.
7.7.3	Modification of 7.7.3 to clarify the protective device reset function and use the defined term “compound” **7.7.3 Thermal protective devices** The requirements of 6.2 shall apply to thermal protective devices. Thermal protection devices shall be used to protect the ~~encapsulation~~ compound from damage caused by local heating, for example, by faulty components or from exceeding the maximum surface temperature. Only non ~~self-~~-resettable thermal protective devices shall be used. These devices have no provision for being reset and open a circuit permanently after being exposed to a temperature higher than their rating for a given maximum period. Adequate thermal connection shall be achieved between the monitored component and the thermal protective device. The

Section 3
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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
	switching capability of the device shall be defined and shall be not less than the maximum possible load of the circuit.
7.7.4	Modification of 7.7.4 to clarify the intent: **7.7.4 Built-in protective devices** Protective devices integral with the encapsulation "mD" apparatus shall be of the enclosed type such that ~~no~~ the compound cannot interfere with the operation of the protective device ~~enter during the encapsulation process~~. ~~The suitability of the protective device for the intended purpose can be confirmed either by~~ Compliance with this requirement shall be determined by either a) a declaration by the manufacturer of the protective device; or b) testing of samples.
8.2.2	Add text to address the US practice of dust-blanketing during temperature testing: This test shall be carried out in accordance with ANSI/ISA-61241-0 with the additional requirement that the apparatus shall be covered with the maximum amount of dust that it can retain. As an alternative, a 12,5 mm thick layer of dust paste may be put on top of the apparatus to simulate the build-up conditions. NOTE The paste should consist of 45 % dust (e.g. wheat flour) and 55 % water by weight. The temperature value should be measured after the paste has dried.
8.2.4.1	Modify text of 3rd paragraph to clarify the ac test frequency and use the fined term "compound": The test voltage shall be 500 V r.m.s. at 48 Hz to 62 Hz for apparatus where the sum of the supply voltages does not exceed 90 V peak. If the sum of the supply voltages exceeds 90 V peak the test voltage shall be 2*U* + 1 000 V, with a minimum of 1 500 V a.c. at 48 Hz to 62 Hz. Where an alternating test voltage would damage electronic components in the ~~encapsulation~~ compound, the test voltage shall be 2 *U* + 1 400 V d.c. with a minimum of 2 100 V d.c.

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Clause	US Differences to IEC 61241-18, 1st Ed, 2004
8.2.5.2	Modify text of 3rd paragraph to correct the units: The tensile force (in Newton) applied shall either be 20 times the value in millimetres of the diameter of the cable or 49 times the mass (in kilograms) ~~5 times the weight~~ of the encapsulation “mD” apparatus, whichever is the lower value. This value can be reduced to 25 % of the required value in the case of permanent installations. The minimum tensile force shall be 1 N and the minimum duration shall be 1 h. The force shall be applied in the least favourable direction.
8.2.6	Delete all text of 8.2.6, 8.2.6.1, and 8.2.6.2 as it is not possible for dust to migrate through compound to form an internal dust cloud.
Annex B	Delete the row of the table dealing with the deleted test of 8.2.6.

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 61779-1

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 61779-1

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Section 3

National Differences

IEC Standard 61779-1 1st Edition 1998 Electrical apparatus for the detection and measurement of flammable gases Part 1: General requirements and test methods			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61779-1: 2000
BR Brazil*		**TBA**	
CA Canada		**NO**	
CH Switzerland	**YES**		EN 61779-1:2000
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 61779-1:2001
DE Germany	**YES**		EN 61779-1:2000
DK Denmark	**YES**		DS/EN 61779-1:2000
FI Finland	**YES**		EN 61779-1:2000
FR France	**YES**		EN 61779-1:2000
GB United Kingdom	**YES**		BS EN 61779-1:2000
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 61779-1:2000
IN India		**N/R**	
IT Italy	**YES**		EN 61779-1:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 61779-1:2000
NO Norway	**YES**		EN 61779-1:2000
NZ New Zealand		**NO**	AS/NZS 61779-1:2000
PL Poland	**YES**		
RO Romania	**YES**		EN 61779-1:2000
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 61779-1:2000
SG Singapore		**NO**	
SI Slovenia	**YES**		EN 61779-1:2000
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 12.13.01 2002
ZA South Africa		**N/R**	

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Please Note IEC 61779-1 to IEC 61779-5:1998 Series has been replaced by IEC 60079-29-1 First Edition (2007)

Section 3
National Differences

Countries that have declared adoption of IEC 61779-1 1st Edition 1998 Without National Differences
AU Australia
CA Canada
JP Japan
NZ New Zealand
SG Singapore

Countries that have declared adoption of IEC 61779-1 1st Edition 1998 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway PL Poland, RO Romania, SE Sweden, SI Slovenia**

The text of the International Standard IEC 61779-1:1998 was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

Introduction **Delete**.

1 General

1.1.1 **Delete** note 1, **renumber** note 2 as note 1 and note 3 as note 2.

1.2 Delete reference to ISO 2738, ISO 4003, ISO 4022 and the IEC 60079 series of standards.

Add:

EN 50073:1999, *Guide for selection, installation, use and maintenance of apparatus for the detection and measurement of combustible gases or oxygen*

EN 50270:1999, *Electromagnetic compatibility : Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen*

prEN 50271, *Electrical apparatus for the detection and measurement of combustible gases, toxic gases or oxygen - Requirements and tests for apparatus using software and/or digital technologies*

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US United States of America National differences to IEC 61779-1 (1998)		
IEC Standard	**Clause**	**Difference**
IEC 61779-1 (1998)	General	The U.S. version of IEC 61779-1 does not include requirements for Group I electrical apparatus.
IEC 61779-1 (1998)	General	Where references are made to other IEC 60079 or IEC 61779 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 61779-1 (1998)	1.2	Remove "Normative" from title and add US national standards and deviations.
IEC 61779-1 (1998)	2.2.11 & 2.2.12	Addition of clauses 2.2.11 & 2.2.12 (stand-alone gas detection) as shown: "2.2.11 Stand-alone gas detection instrument: Gas detection instruments that are utilized with unspecified control apparatus. Such instruments are intended to be interfaced to separate control units, signal processing data acquisition, central monitoring, or other similar systems in which the instrument provides a conditioned electronic signal or output indication to the system of the aforementioned type that typically process information from various locations and sources including, but not limited to, gas detection instruments. 2.2.12 Stand-alone control unit: Gas detection control units intended for use with unspecified stand-alone gas detection instruments."
IEC 61779-1 (1998)	3.2.7	Addition of Clause 3.2.7 (stand-alone gas detection) as shown: 3.2.7 Stand-alone gas detection instruments for use with separate control units

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		3.2.7.1 General Subclause 3.2.7 includes instruments that provide a conditioned electronic signal or output indication intended to be used with separate signal processing, data acquisition, central monitoring or other similar systems which typically process information from various locations and sources including, but not limited to, gas detection instrumentation. 3.2.7.2 Transfer function specification A specification shall be supplied with the instrument that describes the relationship the gas concentration (detected by the instrument) has with the corresponding output signal or indication (transfer function). Such specification shall be detailed to the extent that the accuracy of this transfer function can be verified. As a minimum, the manufacturer shall provide data showing the relationship between the output signal and the gas concentrations corresponding to 0, 10%, 25%, 50%, 75% and 100% of full-scale output indication. Full-scale output shall also be as specified by the manufacturer. 3.2.7.3 Provision for transfer function verification Where necessary, equipment shall be provided by the manufacturer to interpret the output signal or indication, which will enable the accuracy of the transfer function to be verified. 3.2.7.4 Tests Stand alone gas detection instruments shall be tested to the requirements of Clauses 4.4.2 through 4.4.13, 4.4.15, 4.4.16, 4.4.18, 4.4.20, 4.4.21, 4.4.22 and 4.4.25 using the parameters of the transfer function.

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IEC 61779-1 (1998)	3.2.8	Addition of Clause 3.2.8 (stand-alone gas detection) as shown: <u>3.2.8 Separate control units for use with stand-alone gas detection instruments</u> <u>3.2.8.1 General</u> <u>Clause 3.2.8 includes those instruments to be used with stand-alone gas detection instruments (as defined in 3.2.7) to complete a "performance evaluated" combustible gas detection system.</u> <u>3.2.8.2 Tests</u> <u>The control units shall be tested to the requirements of clauses 4.4.2, 4.4.3, 4.4.6, 4.4.7, 4.4.13, 4.4.20, 4.4.21 and 4.4.25 using the parameters of the transfer function pertinent to the specific type of gas detector.</u>
IEC 61779-1 (1998)	3.3	Addition of (f) and (g) as shown: <u>f) the specific marking "CAUTION — READ AND UNDERSTAND INSTRUCTION MANUAL BEFORE OPERATING OR SERVICING."</u> <u>g) for portables instruments indicating 0-100%LFL, the specific marking "CAUTION — OFF-SCALE READINGS MAY INDICATE EXPLOSIVE CONCENTRATION."</u>
IEC 61779-1 (1998)	3.5	Remove Clause 3.5 (Diffusion Sensors)
IEC 61779-1 (1998)	4.2.2	Change “In all cases” to “Unless otherwise noted”.
IEC 61779-1 (1998)	4.3.11	Addition of Clause 4.3.11 (communication options) as shown: <u>4.3.11 Communications options – For instruments having serial or parallel communications options, tests in sub-clauses 4.4.3 and 4.4.16 shall be performed with all communication ports connected to test apparatus which initiates the maximum transaction rate, cabling and activity level specified by the instrument’s manufacturer.</u>

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IEC 61779-1 (1998)	4.3.12	Addition of Clause 4.3.12 (large gas detection systems) as shown: 4.3.12 Gas detection apparatus as part of large systems - For gas detection instruments which are part of large systems, tests in subclauses 4.4.3 and 4.4.16 shall be performed with the maximum system communications transaction rate and activity level which would result from the largest and most complex system configuration permitted by the manufacturer.
IEC 61779-1 (1998)	4.4.3.3	Alarm only apparatus added to requirement as shown: 4.4.3.3 Response to different gases ~~(not applicable to alarm-only apparatus)~~
IEC 61779-1 (1998)	4.4.16	Remove recovery time requirements as shown: The apparatus shall be switched on in clean air and after an interval corresponding to at least two times the warm-up time, as determined in accordance with 4.4.15, without switching off, the apparatus or the sensor(s) shall be subjected to a step change from clean air to the standard test gas, which shall be introduced by means of suitable equipment (see Annex B). ~~a) be subjected to a step change from clean air to the standard test gas, which shall be introduced by means of suitable equipment (see Annex B),~~ ~~following stabilization at the standard test gas, be subjected to a step change back to clean air.~~ Both the times of response t(50) and t(90) shall be measured ~~in each direction~~ (see 2.6.6).
IEC 61779-1 (1998)	4.4.23	Remove entire clause.
IEC 61779-1 (1998)	4.4.26	Add the following clause: 4.4.26 Dielectric strength Following completion of all ~~of the~~ applicable tests ~~of~~ from the preceding clauses, the equipment shall be subjected to dielectric strength tests as specified by ANSI/ISA-S82.02.01-1999 (IEC 61010-1 Mod) or other comparable recognized standard**.**

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IEC 61779-1 (1998)	Annex A	Remove Table A.1 and modify Annex A text to the following: Flammability limits (LFL and UFL) of some flammable gases and vapours are given for guidance in NFPA 497-1997 Recommended Practice for the Classification of Flammable Liquids, Gases, Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas ~~table A.1 only~~ for performing the type testing according to this standard.

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National Differences

IEC Standard 61779-2 1st Edition 1998 Electrical apparatus for the detection and measurement of flammable gases Part 2: Performance requirements for group I apparatus indicating a volume fraction up to 5% methane in air			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61779-2 2000
BR Brazil*		**TBA**	
CA Canada*		**TBA**	
CH Switzerland	**YES**		EN 61779-2:2000
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 61779-2:2001
DE Germany	**YES**		EN 61779-2:2000
DK Denmark	**YES**		DS/EN 61779-2:2000
FI Finland	**YES**		EN 61779-2:2000
FR France	**YES**		EN 61779-2:2000
GB United Kingdom	**YES**		BS EN 61779-2:2000
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 61779-2:2000
IN India		**N/R**	
IT Italy	**YES**		EN 61779-2:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 61779-2:2000
NO Norway	**YES**		EN 61779-2:2000
NZ New Zealand		**NO**	AS/NZS 61779-2 2000
PL Poland	**YES**		
RO Romania	**YES**		EN 61779-2:2000
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 61779-2:2000
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 61779-2:2000
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**N/R**	

(N/R Not Recognised)

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Section 3
National Differences

Countries that have declared adoption of IEC 61779-2 1st Edition 1998 Without National Differences
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JP Japan
NZ New Zealand
ZA South Africa

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Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Endorsement notice

The text of the International Standard IEC 61779-2:1998 was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

Scope

1.1 **Delete** the note.

4 Performance requirements

4.21 **Replace** the subclause by:

4.21 Electromagnetic compatibility

The apparatus shall fulfil the requirements specified in EN 50270.

4.25 **Delete**.

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Section 3
National Differences

IEC Standard 61779-3 1st Edition 1998 Electrical apparatus for the detection and measurement of flammable gases Part 3: Performance requirements for group I apparatus indicating a volume fraction up to 100% methane in air			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61779-3 2000
BR Brazil*		**TBA**	
CA Canada*		**TBA**	
CH Switzerland	**YES**		EN 61779-3:2000
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 61779-3:2001
DE Germany	**YES**		EN 61779-3:2000
DK Denmark	**YES**		DS/EN 61779-3:2000
FI Finland	**YES**		EN 61779-3:2000
FR France	**YES**		EN 61779-3:2000
GB United Kingdom	**YES**		BS EN 61779-3:2000
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 61779-3:2000
IN India		**N/R**	
IT Italy	**YES**		EN 61779-3:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 61779-3:2000
NO Norway	**YES**		EN 61779-3:2000
NZ New Zealand		**NO**	AS/NZS 61779-3 2000
PL Poland	**YES**		
RO Romania	**YES**		EN 61779-3:2000
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 61779-3:2000
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 61779-3:2000
TR Turkey*		**TBA**	
US United States		**YES**	Contact Member Body
ZA South Africa		**N/R**	

(N/R Not Recognised)

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Please Note IEC 61779-1 to IEC 61779-5:1998 Series has been replaced by IEC 60079-29-1 First Edition (2007)

Section 3
National Differences

Countries that have declared adoption of IEC 61779-3 1st Edition 1998 Without National Differences
AU Australia
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 61779-3 1st Edition 1998 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Endorsement notice

The text of the International Standard IEC 61779-3:1998, was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

Scope

Delete the note.

4 Performance requirements

4.21 **Replace** the subclause by:

4.21 Electromagnetic compatibility

The apparatus shall fulfil the requirements specified in EN 50270

4.25 **Delete**.

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 61779-4

INTERNATIONAL
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Section 3
National Differences

IEC Standard 61779-4 1st Edition 1998 Electrical apparatus for the detection and measurement of flammable gases Part 4: Performance requirements for group II apparatus indicating up to 100% lower explosive limit			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61779-4 2000
BR Brazil*		**TBA**	
CA Canada		**NO**	
CH Switzerland	**YES**		EN 61779-4:2000
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 61779-4:2001
DE Germany	**YES**		EN 61779-4:2000
DK Denmark	**YES**		DS/EN 61779-4:2000
FI Finland	**YES**		EN 61779-4:2000
FR France	**YES**		EN 61779-4:2000
GB United Kingdom	**YES**		BS EN 61779-4:2000
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 61779-4:2000
IN India		**N/R**	
IT Italy	**YES**		EN 61779-4:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherland	**YES**		EN 61779-4:2000
NO Norway	**YES**		EN 61779-4:2000
NZ New Zealand		**NO**	AS/NZS 61779-4 2000
PL Poland	**YES**		
RO Romania	**YES**		EN 61779-4:2000
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 61779-4:2000
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 61779-4:2000
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 12.13.01 2002
ZA South Africa		**N/R**	

(N/R Not Recognised)

**** NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.***

Please Note IEC 61779-1 to IEC 61779-5:1998 Series has been replaced by IEC 60079-29-1 First Edition (2007)

Section 3
National Differences

Countries that have declared adoption of IEC 61779-4 1st Edition 1998 Without National Differences
AU Australia
CA Canada
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 61779-4 1st Edition 1998 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Endorsement notice

The text of the International Standard IEC 61779-4:1998 was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

4 Performance requirements

4.21 **Replace** the subclause by:

4.21 Electromagnetic compatibility
The apparatus shall fulfil the requirements specified in EN 50270.

4.25 **Delete**.

Section 3

National Differences

US United States of America

US National differences to IEC 61779-4 (1998)		
IEC Standard	**Clause**	**Difference**
IEC 61779-4 (1998)	General	The U.S. version of IEC 61779-4 does not include requirements for Group I electrical apparatus.
IEC 61779-4 (1998)	General	Where references are made to other IEC 60079 or IEC 61779 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 61779-4 (1998)	4.8	Revise pressure test acceptance criteria as shown: The variation of the indications at ~~80~~ 95 kPa and 110 kPa from the indication at 100 kPa shall not exceed ±5 % of the measuring range or ±30 % of the indication, whichever is the greater.
IEC 61779-4 (1998)	4.16	Remove recovery time requirements as shown: The time of response *t*(50) ~~in either direction~~ shall be not greater than 20 s, and *t*(90) ~~in either direction~~ shall be not greater than 60 s.
IEC 61779-4 (1998)	4.23	Remove entire clause.
IEC 61779-4 (1998)	4.26	Add the following clause: 4.26 Dielectric strength When subjected to the dielectric strength test, the apparatus shall have no breakdown or repeated flashover.

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IECEx Bulletin

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IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 3: National Differences for 61779-5

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IECEx Bulletin –
Section 3: National Differences for 61779-5

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Section 3
National Differences

IEC Standard 61779-5 1st Edition 1998 Electrical apparatus for the detection and measurement of flammable gases Part 5: Performance requirements for group II apparatus indicating a volume fraction up to 100% gas			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 61779-5 2000
BR Brazil*		**TBA**	
CA Canada		**NO**	
CH Switzerland	**YES**		EN 61779-5:2000
CN China*		**TBA**	
CZ Czech Republic	**YES**		ČSN EN 61779-5:2001
DE Germany	**YES**		EN 61779-5:2000
DK Denmark	**YES**		DS/EN 61779-5:2000
FI Finland	**YES**		EN 61779-5:2000
FR France	**YES**		EN 61779-5:2000
GB United Kingdom	**YES**		EN 61779-5:2000
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 61779-5:2000
IN India		**N/R**	
IT Italy	**YES**		EN 61779-5:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 61779-5:2000
NO Norway	**YES**		EN 61779-5:2000
NZ New Zealand		**NO**	AS/NZS 61779-5 2000
PL Poland	**YES**		
RO Romania	**YES**		EN 61779-5:2000
RU Russia*		**TBA**	
SE Sweden	**YES**		EN 61779-5:2000
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 61779-5:2000
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/ISA 12.13.01 2002
ZA South Africa		**N/R**	

(N/R Not Recognised)

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Section 3
National Differences

Countries that have declared adoption of IEC 61779-5 1st Edition 1998 Without National Differences
AU Australia
CA Canada
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 61779-5 1st Edition 1998 With National/Group Differences- Differences follow

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

Endorsement notice

The text of the International Standard IEC 61779-5:1998 was approved by CENELEC as a European Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

4 Performance requirements

4.21 **Replace** the subclause by:

4.21 Electromagnetic compatibility
The apparatus shall fulfil the requirements specified in EN 50270.

4.25 **Delete**.

Section 3
National Differences

US United States of America

US National differences to IEC 61779-5 (1998)		
IEC Standard	**Clause**	**Difference**
IEC 61779-5 (1998)	General	The U.S. version of IEC 61779-5 does not include requirements for Group I electrical apparatus.
IEC 61779-5 (1998)	General	Where references are made to other IEC 60079 or IEC 61779 standards, the referenced requirements found in these standards shall apply as modified by any applicable U.S. National Differences.
IEC 61779-5 (1998)	4.8	Revise pressure test acceptance criteria as shown: The variation of the indications at ~~80~~ <u>95</u> kPa and 110 kPa from the indication at 100 kPa shall not exceed ±7,5 % of the measuring range or ±30 % of the indication, whichever is the greater.
IEC 61779-5 (1998)	4.16	Remove recovery time requirements as shown: The time of response *t*(50) ~~in either direction~~ shall be not greater than 20 s, and *t*(90) ~~in either direction~~ shall be not greater than 60 s.
IEC 61779-5 (1998)	4.23	Remove entire clause.
IEC 61779-5 (1998)	4.26	Add the following clause: <u>4.26 Dielectric strength</u> <u>When subjected to the dielectric strength test, the apparatus shall have no breakdown or repeated flashover.</u>

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

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IECEx Bulletin – Section 3: National Differences for 62013-1

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Section 3
National Differences

IEC Standard 62013-1 2nd Edition 2005 Caplights for use in mines susceptible to firedamp – Part 1: General requirements - Construction and testing in relation to the risk of explosion			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**NO**		EN 62013-1:2006
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 62013-1:2006
DE Germany	**NO**		EN 62013-1:2006
DK Denmark	**NO**		EN 62013-1:2006
FI Finland	**NO**		EN 62013-1:2006
FR France	**NO**		EN 62013-1:2006
GB United Kingdom	**NO**		EN 62013-1:2006
HR Croatia	**NO**		EN 62013-1:2006
HU Hungary	**NO**		EN 62013-1:2006
IN India		**N/R**	
IT Italy	**NO**		EN 62013-1:2006
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 62013-1:2006
NO Norway	**NO**		EN 62013-1:2006
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 62013-1:2006
RO Romania	**NO**		EN 62013-1:2006
RU Russia		**YES**	GOST R 52065-2007
SE Sweden	**NO**		EN 62013-1:2006
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 62013-1:2006
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

(N/R Not Recognised)

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JP Japan
NZ New Zealand
SG Singapore

Countries that have declared adoption of IEC 62013-1 2nd Edition 2005 With National/Group Differences- Differences follow.

RU Differences on following page.

Section 3
National Differences

RU Russia

62013-1 2005 2nd ed.	GOST R 52065-2007	MOD	The text of the standard additionally includes: 1) in Section 1, Paragraph 1: "the marking is also considered"; 2) in Section 1, Note 4: - additional tests to 10.5, 10.6, 10.10.1, 10.10.2, b) by four levels of explosion protection GOST R 52550.0. 3) in Section 2 – reference to GOST R 51330.20, in.4.3 – requirements to solid insulation materials to GOST R 52350.0, and requirements to GOST concerning explosion protection; 4) in 4.1 – additional provision concerning the necessity to meet the requirements to the enclosure of the battery and headlight in accordance with GOST 52350.0, and the requirements to one of the types of protection. 5) 4.8 is excluded; 6) in 5.1- expanded concept of spark protecting devices 7) in 5.2 – the term of "reference explosive gas mixture "is used 8) in .5.5 - the marking shows serial connection of the fuse and the current limiting device; 9) in 6.2,6.7- the requirement to use sealers; 10) in 10.5- testing of fuses using the method accepted by the test organization; 11) in 10.10.1- Note on the frequency of electrodes changing during tests; 12) in 10.10.2- safety coefficient of 1.5 by increasing current; 13) in 11.2 -the marking in accordance with GOST R 52350.0

Section 3

National Differences

IEC Standard 62013-1 1st Edition 1999 Caplights for use in mines susceptible to firedamp – Part 1: General requirements - Construction and testing in relation to the risk of explosion			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 62013-1: 2001
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**YES**		EN 62013-1:2002
CN China*		**TBA**	
CZ Czech Republic	**YES**		EN 62013-1:2002
DE Germany	**YES**		EN 62013-1:2002
DK Denmark	**YES**		DS/EN 62013-1:2002
FI Finland	**YES**		EN 62013-1:2002
FR France	**YES**		EN 62013-1:2002
GB United Kingdom	**YES**		BS EN 62013-1:2002
HR Croatia	**YES**		
HU Hungary	**YES**		MESZ EN 62013-1:2002
IN India		**N/R**	
IT Italy	**YES**		EN 62013-1:2002
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**YES**		EN 62013-1:2002
NO Norway	**YES**		EN 62013-1:2002
NZ New Zealand		**NO**	AS/NZS 62013-1 : 2001
PL Poland	**YES**		
RO Romania	**YES**		EN 62013-1:2002
RU Russia		**YES**	GOST R IEC 52065-2003
SE Sweden	**YES**		EN 62013-1:2002
SG Singapore		**N/R**	
SI Slovenia	**YES**		EN 62013-1:2002
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Section 3
National Differences

Countries that have declared adoption of IEC 62013-1 1st Edition 1999 Without National Differences
AU Australia
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 62013-1 1st Edition 1999 With National/Group Differences- Differences follow.

Group Differences for: **CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia**

The text of the International Standard IEC 62013-1:1999 was approved by CENELEC as a European

Standard with agreed common modifications as given below.

COMMON MODIFICATIONS

Add:

Introduction

In the absence of any suitable category M1 caplight, caplights constructed to this standard may be used for short periods of time in mines which have become temporarily endangered by an explosive
atmosphere of firedamp without having to de-energize them.

1 Scope

Replace the whole text by:

1 Scope

This part of EN 62013 specifies requirements for the construction and testing of caplights for use in mines susceptible to firedamp (Group I - electrical apparatus for explosive gas atmospheres as defined in EN 50014). It deals only with the risk of the caplight becoming a source of ignition. The requirements of EN 50014 do not apply unless specified.

This standard covers category M2 caplights.

Where the caplight is to meet the requirements of EC Directive 94/9, subclause 5.1.1 and clauses 27 and 28 of EN 50014:1997 additionally apply.

Section 3
National Differences

RU Russia

IEC Clause or
Annex Number
Details of Differences IEC 62013-1, 1 edition, 1999

4.8 Replaced by:
Values of resistance and electrical insulation strength shall comply with 1.3.2 GOST 24471.

5.1 The second paragraph is changed to:
Depending on the type of protection of caplights and the level of intrinsically safe circuit such protection*, which exclude ignition of test mixture by thermal exposure*, shall be provided by one or more of the following means a-d, and shall conform to 5.2 and 5.6;
and additionally, *protection which exclude ignition of test mixture by thermal exposure or electric discharge* – shall be provided by of one or more flowing means b)-d) and shall conform to 5.3, 5.4 and 5.6 or to 5.3, 5.5 and 5.6.

5.5 When the over current protection is made by active current-limiting devices, limiting the current to the value, when the electric charge cannot ignite the explosive test mixture, two (*or three*) devices in accordance with 7.1 of GOST R 51330.10 in series shall be used in the circuit. *Depending on the type of protection of caplights* it does not preclude the use of one active device and a fuse, complying with 5.2, in series and *it shall be specified in the marking.*

10.6 Replaced by:

Test to verify the non-ignition of a gas mixture by one strand of the cable between the headpiece and the battery by thermal ignition.

A fully charged caplight battery *with current-limiting devices* is short-circuited by the smallest cross-section strand of the cable conductor in a mixture of methane and air containing 6,5% ± 0,3% by volume of methane. *The length of strand is determined by the testing organization.*

Cable, connecting the headpiece and the battery with one of its strands short-circuited, is considered to pass the test, if there haven't occurred an ignition of explosive test mixture.

It is possible not to carry out this test if and the heat temperature of the strand, short-circuiting a fully charged caplight battery, (according to assessment of temperature class) is lower than the ignition temperature of methane containing mixture.

10.10.2 The second paragraph is changed to:
Increasing the test voltage to obtain *1,5 times current in the circuit;*

Supplemented with the clause:
10.10.3 Spark ignition test in accordance with A.2 GOST R 51330.10 shall be performed with the use of spark safety characteristics $I_в = f(L,E)$.

Section 3
National Differences

Supplemented with the clause:
10.14 Cable test of headlight to resistance to multiple curving with slew shall be carry out in accordance with 5.8 GOST22471.

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Section 3
National Differences

IEC Standard 62013-2 2nd Edition 2005 Caplights for use in mines susceptible to firedamp – Part 1: General requirements - Construction and testing in relation to the risk of explosion			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**NO**		EN 62013-2:2006
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 62013-2:2006
DE Germany	**NO**		EN 62013-2:2006
DK Denmark	**NO**		EN 62013-2:2006
FI Finland	**NO**		EN 62013-2:2006
FR France	**NO**		EN 62013-2:2006
GB United Kingdom	**NO**		EN 62013-2:2006
HR Croatia	**NO**		EN 62013-2:2006
HU Hungary	**NO**		EN 62013-2:2006
IN India		**N/R**	
IT Italy	**NO**		EN 62013-2:2006
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 62013-2:2006
NO Norway	**NO**		EN 62013-2:2006
NZ New Zealand		**NO**	
PL Poland	**NO**		EN 62013-2:2006
RO Romania	**NO**		EN 62013-2:2006
RU Russia		**YES**	GOST R 52066-2007
SE Sweden	**NO**		EN 62013-2:2006
SG Singapore		**NO**	
SI Slovenia	**NO**		EN 62013-2:2006
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

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Section 3
National Differences

Countries that have declared adoption of IEC 62013-2 2nd Edition 2005 Without National Differences
AU Australia
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JP Japan
NZ New Zealand
SG Singapore

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RU Differences on following page.

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National Differences

RU Russia

62013-2 2005 2nd ed.	GOST R 52066-2007	MOD	1) in 3.1- replace the term “illuminance” by “ light intensity”; 2) in 8.1 add the note: battery charging in accordance with technical documentation; 3) in 8.1 add the following: illumination test using a power supply source in accordance with GOST 17616

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National Differences

IEC Standard 62013-2 1st Edition 2000 Caplights for use in mines susceptible to firedamp – Part 1: General requirements - Construction and testing in relation to the risk of explosion			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 62013-2: 2001
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**NO**		EN 62013-2:2000
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 62013-2:2000
DE Germany	**NO**		EN 62013-2:2000
DK Denmark	**NO**		EN 62013-2:2000
FI Finland	**NO**		EN 62013-2:2000
FR France	**NO**		EN 62013-2:2000
GB United Kingdom	**NO**		BS EN 62013-2:2001
HR Croatia	**NO**		
HU Hungary	**NO**		EN 62013-2:2000
IN India		**N/R**	
IT Italy	**NO**		EN 62013-2:2000
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD
NL The Netherlands	**NO**		EN 62013-2:2000
NO Norway	**NO**		
NZ New Zealand		**NO**	AS/NZS 62013-2: 2001
PL Poland	**NO**		
RO Romania	**NO**		EN 62013-2:2000
RU Russia		**YES**	GOST R IEC 52066 2003
SE Sweden	**NO**		EN 62013-2:2000
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 62013-2:2000
TR Turkey*		**TBA**	
US United States		**N/R**	
ZA South Africa		**N/R**	

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details

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National Differences

Countries that have declared adoption of IEC 62013-2 1st Edition 2000 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
NZ New Zealand

Countries that have declared adoption of IEC 62013-2 1st Edition 2000 With National/Group Differences- Differences follow.

RU Russia

IEC Clause or
Annex Number
Details of Differences IEC 62013-2, 1 edition, 2000

4 Supplemented with:

Construction of caplights must be provided for signaling and ability to work from the emergency (power) supply not less that 0,5 hour with the lowering of battery voltage lower than the minimal value, defined in specification for the battery.

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 62086-1

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IECEx Bulletin

Edition 4.0 2011-02

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IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 3: National Differences for 62086-1

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 3
National Differences

IEC Standard 62086-1 1st Edition 2001 Electrical apparatus for explosive gas atmospheres - Electrical resistance trace heating Part 1: General and testing requirements			
Country	**Group differences**	**National differences**	**National Standards**
AU Australia		**NO**	AS/NZS 62086.1:2002
BR Brazil*		**TBA**	
CA Canada		**N/R**	
CH Switzerland	**NO**		EN 62086-1:2005
CN China*		**TBA**	
CZ Czech Republic	**NO**		EN 62086-1:2005
DE Germany	**NO**		EN 62086-1:2005
DK Denmark	**NO**		EN 62086-1:2005
FI Finland	**NO**		EN 62086-1:2005
FR France	**NO**		EN 62086-1:2005
GB United Kingdom	**NO**		EN 62086-1:2005
HR Croatia	**NO**		
HU Hungary	**NO**		EN 62086-1:2005
IN India		**N/R**	
IT Italy	**NO**		EN 62086-1:2005
JP Japan		**NO**	
KR Republic of Korea*		**TBA**	
MY Malaysia		**N/R**	NO NATIONAL STANDARD, BUT MY ADOPTED THE NEW REVISION OF THIS STANDARD I.E. MS IEC 60079-30-1:2009
NL The Netherlands	**NO**		EN 62086-1:2005
NO Norway	**NO**		EN 62086-1:2005
NZ New Zealand		**NO**	AS/NZS 62086.1:2002
PL Poland	**NO**		
RO Romania	**NO**		EN 62086-1:2005
RU Russia		**NO**	GOST R IEC 62086-1 2003
SE Sweden	**NO**		EN 62086-1:2005
SG Singapore		**N/R**	
SI Slovenia	**NO**		EN 62086-1:2005
TR Turkey*		**TBA**	
US United States		**YES**	ANSI/IEEE 515 : 2004
ZA South Africa		**NO**	SANS 62086-1

(N/R Not Recognised)

NOTE: For information relating to National Differences not listed, please contact the IECEx Member Body of that country. Refer to Section 1 for contact details.

Please Note IEC 62086-1 First Edition (2001) has been replaced by IEC 60079-30-1 First Edition (2007)

Section 3
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Countries that have declared adoption of IEC 62086-1: 1st Ed. 2001 Without National Differences
AU Australia
Group: CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, FI Finland, FR France, GB United Kingdom, HR Croatia, HU Hungary, IT Italy, NL The Netherlands, NO Norway, PL Poland, RO Romania, SE Sweden, SI Slovenia
JP Japan
NZ New Zealand
RU Russia
ZA South Africa

Countries that have declared adoption of IEC 62086-1: 1st Ed. 2001 With National/Group Differences- Differences follow

US Differences on following page.

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National Differences

US United States of America

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Differences**
IEC 62086-1 (2001)	1	~~IEC 60079-10~~ ANSI/NFPA 70 Art. 505
IEC 62086-1 (2001)	2	ASTM D 5025-1999, A Laboratory Burner Used for Small-Scale Burning Tests on Plastic Materials ASTM D 5207-2003, Calibration of 20 and 125 mm Test Flames for Small-Scale Burning Tests on Plastic Materials. ANSI/NFPA 70, National Electrical Code® (NEC®) NFPA 325-1994, Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids.
IEC 62086-1 (2001)	4.2	~~70 %~~ 80 % of the surface. For surface heating devices (panels) an integral metallic screen grid or equivalent electrically conductive covering, covering at least 80% of the exposed surface opposite the surface to be heated shall be incorporated into the construction. ~~…an impact load of 7 J or 4 J when tested in accordance with 5.1.5. For the 4 J impact load, the heater shall be marked with the symbol “X”.~~ …an impact load of 13.6 J when tested in accordance with 5.1.5.
IEC 62086-1 (2001)	4.3	Terminations and connections may be identified as an integral part of a trace heater, or may be identified separately, in which case they are regarded as Ex components in accordance with clause 13 of IEC 60079-0. In addition, these Ex components shall be evaluated in accordance with national and international standards relevant to their construction and use.

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IEC 62086-1 (2001)	4.5.3	~~Controlled design applications, which require the use of a temperature control device to limit the maximum pipe temperature, shall comply with a) for zone 1 and either a) or b) for zone 2:~~ ~~a) For zone 1 or zone 2 applications: controlled design applications, which require the use of a temperature control device to limit the maximum surface temperature shall utilise a protective device that will de-energise the system after exceeding the maximum operating temperature and a reset shall only be possible by hand after the previously defined process conditions have returned. In case of an error by, or damage to the sensor, the heating system shall be de-energised before replacing the defective equipment. The adjustment of the protective device shall be secured and sealed to avoid manipulation. The protective device shall operate independently from the temperature monitoring system;~~ ~~b) For zone 2 applications: a single temperature controller with failure annunciation may be used. If so provision of adequate monitoring of such annunciation, such as 24 h surveillance, shall be made.~~

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	4.5.3 Continued	Controlled design – This approach requires the use of a temperature control device to limit the maximum pipe temperature for Zone 2 applications. When using a temperature controller without failure annunication, a separate high-temperature limit controller to de-energize the heating device shall be included in the design with either a manual reset or annunciatioon. Alternately, a single temperature controller with failure annunciation can be used. Controlled design may not be used for Zone 1 applications. Trace heating devices may not be used in Zone 0 applications.

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IEC 62086-1 (2001)	5.1.4	…the distance from the cotton to the point of the flame application is 250 mm. A laboratory burner described in ASTM D 5025-1999 shall be used for the test. The gas flame produced by the burner is to be calibrated as described in ASTM D 5207-2003. The fuel shall be methane, propane, or natural gas, and shall be of a grade suitable for calibration to the ASTM D 5207-2003 procedure. The height of the ~~natural gas~~ flame of the burner shall be adjusted…
IEC 62086-1 (2001)	5.1.5	~~A sample approximately 200 mm in length is placed on a rigid flat steel plate and positioned underneath an intermediate piece of hardened steel in the shape of a horizontal cylinder with a diameter of 25 mm. This cylinder is required to have a length of 25 mm with smoothly rounded edges to a radius of approximately 5 mm when used to test heating pads and heating panels (see figure 2). For the test, the cylinder is laid horizontally on the sample and, in the case of a trace heating cable, its axis is placed across the sample. A trace heating cable having a non-circular cross-section shall be so positioned that the impact is applied along the minor axis, (that is to say the trace heating cable is positioned flat on the steel plate).~~ ~~Other than in tests on electrical trace heating intended for use in applications with low risk of mechanical damage, a hammer with a mass of 1 kg shall be allowed to fall once onto the horizontal cylinder from a height of 700 mm (that is to say with an impact load of 7 J).~~

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IEC 62086-1 2001 A3

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	5.1.5 Continued	~~For trace heating intended for use in applications with low risk of mechanical damage in accordance with 4.2, the height may be reduced to350 mm (that is to say an impact load of 4 J). Electrical resistance trace heating submitted to such a test shall be clearly marked with the symbol "X" in addition to the marking requirements in clause 6, to caution the user as to its reduced mechanical capability.~~ ~~Conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3 while the steel cylinder and hammer are still in place on the sample.~~ [Note: Figure 2 is also deleted] The impact test shall be conducted on a sample that is positioned on a hardened rigid steel plate and the assembly conditioned for a minimum of 4 h at the manufacturer's minimum recommended installation temperature. After conditioning and while still at the minimum recommended installation temperature, a sample shall be subjected to the impact forces of a 50.8 mm diameter cylindrical steel plunger with smoothly rounded edges, having a mass of 1.8 kg and allowed to free fall from a height of 762 mm, resulting in an impact force of 13.6 J. Then the impacted portion of the sample shall be immersed in tap water at 10C to 25C for 5 minutes, and the dielectric test outlined in 5.1.2 shall be conducted. For surface heating devices, both the heating region and cold leads shall be impacted. After testing, the sheath or dielectric insulation shall have no visible cracks when examined with normal vision.

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IEC 62086-1 2001 A4

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	5.1.6	A sample ~~is~~ of the heating device as well as a sample of each type of integral components and non-heating lead (if applicable), shall be placed on a rigid flat steel plate. A crushing force of 1500 N is then applied for 30 s, without shock, by means of a 6 mm diameter steel rod with hemispherical ends and a total length of 25 mm. For the test, the rod is laid flat on the sample and in the case of a trace heating cable it is placed across a specimen at right angles. For cable that is oval or rectangular in shape, the widest surface shall be the surface on which the load is applied. In the case of a pad, it is necessary to ensure that the cylinder rests across the active element. ~~For electrical trace heating intended for use in applications with low risk of mechanical damage, the crushing force may be reduced to 800 N. Electrical resistance trace heating submitted to such a test shall be clearly marked with the symbol "X" in addition to the marking requirements of clause 6, to caution the user of its reduced mechanical capability~~ ~~Conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3 while the horizontal steel rod is still in place on the sample and the load applied.~~ After a minimum time of not less than 30 s and with the proof loading continuously applied, the dielectric test outlined in 5.1.2 shall be conducted. NOTE Trace heating cable samples need only be approximately 200 mm in length.
IEC 62086-1 (2001)	5.1.7	The test apparatus to be used for the cold bend test is shown in figure 3. The test apparatus, with a sample in position, is maintained for a period of 4 h at ~~between – 25 ºC and – 30 °C or at~~ the manufacturer's minimum declared installation temperature. Immediately afterwards, the sample is bent through 90º around one of the mandrels, then bent through 180º in the opposite direction over the second mandrel and then straightened to its original position. ~~This bending cycle is carried out twice.~~ This cycle of operations shall be performed three times. For heating

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		panels the heating region shall be bent around a mandrel equivalent to the manufacturer's minimum bending radius. ~~Conformity is verified by testing the electrical insulation in accordance with 5.1.2 and 5.1.3.~~ When this has been completed, the sample shall be immersed in tap water at 10C to 25C for 5 min, and then the dielectric test outlined in 5.1.2 shall be conducted.
IEC 62086-1 (2001)	5.1.8	~~A sample of trace heating cable at least 3 m in length including terminations shall be placed in a water flow and drain apparatus as shown in figure 4. The rate of water flow shall be regulated to cover the heating cable and terminations completely for a period of at least 30 s every 5 min, after which it is drained off.~~

IEC 62086-1 2001 A5

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	5.1.8 Continued	A sample of the heating device with all integral components shall be placed in a tap water flow and drain apparatus as shown in figure 4. For surface heating devices, a unit with cold leads shall be used. Water flow shall be initiated, and the heating device and integral components shall be completely immersed. At that point, the water flow is stopped, and the heating device is energized. The apparatus is then drained. The total time from the initiation of water flow to the completion of draining shall be no greater than 4.5 min and no less than 2.5 min.
IEC 62086-1 (2001)	5.1.9	The rated output of the heating cable or heating panel or pad shall be verified by one of the following ~~two~~ three methods, as selected by the manufacturer: a) conductance: the measured ac conductance or conductance per unit length, at a specified temperature, shall be within the manufacturer's declared tolerance; ~~a)~~ b) resistance: the measured d.c. resistance per unit length at a specified temperature shall be within the manufacturer's declared tolerance; ~~b)~~ c) thermal: …

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IEC 62086-1 (2001)	5.1.10	~~The thermal stability of the electrical insulating materials of trace heaters shall be verified on a sample or prototype after it has been stored at a temperature of the manufacturer's declared operating temperature + 20 K, but not less than 80 °C, for at least four weeks. (Compliance of the sample or prototype shall be verified by submitting it to the electrical insulation integrity test of 5.1.2.)~~ A sample of heating device shall be placed in a forced-circulation air oven. The oven shall be heated to, and maintained at, a temperature of 25 °C ± 5 °C above the highest exposure temperature declared by the manufacturer for a period of 14 days. The sample shall be removed from the air oven and cooled to room temperature. Heating cable samples shall be wound six close turns around a mandrel having a radius equal to twelve times the radius of the primary bending plane or thickness of the heating cable. Surface heating devices, if flexible, shall be wrapped on a mandrel with a radius equivalent to the manufacturer's minimum recommended bending radius. While still on the mandrel the sample, except at terminations or ends where the conductor is exposed, shall be submerged in tap water at 10 °C to 25 °C for 5 min. While still in the tap water, the dielectric test outlined in 5.1.2 shall be performed. Rigid heating devices shall also be submerged in tap water and tested. Upon completion of the test, the sample shall have no visible cracks when examined with normal vision.

IEC 62086-1 2001 A6

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	5.1.12	This time-current characteristic shall not be more than the value declared by the manufacturer. The samples shall be in the upper 1/3 of the manufacturer's declared power output tolerances.
IEC 62086-1 (2001)	None (1)	**[Procedure from IEEE-515, Section 4.1.3 – no IEC 62086-1 equivalent]** **Water-resistance test** A sample shall be immersed (except at terminations, or ends where the conductors

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		are exposed) in tap water at 10 °C–25 °C for a period of 14 days. Within 1 h after conditioning as above, the sample shall be subjected to the dielectric test outlined in 5.1.2.
IEC 62086-1 (2001)	None (2)	**[Procedure from IEEE-515, Section 4.1.6 – no IEC 62086-1 equivalent]** **Thermal performance benchmark** One of the following tests shall be used to verify power output stability for all parallel heating devices. **Primary Test** Three randomly selected samples representing the maximum output of all cables or surface heating devices under evaluation shall be tested. If the type of cable or surface heating device has different levels of rated voltages and wattages, then three samples each shall be selected that represent 1) the lowest rated voltage level and the maximum rated output, and 2) the maximum rated voltage and the minimum rated output. Samples shall be terminated according to the manufacturer's specifications, such that a heating length of at least 600 mm or representative surface heating device dimensions is provided. The aging temperature of the test shall be the maximum declared maintain temperature of the heating device. The samples shall be conditioned, while energized, at the aging temperature for 120 h ± 24 h. The initial output of the samples is then to be determined by one of the three methods given in 5.1.9, with the exceptions of sample length and number of test temperature points for procedure 5.1.9 b). For this case, the samples shall be evaluated at rated voltage and at the manufacturer's stated reference temperature for the rated output. The output shall be within the manufacturer's stated output range.

IEC 62086-1 2001 A7

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US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	None (2) Continued	The samples shall be attached to a fixture or suitable heat sink as described in 5.1.9 b), and insulated accordingly. The pipe or heat sink temperature shall be set to the specified aging temperature and maintained with ± 3 °C plus 1% of the temperature reading in degrees Celsius. Circulating fluid or external heating may be used to raise the fixture to the aging temperature. The samples shall be operated at rated voltage. The power supply shall be attached to a 15 min cycle timer such that the samples are energized for 12 min and de- energized for 3 min. The samples shall be exposed to this conditioning for 32 weeks (5376 h). For heating devices with an intermittent temperature exposure rating, the samples are exposed to the same conditioning for 32 weeks, except for an 8 h excursion once each week. At the beginning of the 8 h, the samples shall be disengaged from the cycle timer. The pipe or heat sink temperature shall be increased to a temperature equal to the manufacturer's stated intermittent exposure temperature. The time allowed to increase the temperature should be no greater than 1 h. After 7 h from the beginning of the excursion, the pipe or heat sink temperature shall be decreased back to the aging temperature, again allowing no more than 1 h for the operation. Where the intermittent temperature rating is based on the heating device being energized, then the heating device shall be continuously energized during this temperature excursion, except during cool down back to the maximum maintain temperature. Where the intermittent temperature rating is based on the heating device being de-energized, the exposure cycle is conducted in a de-energized condition. At the end of the 8 h excursion, the samples shall be re-engaged to the cycle timer. The excursions should occur on the same day each week. At the end of the 32 weeks of operation at the continuous rated temperature, the output of the samples shall be determined by the same procedure as utilized for the initial readings. The heating device samples shall maintain a power level within plus 20% or minus 25% of the initial measured output.

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US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	None (2) Continued	**Alternative Test** The test apparatus shall consist of a metal platen(s) with the ability to change temperature within specified levels. The platen(s) shall be sized to expose all parts of the heating device samples, which would be exposed under normal installation conditions, to the temperature levels required by this procedure. The test apparatus shall insure that the heating device samples are in intimate contact with the platen. The test apparatus may be supplied with a sample-mounting fixture. Offsets may be built into the fixture or platen(s) to accommodate end termination/power transition fittings/boots, if provided, where their size profile exceeds the heating device profile. The apparatus shall allow energizing of the heating device samples as required during the test procedure. The samples shall be thermally insulated on the side not facing the platen so as to assure effective heat transfer from the platen to the heating device samples. The temperature of the platen(s) shall be uniformly controlled to a maximum tolerance of ± 5 °C for platen temperatures less than100 °C or 5% of the maximum continuous operating temperature if above 100 °C. The platen described above may be a flat metal plate, a metal pipe, or a metal surface typical of the majority of applications for the heating device being tested. The heating device samples shall be randomly selected and shall be a minimum of 0.3 m in length. Where the heating device is irregular in shape, such as a surface heating device, the heating device sample shall consist of at least one heating unit. If the heating devices are part of a heating device product range, with common materials (with materials having the same performance ratings) and construction, which have different levels of rated voltages and power outputs, then three samples each shall be selected that represent:

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		(1) the lowest rated voltage level and the maximum rated power output; (2) the highest rated voltage and the minimum rated power output. Heating device samples may be conditioned, at the maximum rated voltage for up to 150 hours at the manufacturer's declared maximum continuous operating temperature prior to starting the test.

IEC 62086-1 2001 A9

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	None (2) Continued	The heating device samples shall be installed on the sample-mounting fixture or directly applied to the platen. The samples shall be powered at the maximum rated voltage. The temperature of the platen shall be 23±5°C. The initial power output of the samples shall be determined by measuring voltage and current after the device has reached equilibrium. The heating device samples of continuous parallel construction, while still installed on the sample mounting fixture or platen and energized at the maximum rated voltage, shall be temperature cycled by alternately exposing the samples to platen(s) temperatures corresponding to 23 ±5°C and the maximum continuous operating temperature. The samples may be de-energized during the cool down period. The heating device samples of zone type parallel construction shall be temperature cycled in the same manner with the exception that the samples shall be de-energized when not being held at the maximum continuous operating temperature. If the cycle temperature range exceeds 350 C, the lower temperature may be set at 350 C below the maximum continuous operating temperature. The energized samples shall be exposed to each of these temperature extremes for a minimum of 15 minutes and a transition time between extremes shall not exceed 15 minutes with a cycle being one complete exposure at both temperature extremes.

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		A minimum of 1500 cycles shall be performed. Following the temperature cycling, the temperature of the platen(s) shall be raised to the maximum exposure temperature (the higher of the continuous or intermittent value) declared by the manufacturer and held for a period of no less than 250 hours. Where the maximum exposure temperature is declared as "power on", the samples shall be energized at the maximum rated voltage. After completion of the maximum exposure testing, the heating device samples power output shall be measured, using the same method and platen temperature as used during the initial measurements. The heating device samples shall maintain a power level within plus 20% or minus 25% of the initial measured output.

IEC 62086-1 2001 A10

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	None (3)	**[Procedure from IEEE-515, Sections 4.3.1.3 and 4.3.5.3 – no IEC 62086-1 equivalent]** **Chemical exposure tests** The heating device and any integral components shall be exposed (completely immersed except for the connections) to the following chemicals: acetone, ethyl acetate, isooctane, hexane, methanol, methyl ethyl ketone, methylene chloride, and toluene. The exposure duration shall be 24 h. The chemical shall be maintained at a temperature not less than 5 °C below the boiling point. Boiling points shall be obtained from NFPA 325-1994. A separate heater sample including any polymeric sheath (overjacket) shall be stressed by wrapping it six close turns around a mandrel having a radius equal to 12 times the diameter of the primary bending plane of the device, and shall be exposed (completely immersed except for the ends) to each of the above chemicals. The test sample shall be removed from the fluid, allowed to stabilize for at least 1 h, and then

		subjected to the dielectric test outlined in 5.1.2 with the sample (except for the exposed ends) immersed in tap water at 10C to 25C for 5 minutes. The test voltage shall be applied between the water and the metallic covering for heating device constructions with a polymeric jacket, or between the water and the heating device conductors for heater constructions without a polymeric jacket. No dielectric breakdown shall occur. Additionally, stressed samples shall have no visible cracks or other damage when examined. This test confirms the integrity of the heater jacket or sheath after chemical exposure. An additional set of unstressed samples shall be exposed to the same chemicals in a like manner. After the exposure, the test samples shall be removed from the fluid, allowed to stabilize for at least 1 h, and then subjected to the following tests: *Deformation*—The deformation test described in 5.1.6 shall be conducted on the heating device. Proof loads shall be 204 kg (2000 N). *Impact*—The impact test described in 5.1.5 shall be conducted, retaining the same impact areas. The impact force shall be 27.1 J. *Cold bend*—The cold bend test described in 5.1.7 shall be conducted. *Ground path integrity*—Ground path conductance shall be measured based on 5.1.13. After the chemical exposure, the continuous metallic sheath, metallic braid, or other equivalent electrically conductive material shall have a conductivity equal to or greater than the manufacturer's declared value.

IEC 62086-1 2001 A11

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004

IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	5.2.1	~~Each supplied length or item, whether in bulk or individually fabricated items, shall be subjected to the dielectric test specified in 5.1.2.~~ ~~The polymeric sheath (overjacket) used for corrosion resistance over the metallic braiding or continuous metallic covering shall be subjected to the dielectric test of 1000 Va.c. whilst immersed in water. As an alternative to immersing in water, the~~

		~~electrical insulation of trace heaters may be subjected to a dry spark test, with a minimum test voltage of 3 000 V a.c. r.m.s. that has a substantially sinusoidal waveform at 2 500 Hz to 3 500 Hz. For a 3 000 Hz supply, the rate of travel of the trace heater through the dry test device, in metres per second, shall not be more than 3,3 times the length of the electrode measured in centimetres.~~ The primary electrical insulation jacket of the heating device shall withstand a dry-spark test at a minimum of 6000 Vac. The dry-spark test shall have a substantially sinusoidal waveform at 2500 Hz to 3500 Hz. For a 3000 Hz supply, the speed of the wire in meters per second shall not be more than 33 times the length of the electrode in centimeters; this requirement is proportional to frequency. As an alternative to the dry-spark test, the dielectric tests in 5.1.2 may be conducted. After the installation of a metallic covering, metallic braid, or other equivalent electrically conductive material, ground plane, or continuous metal sheath the heating device shall have the dielectric test described in 5.1.2 conducted. Nonmetallic overjackets shall withstand an additional dry-spark test with a minimum test voltage of 3000 Vac. As an alternative to the dry-spark test, the dielectric tests in 5.1.2 may be conducted.
IEC 62086-1 (2001)	6.1	…in accordance with IEC 60079-0 and with the requirements of this document. Tubing bundles shall be marked on the bundle jacket.
IEC 62086-1 (2001)	6.2	For trace heaters with factory fabricated terminations or surfaces where legible printing cannot be applied, the marking shall be on a durable tag/label permanently affixed to the non-heating sheath within 75 mm of the power connection fitting or gland, and shall additionally include the following: ~~a) the operating voltage;~~ ~~b) the rated steady-state current;~~ ~~c) the statement "Bond the metal sheath/braid of this trace heater to a suitable earth terminal".~~

Section 3
National Differences

IEC 62086-1 2001 A12

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	6.2 Continued	The name of the manufacturer, trademark, or other recognised symbol of identification; The catalog number, reference number, or model; The month and year of manufacture, date coding, applicable serial number, or equivalent; The rated voltage for parallel heating devices or maximum operating voltage for series heating devices; The rated power output in watts per unit length at the rated voltage (and at a stated temperature for devices that change output with temperature) or ohms per unit length for series cable, or the operating current or total wattage, as applicable; Agency listing or approval; Markings for hazardous (classified) locations including temperature class or maximum surface temperature (when applicable). In addition, the marking requirements given in ANSI/NFPA 70 Art. 505 apply.
IEC 62086-1 (2001)	6.3	Deleted
IEC 62086-1 (2001)	6.4	Deleted
IEC 62086-1 (2001)	6.5	~~**Heater field termination/splice kits**~~ ~~Kit packaging shall be marked with the information required by IEC 60079-0 and supplemented by the following:~~ ~~a) the intended use(s) of the kit, for instance power termination;~~ ~~b) the statement “Bond the metal sheath/braid of this trace heater to a suitable earth terminal”;~~

Section 3
National Differences

		~~c) the statement "Refer to installation instructions".~~

IEC 62086-1 2001 A13

US National differences to IEC 62086-1 (2001) from ANSI/IEEE 515-2004		
IEC Standard	**Clause**	**Difference**
IEC 62086-1 (2001)	6.5 Continued	**<u>Markings for field assembled components</u>** <u>Field assembled components shall be marked with the following information. In the case of components with small surface areas or surfaces where legible printing cannot be applied, the markings may be placed on the smallest unit container in lieu of the component itself.</u> <u>The name of the manufacturer, trademark, or other recognised symbol of identification.</u> <u>The catalog number, reference number, or model.</u> <u>The month and year of manufacture, date coding, applicable serial number, or equivalent.</u> <u>Agency listing or approval</u> <u>Applicable environmental or area use requirements, such as NEMA 4, Type 4, IP ratings, and hazardous (classified) locations markings including temperature class or maximum surface temperature (when applicable). In addition, the marking requirements given in ANSI/NFPA 70 Art. 505 apply.</u>
IEC 62086-1 (2001)	6.6	~~**Installation marking**~~ ~~The presence of electrically heated pipelines or vessels, or both, shall be evident by the posting of appropriate caution signs or markings at frequent intervals along the pipeline or vessel.~~ **<u>Installation Instructions</u>** <u>The manufacturer shall provide product specific installation instructions for heating</u>

		devices and components. Instructions for components and heaters may be combined where termination/installation instructions are identical. When different instructions are required, they shall be provided on separate documents and be clearly identified as to the products and locations that apply. These shall include the following information: The intended use(s). The statement "Suitable for use with" (or equivalent) and a listing of applicable heating devices, or a listing of applicable connection fittings. The statement "Bond the metallic braid, sheath, or equivalent electrically conductive material of the heating device to a suitable earth terminal."

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin – Section 4: IECEx Certification Bodies (CBs) and Testing Laboratories (TLs)

IECEx Bulletin:2011(E)

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IECEx Bulletin

Edition 4.0 2011-02

IECEx PUBLICATION

IEC System for Certification to Standards relating to Equipment for use in Explosive Atmospheres (IECEx System)

IECEx Bulletin –
Section 4: IECEx Certification Bodies (CBs) and Testing Laboratories (TLs)

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

Section 4:
Ex Certification Bodies & Ex Testing Laboratories

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Section 4:
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United States of America (US)	***FM Approvals LLC*** *Participating in:* **IECEx 02** Scheme- Equipment	**Mr Robert Martell** 1151 Boston-Providence Turnpike P O Box 9102 Norwood MA 02062 UNITED STATES OF AMERICA Ph: +1 781 255 4850 Fax: +1 781 762 9375 E-mail: Robert.Martell@fmapprovals.com And Hazards Lab and Research Centre 743 Reynolds Road West Gloucester, RI 02814 USA
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ExCBs and ExTLs for each IECEx Scheme

On the IECEx website an up to date list of organisations that have successfully completed the IECEx assessment process and are approved to operate within the following IECEx Schemes;

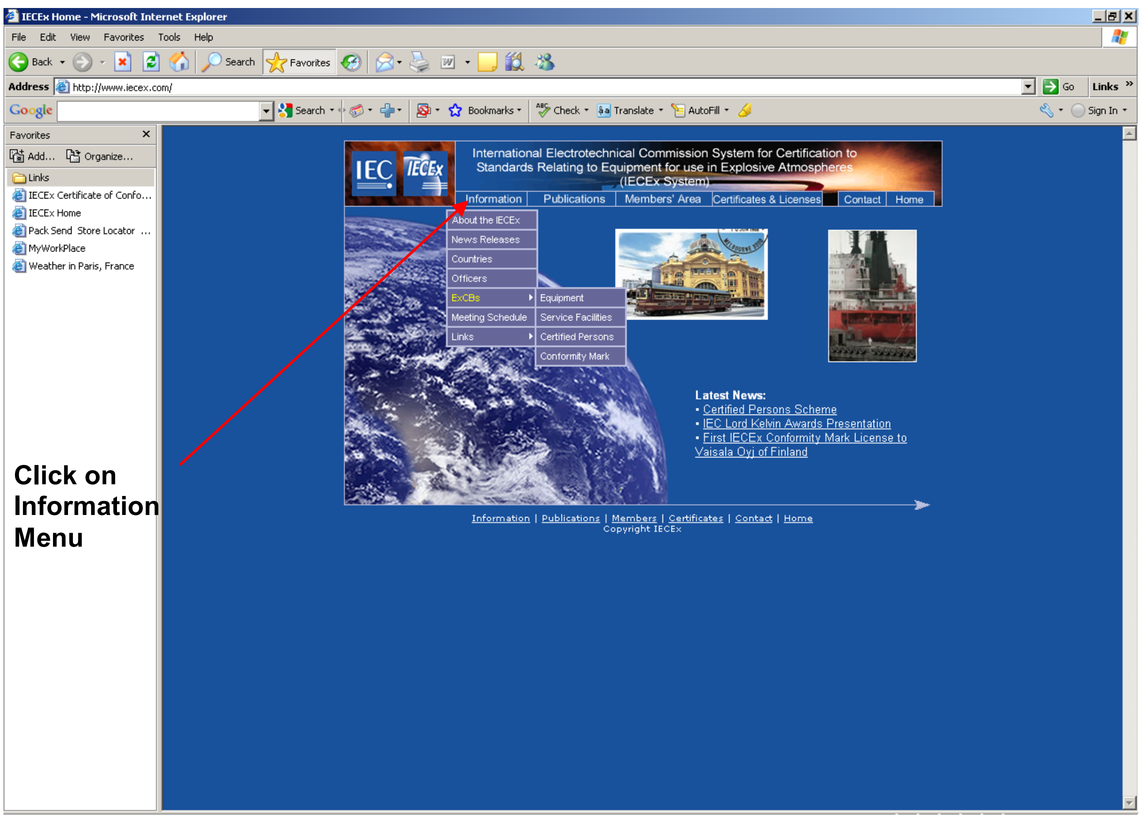
- IECEx Certified Equipment Scheme
- IECEx Certified Service Facility Scheme
- IECEx Certified Persons Scheme
- IECEx Conformity Mark Licensing Scheme

are available at www.iecex.com

To find up to date contact details for the ExCBs (Ex Certification Bodies) and/or ExTLs (Ex Testing Laboratories) use the drop down menu on the left hand side under the Information menu (Image A). Once you have entered the menu click on the specific ExCB/ExTL and that will forward to the contact details for that organisation (Image B).

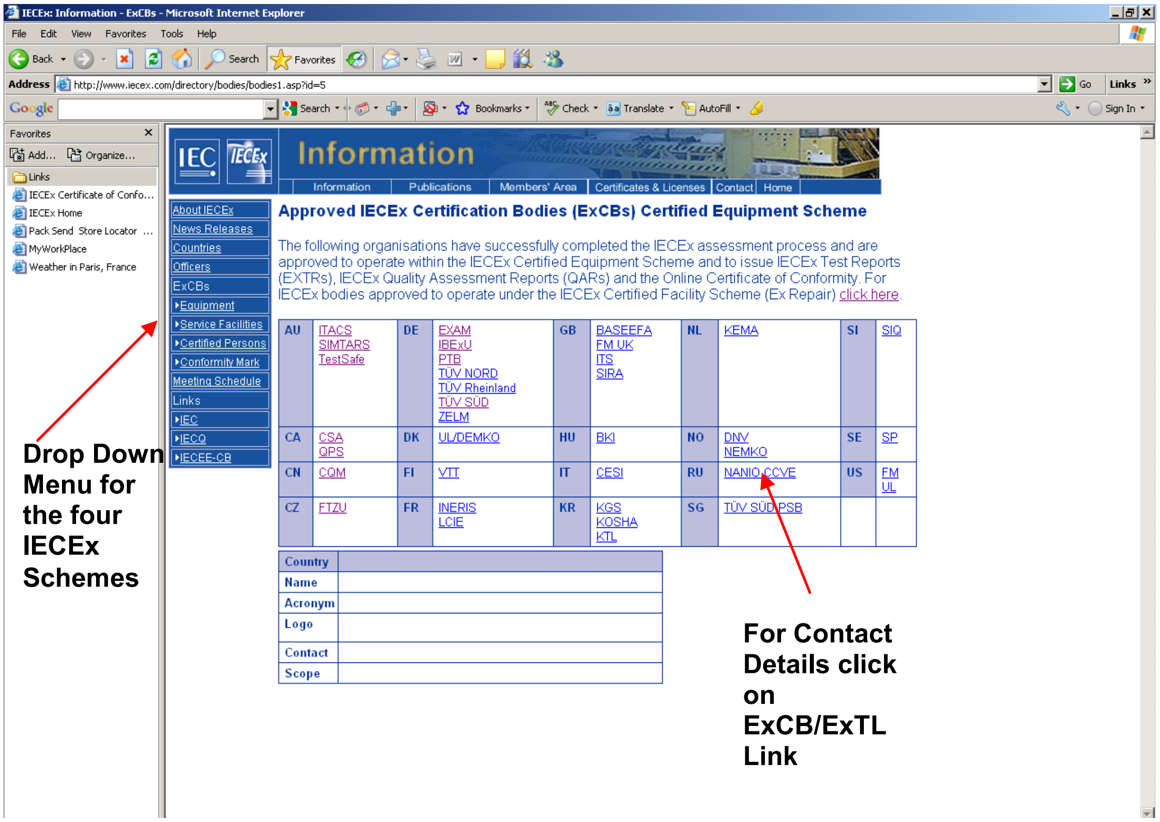
For more information regarding the four IECEx Schemes please visit www.iecex.com/about.htm or refer to Section 1 of this Bulletin (page XX)

Image A



Website Instruction Continued…

Image B



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